

EMCH 574
VIBRATION AND WAVES
course notes

1-dof VIBRATION

SPRING-MASS OSCILLATOR
SPRING-MASS VIBRATOR

VIBRATION
of a
PARTICLE

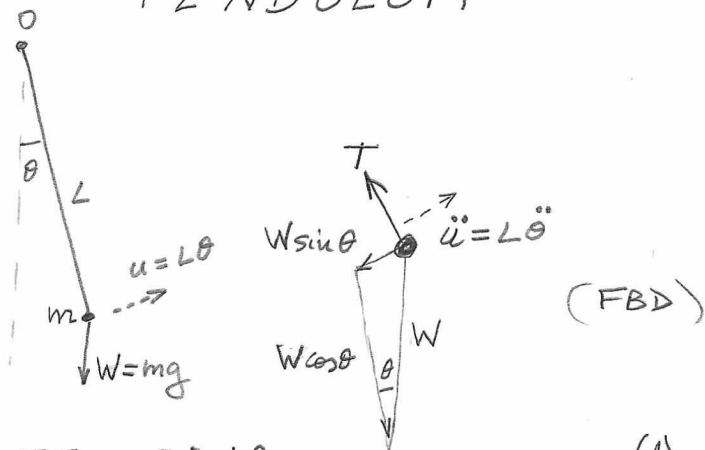
SIMPLE HARMONIC
MOTION

1-dof system

PENDULUM

P1
2024 0824

PENDULUM



Sum of forces: $\Sigma F_u = -W \sin \theta$ (1)

NLM: $\Sigma F_u = m \ddot{u}$ (1a)

$-W \sin \theta = m L \ddot{\theta}$ (2)

Small angle approximation $\sin \theta \approx \theta$ (rad) (3)

$-W \theta = m L \ddot{\theta}$

$m L \ddot{\theta} + W \theta = 0$ EOM (4)

(1) $\rightarrow$ (4):

$m L \ddot{\theta} + m g \theta = 0$

$\ddot{\theta} + \frac{g}{L} \theta = 0$ governing equation (5)

$\theta = e^{st}$

$\ddot{\theta} = s^2 e^{st} = s^2 \theta$ (6)

(6) $\rightarrow$ (5): $(s^2 + \frac{g}{L}) \theta = 0$ (7)

$s^2 + \frac{g}{L} = 0$

characteristic eqⁿ. (8)

$s = \pm i \sqrt{\frac{g}{L}}$

roots

P2
2023/03/1

Define $\omega_n = \sqrt{\frac{g}{L}}$ natural freq. in rad/sec (10).

Solutions $s_1 = i\omega_n$
 $s_2 = -i\omega_n$ (11)

$$\theta(t) = C_1^* e^{i\omega_n t} + C_2^* e^{-i\omega_n t} \quad (12)$$

or $\theta(t) = A^* \cos \omega_n t + B^* \sin \omega_n t$ (13).

Frequency, Hz : $f_n = \frac{\omega_n}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{g}{L}}$ (14)

Period of oscillation $T = \frac{1}{f_n} = 2\pi \sqrt{\frac{L}{g}}$ (15)

Particle motion

$$u = L\theta$$

$$u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t} \quad (16a)$$

or $u(t) = A \cos \omega_n t + B \sin \omega_n t$ (16b)

or $u(t) = C \sin(\omega_n t + \varphi)$ (16c)

4
20230920

From complex exponentials to trigonometric functions

$$C_1 e^{ix} + C_2 e^{-ix} = A \cos x + B \sin x \quad (12a)$$

or

$$C_1 e^{ix} + C_2 e^{-ix} = C \sin(x + \varphi) \quad (12b)$$

PROOF

Recall Euler's identity $e^{ix} = \cos x + i \sin x$

hence $e^{-ix} = \cos x - i \sin x$

$$C_1 e^{ix} = C_1 \cos x + i C_1 \sin x$$

$$C_2 e^{-ix} = C_2 \cos x - i C_2 \sin x$$

$$C_1 e^{ix} + C_2 e^{-ix} = (C_1 + C_2) \cos x + i(C_1 - C_2) \sin x$$

$$\begin{aligned} & \underbrace{C_1 + C_2}_A \cos x + \underbrace{i(C_1 - C_2)}_B \sin x \\ & = A \cos x + B \sin x \end{aligned} \quad \begin{cases} A = C_1 + C_2 \\ B = i(C_1 - C_2) \end{cases}$$

$$A \cos x + B \sin x = C \sin(x + \varphi)$$

$$\begin{aligned} C &= \sqrt{A^2 + B^2} \\ \varphi &= \tan^{-1} \frac{A}{B} \end{aligned} \quad (12c)$$

Proof Recall $\sin(x + \varphi) = \sin \varphi \cos x + \cos \varphi \sin x$

$$C \sin(x + \varphi) = \underbrace{(C \sin \varphi)}_A \cos x + \underbrace{(C \cos \varphi)}_B \sin x$$

$$A^2 + B^2 = C^2 \sin^2 \varphi + C^2 \cos^2 \varphi = C^2 (\sin^2 \varphi + \cos^2 \varphi) = C^2$$

$$C = \sqrt{A^2 + B^2}$$

$$\frac{A}{B} = \frac{C \sin \varphi}{C \cos \varphi} = \tan \varphi \rightarrow \varphi = \tan^{-1} \frac{A}{B}$$

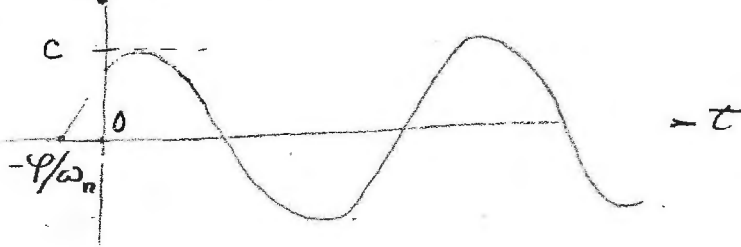
QED.

$$C_1 e^{ix} + C_2 e^{-ix} = C \sin(x + \varphi)$$

(12c) 7/265

5
20230920

Oscillatory motion:



$$u(t) = C \sin(\omega_n t + \varphi) \quad (13c)$$

$$u(t) = A \cos \omega_n t + B \sin \omega_n t \quad (13b)$$

The constants A, B or C, φ must be determined from the initial conditions.

Initial value problem

Assume that the initial displacement, u_0 , and the initial velocity, $\dot{u}_0$, are known. Using Eq. (18), we write

$$(13b) \quad \begin{aligned} u_0 = u(0) &= A \cos \omega_n t + B \sin \omega_n t \Big|_{t=0} = A \\ \dot{u}_0 = \dot{u}(0) &= -\omega_n A \sin \omega_n t + \omega_n B \cos \omega_n t \Big|_{t=0} = \omega_n B \end{aligned} \quad (26)$$

Solving Eq. (26) for A and B yields

$$\begin{aligned} A &= u_0 \\ B &= \frac{\dot{u}_0}{\omega_n} \end{aligned} \quad (27)$$

Substitution of Eq. (27) into Eq. (18) gives the general solution of undamped free vibrations with initial displacement, u_0 , and initial velocity, $\dot{u}_0$, in the form

$$u(t) = u_0 \cos \omega_n t + \frac{\dot{u}_0}{\omega_n} \sin \omega_n t \quad (28)$$

$$2024-10-20 \quad u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t} \quad (1)$$

$$\text{IVP : } u(0) = u_0$$

$$\dot{u}(0) = \dot{u}_0$$

$$\left. \begin{array}{l} u(0) = u_0 \\ \dot{u}(0) = \dot{u}_0 \end{array} \right\} (2)$$

$$\frac{d}{dt}(1) : \quad \dot{u}(t) = i\omega_n C_1 e^{i\omega_n t} - i\omega_n C_2 e^{-i\omega_n t} \quad (3)$$

$$u(0) = C_1 + C_2$$

$$\dot{u}(0) = i\omega_n C_1 - i\omega_n C_2$$

$$(4)$$

$$(4) \rightarrow (2) : \quad \begin{array}{l|l} C_1 + C_2 = u_0 & i\omega_n \\ i\omega_n C_1 - i\omega_n C_2 = \dot{u}_0 & -i\omega_n \end{array} \quad (5)$$

$$\left\{ \begin{array}{l} 2i\omega_n C_1 = i\omega_n u_0 + \dot{u}_0 \\ -2i\omega_n C_2 = -i\omega_n u_0 + \dot{u}_0 \end{array} \right. \quad (6)$$

$$\left\{ \begin{array}{l} C_1 = \frac{u_0}{2} + \frac{\dot{u}_0}{2i\omega_n} \\ C_2 = \frac{u_0}{2} - \frac{\dot{u}_0}{2i\omega_n} \end{array} \right. \quad (7)$$

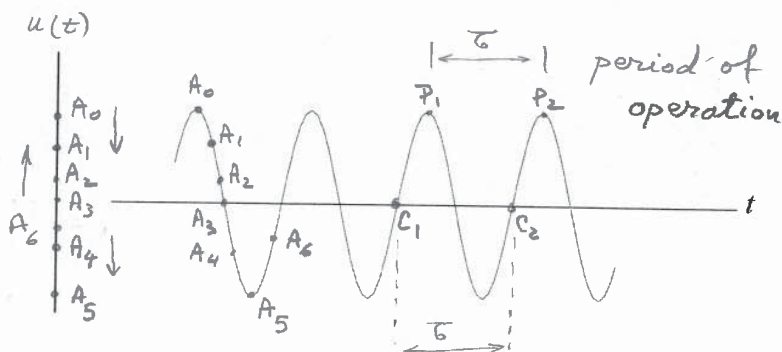
$$(7) \rightarrow (1) : \quad u(t) = \left(\frac{u_0}{2} + \frac{\dot{u}_0}{2i\omega_n} \right) e^{i\omega_n t} + \left(\frac{u_0}{2} - \frac{\dot{u}_0}{2i\omega_n} \right) e^{-i\omega_n t}$$

$$u(t) = u_0 \underbrace{\frac{e^{i\omega_n t} + e^{-i\omega_n t}}{2}}_{\cos} + \frac{\dot{u}_0}{\omega_n} \underbrace{\frac{e^{i\omega_n t} - e^{-i\omega_n t}}{2i}}_{\sin}$$

$$u(t) = u_0 \cos \omega_n t + \frac{\dot{u}_0}{\omega_n} \sin \omega_n t \quad \text{as before}$$

1a

Harmonic oscillation



The particle vibrates up and down going through positions $A_0, A_1, A_2, A_3, A_4, A_5, A_6$

Graphically, this is represented as an oscillatory curve which has peaks and valleys.

Period of oscillation T : the distance in time between two consecutive peaks, P_1 and P_2 , or two consecutive zero crossings with positive slope, C_1 and C_2 .

Frequency f : how often the particle passes through same location. Frequency f is the inverse of period T .

$$f = 1/T \quad \text{Hz}$$

Circular frequency ω

$$\omega = 2\pi f \quad \text{rad/sec}$$

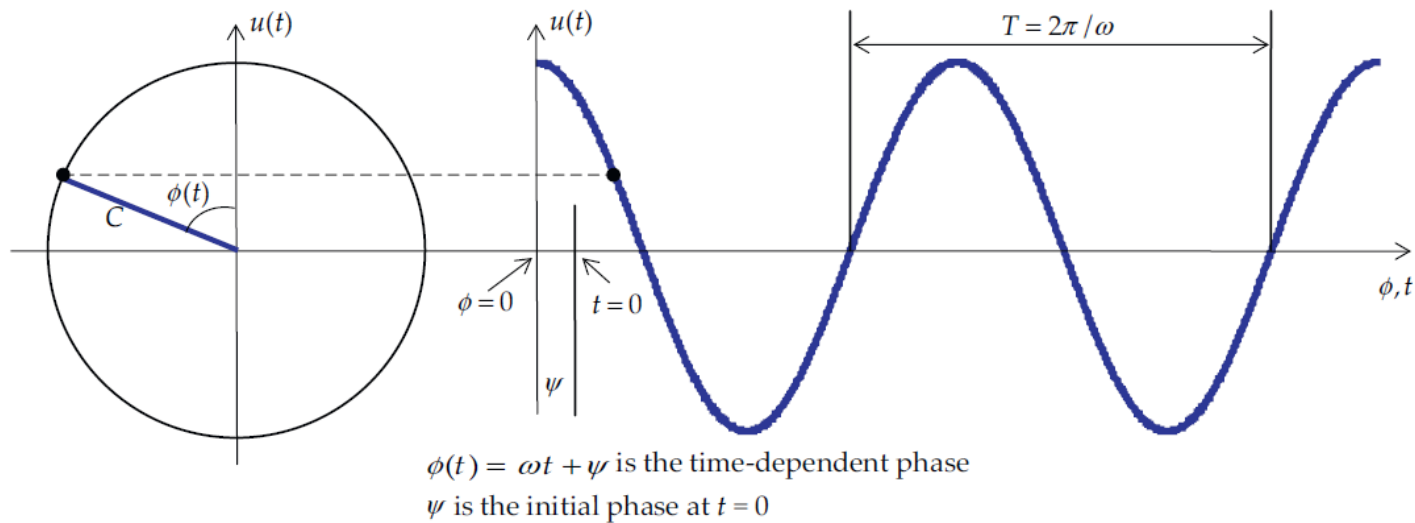


Figure 2 Schematic representation of an oscillatory motion of amplitude C , angular frequency ω , and initial phase ψ

16

$$u(t) = \cos 2\pi \frac{t}{T}$$

or

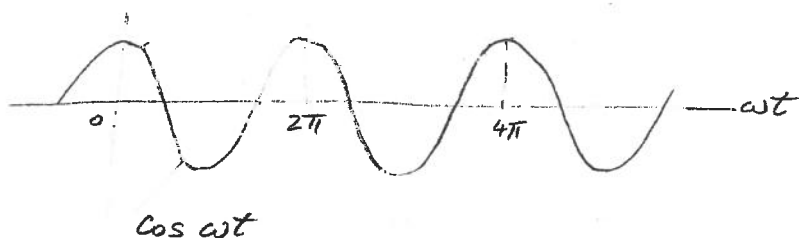
$$u(t) = \cos 2\pi f t$$

or

$$u(t) = \cos \omega t$$

5
FRP

COMPLEX NUMBER REPRESENTATION OF HARMONIC SIGNALS



$$e^{i\omega t} = \cos \omega t + i \sin \omega t \quad (\text{Euler})$$

$$\begin{cases} x(t) = \hat{x} \cos(\omega t + \varphi) \\ x(t) = \operatorname{Re}[\hat{x} e^{i(\omega t + \varphi)}] \end{cases}$$

$$\begin{cases} x(t) = \hat{x} \sin(\omega t + \varphi) \\ x(t) = \operatorname{Im}[\hat{x} e^{i(\omega t + \varphi)}] \end{cases}$$

In general $x(t) = \hat{x} e^{i(\omega t + \varphi)}$

$$\hat{x} e^{i(\omega t + \varphi)} = \hat{x} e^{i\varphi} e^{i\omega t} = X e^{i\omega t}$$

$$X = \hat{x} e^{i\varphi} \quad (\text{phasor})$$

$$|X| = \hat{x}$$

magnitude of $x(t)$

$$|X| = \operatorname{abs}(X)$$

$$\angle X = \varphi$$

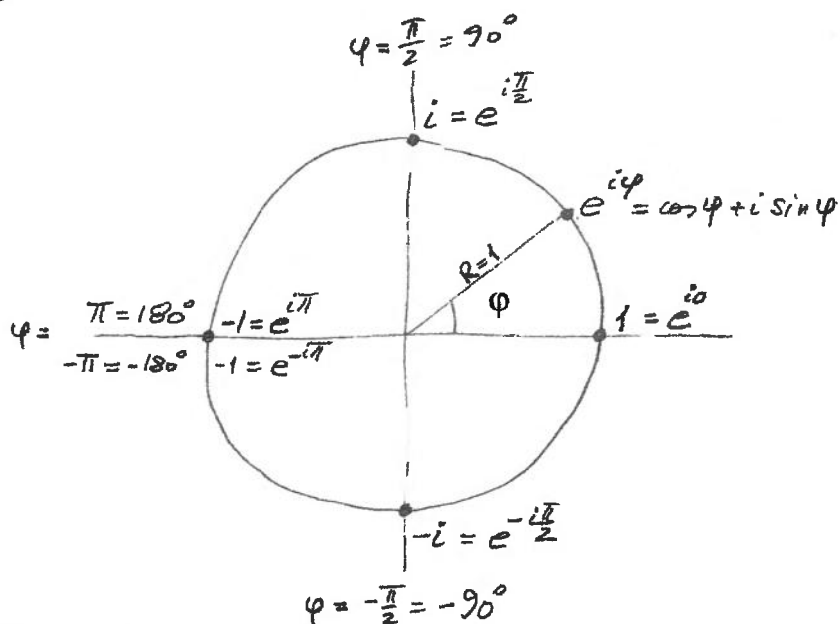
phase of $x(t)$

$$\angle X = \operatorname{angle}(X), \text{ rad}$$

$$\text{rad} \hat{=} \text{deg} (\pi) = 180^\circ$$

333/593

20201029
20220303



NOTES:

(1) $|e^{i\varphi}| = 1$

(2) $\frac{1}{i} = \frac{1}{e^{i\pi/2}} = e^{-i\pi/2} \quad \left(\frac{1}{x} = x^{-1}\right)$

(3) Magnitude and phase of complex number $z = x + iy$

$|z| = \sqrt{x^2 + y^2}$ magnitude $\text{abs}(z)$

$\angle z = \tan^{-1} \frac{y}{x}$ phase $\text{angle}(z)$

(4) Magnitude and phase of fraction $G = \frac{B}{A}$

$|G| = \frac{|B|}{|A|}$ magnitude

$\angle G = \angle B - \angle A$ phase

2022 03 03

Proof

$$B = |B| e^{i\angle B} \quad (1)$$

$$A = |A| e^{i\angle A} \quad (2)$$

$$G = \frac{B}{A} = \frac{|B| e^{i\angle B}}{|A| e^{i\angle A}} = \frac{|B|}{|A|} \cdot \frac{e^{i\angle B}}{e^{i\angle A}} \quad (3)$$

$$\text{But } \frac{1}{e^{i\angle A}} = e^{-i\angle A} \quad \left(\frac{1}{x} = x^{-1} \right) \quad (4)$$

$$(4) \rightarrow (3): \quad G = \frac{|B|}{|A|} e^{i\angle B} e^{-i\angle A} \quad (e^x e^y = e^{x+y})$$

$$G = \frac{|B|}{|A|} e^{i(\angle B - \angle A)} \quad (5)$$

$$\text{Want to express } G \text{ as } G = |G| e^{i\angle G} \quad (6)$$

Comparison of (5) & (6) yields

$$|G| = \frac{|B|}{|A|} \quad \text{magnitude of } G$$

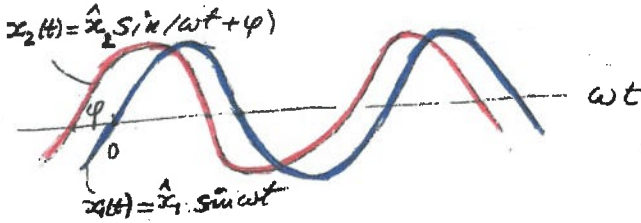
$$\angle G = \angle B - \angle A \quad \text{phase of } G$$

FRF

Phase

$$x_1(t) = \hat{x}_1 \sin \omega t \quad \text{reference signal}$$

$$x_2(t) = \hat{x}_2 \sin(\omega t + \varphi)$$

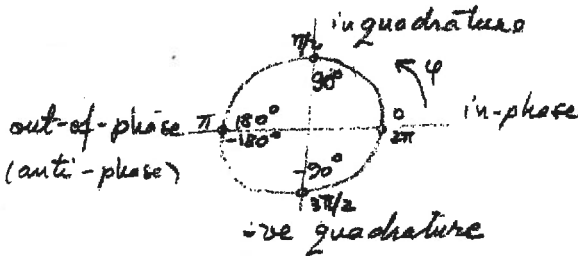


$$\psi_1 = \omega t \quad \text{phase of } x_1 \text{ (reference)}$$

$$\psi_2 = \omega t + \varphi \quad \text{phase of } x_2$$

$$\varphi = \psi_2 - \psi_1$$

- phase difference
- x_2 leads by φ .



Periodic signals

$$\left\{ \begin{array}{l} 2\pi \\ 360^\circ \end{array} \right. \text{ periodicity: } \bullet x(\omega t + 2\pi) = x(\omega t)$$

$$\bullet x(\omega t + \pi) = x(\omega t - \pi)$$

1-dof VIBRATION

1 SPRING-MASS VIBRATING SYSTEMS

Consider a vibrating system consisting of a mass m and a spring of stiffness k as shown in Figure 1.

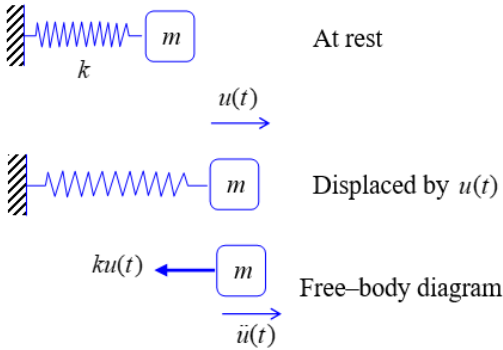


Figure 1 Spring-mass vibration system

1.1 FREE VIBRATION OF SPRING-MASS SYSTEMS

Assume the mass m is displaced from the equilibrium position by a time-dependent displacement $u(t)$.

1.1.1 FREE-BODY DIAGRAM

As shown in Figure 1, free body diagram (FBD) of the mass m gives the sum of x -direction forces as

$$\Sigma F_x = -ku(t) \quad (1)$$

where $ku(t)$ is the reaction force from the spring.

1.1.2 NEWTON'S LAW OF MOTION

Newton's law of motion (NLM) applied to the mass m is stated as

$$\Sigma F_x = m\ddot{u}(t) \quad (2)$$

1.1.3 EQUATION OF MOTION

Substitution of Eq. (1) into Eq. (2) yields as

$$m\ddot{u}(t) = -ku(t) \quad (3)$$

Upon rearrangement, Eq. (3) becomes the equation of motion (EOM), i.e.,

$$m\ddot{u}(t) + ku(t) = 0 \quad (4)$$

Eq. (4) is a linear homogenous ordinary differential equation (ODE).

1.1.4 SOLUTION OF LINEAR ODES

The solution of linear homogenous ODEs is sought in the form e^{st} .

$$u(x) = e^{st} \quad (5)$$

Upon differentiation, Eq. (5) gives

$$\dot{u}(t) = se^{st} = su(t) \quad (6)$$

$$\ddot{u}(t) = s^2 e^{st} = s^2 u(t)$$

Substitution of Eqs. (5), (6) into Eq. (4) yields

$$ms^2 u(t) + ku(t) = 0 \quad (7)$$

After factoring out $u(t)$, Eq. (7) becomes

$$(ms^2 + k) u(t) = 0 \quad (8)$$

A possible solution of Eq. (8) is $u(t) = 0$. However, this solution is trivial and of no interest to us. We are interested in *nontrivial solutions* $u(t) \neq 0$. To obtain a nontrivial solution $u(t) \neq 0$, one needs to find the values of s that make zero the factor multiplying $u(t)$ in Eq. (8), i.e., to have $(ms^2 + k) = 0$.

1.1.5 CHARACTERISTIC EQUATION

It was shown in Section 2.1.4 that a nontrivial solution $u(t) \neq 0$ of Eq. (8) can only be obtained if the factor multiplying $u(t)$ is zero, i.e.,

$$ms^2 + k = 0 \quad (9)$$

Divide Eq. (9) by m to get what is called the *characteristic equation*, i.e.,

$$s^2 + k/m = 0 \quad (\text{characteristic equation}) \quad (10)$$

Solution of the characteristic equation will give us the values of s that permit nontrivial solution $u(t) \neq 0$ of Eq. (8). Upon solution, Eq. (10) gives

$$s^2 = -k/m \quad (11)$$

Denote

$$\omega_n^2 = k/m \quad \omega_n = \sqrt{k/m} \quad (\text{natural frequency}) \quad (12)$$

The natural frequency ω_n is expressed in rad/sec. Substitution of Eq. (12) into Eq. (11) yields

$$s^2 = -\omega_n^2 \quad (13)$$

Eq. (13) has two imaginary roots

$$s_1 = +i\omega_n \quad s_2 = -i\omega_n \quad (14)$$

1.1.6 GENERAL SOLUTION

Both $s_1 = i\omega_n$ and $s_2 = -i\omega_n$ yield solutions of the ODE Eq. (4). These solutions are in the form of complex exponentials $e^{i\omega_n t}$ and $e^{-i\omega_n t}$ respectively. Hence, the general solution of the linear ODE Eq. (4) is a linear superposition of the two individual solutions, i.e.,

$$u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t} \quad (15)$$

where C_1 and C_2 are arbitrary constants that need to be determined from the initial conditions.

Recall Euler's identity

$$e^{i\alpha} = \cos\alpha + i \sin\alpha \quad (16)$$

and hence

$$e^{-i\alpha} = \cos\alpha - i \sin\alpha \quad (17)$$

Eqs. (16), (17) can be used to rewrite Eq. (15) in trigonometric form, i.e.,

$$u(t) = A \cos \omega_n t + B \sin \omega_n t \quad (18)$$

where the constants A , B and C_1 , C_2 are interconnected as

$$A = C_1 + C_2 \quad (19)$$

$$B = i(C_1 - C_2) \quad (20)$$

Further processing allows us to write Eq. (18) as

$$u(t) = C \sin(\omega_n t + \varphi) \quad (21)$$

where

$$C = \sqrt{A^2 + B^2} \quad \varphi = \tan^{-1} A/B \quad (22)$$

1.1.7 INITIAL VALUE PROBLEM (IVP)

The initial value problem (IVP) consists in finding the general-solution constants in terms of the initial conditions of the vibrating system, i.e., initial displacement u_0 and initial velocity, $\dot{u}_0$. The purpose is to find the constants A , B in terms of known u_0 and $\dot{u}_0$. Using Eq. (18), we write

$$\begin{aligned}
 u_0 = u(0) &= A \cos \omega_n t + B \sin \omega_n t \Big|_{t=0} = A \\
 \dot{u}_0 = \dot{u}(0) &= -\omega_n A \sin \omega_n t + \omega_n B \cos \omega_n t \Big|_{t=0} = \omega_n B
 \end{aligned}
 \tag{23}$$

Solving Eq. (23) for A and B yields

$$\begin{aligned}
 A &= u_0 \\
 B &= \frac{\dot{u}_0}{\omega_n}
 \end{aligned}
 \tag{24}$$

Substitution of Eq. (24) into Eq. (18) gives the general solution of undamped free vibrations with initial displacement, u_0 , and initial velocity, $\dot{u}_0$, in the form

$$u(t) = u_0 \cos \omega_n t + \frac{\dot{u}_0}{\omega_n} \sin \omega_n t
 \tag{25}$$

1.1.8 VERTICAL SPRING-MASS ANALYSIS

So far, we have assumed the mass-spring vibrating system is horizontal. However, many spring-mass vibrating systems are vertical. In this section, we show that a vertical spring-mass obeys the same differential equation as a horizontal one which means that the analysis outlined so far applies equally to both situations.

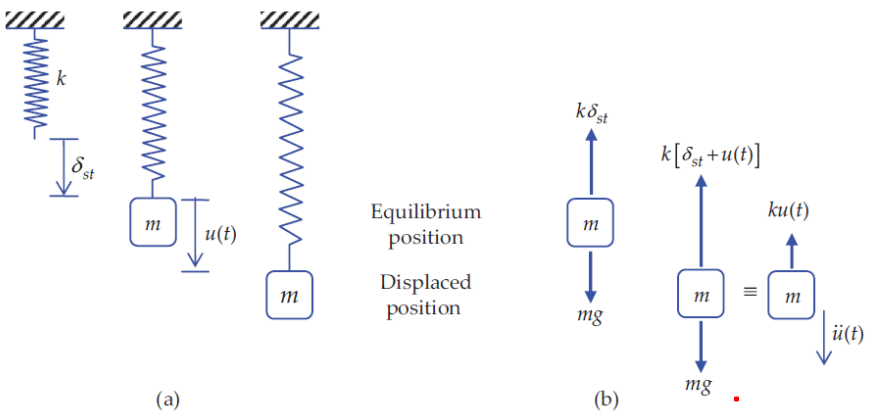


Figure 2 Free vibration of a vertical spring-mass system: (a) physical mechanism; (b) free-body diagram

Consider a vertical spring-mass vibration system consisting of a particle of mass m hanging on an elastic spring of stiffness k as shown in Figure 2.

1.1.8.1 Static analysis

Initially, the mass is at rest and in a state of equilibrium with the static tension in the spring (Figure 2a). In this equilibrium state, the weight $W=mg$ is balanced by the static tension in the spring $k\delta_{st}$ where δ_{st} is the *static displacement* of the spring. The free body diagram of the mass in equilibrium condition is shown in Figure 2b. Force-balance analysis in the vertical direction gives

$$\sum F_y = mg - k\delta_{st} = 0 \quad (\text{static equilibrium}) \quad (26)$$

Solution of Eq. (26) yields the static displacement δ_{st} as

$$\delta_{st} = mg/k \quad (\text{static displacement}) \quad (27)$$

1.1.8.2 Vibration analysis

If the mass is displaced from this equilibrium state and then let free, it will oscillate up and down about the equilibrium position, i.e., it will experience a state of vibration with time-dependent displacement $u(t)$. The free body diagram of the mass in the displaced position gives the sum of forces in the vertical direction as

$$\sum F_y = mg - k[\delta_{st} + u(t)] \quad (\text{vibration motion}) \quad (28)$$

Expansion of Eq. (28) gives

$$\sum F_y = (mg - k\delta_{st}) + ku(t) \quad (\text{vibration motion}) \quad (29)$$

However $(mg - k\delta_{st}) = 0$ as indicated by the equilibrium analysis Eq. (26). Hence, the sum of forces acting on the vibrating mass is simply

$$\sum F_y = ku(t) \quad (\text{sum of forces for vibration analysis}) \quad (30)$$

Newton's law of motion (NLM) for vertical direction is stated as

$$\sum F_y = m\ddot{u} \quad (\text{NLM}) \quad (31)$$

Substitution of Eq. (30) into Eq. (31) gives, upon rearrangement,

$$m\ddot{u}(t) + ku(t) = 0 \quad (32)$$

Eq. (32) is the same as Eq. (4) developed previously for the horizontal spring-mass vibrating system. Its general solution is that given by Eq. (21), i.e.,

$$u(t) = C \sin(\omega_n t + \phi) \quad (21)$$

A rendition of the vertical vibration motion is depicted in Figure 3.

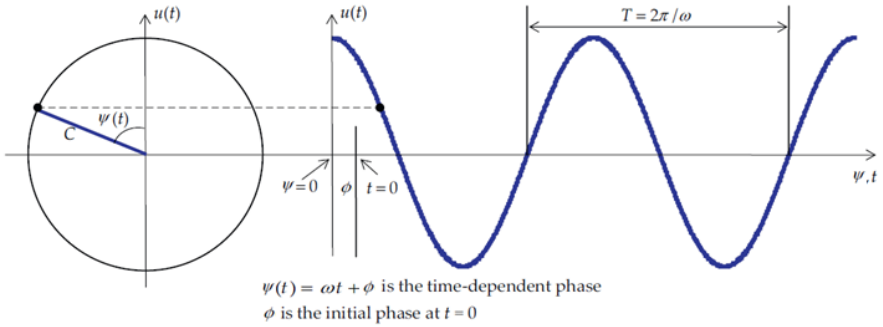


Figure 3 Schematic representation of an oscillatory motion of amplitude C , angular frequency ω , and initial phase ϕ

Note that the above analysis proved that the vibration equation is not modified if static gravity is also included in the analysis. This means that the vibration motion is not affected by the presence of a static force field such as gravity. The analysis can be easily extended to a static force field that is not parallel with the vibration motion, but it has an arbitrary orientation.

1.2 FORCED VIBRATION OF SPRING-MASS SYSTEMS

Consider a spring-mass vibrating system excited by a time-dependent force $F(t)$ as shown in Figure 4.

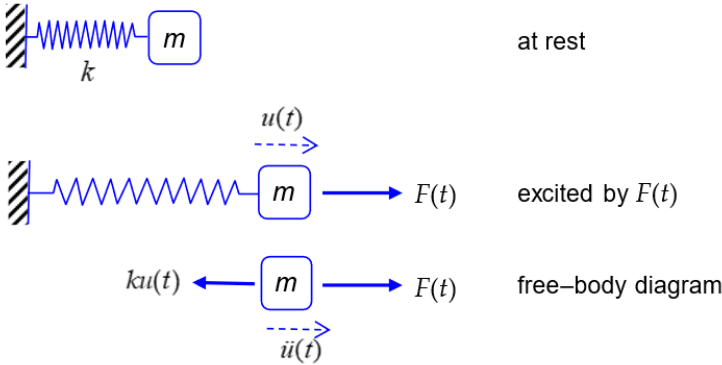


Figure 4 spring-mass system excited by force $F(t)$

1.2.1 EQUATION OF MOTION

Free body diagram (FBD) of the mass m gives the sum of x -direction forces as

$$\Sigma F_x = -ku(t) + F(t) \quad (33)$$

where $ku(t)$ is the reaction force from the spring and $F(t)$ is the excitation force. Newton's law of motion (NLM) applied to the mass m is stated as

$$\Sigma F_x = m\ddot{u}(t) \quad (34)$$

Substitution of Eq. (33) into Eq. (34) yields, upon rearrangement, the equation of motion (EOM) as

$$m\ddot{u}(t) + ku(t) = F(t) \quad (\text{EOM}) \quad (35)$$

Eq. (35) is a nonhomogeneous linear ODE in terms of m , c , k , $F(t)$.

1.2.2 SUDDENLY APPLIED LOAD

When a load is suddenly applied to a vibrating system, the stresses and strains in the system are larger than in the case of a gradually applied load. In other words, *dynamic loading is more severe than static loading of the same amplitude*. This is known as the *dynamic amplification factor*. In the case of a simple vibrating system, the dynamic amplification factor has value of 2, i.e., the loading due to

the oscillations resulting from a sudden application of a weight is twice as severe as the loading that would be obtained if the weight were applied quasi-statically without creating oscillations. (For more complicated vibrating systems, the value of the dynamic amplification factor may be different.) A simple analysis of the dynamic amplification factor for a simple vibrating system is developed next.

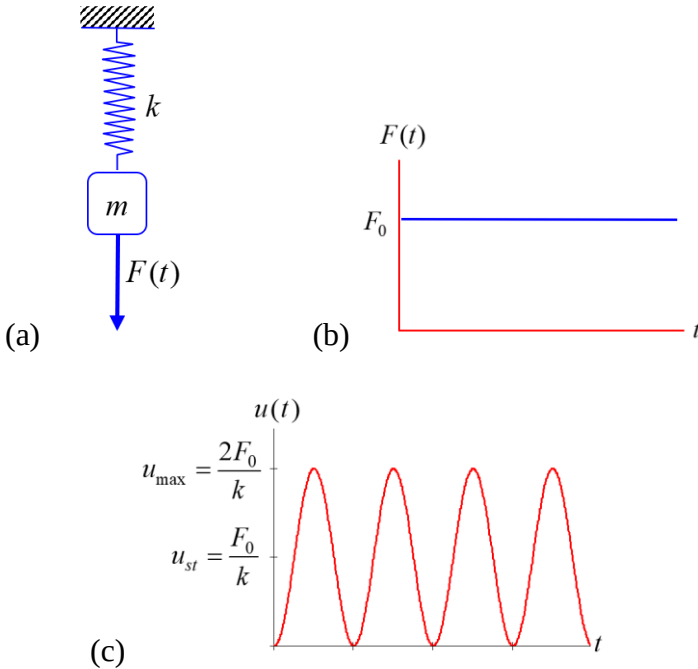


Figure 5 Dynamic amplification factor: (a) simple vibrating system; (b) suddenly applied load; (c) time response of the system

1.2.2.1 Problem setup

Consider the simple undamped vibrating system of Figure 5a. At $t=0$, the system is at rest ($u_0=0$, $\dot{u}_0=0$). A force F_0 is suddenly applied at $t=0$. Free-body diagram for the system at $t>0$ yields the following equation of motion (EOM),

$$m\ddot{u}(t) + ku(t) = F_0 \quad t > 0 \quad (\text{EOM}) \quad (36)$$

Eq. (36) is a nonhomogeneous linear ODE, i.e., with a nonzero right-hand side (RHS). The RHS of Eq. (36) is called the *forcing function*.

In our case, the RHS of Eq. (36) is the constant force F_0 applied suddenly at $t=0$.

The general solution of nonhomogeneous linear ODEs consists of the superposition of the *complementary solution* $u_c(t)$ that satisfies the homogenous ODE and a *particular solution* $u_p(t)$ that satisfies the nonhomogeneous ODE, i.e.,

$$u(t)=u_c(t)+u_p(t) \quad (\text{general solution}) \quad (37)$$

1.2.2.2 Complementary solution

The homogenous variant of Eq. (36) is obtained by making zero the right-hand side (RHS) which gives

$$m\ddot{u}_c(t) + ku_c(t) = 0 \quad (\text{homogeneous ODE}) \quad (38)$$

It is apparent that Eq. (38) is the same as the previously discussed Eq. (4) which has the solution given by Eq. (18) in the form

$$u_c(t) = A\cos\omega_n t + B\sin\omega_n t \quad (\text{complementary solution}) \quad (39)$$

Thus, Eq. (39) is the sought-after complementary solution $u_c(t)$.

1.2.2.3 Particular solution

The particular solution $u_p(t)$ is a solution that satisfies the nonhomogeneous ODE Eq. (36), i.e.,

$$m\ddot{u}_p(t) + ku_p(t) = F_0 \quad (\text{nonhomogeneous ODE}) \quad (40)$$

It is customary to seek the particular solution in the form of the forcing function. In our case, the forcing function is a constant hence we seek the particular solution also as a constant, i.e.,

$$u_p(t) = \hat{u}_p \quad t > 0 \quad (41)$$

Upon differentiation, Eq. (41) gives

$$\ddot{u}_p(t) = 0 \quad (42)$$

Substitution of Eqs. (41), (42) into Eq. (40) gives

$$m \times 0 + k\hat{u}_p = F_0 \quad t > 0 \quad (\text{EOM}) \quad (43)$$

Solution of Eq. (43) gives

$$\hat{u}_p = F_0/k \quad (44)$$

Substitution of Eq. (44) into Eq. (41) gives the particular solution as

$$u_p(t) = F_0/k \quad t > 0 \quad (\text{particular solution}) \quad (45)$$

1.2.2.4 Initial value problem (IVP)

Substitution of Eqs. (39) and (45) into Eq. (37) yields the complete solution

$$u(t) = A \cos \omega_n t + B \sin \omega_n t + F_0/k \quad (\text{complete solution}) \quad (46)$$

The constants A and B are to be determined from the initial conditions of the initial value problem (IVP). In our case, we have “at rest” initial conditions, i.e.,

$$u_0=0, \dot{u}_0=0 \quad (\text{“at rest” initial conditions}) \quad (47)$$

Substitution of Eq. (46) into Eq. (47) yields

$$\begin{aligned} u(0) &= A \cos \omega_n t + B \sin \omega_n t + \frac{F_0}{k} \Big|_{t=0} = A + \frac{F_0}{k} = 0 \\ \dot{u}(0) &= -\omega_n A \sin \omega_n t + \omega_n B \cos \omega_n t \Big|_{t=0} = \omega_n B = 0 \end{aligned} \quad (48)$$

Upon solution of Eq. (48), we get

$$A = -F_0/k \quad B=0 \quad (49)$$

1.2.2.5 Dynamic amplification factor

Substitution of Eq. (49) into Eq. (46) yields the complete solution as

$$u(t) = (1 - \cos \omega_n t) F_0/k \quad t > 0 \quad (50)$$

A plot of Eq. (50) is shown in Figure 5c. It is apparent that the displacement starts from zero, and then climbs to a maximum value, then returns to zero, and continues as an oscillatory motion. It is also apparent that the maximum displacement is

$$u_{\text{dyn}} = u_{\text{max}} = 2F_0/k \quad (\text{maximum dynamic displacement}) \quad (51)$$

The factor of 2 in the maximum displacement is due to the fact that the maximum value of Eq. (50) is obtained when the cosine function takes the value -1 . Eq. (51) illustrates the assertion that *the dynamic displacement u_{dyn} is twice the static displacement u_{st}* defined as

$$u_{\text{st}} = F_0/k \quad (\text{static displacement}) \quad (52)$$

The force in the spring k is proportional to the displacement. Under static conditions, the force in the spring is

$$F_{\text{st}} = ku_{\text{st}} = F_0/k \quad (53)$$

Under dynamic conditions, the maximum force in the spring is

$$F_{\text{dyn}} = ku_{\text{dyn}} = 2F_0/k = 2F_{\text{st}} \quad (54)$$

Eq. (54) proves that *the dynamic amplification factor* is equal to 2. Thus, a structure designed to withstand some static loads may be in trouble if the same loads were applied dynamically.

1.2.3 HARMONIC EXCITATION

If the excitation is harmonic, then the response will be vibrational. Assume the excitation in the form

$$F(t) = \hat{F}e^{i\omega t} \quad (55)$$

where $\hat{F}$ is the excitation amplitude and ω is the excitation frequency. Substitution of Eq. (55) into Eq. (35) yields

$$m\ddot{u}(t) + ku(t) = \hat{F}e^{i\omega t} \quad (56)$$

Upon normalization by m , Eq. (56) becomes

$$\ddot{u}(t) + \omega_n^2 u(t) = \hat{f}e^{i\omega t} \quad (57)$$

where $\hat{f}$ is the normalized excitation amplitude defined as

$$\hat{f} = \hat{F}/m \quad (58)$$

Eq. (57) is a nonhomogeneous linear ODE. The solution of nonhomogeneous linear ODEs consists of the superposition of the *complementary solution* $u_c(t)$ that satisfies the homogenous ODE and a *particular solution* $u_p(t)$ that satisfies the nonhomogeneous ODE, i.e.,

$$u(t) = u_c(t) + u_p(t) \quad (\text{general solution}) \quad (59)$$

1.2.3.1 Complementary solution

The homogenous variant of Eq. (57) is obtained by making zero the right-hand side (RHS) which gives

$$\ddot{u}_c(t) + \omega_n^2 u_c(t) = 0 \quad (\text{homogeneous ODE}) \quad (60)$$

It is apparent that Eq. (60) is the same as the previously discussed Eq. (4) divided by m . Eq. (4) has the solution given by Eq. (21) in the form

$$u_c(t) = C \sin(\omega_n t + \phi) \quad (\text{complementary solution}) \quad (61)$$

Thus, Eq. (61) is the sought-after complementary solution $u_c(t)$.

1.2.3.2 Particular solution

The particular solution is a solution that satisfies the nonhomogeneous Eq. (57).

$$\ddot{u}_p(t) + \omega_n^2 u_p(t) = \hat{f}e^{i\omega t} \quad (62)$$

It is customary to seek the particular solution in the form of the forcing function. In our case, the forcing function is harmonic, hence we assume

$$u_p(t) = \hat{u}_p e^{i\omega t} \quad (63)$$

Upon differentiation, Eq. (63) gives

$$\ddot{u}_p(t) = (i\omega)^2 \hat{u}_p e^{i\omega t} = -\omega^2 \hat{u}_p e^{i\omega t} \quad (64)$$

Substitution of Eqs. (63), (64) into Eq. (57) yields

$$-\omega^2 \hat{u}_p e^{i\omega t} + \omega_n^2 \hat{u}_p e^{i\omega t} = \hat{f} e^{i\omega t} \quad (65)$$

After simplification by $e^{-i\omega t}$, Eq. (65) is solved for $\hat{u}_p$ and gives

$$\hat{u}_p = \frac{1}{-\omega^2 + \omega_n^2} \hat{f} \quad (66)$$

Substitution of Eq. (66) into Eq. (63) given the particular solution $u_p(t)$ as

$$u_p(t) = \frac{1}{-\omega^2 + \omega_n^2} \hat{f} e^{i\omega t} \quad (\text{particular solution}) \quad (67)$$

1.2.3.3 Complete solution

Superposing the complementary solution Eq. (61) and the particular solution Eq. (67) yields the complete solution

$$u(t) = C \sin(\omega_n t + \varphi) + \frac{1}{-\omega^2 + \omega_n^2} \hat{f} e^{i\omega t} \quad (\text{complete solution}) \quad (68)$$

The constants C and φ are to be determined from the initial conditions.

The $C \sin(\omega_n t + \varphi)$ part of Eq. (68) represents the *transient response* whereas the $u_p(t)$ part represents the *forced response*.

2 SPRING-MASS-DAMPER VIBRATING SYSTEMS

Consider the 1-dof damped system consisting of a spring, k , mass, m , and viscous damper, c , as presented in Figure 6.

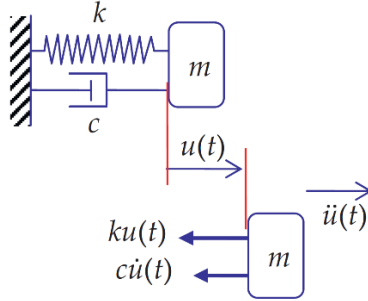


Figure 6 mass-spring-damper vibrating system

2.1 FREE VIBRATION OF SPRING-MASS-DAMPER SYSTEMS

Assume the mass m is displaced from the equilibrium position by a time-dependent displacement $u(t)$.

2.1.1 FREE-BODY DIAGRAM

The free body diagram (FBD) of the mass m is shown in Figure 6. FBD analysis gives the sum of x -direction forces as

$$\Sigma F_x = -c\dot{u}(t) - ku(t) \quad (69)$$

where $ku(t)$ is the reaction force from the spring and $c\dot{u}(t)$ is the reaction force from the damper.

2.1.2 NEWTON'S LAW OF MOTION

Newton's law of motion (NLM) applied to the mass m is stated as

$$\Sigma F_x = m\ddot{u}(t) \quad (70)$$

2.1.3 EQUATION OF MOTION

Substitution of Eq. (69) into Eq. (70) yields as

$$m\ddot{u}(t) = -ku(t) - c\dot{u}(t) \quad (71)$$

Upon rearrangement, Eq. (71) becomes the equation of motion (EOM), i.e.,

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = 0 \quad (72)$$

Eq. (72) is a linear homogenous ordinary differential equation (ODE).

2.1.4 NONTRIVIAL SOLUTION

The solution of a linear homogenous ODE is sought in the form e^{st} . Assume

$$u(x)=e^{st} \quad (73)$$

Upon differentiation, Eq. (73) gives

$$\dot{u}(t) = se^{st} = su(t) \quad (74)$$

$$\ddot{u}(t) = s^2 e^{st} = s^2 u(t)$$

Substitution of Eqs. (73), (74) into Eq. (72) yields

$$ms^2 u(t) + csu(t) + ku(t) = 0 \quad (75)$$

After factoring out $u(t)$, Eq. (75) becomes

$$(ms^2 + cs + k) u(t) = 0 \quad (76)$$

A possible solution of Eq. (76) is $u(t)=0$. However, this solution is trivial and of no interest to us. We are interested in *nontrivial solutions* $u(t) \neq 0$. To obtain a nontrivial solution $u(t) \neq 0$, one needs to find the values of s that make zero the factor multiplying $u(t)$ in Eq. (76), i.e., to have $(ms^2 + cs + k) = 0$.

2.1.5 CHARACTERISTIC EQUATION

It was shown in Section 2.1.4 that a nontrivial solution $u(t) \neq 0$ of Eq. (76) can only be obtained if the factor multiplying $u(t)$ is zero, i.e.,

$$ms^2 + cs + k = 0 \quad (\text{characteristic equation}) \quad (77)$$

Eq. (77) is the *characteristic equation*. Solution of the characteristic equation will give us the values of s that permit nontrivial solution $u(t) \neq 0$ of Eq. (76). The solutions of the characteristic equation are calculated with the quadratic formula as

$$s_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m} \quad (78)$$

2.1.6 CRITICAL DAMPING

Examination of Eq. (78) reveals that the term $c^2 - 4mk$ under the square root sign may be positive or negative depending on the value of damping c in relation to mass m and stiffness k . The value of c that makes this term zero is called *critical damping* c_{cr} . The critical damping value is found by solving the following equation,

$$c_{cr}^2 - 4mk = 0 \quad (79)$$

Solution of Eq. (79) yields

$$c_{cr} = 2\sqrt{mk} \quad (\text{critical damping}) \quad (80)$$

- if $c < c_{cr}$, then the system is said to be *underdamped*.
- if $c = c_{cr}$, then the system is said to be *critically damped*.
- if $c > c_{cr}$, then the system is said to be *overdamped*.

2.1.7 DAMPING RATIO

The damping ratio ζ is the ratio between the system damping c and the critical damping c_{cr} , i.e.,

$$\text{zeta} \quad \zeta = c/c_{cr} \quad (\text{damping ratio}) \quad (81)$$

- if $\zeta < 1$, then the system is said to be *underdamped*.
- if $\zeta = 1$, then the system is said to be *critically damped*.
- if $\zeta > 1$, then the system is said to be *overdamped*.

Eqs. (80), (81) can be used to express the system damping c can be expressed in terms of damping ratio ζ and critical damping c_{cr} as

$$c = \zeta c_{cr} = 2\zeta\sqrt{mk} \quad (82)$$

2.1.8 NATURAL FREQUENCY

Eq. (78) can be processed as follows

$$s_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m} = -\frac{c}{2m} \pm \sqrt{\left(\frac{c}{2m}\right)^2 - \frac{k}{m}} \quad (83)$$

The ratio k/m is recognized as the square of the natural frequency ω_n of the undamped spring-mass system discussed in Section 1. The natural frequency ω_n was defined by Eq. (12) in terms of m and k as

$$\omega_n^2 = k/m \quad \omega_n = \sqrt{k/m} \quad (\text{natural frequency}) \quad (84)$$

2.1.9 ROOTS OF THE CHARACTERISTIC EQUATION

Eq. (83) can be further processed as follows. Use Eqs. (82), (84) to express the ratio $c/2m$ in terms of ζ and ω_n , i.e.,

$$\frac{c}{2m} = \frac{2\zeta\sqrt{mk}}{2m} = \zeta\sqrt{\frac{mk}{m^2}} = \zeta\sqrt{\frac{k}{m}} = \zeta\omega_n \quad (85)$$

Substitution of Eqs. (84), (85) into Eq. (83) yields

$$s_{1,2} = -\zeta\omega_n \pm \sqrt{(\zeta\omega_n)^2 - \omega_n^2} \quad (86)$$

or

$$s_{1,2} = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1} \quad (\text{roots of characteristic eq.}) \quad (87)$$

2.1.10 SOLUTION FOR $\zeta < 1$

If $\zeta < 1$, then the $\zeta^2 - 1 < 0$ and the corresponding term in Eq. (87) becomes imaginary, i.e.,

$$\sqrt{\zeta^2 - 1} = i\sqrt{1 - \zeta^2} \quad \zeta < 1 \quad (88)$$

Substitution of Eq. (88) into Eq. (87) yields

$$s_{1,2} = -\zeta\omega_n \pm i\omega_n \sqrt{1 - \zeta^2} \quad (89)$$

Define the *damped frequency* ω_d as

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \quad (\text{damped frequency}) \quad (90)$$

Substitution of Eq. (90) into Eq. (89) yields the roots of the characteristic equation for $\zeta < 1$ as two complex numbers s_1 and s_2 , i.e.,

$$s_1 = -\zeta\omega_n + i\omega_d, \quad s_2 = -\zeta\omega_n - i\omega_d \quad (\text{roots of char. eqn.}) \quad (91)$$

Both s_1 and s_2 yield solutions of the ODE Eq. (72). These solutions are in the form of complex exponentials $e^{s_1 t}$ and $e^{s_2 t}$ respectively. Hence, the general solution of the linear ODE Eq. (72) is a linear superposition of the two individual solutions, i.e.,

$$u(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t} \quad (92)$$

where C_1 and C_2 are arbitrary constants that need to be determined from the initial conditions. Substitution of Eq. (91) into Eq. (92) gives

$$u(t) = C_1 e^{(-\zeta\omega_n + i\omega_d)t} + C_2 e^{(-\zeta\omega_n - i\omega_d)t} \quad (93)$$

Recall the formula

$$e^{a+b} = e^a e^b \quad (94)$$

The use of Eq. (94) into Eq. (93) allows us to factor out the term $e^{-\zeta\omega_n t}$ such that Eq. (93) can be written as

$$u(t) = e^{-\zeta\omega_n t} \left(C_1 e^{i\omega_d t} + C_2 e^{-i\omega_d t} \right) \quad (95)$$

Examination of Eq. (95) reveals two components, a decaying exponential and a harmonic response, i.e.,

$$e^{-\zeta\omega_n t} \quad (\text{exponential decay function}) \quad (96)$$

$$C_1 e^{i\omega_d t} + C_2 e^{-i\omega_d t} \quad (\text{harmonic function}) \quad (97)$$

As previously shown in Section 1.1.6, a harmonic function as shown in Eq. (97) can be expressed as

$$u(t) = C \sin(\omega_d t + \varphi) \quad (98)$$

Substitution of Eq. (98) into Eq. (95) yields the free vibration of damped system with $\zeta < 1$ as

$$u(t) = C e^{-\zeta\omega_n t} \sin(\omega_d t + \varphi) \quad \zeta < 1 \quad (99)$$

where ζ , ω_n , ω_d are given by Eqs. (81), (84), (90), respectively and the constants C , φ are to be determined from the initial conditions. A schematic representation of Eqs. (96), (98), (99) is shown in Figure 7. It can be noticed that the carrier harmonic signal $C \sin(\omega_d t + \varphi)$ is being modulated by the exponential decay function such that, after some time, the vibration dampens out and asymptotically approaches zero.

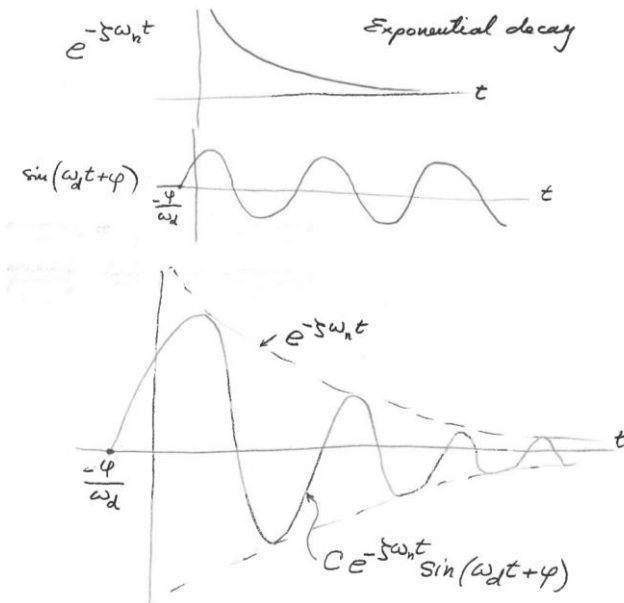


Figure 7 Free vibration of a spring-mass-damper system

2.1.11 INITIAL VALUE PROBLEM FOR $\zeta < 1$

The initial value problem (IVP) consists in finding the general-solution constants in terms of the initial conditions of the vibrating system, i.e., initial displacement u_0 and initial velocity, $\dot{u}_0$. The purpose is to find the constants A, B in terms of known u_0 and $\dot{u}_0$. Recall the solution expressed either as Eq. (92) or as Eq. (95), i.e.,

$$u(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t} \quad (92)$$

$$u(t) = e^{-\zeta \omega_n t} \left(C_1 e^{i \omega_d t} + C_2 e^{-i \omega_d t} \right) \quad (95)$$

and write the initial conditions as

$$u(0) = e^{-\zeta \omega_n t} (C_1 e^{i \omega_d t} + C_2 e^{-i \omega_d t}) \Big|_{t=0} = C_1 + C_2 = u_0 \quad (100)$$

$$\dot{u}(0) = s_1 C_1 e^{s_1 t} + s_2 C_2 e^{s_2 t} \Big|_{t=0} = s_1 C_1 + s_2 C_2 = \dot{u}_0 \quad (101)$$

Solve Eqs. (100), (101) to get

$$\begin{aligned} C_1 &= \frac{-s_2 u_0 + \dot{u}_0}{s_1 - s_2} \\ C_2 &= \frac{s_1 u_0 - \dot{u}_0}{s_1 - s_2} \end{aligned} \quad (102)$$

Eq. (91) is used to get numerical values for the roots s_1, s_2 which are then substituted into Eq. (102) to calculate the constants C_1, C_2 . Then, like before in the case of undamped vibration, calculate the trigonometric part of Eq. (95) as

$$A = C_1 + C_2 \quad B = i(C_1 - C_2) \quad (103)$$

and write

$$C_1 e^{i \omega_d t} + C_2 e^{-i \omega_d t} = A \cos \omega_d t + B \sin \omega_d t \quad (104)$$

Substitution of Eq. (104) into Eq. (95) yields

$$u(t) = e^{-\zeta \omega_n t} \left(A \cos \omega_d t + B \sin \omega_d t \right) \quad (105)$$

Further, one can calculate

$$C = \sqrt{A^2 + B^2} \quad \varphi = \tan^{-1} A/B \quad (106)$$

and substitute into Eq. (105) to get Eq. (99), i.e.,

$$u(t) = C e^{-\zeta \omega_n t} \sin(\omega_d t + \varphi) \quad (\text{damped oscillatory response}) \quad (99)$$

Note that s_1 and s_2 are complex numbers, hence C_1, C_2 would also be complex numbers. However, A, B, C, φ should be real numbers.

2.1.12 SOLUTION FOR $\zeta > 1$

If $\zeta > 1$, then the $\zeta^2 - 1 > 0$ and Eq. (87) yields

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \quad (107)$$

Eq. (107) generates two real roots,

$$\begin{aligned} s_1 &= -\left(\zeta + \sqrt{\zeta^2 - 1}\right)\omega_n \\ s_2 &= -\left(\zeta - \sqrt{\zeta^2 - 1}\right)\omega_n \end{aligned} \quad (108)$$

Both roots are negative because $\zeta > \sqrt{\zeta^2 - 1}$. The general solution is a linear combination of the solutions for s_1, s_2 , i.e.,

$$u(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t} = C_1 e^{-\left(\zeta + \sqrt{\zeta^2 - 1}\right)\omega_n t} + C_2 e^{-\left(\zeta - \sqrt{\zeta^2 - 1}\right)\omega_n t} \quad (109)$$

Eq. (109) represents the response of an overdamped spring-mass-damper system.

To solve the initial value problem (IVP) for u_0 and $\dot{u}_0$ initial conditions, use Eq. (109) to get

$$\begin{aligned} u(0) &= C_1 e^{s_1 t} + C_2 e^{s_2 t} \Big|_{t=0} = C_1 + C_2 = u_0 \\ \dot{u}(0) &= C_1 s_1 e^{s_1 t} + C_2 s_2 e^{s_2 t} \Big|_{t=0} = s_1 C_1 + s_2 C_2 = \dot{u}_0 \end{aligned} \quad (110)$$

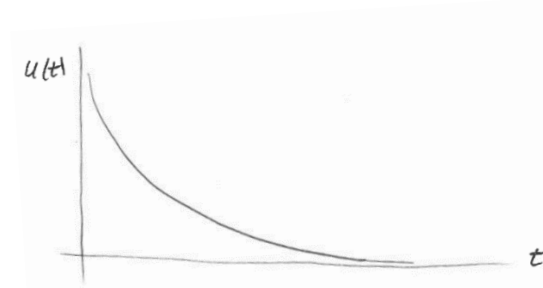


Figure 8 Response of an overdamped spring-mass-damper system

Upon solution, Eq. (110) yields

$$\begin{aligned} C_1 &= \frac{-s_2 u_0 + \dot{u}_0}{s_1 - s_2} \\ C_2 &= \frac{s_1 u_0 - \dot{u}_0}{s_1 - s_2} \end{aligned} \quad (111)$$

Eq. (111) is used to calculate the constants C_1 , C_2 which are real numbers. These C_1 , C_2 values are then used in Eq. (109) to compute the overdamped IVP solution. A plot of this overdamped response is shown in Figure 8. It is a uniformly decaying function that asymptotically reaches zero.

2.1.13 SOLUTION FOR $\zeta=1$

If $\zeta=1$, then $c = c_{cr} = 2\sqrt{mk}$ and Eq. (72) becomes

$$m\ddot{u}(t) + 2\sqrt{mk} \dot{u}(t) + ku(t) = 0 \quad (112)$$

Upon division by m , Eq. (112) becomes

$$\ddot{u}(t) + 2\sqrt{\frac{k}{m}} \dot{u}(t) + \frac{k}{m} u(t) = 0 \quad (113)$$

Recall the natural frequency definition $\omega_n = \sqrt{k/m}$ as given by Eq. (84). Upon substitution of $\sqrt{k/m} = \omega_n$, Eq. (113) becomes

$$\ddot{u}(t) + 2\omega_n \dot{u}(t) + \omega_n^2 u(t) = 0 \quad (114)$$

The roots of the characteristic equation are $s_1=s_2=\omega_n$ and Eq. (114) accepts two solutions

$$u_1(t) = e^{-\omega_n t} \quad u_2(t) = t e^{-\omega_n t} \quad (115)$$

Proof

The proof of Eq. (115) is done by direct substitution.

(a) Assume the first solution $u_1(t) = e^{-\omega_n t}$ and calculate its derivatives:

$$\begin{aligned} u_1(t) &= e^{-\omega_n t} \\ \dot{u}_1(t) &= -\omega_n e^{-\omega_n t} = -\omega_n u_1(t) \\ \ddot{u}_1(t) &= \omega_n^2 e^{-\omega_n t} = \omega_n^2 u_1(t) \end{aligned} \quad (116)$$

Substitution of Eq. (116) into Eq. (114) gives

$$\omega_n^2 u_1(t) - 2\omega_n \omega_n u_1(t) + \omega_n^2 u_1(t) = 2\omega_n^2 u_1(t) - 2\omega_n^2 u_1(t) = 0 \quad (117)$$

Thus, we have proven that the first solution $u_1(t) = e^{-\omega_n t}$ verifies the ODE Eq. (114).

(b) Assume the second solution $u_2(t) = te^{-\omega_n t}$ and calculate its derivatives:

$$u_2(t) = te^{-\omega_n t}$$

$$\dot{u}_2(t) = e^{-\omega_n t} - \omega_n te^{-\omega_n t} \quad (118)$$

$$\ddot{u}_2(t) = -\omega_n e^{-\omega_n t} + \omega_n^2 te^{-\omega_n t} - \omega_n e^{-\omega_n t} = \omega_n^2 te^{-\omega_n t} - 2\omega_n e^{-\omega_n t}$$

Substitution of Eq. (118) into Eq. (114) gives

$$\omega_n^2 te^{-\omega_n t} - 2\omega_n e^{-\omega_n t} + 2\omega_n (e^{-\omega_n t} - \omega_n te^{-\omega_n t}) + \omega_n^2 te^{-\omega_n t} =$$

$$\cancel{\omega_n^2 te^{-\omega_n t}} - \cancel{2\omega_n e^{-\omega_n t}} + \cancel{2\omega_n e^{-\omega_n t}} - \cancel{2\omega_n^2 te^{-\omega_n t}} + \cancel{\omega_n^2 te^{-\omega_n t}} = 0 \quad (119)$$

Thus, we have proven that the second solution $u_2(t) = te^{-\omega_n t}$ also verifies the ODE Eq. (114).

The general solution is a linear combination of the two solutions, i.e.,

$$u(t) = C_1 u_1(t) + C_2 u_2(t) = (C_1 + C_2 t) e^{-\omega_n t} \quad (120)$$

Eq. (120) represents the response of a critically damped spring-mass-damper system.

To solve the initial value problem (IVP) for u_0 and $\dot{u}_0$ initial conditions, use Eq. (120) to get

$$u(0) = (C_1 + C_2 t) e^{-\omega_n t} \Big|_{t=0} = C_1 = u_0$$

$$\dot{u}(0) = [C_2 - \omega_n (C_1 + C_2 t)] e^{-\omega_n t} \Big|_{t=0} = C_2 - \omega_n C_1 = \dot{u}_0 \quad (121)$$

Upon solution, Eq. (121) yields

$$C_1 = u_0 \quad C_2 = \dot{u}_0 + \omega_n u_0 \quad (122)$$

Substitution of Eq. (122) into Eq. (120) yields the critically damped IVP solution

$$u(t) = (u_0 + \dot{u}_0 t + \omega_n u_0 t) e^{-\omega_n t} \quad (123)$$

A plot of this critically damped response is shown in Figure 9. It is a uniformly decaying function that asymptotically reaches zero.

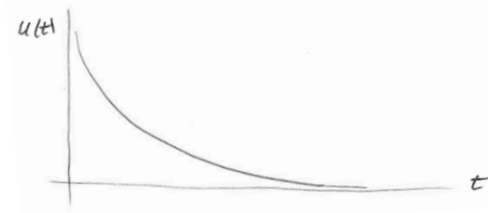


Figure 9 Response of a critically damped spring-mass-damper system

2.1.14 COMPARISON OF CRITICALLY DAMPED AND OVERDAMPED RESPONSES

The plots of Figure 9 and Figure 8 are apparently similar in appearance. However, they are quite different in detail. For illustration, Figure 10 shows a comparison of critically damped $\zeta=1$ and an overdamped $\zeta=3$ responses of spring-mass-damper systems with same natural frequency ω_n

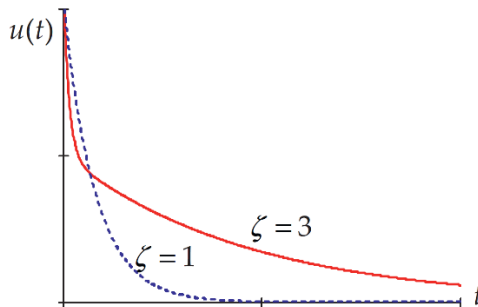


Figure 10 Comparison of critically damped $\zeta=1$ and an overdamped $\zeta=3$ responses of spring-mass-damper systems with same natural frequency ω_n and same initial conditions.

Examination of the plots in Figure 10 reveals the following observations. At first, the overdamped response ($\zeta=3$) decays more rapidly at first but then it takes longer to settle down. The critically

damped response ($\zeta=1$) decays a little slower at first, but then it settles down more rapidly.

2.2 FORCED VIBRATION OF SPRING-MASS-DAMPER SYSTEMS

Consider a mass-spring-damper vibrating system excited by a time-dependent force $F(t)$ as shown in Figure 11.

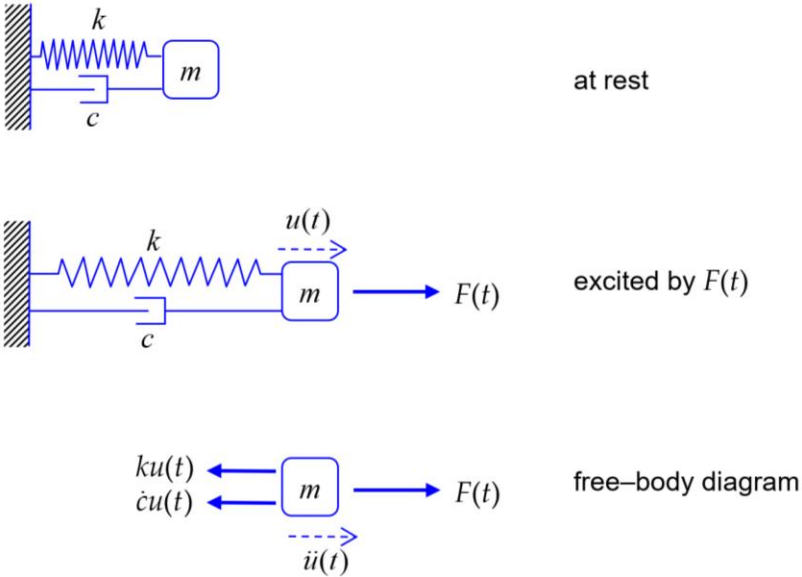


Figure 11 spring-mass-damper system excited by force $F(t)$

2.2.1 EQUATION OF MOTION

Free body diagram (FBD) of the mass m gives the sum of x -direction forces as

$$\Sigma F_x = -c\dot{u}(t) - ku(t) + F(t) \quad (124)$$

where $ku(t)$ is the reaction force from the spring, $c\dot{u}(t)$ is the reaction force from the damper, and $F(t)$ is the excitation force. Newton's law of motion (NLM) applied to the mass m is stated as

$$\Sigma F_x = m\ddot{u}(t) \quad (125)$$

Substitution of Eq. (124) into Eq. (125) yields, upon rearrangement, the equation of motion (EOM) as

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = F(t) \quad (\text{EOM}) \quad (126)$$

2.2.2 HARMONIC EXCITATION

If the excitation is harmonic, then the response will be vibrational. Assume the excitation in the form

$$F(t) = \hat{F}e^{i\omega t} \quad (127)$$

where $\hat{F}$ is the excitation amplitude and ω is the excitation frequency. Substitution of Eq. (127) into Eq. (126) yields

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = \hat{F}e^{i\omega t} \quad (128)$$

Upon division by m , Eq. (128) becomes

$$\ddot{u}(t) + \frac{c}{m}\dot{u}(t) + \frac{k}{m}u(t) = \frac{\hat{F}}{m}e^{i\omega t} \quad (129)$$

Recall Eq. (84) which gives the ratio k/m in terms of natural frequency ω_n as

$$\frac{k}{m} = \omega_n^2 \quad (130)$$

Recall Eq. (85) which gives the ratio c/m in terms of damping ratio ζ and natural frequency ω_n as

$$\frac{c}{m} = 2\zeta\omega_n \quad (131)$$

Define the normalized excitation amplitude $\hat{f}$ as

$$\hat{f} = \hat{F}/m \quad (132)$$

Substitution of Eqs. (130), (131), (132) into Eq. (129) yields

$$\ddot{u}(t) + 2\zeta\omega_n\dot{u}(t) + \omega_n^2u(t) = \hat{f}e^{i\omega t} \quad (133)$$

Eq. (133) is a nonhomogeneous linear ODE. The solution of nonhomogeneous linear ODEs consists of the superposition of the *complementary solution* $u_c(t)$ that satisfies the homogenous ODE and a *particular solution* $u_p(t)$ that satisfies the nonhomogeneous ODE, i.e.,

$$u(t) = u_c(t) + u_p(t) \quad (\text{general solution}) \quad (134)$$

2.2.2.1 Complementary solution

The homogenous variant of Eq. (133) is obtained by making zero the right-hand side (RHS) which gives

$$\ddot{u}_c(t) + 2\zeta\omega_n\dot{u}_c(t) + \omega_n^2u_c(t) = 0 \quad (\text{homogeneous ODE}) \quad (135)$$

It is apparent that Eq. (135) is the same as the previously discussed Eq. (76) divided by m . Eq. (76) has the solution given by Eq. (99) in the form

$$u_c(t) = Ce^{-\zeta\omega_n t} \sin(\omega_d t + \varphi) \quad (\text{complementary solution}) \quad (136)$$

where ω_d is the damped frequency given by Eq. (90). Thus, Eq. (136) is the sought-after complementary solution $u_c(t)$.

2.2.2.2 Particular solution

The particular solution is a solution that satisfies the nonhomogeneous Eq. (133).

$$\ddot{u}_p(t) + 2\zeta\omega_n\dot{u}_p(t) + \omega_n^2 u_p(t) = \hat{f}e^{i\omega t} \quad (137)$$

It is customary to seek a particular solution in the form of the forcing function. In our case, the forcing function is harmonic, hence we assume

$$u_p(t) = \hat{u}_p e^{i\omega t} \quad (138)$$

Upon differentiation, Eq. (138) gives

$$\begin{aligned} \dot{u}_p(t) &= i\omega\hat{u}_p e^{i\omega t} = i\omega\hat{u}_p e^{i\omega t} \\ \ddot{u}_p(t) &= (i\omega)^2 \hat{u}_p e^{i\omega t} = -\omega^2 \hat{u}_p e^{i\omega t} \end{aligned} \quad (139)$$

Substitution of Eqs. (138), (139) into Eq. (137) yields

$$-\omega^2 \hat{u}_p e^{i\omega t} + 2i\zeta\omega_n \omega \hat{u}_p e^{i\omega t} + \omega_n^2 \hat{u}_p e^{i\omega t} = \hat{f}e^{i\omega t} \quad (140)$$

After simplification by $e^{-i\omega t}$, Eq. (65) is solved for $\hat{u}_p$ and gives

$$\hat{u}_p = \frac{1}{-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2} \hat{f} \quad (141)$$

Substitution of Eq. (141) into Eq. (138) given the particular solution $u_p(t)$ as

$$u_p(t) = \frac{1}{-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2} \hat{f}e^{i\omega t} \quad (\text{particular solution}) \quad (142)$$

2.2.2.3 Complete solution

Superposing the complementary solution Eq. (136) and the particular solution Eq. (142) yields the complete solution

$$u(t) = Ce^{-\zeta\omega_n t} \sin(\omega_d t + \varphi) + \frac{1}{-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2} \hat{f}e^{i\omega t} \quad (143)$$

The constants C and φ are to be determined from the initial conditions.

The $Ce^{-\zeta\omega_n t} \sin(\omega_d t + \varphi)$ part of Eq. (143) represents the *transient response* whereas the $u_p(t)$ part represents the *steady-state response*. The transient response will quickly vanish due to damping and, after some time, only the steady-state response will persist. For this reason, the subsequent analysis will only deal with the steady-state response.

2.2.2.4 Steady-state response

Examination of Eq. (143) reveals that complementary solution $Ce^{-\zeta t} e^{i(\omega_d t + \psi)}$ is only of interest at the beginning of the forced vibration, as it soon dies out due to damping. In practice, the response to harmonic excitation is of interest under *steady-state conditions*, i.e., after a machinery has run for some time and the transient response has been damped out. The *steady-state response* is given by the $u_p(t)$ part of Eq. (143), i.e.,

$$u_{ss}(t) = \frac{1}{-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2} \hat{f}e^{i\omega t} \quad (\text{steady-state response}) \quad (144)$$

The amplitude of the steady-state response is

$$\hat{u}_{ss}(\omega) = \frac{1}{-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2} \hat{f} \quad (145)$$

Eq. (145) indicates that the amplitude of the steady-state response is a frequency-dependent complex number whose magnitude and phase changes as excitation frequency varies as discussed in the next section.

To write the steady-state response in term of physical excitation, recall Eqs. (132), (145), i.e.,

$$\hat{f} = \hat{F}/m \quad (146)$$

$$\hat{u}_{ss}(\omega) = \frac{1}{-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2} \hat{f} \quad (147)$$

Substitution of Eq. (146) into Eq. (147) yields the steady-state response in term of physical excitation $\hat{F}$ as

$$\hat{u}_{ss}(\omega) = \frac{1}{-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2} \frac{\hat{F}}{m} \quad (148)$$

2.2.2.5 Quasi-static response

Quasi-static conditions are those when the frequency of excitation is so small, and the motion is so slow that it appears as “static” to an external observer. Taking Eq. (148) to the limit $\omega \rightarrow 0$ yields the quasi-static displacement u_{qst} as

$$u_{qst} = \lim_{\omega \rightarrow 0} \hat{u}_{ss}(\omega) = \frac{\hat{F}}{m\omega_n^2} = \frac{\hat{F}}{k} \quad (\text{quasi-static displacement}) \quad (149)$$

2.2.3 FREQUENCY RESPONSE FUNCTION

The frequency response function (FRF) is the frequency-domain relationship between the harmonic excitation and the response of a vibrating system. FRF is expressed as the ratio between the steady-state response amplitude and the excitation amplitude. The FRF of a vibrating system is customarily normalized with the quasi-static deflection such that it approaches the value 1 as the frequency approaches zero.

2.2.3.1 FRF definition

Recall from Eq. (149) the definition of the quasi-static deflection u_{qst} as the excitation amplitude divided by stiffness, i.e.,

$$u_{qst} = \hat{F}/k \quad (\text{quasi-static displacement}) \quad (150)$$

Define the FRF function as the steady-state response amplitude normalized by the quasi-static displacement, i.e.,

$$FRF(\omega) = \frac{\hat{u}_{ss}(\omega)}{u_{qst}} \quad (\text{frequency response function}) \quad (151)$$

Substitute Eqs. (148), (150) into Eq. (151) to get

$$FRF(\omega) = \frac{\hat{u}_{ss}(\omega)}{u_{qst}} = \frac{1}{-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2} \frac{\cancel{F}}{m} \frac{k}{\cancel{F}} \quad (152)$$

$$= \frac{k/m}{-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2}$$

Recall Eq. (84) which gives the ratio k/m in terms of natural frequency ω_n as

$$k/m = \omega_n^2 \quad (153)$$

Substitution of Eq. (153) into Eq. (152) yields the FRF formula

$$FRF(\omega) = \frac{\omega_n^2}{-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2} \quad (\text{frequency response function}) \quad (154)$$

FRF can be used in Eq. (151) to calculate the physical response amplitude at any excitation frequency as

$$\hat{u}(\omega) = u_{qst} FRF(\omega) \quad (\text{frequency response function}) \quad (155)$$

where u_{st} is given by Eq. (150).

2.2.3.2 FRF variation with frequency

Figure 12 shows the frequency response function $FRF(\omega)$ vs. the normalized frequency ω/ω_n for a range of damping values ζ . Figure 12a presents a plot of the normalized FRF magnitude vs. normalized frequency. Starting from the normalized quasi-static value $FRF(0)=1$, the magnification factor increases as the frequency increases until it reaches a maximum and then decreases rapidly towards zero. Figure 12b gives a plot of FRF phase vs. frequency. It can be observed that the phase is always negative, i.e., the response is lagging behind excitation.

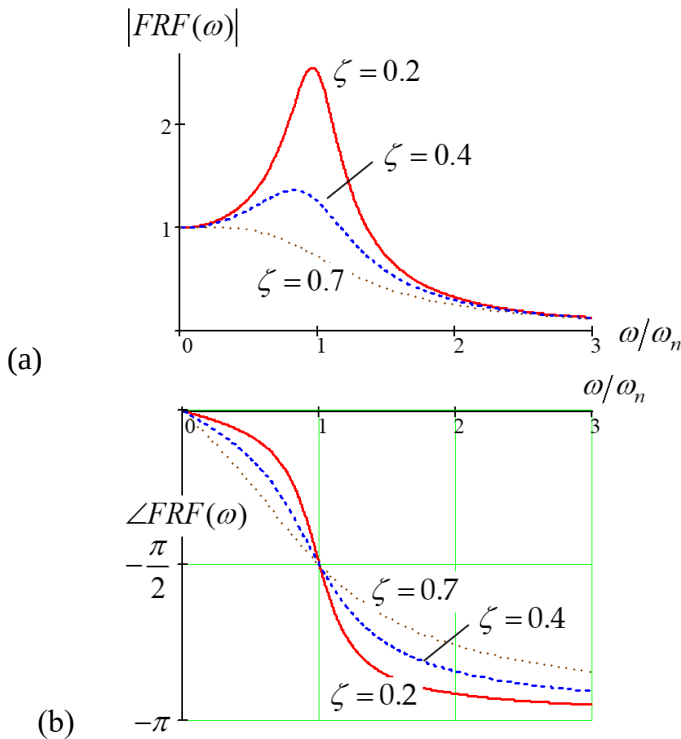


Figure 12 Frequency response function for a damped 1-dof system: (a) magnitude; (b) phase

2.2.3.3 Magnification factor

The magnitude of the frequency response function is also known as *magnification factor* $M(\omega)$ calculated as

$$M(\omega) = |FRF(\omega)| = \left| \frac{\omega_n^2}{-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2} \right| \quad (156)$$

Figure 12a presents a plot of the magnification factor vs. normalized frequency. Starting from the quasi-static value $M_{st}=1$, the magnification factor increases as the frequency increases until it reaches a maximum and then decreases rapidly towards zero.

2.2.3.4 Resonance frequency

Resonance frequency is defined as the frequency ω_r at which the frequency response is maximum. Regarding the exact location of the resonance frequency, examination of Figure 12a and Eq. (154)

reveals that the point of maximum response is not at $\omega = \omega_n$ as it would be intuitively expected. The point of maximum response can be calculated by seeking the maximum of the magnification factor Eq. (156). Upon calculation, one finds that the maximum response is found at the *resonance frequency* ω_r given by the formula

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2} \quad (\text{resonance frequency}) \quad (157)$$

Note that the resonance frequency ω_r given by Eq. (157) is different from the natural frequency ω_n given by Eq. (84) as well as from the damped frequency ω_d given by Eq. (90). At low damping ratios $\zeta \ll 1$, the damped resonance frequency ω_r and the damped natural frequency, ω_d , are only marginally different from the natural frequency, ω_n . However, as the damping increases, these three frequencies ω_r , ω_d , ω_n may differ considerably.

The magnification peaks at resonance and its value M_r is given by the formula

$$M_r = \frac{1}{2\zeta \sqrt{1 - \zeta^2}} \quad (\text{peak magnification}) \quad (158)$$

Figure 12a indicates that the amplitude at resonance increases as the damping decreases. The resonance response of lightly damped systems can be very large and can endanger the safe operation of a dynamic system if not properly controlled. However, as the damping increases, the resonance response decreases and, for $\zeta \geq \sqrt{2}$, no peak exists because in this case Eq. (157) would give imaginary results.

2.2.3.5 FRF phase

The phase of the frequency response function can be calculated as

$$\phi(p) = \text{angle}(FRF(p)) = \text{angle}\left(\frac{1}{-p^2 + i2\zeta p + 1}\right) \quad (159)$$

where $p = \omega/\omega_n$. The phase angle $\phi(p)$ represents the phase difference between response and excitation. Figure 12b also indicates that the phase angle increases with frequency, i.e., the lag between excitation and response increases with frequency, which is consistent with common experience. As the excitation frequency sweeps the entire range from 0 through ∞ , the phase angle goes from 0° through -180° ($-\pi$ radians).

2.2.3.6 Phase resonance

For $\omega = \omega_n$, Figure 12b also indicates that the phase angle is -90° ($-\pi/2$ radians). On the unit circle this angle is called *quadrature* because it represents a quarter of the circle. This indicates that when the excitation frequency, ω , matches the natural frequency, ω_n , the response is in *quadrature* with the excitation (i.e., it lags by 90° behind the excitation).

This fact is especially useful in experimental work on lightly damped systems because it permits the estimation of the natural frequency ω_n by locating the -90° point on a plot of the phase response vs. frequency.

2.2.3.7 Response amplitude at phase resonance

For $\omega = \omega_n$, Eq. (154) gives a frequency response value that is purely imaginary, i.e.,

$$FRF(\omega) \Big|_{\omega=\omega_n} = \frac{\omega_n^2}{-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2} \Big|_{\omega=\omega_n} = \frac{1}{2i\zeta} = \frac{-i}{2\zeta} \quad (160)$$

The phase of the imaginary number $-i/2\zeta$ is -90° ($-\pi/2$ radians) which confirms the observation made by examining the plot in Figure 12b.

The magnification factor at $\omega = \omega_n$ denoted as M_{90} , is calculated by taking the magnitude of Eq. (160), i.e.,

$$M_{90} = |FRF(\omega_n)| = \frac{1}{2\zeta} \quad (161)$$

For lightly damped systems with $\zeta \ll 1$, the M_{90} value as given by Eq. (161) may be very large and the phenomenon is similar to a resonance. This is why the frequency $\omega = \omega_n$ is also known as *phase resonance frequency*.

2.2.3.8 Physical response at $\omega = \omega_n$

It is of practical interest to calculate the magnitude of the physical response at $\omega = \omega_n$. To calculate it, recall the EOM Eq. (128),

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = \hat{F}e^{-i\omega t} \quad (128)$$

Assume steady-state harmonic motion with the excitation frequency ω , i.e.,

$$u(t) = \hat{u}e^{i\omega t} \quad (162)$$

Substitute Eq. (162) into Eq. (128) and cancel the common factor $e^{-i\omega t}$ to get

$$-\omega^2 m \hat{u} + i\omega c \hat{u} + k \hat{u} = \hat{F} \quad (163)$$

Solution of Eq. (163) gives

$$\hat{u} = \frac{\hat{F}}{-\omega^2 m + i\omega c + k} \quad (164)$$

At $\omega = \omega_n$, Eq. (164) yields

$$\hat{u}_{90} = \frac{\hat{F}}{-\omega_n^2 m + i\omega_n c + k} \quad (165)$$

where the notation u_{90} was used to signify that, at $\omega = \omega_n$, the phase is -90° . Recall Eq. (84) that defines the natural frequency ω_n in terms of stiffness k and mass m and rearrange as

$$\omega_n^2 m = k \quad (166)$$

Substitution of Eq. (166) into Eq. (165) gives

$$\hat{u}_{90} = \frac{\hat{F}}{\cancel{k} + i\omega_n c + \cancel{k}} = \frac{\hat{F}}{i c \omega_n} \quad (167)$$

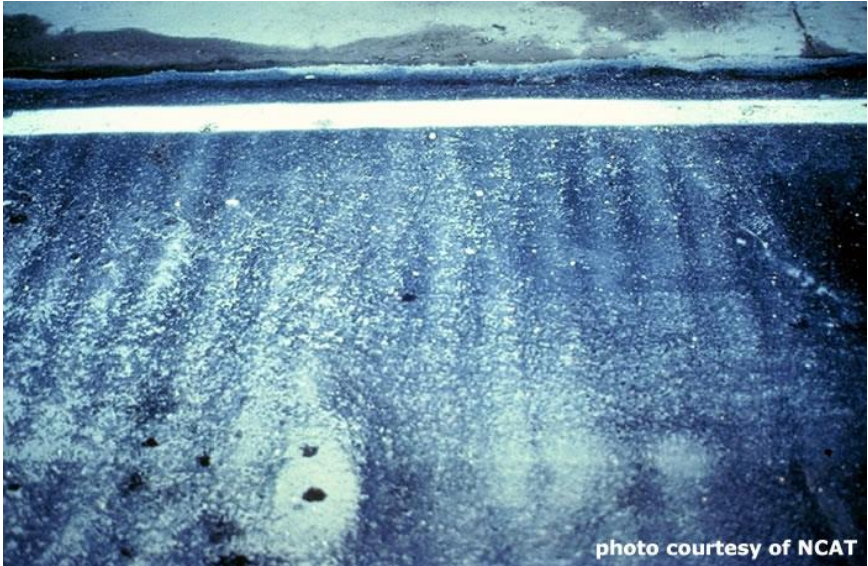
The magnitude of Eq. (167) is

$$|\hat{u}_{90}| = \frac{\hat{F}}{c \omega_n} \quad (168)$$

Substitution of Eq. (167) into Eq. (162) gives the response $u_{90}(t)$ at phase resonance as

$$u_{90}(t) = \hat{u}e^{i\omega t} \quad (169)$$

1 CAR RIDING ON A WAVY ROAD



<https://pavementinteractive.org/reference-desk/pavement-management/pavement-distresses/surface-waves/>

https://ftp.dot.state.tx.us/pub/txdot-info/library/pubs/bus/bridge/span_tables_i_girders.pdf

- Short waves (also called ripples or auger shadows) are about 0.3–0.9 m (1–3 ft) apart with 0.45–0.60 m (1.5–2 ft) being the most common separation.
- Long waves are farther apart than short waves. Can be in excess of 30 m (100 ft) apart.
- Long waves of ~ 100 ft may also appear on multispan bridges.

Consider a car driving over a wavy road. The car's velocity is v . The car can be modeled as a spring-mass-damper system with mass m , stiffness k , damping c (Figure 1). The road surface is described by a function $u_b(x)$. While traveling over the wavy road, the car wheel performs a vertical motion $u_w(t)$.

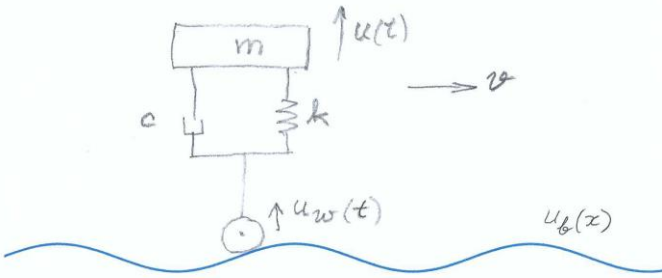


Figure 1 Car riding on a wavy road

The analysis of this problem involves several steps:

- differential equation development
- solution of the differential equation
- identification of salient features in the solution
- guidelines for successfully negotiating the wavy road

1.1 WHEEL MOTION IN TERMS OF ROAD PROFILE

We start by stating the generic expressions for the road profile $u_b(x)$ and the wheel motion $u_w(t)$, i.e.,

$$u_b(x) \quad (\text{road profile}) \quad (1)$$

$$u_w(t) \quad (\text{wheel motion}) \quad (2)$$

The first step in the modeling process is to establish a relationship between the road profile $u_b(x)$ and the vertical motion $u_w(t)$ of the wheel. As the car rides on the wavy surface, the wheel goes up and down. At time t , the wheel vertical position will be equal to the height of the road surface at that instance.

$$u_w(t) = u_b(x) \quad (\text{wheel follows the road}) \quad (3)$$

But one needs to know the value of x at time t , i.e., the function $x(t)$, in order to convert time t to horizontal position x along the hump. This is achieved using the car velocity v , i.e.,

$$x(t) = vt \quad (4)$$

Now, one substitutes Eq. (4) into Eq. (3) and finds vertical position of the wheel at time t , i.e.,

$$u_w(t) = u_b(vt) \quad (5)$$

1.2 FREE BODY DIAGRAM

The free body diagram (FBD) is shown in Figure 2. The vertical force F_y is due to the spring and damper reactions to being compressed as the wheel is moving up. However, the mass also goes up as the spring and damper push up onto it. The net compression of the spring is calculated as the difference between the wheel motion and the mass motion, i.e.,

$$\Delta u = u_w - u \quad (6)$$

Eq. (6) is used to calculate the spring compression force as

$$F_k = k\Delta u = k(u_w - u) \quad (\text{spring compression force}) \quad (7)$$

Similarly, the damper compression force is calculated as

$$F_c = c \Delta \dot{u} = c(\dot{u}_w - \dot{u}) \quad (\text{damper compression force}) \quad (8)$$

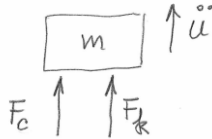


Figure 2 Free body diagram (FBD)

The net vertical force acting on the mass is

$$F_y = F_c + F_k = c(\dot{u}_w - \dot{u}) + k(u_w - u) \quad (9)$$

Rearrangement of Eq. (9) into mass and wheel parts yields

$$F_y = -(c\dot{u} + ku) + (c\dot{u}_w + ku_w) \quad (\text{net vertical force}) \quad (10)$$

1.3 NEWTON'S LAW OF MOTION

Newton's law of motion (NLM) is written as

$$F_y = m\ddot{u} \quad (\text{NLM}) \quad (11)$$

Substitution of Eq. (7) into Eq. (11) yields

$$-(c\dot{u} + ku) + (c\dot{u}_w + ku_w) = m\ddot{u} \quad (12)$$

1.4 EQUATION OF MOTION

NLM equation Eq. (12) can be rearranged to yield the equation of motion (EOM), i.e.,

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = c\dot{u}_w(t) + ku_w(t) \quad (\text{EOM}) \quad (13)$$

EOM given by Eq. (14) is a linear second order ordinary differential equation (ODE) with nonzero right hand side (RHS).

1.5 NORMALIZED EOM

Division of Eq. (13) by m yields the normalized EOM, i.e.,

$$\ddot{u}(t) + 2\zeta\omega_n\dot{u}(t) + \omega_n^2u(t) = 2\zeta\omega_n\dot{u}_w(t) + \omega_n^2u_w(t) \quad (\text{EOM}) \quad (14)$$

where, as defined in the analysis of spring-mass-damper systems,

$$\omega_n = \sqrt{k/m} \quad (\text{natural frequency}) \quad (15)$$

$$k = m\omega_n^2 \quad (\text{physical stiffness}) \quad (16)$$

$$\zeta = c/c_{cr} \quad (\text{damping ratio}) \quad (17)$$

$$c_{cr} = 2\sqrt{mk} \quad (\text{critical damping}) \quad (18)$$

$$c = 2\zeta\omega_n \quad (\text{physical damping}) \quad (19)$$

1.6 SOLUTION OF A NONHOMOGENEOUS LINEAR ODE

When the RHS of a linear ODE is zero, then the ODE is called homogeneous. If the RHS is nonzero, then the ODE is called nonhomogeneous. Our Eq. (14) is nonhomogeneous. The general solution of nonhomogeneous linear ODEs consists of the superposition of the *complementary solution* $u_c(t)$ that satisfies the homogenous ODE and a *particular solution* $u_p(t)$ that satisfies the nonhomogeneous ODE, i.e.,

$$u(t) = u_c(t) + u_p(t) \quad (\text{general solution}) \quad (20)$$

1.7 COMPLEMENTARY SOLUTION

The complementary solution $u_c(t)$ satisfies the homogeneous ODE, i.e.,

$$\ddot{u}_c(t) + 2\zeta\omega_n\dot{u}_c(t) + \omega_n^2u_c(t) = 0 \quad (\text{homogeneous ODE}) \quad (21)$$

The solution of Eq. (21) is known from the free vibration analysis of mass-spring-damper system and has the form

$$u_c(t) = Ce^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \quad (\text{complementary solution}) \quad (22)$$

where ω_d is the damped frequency given by

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \quad (\text{damped frequency}) \quad (23)$$

1.8 PARTICULAR SOLUTION

The particular solution $u_p(t)$ is a function that satisfies the nonhomogeneous equation, i.e.,

$$\ddot{u}_p(t) + 2\zeta\omega_n\dot{u}_p(t) + \omega_n^2 u_p(t) = 2\zeta\omega_n\dot{u}_w(t) + \omega_n^2 u_w(t) \quad (24)$$

In order to solve Eq. (24) one needs to know the actual expression of $u_w(t)$. In view of the relation $u_w(t) = u_b(vt)$ given by Eq. (5), this means that one needs to know the expression of the road profile $u_b(x)$. Once the road surface profile is specified one can proceed and solve Eq. (24) to find $u_p(t)$.

1.9 WAVY ROAD PROFILE

We want to have a hump profile that permits a shockless smooth transition of the car wheel from a flat road surface to the hump surface, i.e., a function that has the value and the tangent both equal to zero at the beginning and the end of the hump.

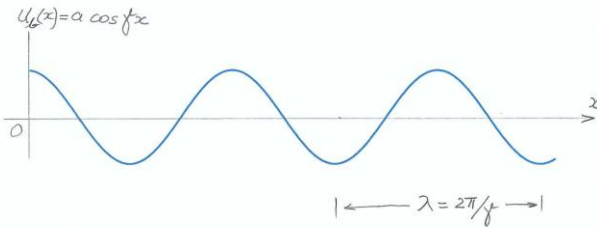


Figure 3 wavy road profile $u_b(x) = a \cos \gamma x$

A function that describes the wavy road profile is the cosine function (Figure 3) with the mathematical representation

$$u_b(x) = a \cos \gamma x \quad (\text{wavy road surface}) \quad (25)$$

where γ is the wavy-road wavenumber. The wavy-road wavenumber can be calculated from the wavy road wavelength λ which is the spacing between consecutive road-surface crests (Figure 3). The wavy-road wavenumber γ can be calculated in terms of wavy-road wavelength λ as

$$\gamma = 2\pi/\lambda \quad (\text{wavy-road wavenumber}) \quad (26)$$

1.10 WHEEL MOTION OVER WAVY ROAD

Substitution of Eq. (4) $x(t) = vt$ into Eq. (25) gives

$$u_b(vt) = a \cos \gamma vt \quad (27)$$

Recall Eq. (5) $u_w(t) = u_b(vt)$ and get the wheel motion $u_w(t)$ as

$$u_w(t) = a \cos \gamma vt \quad (28)$$

The expression in Eq. (28) can be brought to the simpler form by defining the *wavy road frequency* ω as

$$\omega = \gamma v \quad (\text{wavy road frequency } \omega, \text{ rad/s}) \quad (29)$$

Note that the wavy road frequency ω is the frequency of the excitation applied by the wavy road to the car through the wheel. It is **not** the natural frequency of the car ω_n . However, when the wavy road frequency approaches the car frequency, the car response increases significantly. Substitution of Eq. (29) into Eq. (28) yields

$$u_w(t) = a \cos \omega t \quad (30)$$

The wheel motion expression of Eq. (30) will go into Eq. (24) to allow us to solve for the particular solution $u_p(t)$.

The wavy-road frequency ω can be also calculated from the car speed v and the wavy-road wavelength λ which is the spacing between consecutive road-surface crests. To this purpose, substitute Eq. (26) $\gamma = 2\pi/\lambda$ into Eq. (29) to get

$$\omega = 2\pi v/\lambda \quad (\text{wavy road frequency } \omega, \text{ rad/s}) \quad (31)$$

or, in Hz,

$$f = v/\lambda \quad (\text{wavy road frequency } f, \text{ Hz}) \quad (32)$$

1.11 PARTICULAR SOLUTION FOR WAVY ROAD SURFACE

Substitution of Eq. (30) into Eq. (24) would yield the nonhomogeneous ODE that needs to be solved for $u_p(t)$. First, we need to simplify the mathematical handling by invoking Euler's identity that connects a complex exponential function $e^{i\alpha}$ to the corresponding trigonometric functions $\cos\alpha$ and $\sin\alpha$, i.e.,

$$e^{i\alpha} = \cos\alpha + i\sin\alpha \quad (33)$$

Eq. (33) can be resolved to express the trigonometric function $\cos\alpha$ as the real part of the complex exponential function $e^{i\alpha}$, i.e.,

$$e^{i\alpha} = \cos\alpha = \text{Re}(e^{i\alpha}) \quad (34)$$

With the convention of Eq. (34) and the tacit understanding that when we write $e^{i\alpha}$ we actually mean its real part $\cos\alpha$, one can write the wheel motion of Eq. (30) as

$$u_w(t) = a e^{i\omega t} \quad (35)$$

The time derivative of Eq. (35) is simply

$$\dot{u}_w(t) = a i\omega e^{i\omega t} \quad (36)$$

Substitution of Eqs. (35), (36) into Eq. (24) yields

$$\begin{aligned} \ddot{u}_p(t) + 2\zeta\omega_n\dot{u}_p(t) + \omega_n^2 u_p(t) \\ = 2\zeta\omega_n a i\omega e^{i\omega t} + \omega_n^2 a e^{i\omega t} \end{aligned} \quad (37)$$

Factoring out of $a e^{i\omega t}$ yields

$$\ddot{u}_p(t) + 2\zeta\omega_n\dot{u}_p(t) + \omega_n^2 u_p(t) = (\omega_n^2 + i2\zeta\omega_n\omega) a e^{i\omega t} \quad (38)$$

The particular solution is sought in the form of the RHS, i.e.,

$$u_p(t) = D a e^{i\omega t} \quad D \in \mathbb{C} \quad (39)$$

where D is a complex constant that needs to be determined. Differentiation of Eq. (39) yields

$$\dot{u}_p(t) = D(i\omega) a e^{i\omega t} \quad (40)$$

$$\ddot{u}_p(t) = D(i\omega)^2 a e^{i\omega t} = -D\omega^2 a e^{i\omega t} \quad (41)$$

Substitution of Eqs. (39), (40), (41) into Eq. (38) and division by $a e^{i\omega t}$ on both sides yields

$$-D\omega^2 + i2\zeta\omega_n D + \omega_n^2 D = \omega_n^2 + i2\zeta\omega_n\omega \quad (42)$$

Eq. (42) is solved for D to get

$$D = \frac{\omega_n^2 + i2\zeta\omega_n\omega}{-\omega^2 + i2\zeta\omega_n\omega + \omega_n^2} \quad (43)$$

Substitution of Eq. (43) into Eq. (39) yields the final form of the particular solution as

$$u_p(t) = \frac{\omega_n^2 + i2\zeta\omega_n\omega}{-\omega^2 + i2\zeta\omega_n\omega + \omega_n^2} a e^{i\omega t} \quad (\text{particular solution}) \quad (44)$$

1.12 FREQUENCY RESPONSE FUNCTION FRF

Define the frequency response function $FRF(\omega; \omega_n)$ as

$$FRF(\omega; \omega_n) = \frac{\omega_n^2 + i2\zeta\omega_n\omega}{-\omega^2 + i2\zeta\omega_n\omega + \omega_n^2} \quad (45)$$

or, in terms of Hz frequencies f and f_n ,

$$FRF(f; f_n) = \frac{f_n^2 + i2\zeta f_n f}{-f^2 + i2\zeta f_n f + f_n^2} \quad (\text{freq. response function}) \quad (46)$$

where f is the independent variable and f_n is a parameter. As the hump frequency f approaches the car natural frequency f_n , the denominator in Eq. (46) becomes very small and FRF becomes very large as illustrated in Figure 4. It is apparent that for safe driving the hump frequency f must be kept below the car's natural frequency f_n .

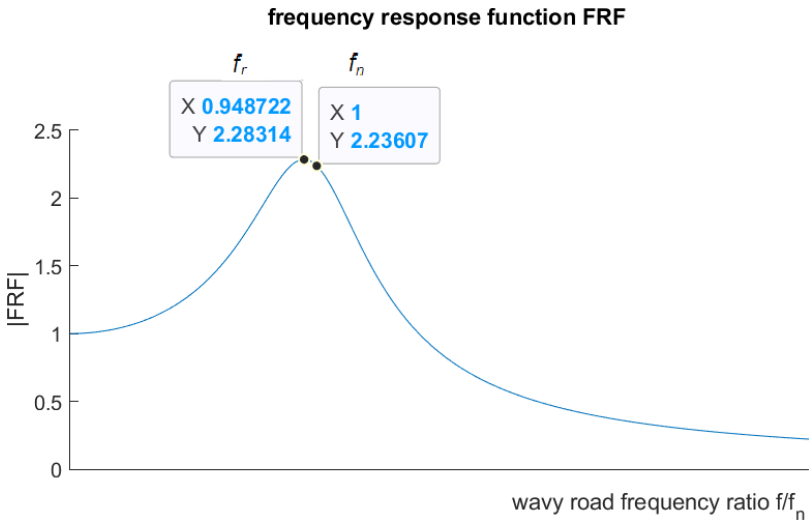


Figure 4 Frequency response function vs. wavy road frequency f

The frequency response function FRF can be used to write the compactly the particular solution $u_p(t)$. Substitute Eq. (45) into Eq. (44) to get

$$u_p(t) = FRF(\omega; \omega_n) a e^{i\omega t} \quad (47)$$

1.13 COMPLETE SOLUTION

Substitution of Eqs. (22), (47) into Eq. (20) yields the complete solution as

$$u(t) = Ce^{-\zeta\omega_n t} \sin(\omega_d t + \phi) + FRF(\omega; \omega_n) a e^{i\omega t} \quad (48)$$

1.14 STEADY-STATE RESPONSE

The $Ce^{-\zeta\omega_n t} \sin(\omega_d t + \phi)$ part of Eq. (48) represents the *transient response* whereas the FRF part represents the *steady-state response*. The transient response will quickly vanish due to damping and, after some time, only the steady-state response will persist. For this reason, the subsequent analysis will only deal with the steady-state response given by

$$u(t) \approx u_{ss}(t) = a FRF(\omega; \omega_n) e^{i\omega t} \quad (\text{steady-state response}) \quad (49)$$

or

$$u(t) = \hat{u}(f; f_n) e^{i\omega t} \quad (50)$$

where $\hat{u}$ is the amplitude of car up and down motion given by

$$\hat{u}(f; f_n) = a FRF(f; f_n) \quad (\text{amplitude of car motion}) \quad (51)$$

1.15 MATCHED-FREQUENCY FRF

The car response depends on the interplay of wavy road frequency f and the car natural frequency f_n through the frequency response function $FRF(f; f_n)$ given by Eq. (46). At matched-frequency condition, the car and road frequencies coincide, i.e.,

$$f = f_n \quad (\text{critical condition}) \quad (52)$$

This condition corresponds to the phase resonance condition discussed in connection with harmonic response, but the FRF phase is not -90° . When the two frequencies are matched $f = f_n$, the FRF of Eq. (46) simplifies as

$$FRF(f; f_n) \Big|_{f=f_n} = \frac{f_n^2 + i2\zeta f_n f}{-f^2 + i2\zeta f_n f + f_n^2} \Big|_{f=f_n} \quad \left(\begin{array}{c} \text{matched-freq.} \\ FRF \end{array} \right) \quad (53)$$

$$FRF_n = \frac{1 + i2\zeta}{i2\zeta} = 1 - \frac{i}{2\zeta}$$

Figure 4 shows the matched-frequency location f_n and the corresponding $|FRF_n| = \text{abs}(FRF_n)$. Note that the matched frequency condition occurs after the peak of the $|FRF|$ plot.

1.16 MAXIMUM RESPONSE

Maximum car response is achieved when the FRF amplitude peaks. We define the resonance frequency f_r as the excitation frequency at which $|FRF|$ curve peaks. In lieu of a closed form solution, the f_r frequency can be determined graphically with a datatip on the $|FRF|$ plot and finding the peak value $|FRF_{max}|$. Figure 4 shows the $|FRF|$ peak location f_r and the corresponding peak value $|FRF_{max}|$. Note that the peak condition $|FRF_{max}|$ occurs before the matched frequency condition $|FRF_n|$.

1.17 FRF AS FUNCTION OF CAR SPEED

For any wavy-road frequency f , one can use Eq. (32) $f=v/\lambda$ to calculate the corresponding car speed v as

$$v=f\lambda \quad (\text{car speed for given } f) \quad (54)$$

Thus, the frequency response function can be plotted vs. car speed v as illustrated in Figure 5.

1.18 MATCHED-FREQUENCY CAR SPEED

The matched-frequency car speed v_n is calculated from Eq. (32) $f=v/\lambda$. Make $f=f_n$ and $v=v_n$ and then solve for v to get

$$v_n=f_n\lambda \quad (\text{matched-frequency car speed}) \quad (55)$$

where λ is the wavy road wavelength, i.e., the spacing between consecutive road-surface crests and f_n is the car natural frequency. One can use a datatip to locate v_n on the $|FRF|$ - v plot and verify that $|FRF_n|$ value is indeed that predicted by Eq. (53) as illustrated in Figure 5.

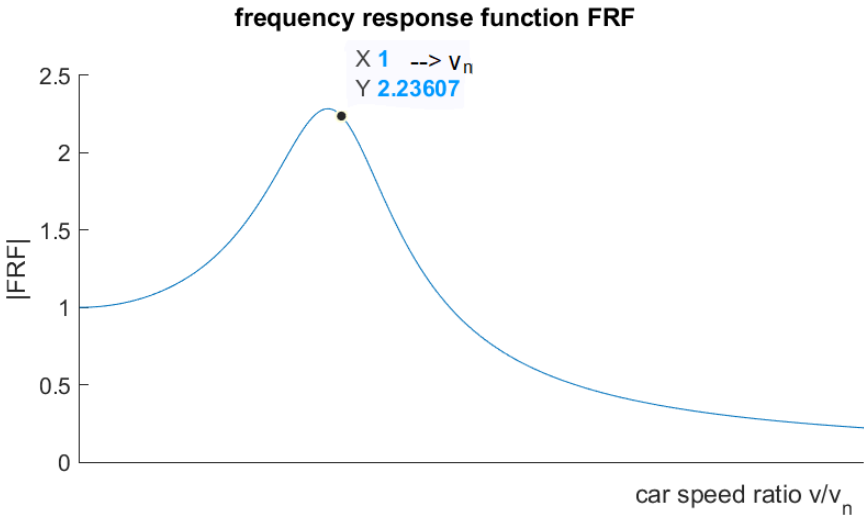


Figure 5 Frequency response function vs. car speed v

1.19 CRITICAL SPEED

The car speed that produces a peak of the $|FRF|$ plot is called the critical speed v_{cr} . The critical speed v_{cr} can be located with a datatip on the $|FRF|$ vs. car speed v plot. The critical speed v_{cr} and the resonance wavy-road frequency f_r should verify the relationship

$$v_{cr} = f_r \lambda \quad (\text{critical car speed} - f_r \text{ relationship}) \quad (56)$$

Eq. (56) is derived from Eq. (32) $f = v/\lambda$ because both f_r and v_{cr} represent that same physical phenomenon, viz. the peak of the $|FRF|$ plot.

As the critical speed v_{cr} is approached, the car response increases amplitude at a high rate. It is therefore advisable to drive the car slower and not to exceed $v_{\max}$ which gives an acceptable response amplitude.

1.20 NUMERICAL EXAMPLE

Consider a car traveling on a wavy road. The road surface undulates with an amplitude $a=4$ -in and a wavelength $\lambda=100$ ft that follows a cosine shape. The car is modeled as a spring-mass-damper (Figure 1) with natural frequency $f_n=0.95$ Hz and damping ratio $\zeta=25\%$. The maximum car speed is $v_{max}=120$ mph. Do the following:

- (1) Display input data
- (2) Calculate wavy road wavenumber γ
- (3) Plot the wavy road profile $u_b(x)$ for $N_\lambda=2$ wavelengths
- (4) calculate wavy-road frequency f_{max} for a maximum car speed of v_{max} and define the frequency range $f=\text{linspace}(0, f_{max}, N_f)$ with $N_f=2000$
- (5) plot $|FRF(f; f_n)|$ for. Use datatips on the plot to locate the peak response and the matched-frequency response.
- (6) Calculate the matched-frequency response value $|FRF_n|$. Compare with the matched-frequency readings at item (5)
- (7) Calculate the car speed v_n needed to make wavy-road frequency f match the car natural frequency f_n
- (8) Use the conversion $f=v/\lambda$ to plot $|FRF(v; f_n)|$. Use datatips on the plot to find:
 - (a) v_n and $|FRF_n|$ and discuss them in comparison with results of items (6), (7)
 - (b) v_{cr} and $|FRF_{max}|$
 - (c) the limit speed v_{limit} for which the motion amplification is limited to $|FRF_{lim}|=1.5$
- (9) Find the road wavelength λ_{new} that will make $v_{cr}=30$ mph
- (10) Repeat items (5) through (8) for the case of a car with worn-out shocks that can only give $\zeta=10\%$ and comment on the comparison between the two cases

Solution

Example: car on wavy road

(1) input data

- amplitude of road undulations $a=4$ in
- wavy road wavelength $L=100$ ft
- car natural freq $f_n=0.95$ Hz
- car damping ratio $\zeta=25\%$
- max car speed $v_{\max}=120$ mph

(2) wavy road wavenumber $g=0.20614$ rad/m

(3) see plot of wavy road profile

(4) $f_{\max}=1.76$ Hz

(5) please see plot of FRF($f;f_n$)

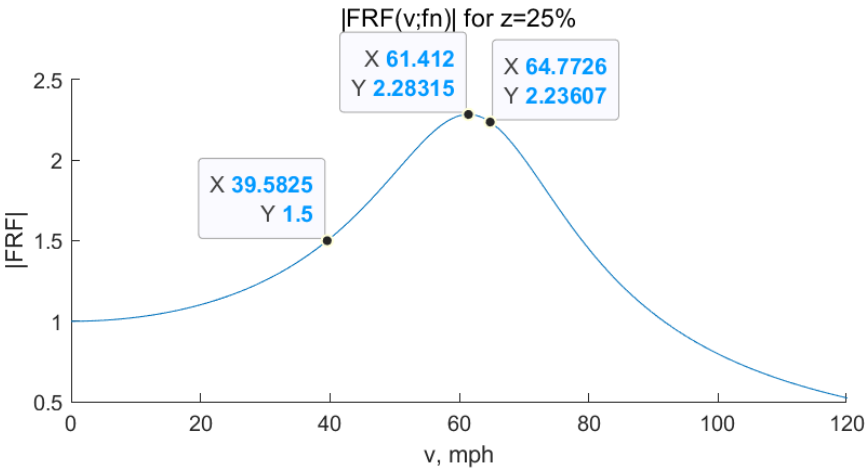
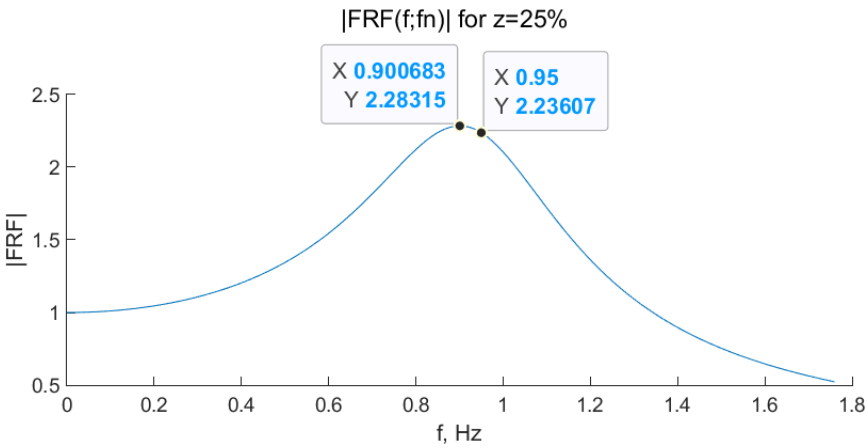
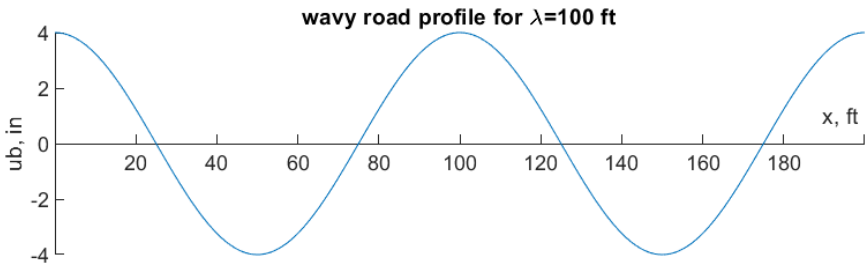
(6) for $f=f_n$, $FRF_n=2.2361$

(7) $v_n=64.7727$ mph

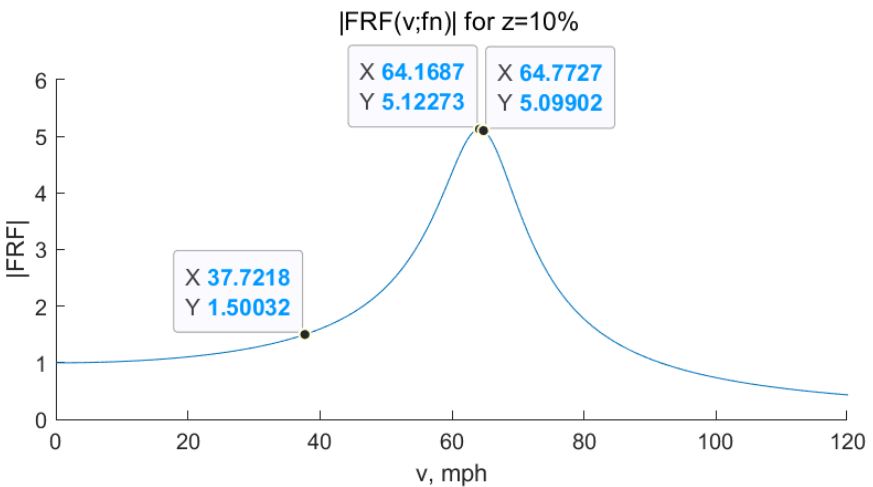
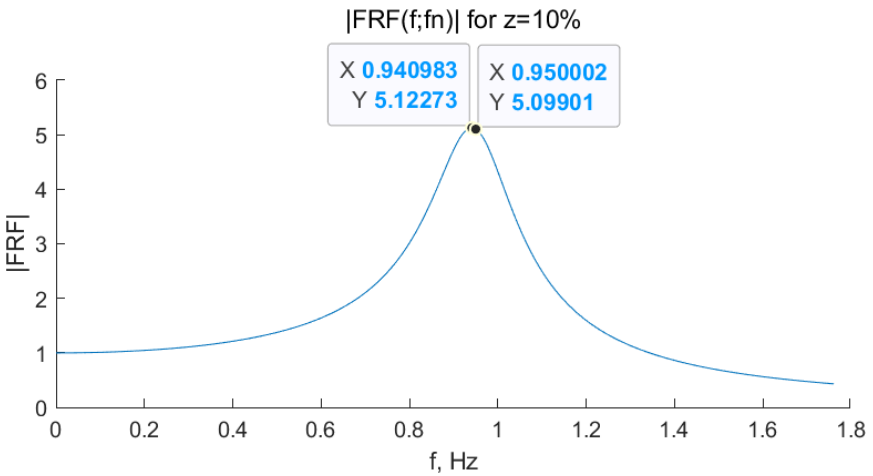
(8) please see plot of FRF($v;f_n$)

(9) new $v_n=30$ mph

$L_{\text{new}}=46.3158$ ft



- (10) repeat for worn-out shocks
- car damping ratio $\zeta=10\%$
- (5) please see plot of $\text{FRF}(f;f_n)$
- (6) for $f=f_n$, $\text{FRF}_n=5.099$
- (7) $v_n=64.7727$ mph
- (8) please see plot of $\text{FRF}(v;f_n)$



20230924

WAVES

1
20230924

Waves

- Disturbance waves



- Continuous waves



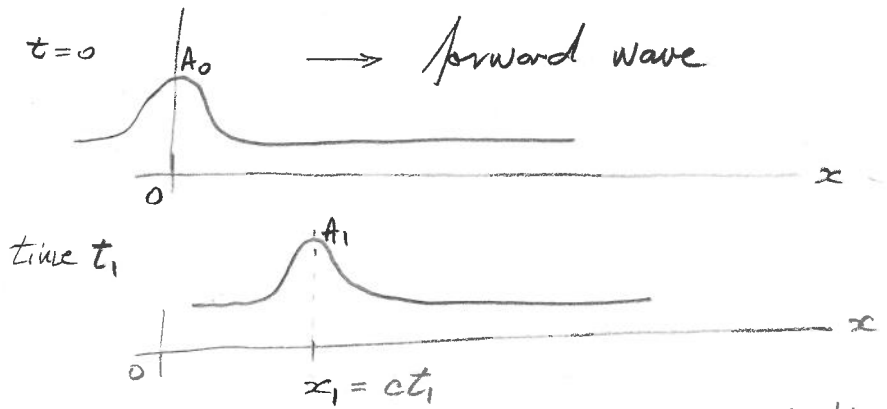
Examples:

- Sound waves
- Ocean waves
- bow waves from a boat
- pool waves from a splash or a rock
- radio waves
- gravitational waves
- rope waves
- car pile up wave
- aerospace shock wave: "sonic boom"

Waves carry information/energy

Waves do NOT carry matter

Wave speed.



The wave has traveled a distance x_1 in time t_1 .
the wave speed is c such that $x_1 = ct_1$.

$$u(x,t) = f(x-ct) \text{ forward wave} \quad (1)$$

The shape of the wave is described by the function $f(z)$ where $z(x,t) = x-ct$ (wave phase)

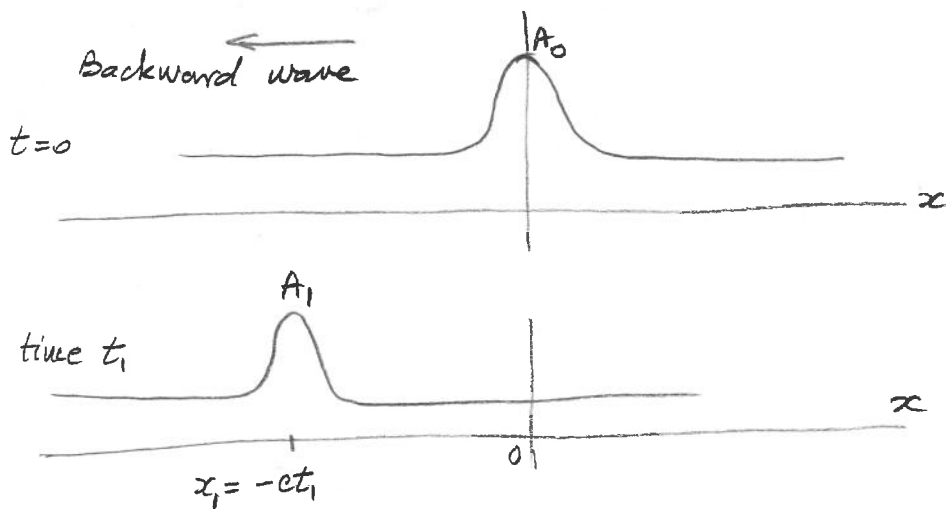


The crest of the wave, point A, is located at $z=0$. One can trace the movement of the wave by tracing the location of A.
The condition $z(x,t)=0$ yields $x=ct$

$$x-ct=0 \rightarrow x=ct \quad (2)$$

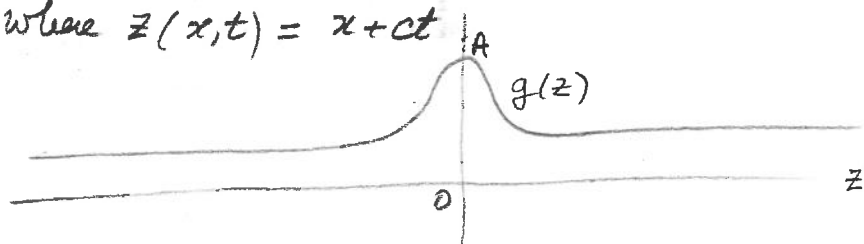
3
20230924

Backward Waves



$-u(x,t) = g(x+ct)$ backward wave (3).

The shape of the wave at $t=0$ is $g(z)$,
where $z(x,t) = x+ct$



The location of the crest A at time t_1 is
found by imposing $z(x_1, t_1) = 0$.

$$z(x_1, t_1) = 0 \rightarrow x_1 + ct_1 = 0$$

$$x_1 = -ct_1$$

3a

Equivalent expressions

$$u(x, t) = f(x - ct) \quad (\text{space emphasis})$$

where ct is the distance traveled

$f(z)$ is the ^{by the wave} shape of the wave at $t=0$

$$u(x, t) = f\left(\frac{x}{c} - t\right) \quad (\text{time emphasis})$$

where $\frac{x}{c}$ is the time it takes the

wave to travel

$f(z)$ is the motion of a particle at $x=0$

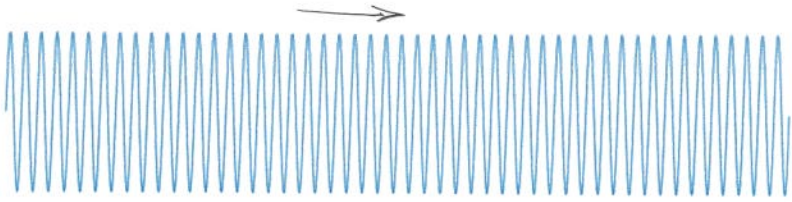
For backward waves:

$$-u(x, t) = g(x + ct) \quad (\text{space emphasis})$$

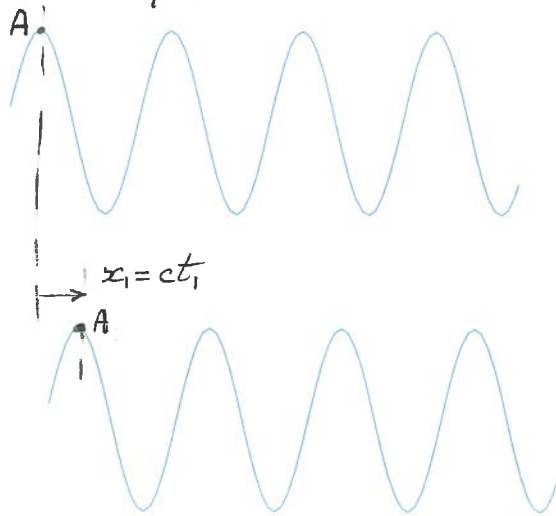
$$-u(x, t) = g\left(\frac{x}{c} + t\right) \quad (\text{time emphasis})$$

4

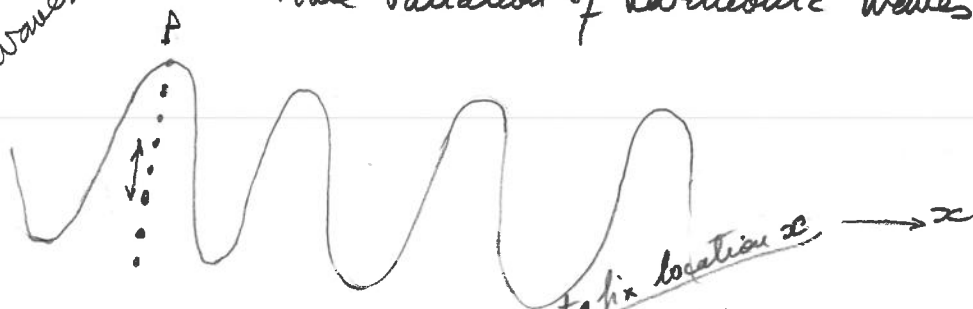
Continuous Harmonic Waves (Oscillation propagating in space)



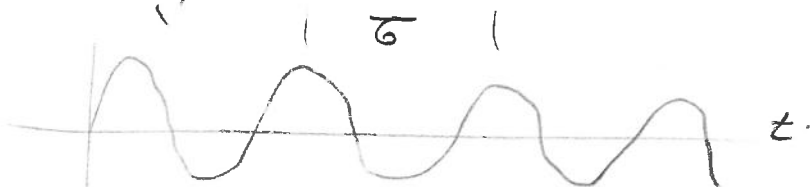
Propagation of harmonic waves.



Time variation of harmonic waves



If one observes a particle at a fix location x over time, one will see it moving up and down in an oscillatory manner.



This time oscillation has a period τ seconds.

The frequency of oscillation f is the inverse of the period τ

$$f = \frac{1}{\tau}$$

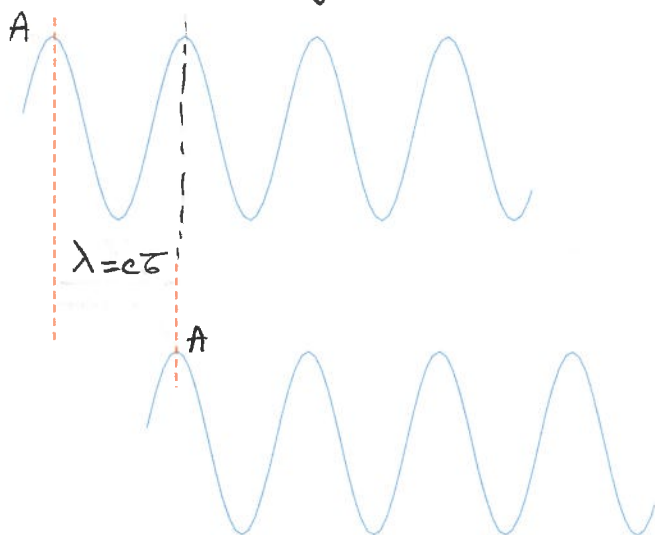
The units of frequency are Hz.

The circular frequency ω is given by

$$\omega = 2\pi f.$$

$$\cos 2\pi f t = \cos \omega t$$

Wavelength = distance traveled by the wave during a period of oscillation T



The wave number γ is defined as the ratio between circular frequency ω and wavespeed c :

$\gamma = \frac{\omega}{c}$ (wavenumber)
The wavenumber γ is related to wavelength λ as

$$\gamma = \frac{2\pi}{\lambda} \quad \text{rad/m}$$

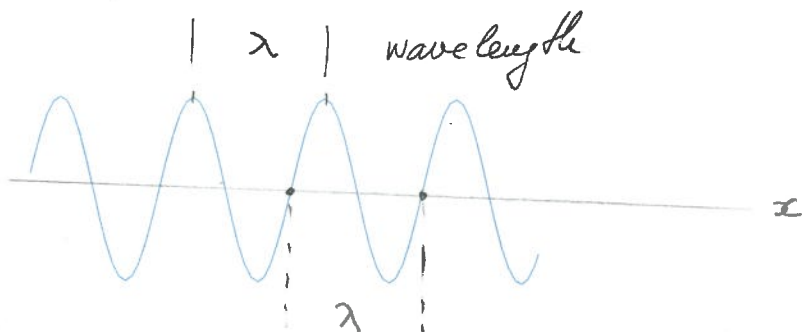
Proof:

$$\lambda = cT = c \frac{1}{f} = 2\pi c \frac{1}{2\pi f} = 2\pi \frac{c}{\omega} = \frac{2\pi}{\gamma}$$

$$\gamma = \frac{2\pi}{\lambda} = \frac{2\pi}{cT} = \frac{2\pi}{c \frac{1}{f}} = \frac{2\pi f}{c} = \frac{\omega}{c}$$

6a

Wavelength measurement.



wavelength = distance between two consecutive peaks
= distance between two consecutive zero crossings with same slope

$$u(x, t) = \cos(kx - \omega t)$$

$$\begin{aligned} z(x, t) &= (kx - \omega t) = \left(\frac{\omega}{c} x - \omega t\right) = \omega \left(\frac{x}{c} - t\right) \\ (\text{phase}) & \\ &= \frac{\omega}{c} (x - ct) \end{aligned}$$

$$(kx - \omega t) = \left(\frac{2\pi}{\lambda} x - 2\pi f t\right) = 2\pi \left(\frac{x}{\lambda} - \frac{t}{T}\right)$$

$$u(x, t) = \cos 2\pi \left(\frac{x}{\lambda} - \frac{t}{T}\right)$$

$$u(x, t) = \cos \frac{\omega}{c} (x - ct) \quad \text{forward wave}$$

$$\cos \omega \left(\frac{x}{c} - t\right) \quad \text{alternative form of forward wave}$$

$$u(x, t) = e^{i(kx - \omega t)}$$

complex-exponential representation of a continuous harmonic wave

Implied convention: take real part:

$$u(x, t) = \text{Re } e^{i(kx - \omega t)} = \cos(kx - \omega t)$$

STANDING WAVES

STANDING WAVES

The concept of *standing waves* bridges the gap between wave analysis and vibration analysis. The difference between vibration and wave propagation resides mainly in the presence or absence of boundaries. In a boundless medium, a disturbance source performing continuous oscillations will produce sustained harmonic waves continuously propagating outwards towards infinity. However, if the medium is bounded, then waves will be reflected by the boundary back towards the source and will interfere with the waves emanating from the source, resulting in a complicated process that may, under certain conditions, results in *standing waves*, or structural vibration.

1 STANDING WAVE FORMULA

Consider two identical continuous harmonic waves propagating in opposite directions, i.e.,

$$^+u(x, t) = \frac{1}{2} e^{i(\gamma x - \omega t)} \quad (\text{forward wave}) \quad (1)$$

$$^-u(x, t) = \frac{1}{2} e^{-i(\gamma x + \omega t)} \quad (\text{backward wave}) \quad (2)$$

The total motion is obtained via the superposition of these two waves, i.e.,

$$\begin{aligned} u(x, t) &= ^+u(x, t) + ^-u(x, t) = \frac{1}{2} e^{i(\gamma x - \omega t)} + \frac{1}{2} e^{-i(\gamma x + \omega t)} \\ &= \frac{1}{2} \left(e^{i\gamma x} + e^{-i\gamma x} \right) e^{-i\omega t} \end{aligned} \quad (3)$$

Recall Euler's formula $\cos \alpha = \frac{1}{2} \left(e^{i\alpha} + e^{-i\alpha} \right)$; hence, Eq. (3) becomes

$$u(x, t) = \left(\cos \gamma x \right) e^{-i\omega t} \quad (\text{standing wave}) \quad (4)$$

Equation (4) represents a *standing wave* of space-wise shape $\cos \gamma x$ and harmonic time-varying amplitude $e^{-i\omega t}$.

2 NODES OF A STANDING WAVE

The standing wave has specific *nodes* and *anti-nodes* (bows) that have space-wise fixed positions (Figure 1). As time progresses, the

amplitude of the bows increases and decreases but the nodes always remain fixed.

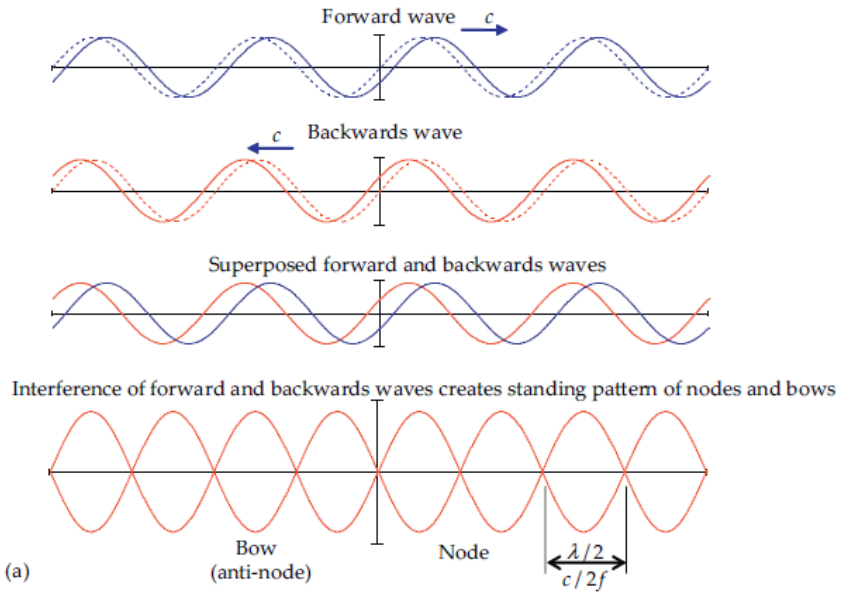


Figure 1 Standing waves: interference of forward and backward waves results in standing waves

The locations of the nodes, x_n , are given by the requirement that γx_n is an odd multiple of $\pi/2$ such that the cosine function vanishes, i.e.,

$$\gamma x_n = (2n-1)\pi/2 \quad n=1,2,3, \dots \quad (5)$$

Hence, the spacing of the nodes Δx is calculated as follows. First, solve Eq. (5) for x_n and write

$$x_n = (2n-1)\pi/2\gamma \quad n=1,2,3, \dots \quad (6)$$

Then, use Eq. (6) to calculate two consecutive x values, say $x_1 = \pi/2\gamma$ and $x_2 = 3\pi/2\gamma$. The difference between them is

$$\Delta x = x_2 - x_1 = 3\pi/2\gamma - \pi/2\gamma = \pi/\gamma \quad (7)$$

Eq. (7) shows that the nodes of standing waves are spaced at a pitch of $\Delta x = \pi/\gamma$. Recall the relationship between wavelength λ and wavenumber γ , i.e.,

$$\lambda = 2\pi/\gamma \quad (\text{wavelength-wavenumber relationship}) \quad (8)$$

In view of Eq. (8), the spacing between nodes becomes

$$\Delta x = \lambda/2 \quad (\text{node spacing}) \quad (9)$$

Equation (9) indicates that the nodes of a standing wave are spaced at half-wavelength intervals $\Delta x = \lambda/2$.

3 STANDING WAVE IN FINITE-LENGTH MEDIA

Standing waves can be created in finite-length media under certain conditions as discussed in later sections. For example, we will show that the differential equation for air vibrating in a tube has the same form as the equation of waves traveling in the tube. Similarly, the general solution for air vibrating in a tube to generate musical sounds is the same as that of a continuous harmonic wave propagating in the tube.

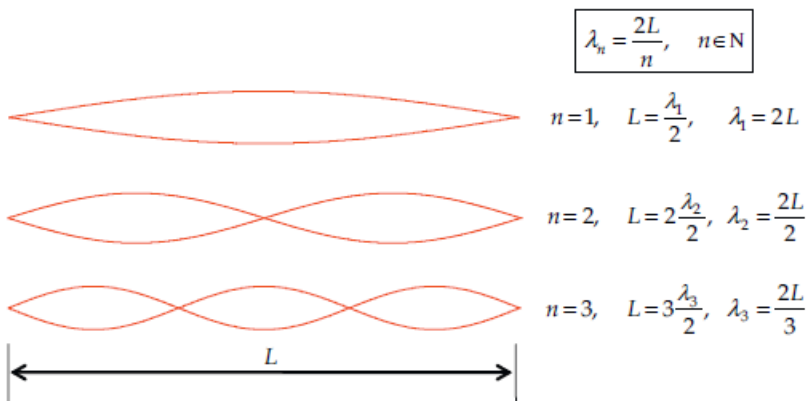


Figure 2 Standing waves aka vibration: multiple resonances of air vibrating in a tube of length L closed at both ends

PLANE

SPHERICAL

&

CIRCULAR

WAVEFRONTS.

5.9 PLANE, SPHERICAL, AND CIRCULAR WAVE FRONTS

In this section we will do a review of the main ways in which waves may propagate by looking at the shape of their wave fronts. We will investigate plane waves, spherical waves, and circular waves. Both generic waves and harmonic waves will be reviewed. Recall the wave equation

$$\nabla^2 \Phi(\vec{r}, t) = \frac{1}{c^2} \ddot{\Phi}(\vec{r}, t) \quad (\text{wave equation}) \quad (410)$$

where $\Phi(\vec{r}, t)$ is a generic disturbance that propagates in space as a wave.

5.9.1 Plane Waves

Plane waves are those having a planar wave front. Plane waves may be generated by a planar source (Figure 5.24a). All the particles in the wave front are contained in a plane with normal unit vector $\vec{n} = \{n_x, n_y, n_z\}$. The planar wave front moves along $\vec{n}$ with wave speed c . As the wave front propagates, it traces a path s (Figure 5.24b). The wave motion of the particles inside the planar wave front is self-similar and the Laplacian of Φ along the path s in the direction $\vec{n}$ is

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial s^2} \quad (411)$$

Substitution of Eq. (411) into Eq. (410) yields the wave equation for plane waves, i.e.,

$$\frac{\partial^2 \Phi}{\partial s^2} = \frac{1}{c^2} \frac{\partial^2 \Phi}{\partial t^2} \quad (\text{plane wave equation}) \quad (412)$$

5.9.1.1 Generic Plane Waves

The d'Alembert solution of Eq. (412) is similar to Eq. (20), i.e.,

$$\Phi(s, t) = f(s - ct) + g(s + ct) \quad (413)$$

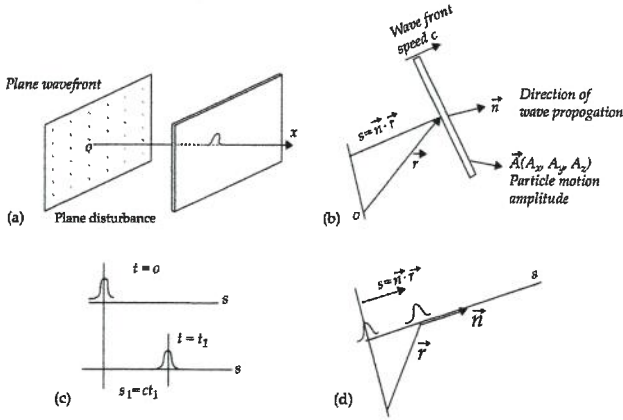


FIGURE 5.24 Plane waves: (a) planar disturbance generating a planar wave front; (b) the geometry of a planar wave front of normal $\vec{n}$; (c) generic wave shape; (d) wave propagation path s measured along $\vec{n}$.

We only retain the term $f(s - ct)$, which represents a forward-propagating wave (Figure 5.24c), i.e.,

$$\Phi(s, t) = f(s - ct) \quad (\text{generic plane wave}) \quad (414)$$

As indicated in Figure 5.24d, the path s is the projection of the position vector $\vec{r}$ onto the propagation direction $\vec{n}$, i.e.,

$$s = \vec{n} \cdot \vec{r}, \quad \vec{n} = \{n_x, n_y, n_z\} \quad (415)$$

Substitution of Eq. (415) into Eq. (414) yields

$$\Phi(\vec{r}, t) = f(\vec{n} \cdot \vec{r} - ct) = f(n_x x + n_y y + n_z z - ct) \quad (416)$$

The particle motion of a plane wave is given by

$$\vec{u}(\vec{r}, t) = \vec{A} f(\vec{n} \cdot \vec{r} - ct) = \vec{A} f(n_x x + n_y y + n_z z - ct) \quad (417)$$

where $\vec{u}$ is the particle motion and $\vec{A}$ is the motion amplitude. Both $\vec{u}$ and $\vec{A}$ have components in the x, y, z directions, i.e.,

$$\vec{u} = \{u_x, u_y, u_z\}, \quad \vec{A} = \{A_x, A_y, A_z\} \quad (418)$$

5.9.1.2 Harmonic Plane Waves

The separation of variables solution to Eq. (412) is similar to Eq. (59), i.e.,

$$\Phi(s, t) = B e^{-i(\gamma s + \omega t)} + C e^{i(\gamma s - \omega t)} \quad (419)$$

where $\gamma = \omega/c$ is the wave number in the propagation direction $\vec{n}$ along the path s . We only retain the term $e^{i(\gamma s - \omega t)}$, which represents a forward-propagating wave, i.e.,

$$\Phi(s, t) = C e^{i(\gamma s - \omega t)} \quad (\text{harmonic plane wave}) \quad (420)$$

Recall Eq. (415) and write it explicitly as

$$s = \vec{n} \cdot \vec{r} = n_x x + n_y y + n_z z \quad (421)$$

Substitution of Eq. (421) into the exponent of Eq. (420) gives

$$\gamma s = \gamma(\vec{n} \cdot \vec{r}) = \gamma(n_x x + n_y y + n_z z) = \gamma n_x x + \gamma n_y y + \gamma n_z z \quad (422)$$

Define the wave number vector $\vec{\gamma}$ in the direction $\vec{n}$ as

$$\vec{\gamma} = \{\xi, \eta, \zeta\} \quad (423)$$

The components ξ, η, ζ of the wave number vector $\vec{\gamma}$ are

$$\xi = \gamma n_x, \quad \eta = \gamma n_y, \quad \zeta = \gamma n_z \quad (424)$$

Combination of Eqs. (422), (423) yields

$$\gamma s = \vec{\gamma} \cdot \vec{r} \quad (425)$$

Substitution of Eq. (425) into Eq. (420) gives

$$\Phi(\vec{r}, t) = C e^{i(\vec{\gamma} \cdot \vec{r} - \omega t)} = C e^{i(\xi x + \eta y + \zeta z - \omega t)} \quad (\text{harmonic plane wave}) \quad (426)$$

The particle motion in a harmonic plane wave has the general expression

$$\vec{u}(\vec{r}, t) = \vec{A} e^{i(\vec{\gamma} \cdot \vec{r} - \omega t)} = \vec{A} e^{i(\xi x + \eta y + \zeta z - \omega t)} \quad (\text{harmonic plane wave}) \quad (427)$$

where $\vec{u}$ is the particle motion and $\vec{A}$ is the motion amplitude as given by Eq. (418).

5.9.2 Spherical Waves

Spherical waves are those for which the wave front is the surface of a sphere. Spherical waves emanate from a point source in the 3-D space (Figure 5.25a). Spherical waves

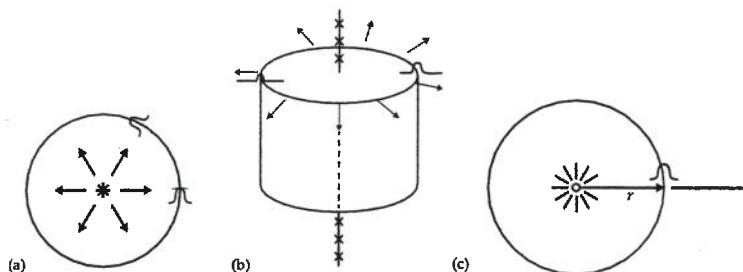


FIGURE 5.25 (a) spherical waves in 3-D; (b) cylindrical waves in 3-D; (c) circular waves in 2-D.

propagate with spherical symmetry, i.e., they only depend on the radial distance r from the point source situated at the origin O , and Eq. (410) depends only on the scalar r , i.e.,

$$\nabla^2 \Phi(r, t) = \frac{1}{c^2} \ddot{\Phi}(r, t) \quad (\text{wave equation}) \quad (428)$$

Recall the Laplacian of Φ in spherical coordinates, i.e.,

$$\nabla^2 \Phi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \Phi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \Phi}{\partial \phi^2} \quad (\text{Laplacian in spherical coord.}) \quad (429)$$

For spherical symmetry, Eq. (429) depends on r only, i.e.,

$$\nabla^2 \Phi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi}{\partial r} \right) \quad (\text{spherical symmetry}) \quad (430)$$

Eq. (430) can be written as

$$\nabla^2 \Phi = \frac{1}{r} \frac{\partial^2 (r\Phi)}{\partial r^2} \quad (431)$$

Proof of Eq. (431): expand the left-hand side of Eq. (431) starting from Eq. (430), i.e.,

$$\nabla^2 \Phi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \Phi}{\partial r} \right) = \frac{1}{r^2} \left(2r \frac{\partial \Phi}{\partial r} + r^2 \frac{\partial^2 \Phi}{\partial r^2} \right) = \frac{1}{r} \left(2 \frac{\partial \Phi}{\partial r} + r \frac{\partial^2 \Phi}{\partial r^2} \right) \quad (432)$$

Expand the right-hand side of Eq. (431), i.e.,

$$\begin{aligned} \frac{1}{r} \frac{\partial^2(r\Phi)}{\partial r^2} &= \frac{1}{r} \frac{\partial}{\partial r} \left[\frac{\partial(r\Phi)}{\partial r} \right] = \frac{1}{r} \frac{\partial}{\partial r} \left[\Phi + r \frac{\partial \Phi}{\partial r} \right] \\ &= \frac{1}{r} \left[\frac{\partial \Phi}{\partial r} + \frac{\partial}{\partial r} \left(r \frac{\partial \Phi}{\partial r} \right) \right] = \frac{1}{r} \left[\frac{\partial \Phi}{\partial r} + \frac{\partial \Phi}{\partial r} + r \frac{\partial^2 \Phi}{\partial r^2} \right] = \frac{1}{r} \left(2 \frac{\partial \Phi}{\partial r} + r \frac{\partial^2 \Phi}{\partial r^2} \right) \end{aligned} \quad (433)$$

Since Eqs. (432) and (433) have the same final expression, the proof is complete. QED.

Substitution of Eq. (431) into Eq. (428) and multiplication by r yields the wave equation in spherical waves, i.e.,

$$\frac{\partial^2(r\Phi)}{\partial r^2} = \frac{1}{c^2} \frac{\partial^2(r\Phi)}{\partial t^2} \quad (\text{spherical wave equation}) \quad (434)$$

where Φ depends only on the radial distance and time, i.e., $\Phi = \Phi(r, t)$.

5.9.2.1 Generic Spherical Waves

The d'Alembert solution of Eq. (434) is similar to Eq. (20), i.e.,

$$r\Phi(r, t) = f(r - ct) + g(r + ct) \quad (435)$$

The term $g(r + ct)$ must be discarded since it represents a wave propagating inward (e.g., a wave from a multitude of sources placed spherically at infinity). We only retain the term $f(r - ct)$, which represents a wave propagating outward from a point source at the origin. Hence, solution of Eq. (435) yields

$$\Phi(r, t) = \frac{1}{r} f(r - ct) \quad (\text{generic spherical wave}) \quad (436)$$

The particle displacement for a generic spherical wave is

$$\vec{u}(\vec{r}, t) = \frac{\vec{A}}{r} f(r - ct) \quad (\text{generic spherical wave}) \quad (437)$$

where $\vec{u}$ and $\vec{A}$ have components in the r, θ, ϕ directions, i.e.,

$$\vec{u} = \{u_r, u_\theta, u_\phi\}, \quad \vec{A} = \{A_r, A_\theta, A_\phi\} \quad (438)$$

5.9.2.2 Harmonic Spherical Waves

The separation of variables solution to Eq. (434) is similar to Eq. (59), i.e.,

$$r\Phi(r, t) = B e^{-ik(r+\omega t)} + C e^{i(kr-\omega t)} \quad (439)$$

The term $e^{-i(kr+\omega t)}$ must be discarded since it represents a wave propagating inward (e.g., a wave from a multitude of sources placed spherically at infinity). We only retain the term

$e^{i(\gamma r - \omega t)}$, which represents a wave propagating outward from a point source at the origin. Equation (439) yields

$$\Phi(r, t) = \frac{1}{r} C e^{i(\gamma r - \omega t)} \quad (\text{harmonic spherical wave}) \quad (440)$$

The particle displacement for a harmonic spherical wave is

$$\vec{u}(\vec{r}, t) = \frac{\vec{A}}{r} e^{i(\gamma r - \omega t)} \quad (\text{harmonic spherical wave}) \quad (441)$$

where $\vec{u}$ and $\vec{A}$ have components in the r, θ, ϕ directions as indicated by Eq. (438).

5.9.3 Circular and Cylindrical Waves

Cylindrical waves in 3-D (Figure 5.25b) and circular waves in 2-D (Figure 5.25c) share the same Laplacian in polar coordinates and will be treated together. Recall the expression of the Laplacian in polar coordinates, i.e.,

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial r^2} + \frac{1}{r} \frac{\partial \Phi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \Phi}{\partial \theta^2} + \frac{\partial^2 \Phi}{\partial z^2} \quad (\text{Laplacian in polar coordinates}) \quad (442)$$

For z -invariant circular symmetry, Eq. (442) depends on r only, i.e.,

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial r^2} + \frac{1}{r} \frac{\partial \Phi}{\partial r} \quad (\text{Laplacian in circular symmetry}) \quad (443)$$

Substitution of Eq. (443) into Eq. (410) yields

$$\frac{\partial^2 \Phi(r, t)}{\partial r^2} + \frac{1}{r} \frac{\partial \Phi(r, t)}{\partial r} = \frac{1}{c^2} \ddot{\Phi}(r, t) \quad (444)$$

Equation (444) accepts an exact solution in terms of Bessel functions; this Bessel function solution will be discussed in later chapters. Here, we will derive an approximate solution that allows us to understand the asymptotic behavior of circular and cylindrical waves at large distances from the source ($r \rightarrow \infty$). It can be shown that Eq. (443) can be rearranged as

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial r^2} + \frac{1}{r} \frac{\partial \Phi}{\partial r} = \frac{1}{\sqrt{r}} \frac{\partial^2 (\Phi \sqrt{r})}{\partial r^2} + \frac{1}{4r^2} \Phi \quad (445)$$

Proof of Eq. (445): calculate the first derivative of $\Phi \sqrt{r}$, i.e.,

$$\frac{\partial (\Phi \sqrt{r})}{\partial r} = \frac{\partial (\Phi r^{1/2})}{\partial r} = \frac{\partial \Phi}{\partial r} r^{1/2} + \Phi \frac{1}{2} r^{-1/2} \quad (446)$$

Calculate the second derivative by differentiating Eq. (446), i.e.,

$$\begin{aligned}
\frac{\partial^2 (\Phi \sqrt{r})}{\partial r^2} &= \frac{\partial}{\partial r} \left[\frac{\partial (\Phi r^{1/2})}{\partial r} \right] = \frac{\partial}{\partial r} \left(\frac{\partial \Phi}{\partial r} r^{1/2} + \Phi \frac{1}{2} r^{-1/2} \right) \\
&= \frac{\partial^2 \Phi}{\partial r^2} r^{1/2} + \frac{\partial \Phi}{\partial r} \frac{1}{2} r^{-1/2} + \frac{\partial \Phi}{\partial r} \frac{1}{2} r^{-1/2} - \Phi \frac{1}{4} r^{-3/2} \\
&= \frac{\partial^2 \Phi}{\partial r^2} r^{1/2} + \frac{\partial \Phi}{\partial r} r^{-1/2} - \Phi \frac{1}{4} r^{-3/2} = r^{1/2} \left(\frac{\partial^2 \Phi}{\partial r^2} + \frac{\partial \Phi}{\partial r} r^{-1} - \frac{1}{4} \Phi r^{-2} \right)
\end{aligned} \tag{447}$$

Divide Eq. (447) by $r^{1/2} = \sqrt{r}$ to get

$$\frac{1}{\sqrt{r}} \frac{\partial^2 (\Phi \sqrt{r})}{\partial r^2} = \frac{\partial^2 \Phi}{\partial r^2} + \frac{1}{r} \frac{\partial \Phi}{\partial r} - \frac{1}{4} \Phi r^{-2} = \nabla^2 \Phi - \frac{1}{4} \Phi r^{-2} \tag{448}$$

Rearrangement of Eq. (448) yields Eq. (445). QED.

At large values of r , i.e., far away from the source, the second term in Eq. (446) becomes negligible with respect to the first term and we can write

$$\nabla^2 \Phi|_{r \gg 1} = \frac{1}{\sqrt{r}} \frac{\partial^2 (\Phi \sqrt{r})}{\partial r^2} + \frac{1/\sqrt{r}}{r^2} \Phi \approx \frac{1}{\sqrt{r}} \frac{\partial^2 (\Phi \sqrt{r})}{\partial r^2} \quad (\text{for away from the source}) \tag{449}$$

Substitution of Eq. (449) into Eq. (410) yields an approximate expression of the wave equation for circular wave fronts at relatively large distances from the source, i.e.,

$$\frac{\partial^2 (\Phi \sqrt{r})}{\partial r^2} \approx \frac{1}{c^2} \frac{\partial^2 (\Phi \sqrt{r})}{\partial t^2} \quad (\text{circular wave equation}) \tag{450}$$

where Φ depends only on the radial distance and time, i.e., $\Phi = \Phi(r, t)$.

5.9.3.1 Generic Circular and Cylindrical Waves

The d'Alembert solution of Eq. (450) is similar to Eq. (20), i.e.,

$$\Phi(r, t) \sqrt{r} = f(r - ct) + g(r + ct) \tag{451}$$

As before, we only retain the term $f(r - ct)$, which represents a wave propagating outward from a point source at the origin of the circle. Hence, solution of Eq. (451) yields

$$\Phi(r, t) = \frac{1}{\sqrt{r}} f(r - ct) \quad (\text{generic circular wave}) \tag{452}$$

The particle displacement for a generic circular wave is

$$\vec{u}(\vec{r}, t) = \frac{\vec{A}}{\sqrt{r}} f(r - ct) \quad (\text{generic circular wave}) \tag{453}$$

where $\vec{u}$ and $\vec{A}$ have components in the r, θ directions, i.e.,

$$\vec{u} = \{u_r, u_\theta\}, \quad \vec{A} = \{A_r, A_\theta\} \quad (\text{circular wave displacements}) \quad (454)$$

For a cylindrical wave, the vectors $\vec{u}$ and $\vec{A}$ have components in the r, θ, z directions, i.e.,

$$\vec{u} = \{u_r, u_\theta, u_z\}, \quad \vec{A} = \{A_r, A_\theta, A_z\} \quad (\text{cylindrical wave displacements}) \quad (455)$$

However, the wave front of a cylindrical wave still propagates according to Eq. (453) since the cylindrical wave is self-similar along the z axis; hence, we write

$$\vec{u}(\vec{r}, t) = \frac{\vec{A}}{\sqrt{r}} f(r - ct) \quad (\text{generic cylindrical wave}) \quad (456)$$

5.9.3.2 Harmonic Circular and Cylindrical Waves

The separation of variables solution to Eq. (450) is similar to Eq. (59), i.e.,

$$\Phi(r, t)\sqrt{r} = B e^{-i(\gamma r + \omega t)} + C e^{i(\gamma r - \omega t)} \quad (457)$$

As before, we only retain the term $e^{i(\gamma r - \omega t)}$, which represents a wave propagating outward from a point source at the origin of the circle. Equation (439) yields

$$\Phi(r, t) = \frac{1}{\sqrt{r}} C e^{i(\gamma r - \omega t)} \quad (\text{harmonic circular wave}) \quad (458)$$

The particle displacement for a harmonic circular wave is

$$\vec{u}(\vec{r}, t) = \frac{\vec{A}}{\sqrt{r}} e^{i(\gamma r - \omega t)} \quad (\text{harmonic circular wave}) \quad (459)$$

where $\vec{u}$ and $\vec{A}$ have components in the r, θ directions as indicated by Eq. (454).

For a cylindrical wave, the vectors $\vec{u}$ and $\vec{A}$ have components in the r, θ, z directions as indicated by Eq. (455) and we write

$$\vec{u}(\vec{r}, t) = \frac{\vec{A}}{\sqrt{r}} e^{i(\gamma r - \omega t)} \quad (\text{harmonic cylindrical wave}) \quad (460)$$

MUSICAL INSTRUMENTS

Musical instruments:

- string instruments
- percussion instrument
- wind instruments

- Musical instruments make music
- Music consists of musical sounds
- Musical sounds are written as musical notes
- Musical notes are associated with certain frequencies

MUSICAL NOTES

16.351	C ₀	—
18.354	D ₀	—
20.601	E ₀	—
21.827	F ₀	—
24.499	G ₀	—
27.500	A ₀	—
30.868	B ₀	—
32.703	C ₁	—
36.708	D ₁	—
41.203	E ₁	—
43.654	F ₁	—
48.939	G ₁	—
55.000	A ₁	—
61.735	B ₁	—
65.406	C ₂	—
73.416	D ₂	—
82.407	E ₂	—
87.307	F ₂	—
97.999	G ₂	—
110.00	A ₂	—
123.47	B ₂	—
130.81	C ₃	—
146.83	D ₃	—
164.81	E ₃	—
174.61	F ₃	—
196.00	G ₃	—
220.00	A ₃	—
246.94	B ₃	—
261.63	C ₄	MIDDLE C
293.66	D ₄	—
329.63	E ₄	—
349.23	F ₄	—
391.99	G ₄	—
440.00	A ₄	—
493.88	B ₄	—
523.25	C ₅	—
587.33	D ₅	—
659.26	E ₅	—
698.46	F ₅	—
783.99	G ₅	—
880.00	A ₅	—
987.77	B ₅	—
1046.5	C ₆	—
1174.7	D ₆	—
1318.5	E ₆	—
1396.9	F ₆	—
1568.0	G ₆	—
1760.0	A ₆	—
1975.5	B ₆	—
2093.0	C ₇	—
2349.3	D ₇	—
2637.0	E ₇	—
2793.8	F ₇	—
3136.0	G ₇	—
3520.0	A ₇	—
3951.1	B ₇	—
4186.0	C ₈	—
4698.6	D ₈	—
5274.0	E ₈	—
5587.7	F ₈	—
6271.9	G ₈	—
7040.0	A ₈	—
7902.1	B ₈	—
8372.0	C ₉	—
9397.3	D ₉	—
10548	E ₉	—
11175	F ₉	—
12544	G ₉	—
14080	A ₉	—
15804	B ₉	—
16744	C ₁₀	—

Scale of equal
temperament
showing only the
progression of full
notes

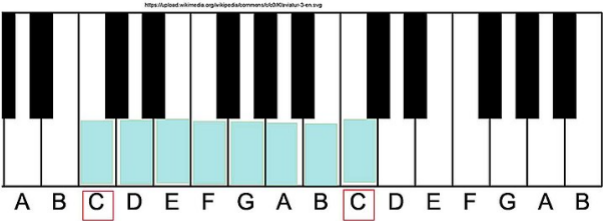
D3=146.83

1 MUSICAL SCALE AND MUSICAL NOTES

https://en.wikipedia.org/wiki/Major_scale

The **major scale** is one of the most commonly used musical scales. It is made up of seven notes: the eighth note duplicates the first note at double its frequency so that it is called a higher octave of the same note (from Latin "octavus", the eighth).

On a keyboard, the major scales begin with white keys. The simplest major scale to write is C major, the only major scale not requiring sharps or flats. The notes are C, D, E, F, G, A, B.



Also commonly used is the [solfege](#) naming convention in which each scale degree is denoted by a syllable. In the major scale, the solfège syllables are do, re, mi, fa, sol, la, si.

Table 1: Some musical notes and their Hz frequencies according to the scale of equal temperament

do	re	mi	fa	sol	la	si
C ₂ 65.406	D ₂ 73.416	E ₂ 82.407	F ₂ 87.307	G ₂ 97.999	A ₂ 110.00	B ₂ 123.47
C ₃ 125.81	D ₃ 146.83	E ₃ 164.81	F ₃ 174.61	G ₃ 196.00	A ₃ 220.00	B ₃ 246.94
C ₄ 261.63	D ₄ 293.66	E ₄ 329.63	F ₄ 349.23	G ₄ 391.99	A ₄ 440.00	B ₄ 493.88
C ₅ 523.25	D ₅ 587.33	E ₅ 659.26	F ₅ 698.46	G ₅ 783.99	A ₅ 880.00	B ₅ 987.77

The **scale of equal temperament** has the octave divided in twelve intervals which are equally spaced (equally tempered) on a logarithmic scale with a ratio equal to $2^{1/12} \approx 1.05946$.

https://en.wikipedia.org/wiki/12_equal_temperament

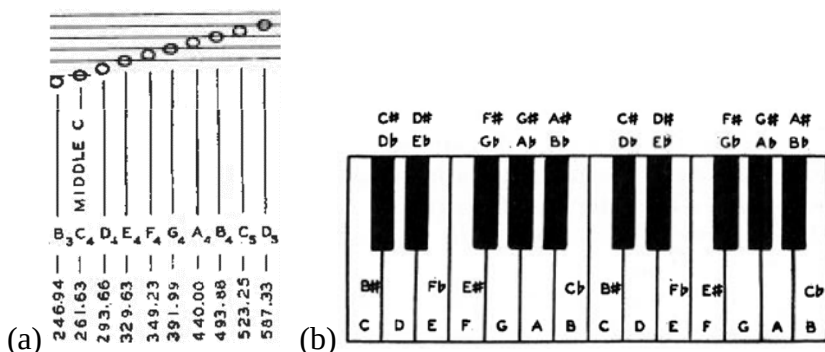


Figure 1: Scale of equal temperament: (a) whole tones starting with C₄ of frequency 261.63 Hz; (b) piano keys resulting in 12 intervals per octave when counting the 12 semitones created by both white and black keys

The *scale of just intonation* has the note frequencies represented by integer multiples of the fundamental note (other ratios from the harmonic series may also be used). Figure 2 shows a scale of just intonation based on fundamental C₂ note of frequency 66 Hz. One observes that the frequencies of the notes increase in the ratio 1,2,3,4,5,6,7,8,9,10. The scale of just intonation is said to give a pleasant sound.

The frequencies of the just scale and of the equal-temperament scale are not exactly coincidental, but the difference is $< \pm 2\%$.

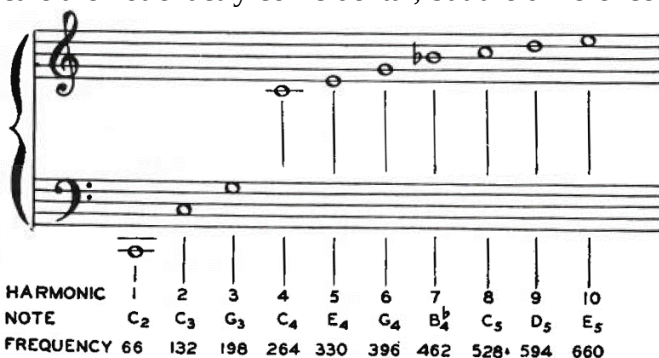


Figure 2: Scale of just intonation based on C₂ fundamental with frequency of 66 Hz. The note frequencies increase in proportion with the harmonic series 1,2,3,4,5,6,7,8,9,10

[1]Olsen, H F (1967) *Music, Physics, and Engineering*, Dover Publications, Inc.

Cents are a unit of measurement used in music to describe the difference in pitch between two notes. One cent is equal to 1/100th of a semitone. This means that there are 100 cents in a semitone, and 1200 cents in an octave.





Flat

The flat symbol lowers the pitch of a note by one **semitone**.



Sharp

The sharp symbol raises the pitch of a note by one semitone.

ANSI S1.1-1994
(ASA 111-1994)

Revision of ANSI S1.1-1960 (R1976)

Reaffirmed by
ANSI on
24 May-1999

AMERICAN NATIONAL STANDARD **ACOUSTICAL TERMINOLOGY**

ANSI S1.1-1994 (ASA 111-1994)

Accredited Standards Committee S1, Acoustics

Standards Secretariat
Acoustical Society of America
120 Wall Street, 32nd Floor
New York, New York 10005-3993

vowel (or consonant) articulation, word intelligibility, sentence intelligibility, etc.

12.44 speech interference level. One-fourth of the sum of the band sound pressure levels for octave bands with nominal midband frequencies of 500, 1000, 2000, and 4000 Hz. Unit, decibel; abbreviation, SIL; symbol, L_{SI} .

12.45 central masking. Masking that occurs when the signal is presented to one ear and the masking sound to the other.

NOTE—The site of central masking is in the higher auditory pathways of the brain.

12.46 backward masking. Condition in which the signal appears after the masking sound.

12.47 forward masking. Condition in which the signal appears before the masking sound.

12.48 remote masking; downward masking. Phenomenon in which an intense band of noise raises the threshold of hearing for sounds lower in frequency than that of the noise.

12.49 acoustic trauma. Ear injury caused by a sudden and intense acoustic stimulus that causes a degree of permanent or temporary hearing loss.

12.50 noise induced temporary threshold shift. Temporary hearing loss occurring as a result of noise exposure. Abbreviation, NITTS.

12.51 noise induced permanent threshold shift. Permanent hearing loss resulting from noise exposure. Abbreviation, NIPTS.

13 Musical Acoustics

13.01 tone. (a) Sound wave capable of exciting an auditory sensation having pitch. (b) Sound sensation having pitch.

13.02 simple tone; pure tone. (a) Sound wave, the instantaneous sound pressure of which is a simple sinusoidal function of time. (b) Sound sensation characterized by its singleness of pitch.

NOTES

(1) Whether or not a listener hears a tone as simple or complex depends upon ability, experience, and listening attitude.

(2) See also 3.40.

13.03 complex tone. (a) Sound wave containing simple harmonic components of different frequencies. (b) Sound sensation characterized by more than one pitch.

13.04 fundamental. Component of a periodic wave having the lowest frequency (see 3.21).

13.05 partial. (a) Simple sinusoidal component of a complex tone. (b) Component of a sound sensation which may be distinguished as a simple tone that cannot be further analyzed by the ear and which contributed to the timbre of the complex sound (see 12.09).

NOTES

1 The frequency of a partial may be either higher or lower than the basic frequency and may not be an integral multiple or submultiple of the basic frequency (for definition of basic frequency, see 3.11). If the frequency is not a multiple or submultiple, the partial is inharmonic.

2 When a system is maintained in steady forced vibration at a basic frequency equal to one of the frequencies of the normal modes of vibration of the system, the partials of the resulting complex tone are not necessarily identical in frequency with those of the other normal modes of vibration.

13.06 harmonic. Partial whose frequency is an integral multiple of the fundamental frequency.

NOTES

1 The term "overtone" has frequently been used in place of "harmonic," the n th harmonic being called the $(n-1)$ overtone. The term "overtone" is now deprecated to reduce ambiguity in the numbering of the components of a complex tone.

2 See also 3.22.

13.07 harmonic series of sounds. One in which each basic frequency in the series is an integral multiple of a fundamental frequency.

13.08 note. Conventional designation indicating pitch and duration; also the tone sensation itself or the sound pressure causing the sensation. The word serves when no distinction is desired among the symbol, the sensation, and the physical stimulus.

NOTE—Letter symbols from A to G are conventionally used to designate basic notes. Subscripts consisting of integers designate successive octave increases in frequency, starting at C_0 . See Table 13.2.

13.09 vibrato. Family of tonal effects in music that depend upon periodic variations of one or more characteristics of the sound wave.

Table 13.1 – Interval comparisons in different mathematical tuning systems

	Pythagorean			Equally Tempered		Just		
	Origin	Ratio	Cents	Ratio	Cents	Origin	Ratio	Cents
Unison	1:1	1.000	0.00	1.000	0	1:1	1.000	0.00
Minor Second	2 ⁸ :3 ⁵	1.054	90.22	1.059	100	16:15	1.067	111.73
Lesser Major Second	3 ² :2 ³	1.125	203.91	1.122	200	10:9	1.111	182.40
Greater Major Second						9:8	1.125	203.91
Minor Third	2 ⁵ :3 ³	1.185	294.13	1.189	300	6:5	1.200	315.64
Major Third	3 ⁴ :2 ⁶	1.266	407.82	1.260	400	5:4	1.250	386.31
Perfect Fourth	2 ² :3	1.333	498.04	1.335	500	4:3	1.333	498.04
Augmented Fourth	3 ⁶ :2 ⁹	1.424	611.73	1.414	600	45:32	1.406	590.22
Diminished Fifth	2 ¹⁰ :3 ⁶	1.405	588.27	1.414	600	64:45	1.422	609.78
Perfect Fifth	3:2	1.500	701.96	1.498	700	3:2	1.500	701.96
Minor Sixth	2 ⁷ :3 ⁴	1.580	792.18	1.587	800	8:5	1.600	813.69
Major Sixth	3 ³ :2 ⁴	1.688	905.87	1.682	900	5:3	1.667	884.36
Minor Seventh	2 ⁴ :3 ²	1.778	996.09	1.782	1000	7:4	1.750	968.83
Harmonic Minor Seventh						16:9	1.778	996.09
Grave Minor Seventh						9:5	1.800	1017.60
Major Seventh	3 ⁵ :2 ⁷	1.898	1109.78	1.888	1100	15:8	1.875	1088.27
Octave	2:1	2.000	1200.00	2.000	1200	2:1	2.000	1200.00

NOTE—When the particular characteristics are known, the term "vibrato" should be modified accordingly, e.g., frequency vibrato, amplitude vibrato, phase vibrato, and so forth.

13.10 interval. Between two sounds, their spacing in pitch or frequency, whichever is indicated by the context. The frequency spacing is expressed by the ratio of the frequencies or by a specified logarithm of this ratio.

13.11 octave. Unit of logarithmic frequency interval between two frequencies when the base of the logarithm is two.

13.12 semitone; half-step. Unit of logarithmic frequency interval between two frequencies when the base of the logarithm is the twelfth root of two.

NOTE—In equally tempered semitones, the interval between any two frequencies is 12 times the logarithm to the base 2 (or ≈ 39.86314 times the logarithm to the base 10) of the frequency ratio.

13.13 cent. Unit of logarithmic frequency interval when the base of the logarithm is the 1200th root of two.

NOTE—One octave is 1200 cents.

13.14 scale. Series of notes (symbols, sensations, or stimuli) arranged from low to high by a specified scheme of intervals, suitable for musical purposes.

13.15 Pythagorean scale. Musical scale in which the frequency intervals are represented by ratios of integral powers of the numbers 2 and 3. (See Table 13.1.)

13.16 just scale. Musical scale formed by octave rearrangement of the notes of three consecutive triads, each having the frequency ratio 4:5:6 or 10:12:15. (See Table 13.1.)

NOTE—For consecutive triads, the highest note of one triad is the lowest note of the next.

13.17 equally tempered scale. Musical scale formed by division of the octave into a number (usually twelve) of equal intervals. (See Table 13.1.)

13.18 standard tuning frequency; standard musical pitch. Frequency for the note A₄, 440 Hz. (See Table 13.2.)

Table 13.2 – Frequencies in hertz (Hz) and frequency levels in semitones counted (sc) up from C₀ for the usual equally tempered scale. Subscripts by octaves above C₀. C₀ approximately equals 16.352 Hz so that A₄ = 440 Hz exactly

Frequency			Frequency			Frequency		
Note	Level, sc	Frequency, Hz	Note	Level, sc	Frequency, Hz	Note	Level, sc	Frequency, Hz
B ₂	35	123.47	B ₅	71	987.77	B ₈	107	7902.1
	34	116.54		70	932.33		106	7458.6
A ₂	33	110.00	A ₅	69	880.00	A ₈	105	7040.0
	32	103.83		68	830.61		104	6644.9
G ₂	31	97.999	G ₅	67	783.99	G ₈	103	6271.9
	30	92.499		66	739.99		102	5919.9
F ₂	29	87.307	F ₅	65	698.46	F ₈	101	5587.7
E ₂	28	82.407	E ₅	64	659.26	E ₈	100	5274.0
	27	77.782		63	622.25		99	4978.0
D ₂	26	73.416	D ₅	62	587.33	D ₈	98	4698.6
	25	69.296		61	554.37		97	4434.9
C ₂	24	65.406	C ₅	60	523.25	C ₈	96	4186.0
B ₁	23	61.735	B ₄	59	493.88	B ₇	95	3951.1
	22	58.270		58	466.16		94	3729.3
A ₁	21	55.000	A ₄	57	440.00	A ₇	93	3520.0
	20	51.913		56	415.30		92	3322.4
G ₁	19	48.999	G ₄	55	392.00	G ₇	91	3136.0
	18	46.249		54	369.99		90	2960.0
F ₁	17	43.654	F ₄	53	349.23	F ₇	89	2793.8
E ₁	16	41.203	E ₄	52	329.63	E ₇	88	2637.0
	15	38.891		51	311.13		87	2489.0
D ₁	14	36.708	D ₄	50	293.66	D ₇	86	2349.3
	13	34.648		49	277.18		85	2217.5
C ₁	12	32.703	C ₄	48	261.63	C ₇	84	2093.0
B ₀	11	30.868	B ₃	47	246.94	B ₆	83	1975.5
	10	29.135		46	233.08		82	1864.7
A ₀	9	27.500	A ₃	45	220.00	A ₆	81	1760.0
	8	25.957		44	207.65		80	1661.2
G ₀	7	24.500	G ₃	43	196.00	G ₆	79	1568.0
	6	23.125		42	185.00		78	1480.0
F ₀	5	21.827	F ₃	41	174.61	F ₆	77	1396.9
E ₀	4	20.602	E ₃	40	164.81	E ₆	76	1318.5
		19.445		39	155.56		75	1244.5
D ₀	2	18.354	D ₃	38	146.83	D ₆	74	1174.7
	1	17.324			138.59		73	1108.7
C ₀	0	16.352	C ₃	36	130.81	C ₆	72	1046.5

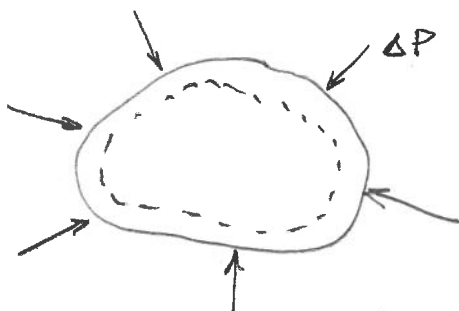
NOTE—Tuning or retuning of musical instruments is recommended within plus or minus 0.5 Hz at the standard tuning frequency, when the instruments are played where the ambient temperature is 22 °C (72 °F).

13.19 musical frequency level. In semitones, 12 times the logarithm to the base 2 of the ratio of the frequency of a sound to the reference frequency of 16.352 Hz, so chosen that $A_4 = 440$ Hz exactly.

FLUID WAVES & VIBRATION

FLUID ELASTICITY

Elasticity of "incompressible" fluids manifests itself as a change of volume ΔV due to a change of pressure ΔP



If the initial volume is V , the volumetric strain is defined as

$$\epsilon_v = \lim_{\Delta V \rightarrow 0} \frac{\Delta V}{V}$$

volumetric strain (1)

The bulk modulus K relates the pressure change Δp and the volume change ΔV as

$$\Delta P = -K \frac{\Delta V}{V} \quad (2)$$

The negative sign is needed because an increase of pressure produces a decrease of volume ($\Delta p > 0 \rightarrow \Delta V < 0$).

2
FB
2023/10/11

Water Bulk Modulus

$$K_{\text{Water}} \approx 2.25 \text{ GPa}$$

steel $E = 210 \text{ GPa}$

(3)

Bulk modulus of water varies with water temperature and water salinity

Air Bulk Modulus

$$= 0.142 \text{ MPa} = 0.000142 \text{ GPa}$$

$$K_{\text{air}} \approx 142 \text{ kPa (dry air @ } 20^\circ\text{C)} \quad (4)$$

Air bulk modulus can be related directly to air pressure P through the formula

$$K_{\text{air}} = \gamma P \quad (5)$$

where γ is the "adiabatic index" also known as "isentropic expansion factor". For dry air,

$$\gamma = 1.4 \text{ (dry air @ } 20^\circ\text{C)} \quad \text{adiabatic index} \quad (6)$$

Air bulk modulus changes with altitude as pressure decreases with altitude.

Fluid Waves and Vibration

1 WAVE EQUATION IN FLUIDS

Assume an infinite tube of cross section A containing fluid of density ρ and bulk modulus K (Figure 1a). The fluid in the tube undergoes time-varying axial displacement $u(x,t)$ under the influence of a pressure disturbance $p(x,t)$. Consider an infinitesimal element dx as shown in Figure 1b. The pressure disturbance $p(x,t)$ and the axial displacement $u(x,t)$ change by infinitesimal increments

$$dp = p'(x,t)dx \quad (1)$$

$$du = u'(x,t)dx \quad (2)$$

where the dash was used to designate derivative wrt space x , i.e.,

$$(\quad)' = \frac{\partial}{\partial x}(\quad) \quad (3)$$

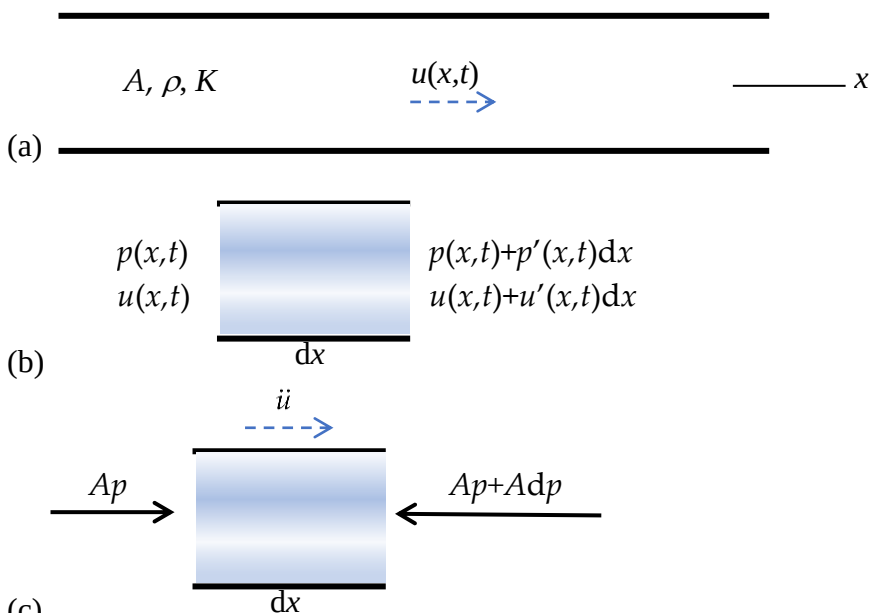


Figure 1 Fluid waves in an infinite tube: (a) schematic overview; (b) infinitesimal axial element; (c) free-body diagram

1.1 FREE-BODY DIAGRAM AND NEWTON'S LAW OF MOTION

The free-body diagram of the infinitesimal element of length dx is shown in Figure 1c. The forces acting on the infinitesimal element are Ap on the left and $Ap+Adp$ on the right. The net force in the positive x direction is

$$\text{net force} = -(Ap+Adp)+Ap=-Adp \quad (4)$$

The mass of the infinitesimal element is $\rho A dx$. The acceleration of the infinitesimal element is $\ddot{u}$, where a dot over a variable represents derivative with respect to time, i.e.,

$$\dot{(\quad)} = \frac{\partial}{\partial t}(\quad) \quad (5)$$

The two dots in $\ddot{u}$ represent the second derivative wrt time, i.e., the acceleration.

Newton's law of motion applied to the infinitesimal element states that net force should equal mass times acceleration, i.e.,

$$-Ap'dx = (\rho A dx)\ddot{u} \quad (6)$$

Upon simplification, Eq. (6) becomes

$$-p' = \rho \ddot{u} \quad (7)$$

1.2 FLUID ELASTICITY EQUATION

Recall the fluid elasticity equation, i.e.,

$$\Delta P = -K \Delta V/V \quad (8)$$

where K is the bulk modulus of the fluid. For dry air at 20°C , $K=142\text{kPa}$.

The volume of the infinitesimal element is $V=A dx$. The change in volume is $\Delta V=A du=Au'dx$. The change in pressure ΔP is the pressure disturbance p . Upon substitution in Eq. (8), one gets

$$p = -KAu'dx/Adx \quad (9)$$

After simplification, Eq. (9) becomes

$$p = -Ku' \quad (10)$$

Upon differentiation wrt x and sign interchange, Eq. (10) becomes

$$-p' = Ku'' \quad (11)$$

1.3 WAVE EQUATION

Substitution of Eq. (11) into Eq. (7) yields the governing equation

$$Ku'' = \rho \ddot{u} \quad (\text{governing equation}) \quad (12)$$

Upon division by ρ , Eq. (12) becomes

$$\frac{K}{\rho} u'' = \ddot{u} \quad (13)$$

Define the wavespeed c in the fluid as

$$c = \sqrt{\frac{K}{\rho}} \quad c^2 = \frac{K}{\rho} \quad (\text{wavespeed in the fluid}) \quad (14)$$

The wavespeed c is often referred to as the *sound speed*. Substitution of Eq. (14) into Eq. (13) yields the *wave equation*, i.e.,

$$c^2 u'' = \ddot{u} \quad (\text{wave equation}) \quad \text{PDE} \quad (15)$$

1.4 SOUND SPEED IN AIR AND WATER

Eq. (14) can be used to calculate the wavespeed aka sound speed in air and in water.

1.4.1 SOUND SPEED IN AIR

For dry air at 20°C, the bulk modulus is $K_{air}=142$ kPa and the air density is $\rho_{air}=1.204$ kg/m³. Substitution into Eq. (14) yields

$$c_{air} = \sqrt{\frac{K}{\rho}} = \sqrt{\frac{142 \times 10^3}{1.204}} = 343 \text{ m/s} \quad (\text{sound speed in air}) \quad (16)$$

Sound speed in air varies with temperature. For example, it takes the value $c=331$ m/s at 0°C. An often use round-number approximation of air density and sound speed are $\rho_{air} \approx 1.2$ kg/m³, $c_{air} \approx 340$ m/s.

1.4.2 SOUND SPEED IN WATER

Commonly used numerical values for water density and bulk modulus are $\rho_{water}=1000$ kg/m³ and $K_{water} \approx 2.25$ GPa. Substitution into Eq. (14) yields

$$c_{water} = \sqrt{\frac{K_{water}}{\rho_{water}}} = \sqrt{\frac{2.25 \times 10^9}{1 \times 10^3}} \approx 1500 \text{ m/s} \quad \left(\begin{array}{l} \text{sound speed} \\ \text{in water} \end{array} \right) \quad (17)$$

Sound speed in water varies with temperature and salinity.

1.5 BULK MODULUS AS FUNCTION OF SOUND SPEED AND DENSITY

Eq. (14) can be used to express the bulk modulus K as function of sound speed c and fluid density ρ , i.e.,

$$K = \rho c^2 \quad (\text{bulk modulus}) \quad (18)$$

2 SOLUTION OF WAVE EQUATION

The wave equation Eq. (15) accepts two types of solutions:

- d'Alembert solutions that emphasize the propagating character of the wave
- harmonic solutions that emphasize the oscillatory character of the wave

In this section we will develop the harmonic solution of the wave equation. Assume the time and space behaviors of the wave can be described independently by two separate functions $U(x)$ and $T(t)$, i.e.,

$$u(x,t)=U(x)T(t) \quad (19)$$

Differentiation of Eq. (19) wrt x and t yields the following expressions

$$\begin{aligned} u &= U(x)T(t) \\ u' &= U'(x)T(t) \quad \dot{u} = U(x)\dot{T}(t) \\ u'' &= U''(x)T(t) \quad \ddot{u} = U(x)\ddot{T}(t) \end{aligned} \quad (20)$$

Assume that the time behavior is vibratory with angular frequency ω , i.e.,

$$T(t)=e^{-i\omega t} \quad (21)$$

Substitution of Eq. (21) into Eq. (20) yields

$$u(x,t)=U(x) e^{-i\omega t} \quad (22)$$

Function $U(x)$ represents the amplitude of vibration at any location x along the tube. Substitution of Eq. (22) into Eq. (20) gives

$$u'' = U''(x)e^{-i\omega t} \quad \ddot{u} = -\omega^2 U(x)e^{-i\omega t} \quad (23)$$

Recall the wave equation Eq. (15), i.e.,

$$c^2 u'' = \ddot{u} \quad (\text{wave equation}) \quad (15)$$

Substitution of Eq. (23) into Eq. (15) gives

$$c^2 U''(x) e^{-i\omega t} = -\omega^2 U(x) e^{-i\omega t} \quad (24)$$

Upon simplification by $e^{-i\omega t}$, Eq. (24) becomes

$$c^2 U''(x) = -\omega^2 U(x) \quad (25)$$

Recall the wavenumber γ defined as

$$\gamma = \omega/c \quad (\text{wavenumber}) \quad (26)$$

Substitution of Eq. (26) into Eq. (25) yields

$$U''(x) + \gamma^2 U(x) = 0 \quad (27)$$

Eq. (27) is a linear 2nd order ordinary differential equation (ODE) in space. Assume exponential solution of the form

$$U(x)=e^{sx} \quad (28)$$

Upon differentiation, Eq. (28) gives

$$U''(x)=s^2e^{sx}=s^2U(x) \quad (29)$$

2.1 CHARACTERISTIC EQUATION

Substitution of Eq. (29) into Eq. (27) and simplification by $U(x)$ yields the characteristic equation

$$s^2+\gamma^2=0 \quad (\text{characteristic equation}) \quad (30)$$

The solutions of the characteristic equation Eq. (30) are

$$s_1=+i\gamma \quad s_2=-i\gamma \quad (31)$$

Substitution of s_1 and s_2 into Eq. (28) gives two independent solutions, $e^{i\gamma x}$ and $e^{-i\gamma x}$. The complete solution of Eq. (27) is a linear superposition of the two individual solutions, i.e.,

$$U(x)=A_1e^{i\gamma x}+A_2e^{-i\gamma x} \quad (32)$$

Using Euler's formulae, one can write the complex exponentials of Eq. (32) in terms of trigonometric functions and thus obtain the solution of Eq. (27) as

$$U(x)=C_1\sin\gamma x+C_2\cos\gamma x \quad (33)$$

2.2 WAVE SOLUTION

Substitution of Eq. (32) into Eq. (22) gives

$$u(x,t)=(A_1e^{i\gamma x}+A_2e^{-i\gamma x})e^{-i\omega t} \quad (34)$$

2.2.1 WAVE SOLUTION IN TERMS OF COMPLEX EXPONENTIALS

Rearrangement of the exponential terms in Eq. (34) yields the general solution of the wave equation as

$$u(x,t)=A_1e^{i(\gamma x-\omega t)}+A_2e^{-i(\gamma x+\omega t)} \quad (\text{wave solution}) \quad (35)$$

Eq. (35) reflects the wave aspects of the phenomenon. The first term is a forward wave whereas the second term is a backward wave. The wave propagation happens in the x direction.

2.2.2 WAVE SOLUTION IN TERMS OF TRIGONOMETRIC FUNCTIONS

Substitution of Eq. (33) into Eq. (22) gives the general solution as

$$u(x,t)=U(x)e^{-i\omega t}=(C_1\sin\gamma x+C_2\cos\gamma x)e^{-i\omega t} \quad (36)$$

If one absorbs the time variation $e^{-i\omega t}$ into the trigonometric functions of Eq. (36), then one can write the general solution of the wave equation in terms of trigonometric functions, i.e.,

$$u(x,t)=C_1\sin(\gamma x-\omega t)+C_2\cos(\gamma x-\omega t) \quad (37)$$

Eqs. (35), (36), (37) may be used interchangeably as convenient.

2.3 VIBRATION SOLUTION

The vibration solution involves the displacement vibration and the pressure vibration.

2.3.1 DISPLACEMENT VIBRATION

Recall Eq. (36), i.e.,

$$u(x,t)=U(x)e^{-i\omega t}=(C_1\sin\gamma x+C_2\cos\gamma x)e^{-i\omega t} \quad (36)$$

Eq. (36) is a formula that reflects the vibration aspects of the phenomenon. The constants C_1 and C_2 are to be determined by matching the boundary conditions at the two ends of a finite-length tube.

2.3.2 PRESSURE VIBRATION

Recall Eq. (10) that relates the pressure $p(x,t)$ to the displacement $u(x,t)$ as $p=-Ku'$ as well as Eq. (18) which relates bulk modulus K to the wavespeed c and density ρ as $K=\rho c^2$. Hence, one writes

$$p=-\rho c^2 u' \quad (38)$$

Substitution of Eq. (36) into Eq. (38) yields the pressure vibration expression

$$p(x,t)=-\rho c^2 U'(x)e^{-i\omega t} \quad (39)$$

Eq. (39) indicates that the pressure vibration is derived from the displacement vibration through differentiation.

3 AIR VIBRATION IN A TUBE

Consider a uniform tube of finite length L (Figure 2). Let's examine under what conditions the air inside the tube may undergo sustained vibration.

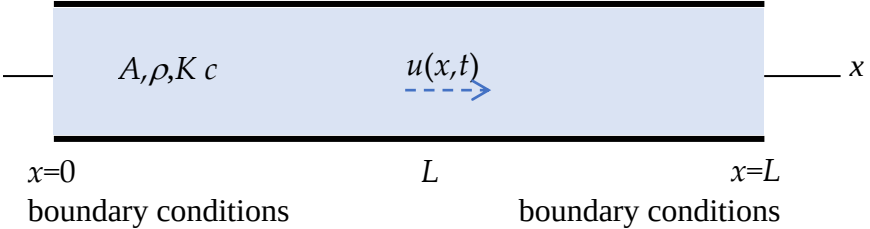


Figure 2 Air vibration in a tube of finite length L and generic boundary conditions at $x=0$ and $x=L$.

Recall Eqs. (33), (36) giving the harmonic solution of the wave equation in terms of trigonometric functions, i.e.,

$$U(x) = C_1 \sin \gamma x + C_2 \cos \gamma x \quad (33)$$

$$u(x, t) = U(x) e^{-i\omega t} = (C_1 \sin \gamma x + C_2 \cos \gamma x) e^{-i\omega t} \quad (36)$$

The constants C_1 and C_2 will now be determined by matching the boundary conditions at the two ends of the tube which may be either open or closed. At an open end, the sound-wave pressure p will go to zero, whereas at a closed end sound-wave particle displacement u will become zero.

3.1 AIR VIBRATION IN AN TUBE CLOSED AT BOTH ENDS

Consider a tube of length L closed at both ends as shown in Figure 3. The tube has a small opening through which the sound produced in the tube may get out such as in a tubular organ or a whistle. https://en.wikipedia.org/wiki/Steam_whistle

At a closed end, the displacement of the air particle is zero. Hence, the boundary conditions for a tube closed at both ends are

$$u(0, t) = 0 \quad (\text{left closed-end boundary condition}) \quad (40)$$

$$u(L, t) = 0 \quad (\text{right closed-end boundary condition}) \quad (41)$$

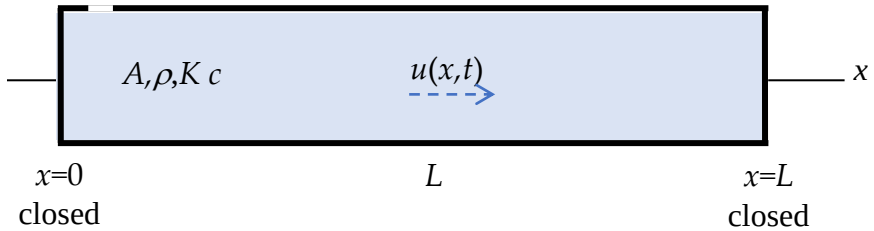


Figure 3 Air vibration in a tube of finite length L closed at both ends

Recall the displacement expression of Eq. (36), i.e.,

$$u(x, t) = (C_1 \sin \gamma x + C_2 \cos \gamma x) e^{-i\omega t} \quad (36)$$

Insertion of Eq. (36) into the boundary conditions Eqs. (40), (41), yields

$$u(0, t) = C_2 e^{-i\omega t} = 0 \quad (42)$$

$$u(L, t) = (C_1 \sin \gamma L + C_2 \cos \gamma L) e^{-i\omega t} = 0 \quad (43)$$

Eq. (42) gives $C_2 = 0$. Introduction of this value $C_2 = 0$ into Eq. (43) yields, upon simplification,

$$C_1 \sin \gamma L = 0 \quad (44)$$

The trivial solution of Eq. (44) is $C_1 = 0$ which means no vibration. To find a nontrivial solution $C_1 \neq 0$, one needs to have $\sin \gamma L = 0$, i.e.,

$$\sin \gamma L = 0 \quad (45)$$

If Eq. (44) is satisfied, i.e., $\sin \gamma L = 0$, then the value of C_1 becomes arbitrary since any C_1 value will satisfy Eq. (44) as long as $\sin \gamma L = 0$. This means that one can take any value for C_1 and still satisfy the conditions of the problem, which means that the problem has one degree of indeterminacy.

3.1.1 EIGENVALUE EQUATION

Introduce the notation

$$z = \gamma L \quad (46)$$

Substitution of Eq. (46) into Eq. (45) yields the eigenvalue equation,

$$\sin z = 0 \quad (\text{eigenvalue equation}) \quad (47)$$

Equation (47) defines the values of z for which nontrivial solutions of Eq. (44) exist. These values of z are the *eigenvalues* of the system. Solution of Eq. (47) gives

$$z=\pi, 2\pi, 3\pi, \dots \quad (48)$$

$$z_j=j\pi \quad j=1,2,3,\dots \quad (49)$$

3.1.2 WAVENUMBERS

Substitution of Eq. (49) into Eq. (46) yields the wavenumbers

$$\gamma_j=j\pi/L \quad j=1,2,3,\dots \quad (50)$$

3.1.3 NATURAL FREQUENCIES

Recall the relationship between wavenumber γ , wavespeed c and angular frequency ω , i.e.,

$$\gamma=\omega/c \quad (51)$$

Eq. (51) can be solved for ω to yield

$$\omega=\gamma c \quad (52)$$

Substitution of Eq. (50) into Eq. (52) yields the *natural angular frequencies*

$$\omega_j=jc\pi/L \quad j=1,2,3,\dots \quad (53)$$

Since $\omega=2\pi f$, the corresponding Hz frequencies f_j are given by the formula

$$f_j=jc/2L \quad j=1,2,3,\dots \quad (54)$$

We note that Eqs. (50), (53), (54) predict a denumerable infinite set of wavenumbers and natural frequencies. The first frequency, f_1 , is called the *fundamental frequency*, whereas the other frequencies (f_j $j=2,3,4,\dots$) are called *overtones*. The overtones are multiples of the fundamental frequency. The fundamental frequency is also referred to as the *fundamental harmonic*, whereas the overtones are also referred to as *higher harmonics*.

3.1.4 WAVELENGTHS

Recall the relationship between wavelength λ and wavenumber γ , i.e.,

$$\lambda=2\pi/\gamma \quad (55)$$

Substitution of Eq. (50) into Eq. (55) yields

$$\lambda_j = \frac{2\pi}{\gamma_j} = \frac{2\pi}{j\pi/L} = \frac{2L}{j} \quad j=1,2,3,\dots \quad (56)$$

It is apparent from Eq. (56) that higher frequencies have shorter wavelengths. The vibration of the air inside the tube can be interpreted in terms of standing waves fitting the boundary

conditions. In the present, case the standing waves are such that the tube length L accommodates multiples of the half wavelengths, $\lambda_j/2$, $j=1,2,3,\dots$, i.e.,

$$L=j\lambda_j/2 \quad j=1,2,3,\dots \quad (57)$$

The higher the vibration frequency, the shorter the wavelength of these standing waves. The wavenumbers, resonant frequencies, and wavelength multiplicity for the vibration of air in a tube of length L closed at both ends are listed in Table 1.

3.1.5 MODESHAPES

Recall Eq. (33), i.e.,

$$U(x)=C_1\sin\gamma x+C_2\cos\gamma x \quad (33)$$

where the constants C_1 and C_2 are given by the solution of Eqs. (42) and (43). Eq. (42) gave $C_2=0$ whereas Eq. (43) yielded Eq. (44), i.e., $C_1\sin\gamma L=0$. The nontrivial solution of Eq. (44) was only possible for special values of γL that made $\sin\gamma L=0$. These special values were called eigenvalues. Associated with the eigenvalues are the wavenumbers $\gamma_j=j\pi/L$. For each wavenumber $\gamma_j=j\pi/L$, one gets $\sin\gamma_j L=0$ in Eq. (44) and hence the value of C_1 becomes arbitrary since any C_1 value will satisfy Eq. (44) as long as $\sin\gamma L=0$. This means that one can take any value for C_1 and still satisfy the conditions of the problem, which means that the problem has one degree of indeterminacy. Here, we take $C_1=1$. Thus, with $C_1=1$ and $C_2=0$, Eq. (33) becomes

$$U_j(x)=\sin\gamma_j x \quad j=1,2,3,\dots \quad (\text{modeshapes}) \quad (58)$$

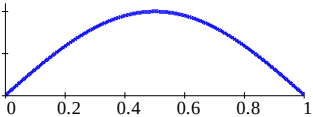
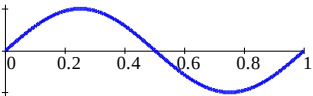
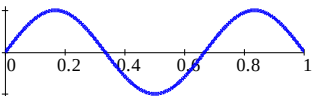
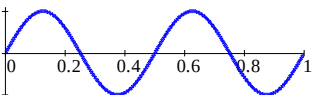
Eq. (58) defines *natural modes of vibration (modeshapes)*. The modeshapes of Eq. (58) may be scaled up or down arbitrarily. Whatever the scale, the modeshapes will still satisfy the boundary conditions of the problem.

Table 1 gives the first four natural frequencies and corresponding modeshapes. One notes that the air motion is zero at both ends of the tube which is consistent with the closed-end condition.

3.1.6 PRESSURE MODESHAPES

Substitution of Eq. (58) into Eqs. (10), (18) yields the pressure modeshapes $p_j(x)=-\rho c^2 U'_j(x)$.

Table 1 Modeshapes of air vibration in a tube of length L closed at both ends

Mode #	Eigenvalue $z = \gamma L$ $z = j\pi$	Natural frequency	Modeshape		Wavelength multiplicity
1	$z_1 = \gamma_1 L = \pi$	$f_1 = \frac{c}{2L}$		$U_1 = \sin \pi \frac{x}{L}$	$L = \frac{\lambda_1}{2}$
2	$z_2 = \gamma_2 L = 2\pi$	$f_2 = 2 \frac{c}{2L}$		$U_2 = \sin 2\pi \frac{x}{L}$	$L = 2 \frac{\lambda_2}{2}$
3	$z_3 = \gamma_3 L = 3\pi$	$f_3 = 3 \frac{c}{2L}$		$U_3 = \sin 3\pi \frac{x}{L}$	$L = 3 \frac{\lambda_3}{2}$
4	$z_4 = \gamma_4 L = 4\pi$	$f_4 = 4 \frac{c}{2L}$		$U_4 = \sin 4\pi \frac{x}{L}$	$L = 4 \frac{\lambda_4}{2}$

3.2 AIR VIBRATION IN AN TUBE CLOSED AT ONE END

Consider a tube of length L closed at one end and open at the other end as shown in Figure 4.

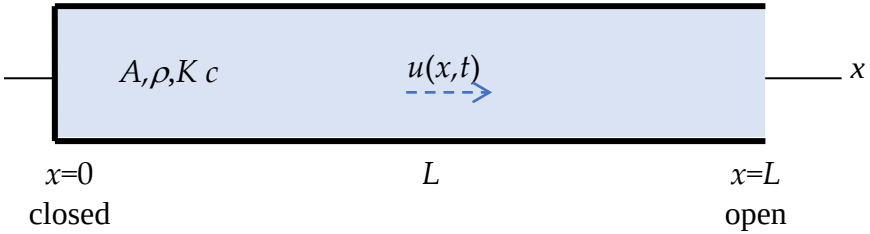


Figure 4 Air vibration in a tube of finite length L closed at one end and open at the other end

At the closed end, the displacement is zero. At the open end, the pressure disturbance p goes to zero. Hence, the boundary conditions for this case are

$$u(0,t)=0 \quad (\text{closed-end boundary condition at the left}) \quad (59)$$

$$p(L,t)=0 \quad (\text{open-end boundary condition at the right}) \quad (60)$$

Recall the displacement solution of Eq. (36), i.e.,

$$u(x,t)=(C_1 \sin \gamma x + C_2 \cos \gamma x) e^{-i\omega t} \quad (36)$$

Recall Eq. (10) that connects pressure to displacement derivative, i.e.,

$$p = -K u' \quad (10)$$

Substitution of Eq. (36) into Eq. (10) yields

$$p(x,t) = -K(C_1 \gamma \cos \gamma x - C_2 \gamma \sin \gamma x) e^{-i\omega t} \quad (61)$$

Insertion of Eqs. (36), (61) into the boundary conditions Eqs. (59), (60), yields

$$u(0,t) = C_2 e^{-i\omega t} = 0 \quad (62)$$

$$p(L,t) = -K(C_1 \gamma \cos \gamma L - C_2 \gamma \sin \gamma L) e^{-i\omega t} = 0 \quad (63)$$

Eq. (62) gives $C_2=0$. Substitution of $C_2=0$ into Eq. (63) yields, upon simplification,

$$C_1 \cos \gamma L = 0 \quad (64)$$

The trivial solution of Eq. (64) is $C_1=0$ which means no vibration.

To find a nontrivial solution $C_1 \neq 0$, one needs to have $\cos \gamma L = 0$, i.e.,

$$\cos \gamma L = 0 \quad (65)$$

If Eq. (65) is satisfied, i.e., $\cos \gamma L = 0$, then the value of C_1 becomes arbitrary since any C_1 value will satisfy Eq. (65) as long as $\cos \gamma L = 0$. This means that one can take any value for C_1 and still satisfy the conditions of the problem, which means that the problem has one degree of indeterminacy.

3.2.1 EIGENVALUE EQUATION

Recall the notation of Eq. (46)

$$z = \gamma L \quad (46)$$

Substitution of Eq. (46) into Eq. (65) yields the eigenvalue equation,

$$\cos z = 0 \quad (\text{eigenvalue equation}) \quad (66)$$

Equation (66) defines the values of z for which nontrivial solutions of Eq. (64) exist. These values of z are the *eigenvalues* of the system. Solution of Eq. (66) gives

$$z = \pi/2, 3\pi/2, 5\pi/2, \dots \quad (67)$$

$$z_j = (2j-1)\pi/2 \quad j=1,2,3,\dots \quad (68)$$

3.2.2 WAVENUMBERS

Substitution of Eq. (68) into Eq. (46) yields the wavenumbers

$$\gamma_j = (2j-1)\pi/2L \quad j=1,2,3,\dots \quad (69)$$

3.2.3 NATURAL FREQUENCIES

Recall Eq. (52) that gives angular frequency ω in terms of wavenumber γ , and wavespeed c , i.e.,

$$\omega = \gamma c \quad (52)$$

Substitution of Eq. (69) into Eq. (52) yields the *natural angular frequencies*

$$\omega_j = (2j-1)c\pi/2L \quad j=1,2,3,\dots \quad (70)$$

Since $\omega = 2\pi f$, the corresponding Hz frequencies f_j are given by the formula

$$f_j = (2j-1)c/4L \quad j=1,2,3,\dots \quad (71)$$

We note that Eqs. (69), (70), (71) predict a denumerable infinite set of wavenumbers and natural frequencies. The first frequency, f_1 , is called the *fundamental frequency*, whereas the other frequencies (f_j $j=2,3,4,\dots$) are called *overtones*. The overtones are multiples of the

fundamental frequency. The fundamental frequency is also referred to as the *fundamental harmonic*, whereas the overtones are also referred to as *higher harmonics*.

3.2.4 WAVELENGTHS

Recall Eq. (55) giving the relationship between wavelength λ and wavenumber γ , i.e.,

$$\lambda = 2\pi/\gamma \quad (55)$$

Substitution of Eq. (69) into Eq. (55) yields

$$\lambda_j = \frac{2\pi}{\gamma_j} = \frac{2\pi}{(2j-1)\pi/2L} = \frac{4L}{(2j-1)} \quad j = 1, 2, 3, \dots \quad (72)$$

It is apparent from Eq. (72) that higher frequencies have shorter wavelengths. The vibration of the air inside the tube can be interpreted in terms of standing waves fitting the boundary conditions. In the present, case the standing waves are such that the tube length L accommodates odd multiples of the quarter wavelengths, $\lambda_j/4$, $j=1,2,3,\dots$, i.e.,

$$L = (2j-1)(\lambda_j/4) \quad j=1,2,3,\dots \quad (73)$$

The higher the vibration frequency, the shorter the wavelength of these standing waves. The wavenumbers, resonant frequencies, and wavelength multiplicity for the vibration of air in a tube of length L closed at one end and open at the other end are listed in Table 2.

3.2.5 MODESHAPES

Recall Eq. (33), i.e.,

$$U(x) = C_1 \sin \gamma x + C_2 \cos \gamma x \quad (33)$$

where the constants C_1 and C_2 are given by the solution of Eqs. (62) and (63). Eq. (62) gave $C_2=0$ whereas Eq. (63) yielded Eq. (64) i.e., $C_1 \cos \gamma L = 0$. The nontrivial solution of Eq. (64) was only possible for special values of γL that made $\cos \gamma L = 0$. These special values were called eigenvalues. Associated with the eigenvalues are the wavenumbers $\gamma_j = (2j-1)\pi/2L$. For each wavenumber $\gamma_j = (2j-1)\pi/2L$, one gets $\cos \gamma_j L = 0$ in Eq. (64) and hence the value of C_1 becomes arbitrary since any C_1 value will satisfy Eq. (64) as long as $\cos \gamma L = 0$. This means that one can take any value for C_1 and still satisfy the conditions of the problem, which means that the problem has one

degree of indeterminacy. Here, we take $C_1=1$. Thus, with $C_1=1$ and $C_2=0$, Eq. (33) becomes

$$U_j(x)=\sin\gamma_j x \quad j=1,2,3,\dots \quad (\text{modeshapes}) \quad (74)$$

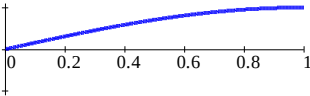
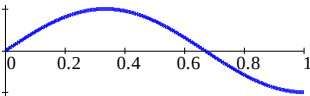
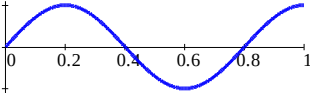
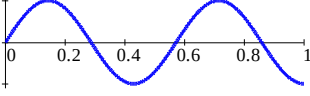
Eq. (74) defines a *natural mode of vibration (modeshape)*. The modeshapes of Eq. (74) may be scaled up or down arbitrarily. Whatever the scale, the modeshapes will still satisfy the boundary conditions of the problem.

Table 2 gives the first four natural frequencies and corresponding modeshapes. One notes that the air motion is zero at the closed end and maximum at the open end of the tube, as expected.

3.2.6 PRESSURE MODESHAPES

Substitution of Eq. (74) into Eqs. (10), (18) yields the pressure modeshapes $p_j(x) = -\rho c^2 U'_j(x)$.

Table 2 Modeshapes of air vibration in a tube of length L closed on the left and open on the right

Mode #	Eigenvalue $z=jL$ $z_j=(2j-1)\pi/2$	Natural frequency	Modeshape		Wavelength multiplicity
1	$z_1=\gamma_1 L=\pi/2$	$f_1 = \frac{c}{4L}$		$U_1 = \sin \frac{\pi}{2} \frac{x}{L}$	$L = \frac{\lambda_1}{4}$
2	$z_2=\gamma_2 L=3\pi/2$	$f_2 = 3 \frac{c}{4L}$		$U_2 = \sin 3 \frac{\pi}{2} \frac{x}{L}$	$L = 3 \frac{\lambda_2}{4}$
3	$z_3=\gamma_3 L=5\pi/2$	$f_3 = 5 \frac{c}{4L}$		$U_3 = \sin 5 \frac{\pi}{2} \frac{x}{L}$	$L = 5 \frac{\lambda_3}{4}$
4	$z_4=\gamma_4 L=7\pi/2$	$f_3 = 7 \frac{c}{4L}$		$U_4 = \sin 7 \frac{\pi}{2} \frac{x}{L}$	$L = 7 \frac{\lambda_4}{4}$

3.3 AIR VIBRATION IN A TUBE OPEN AT BOTH ENDS

Consider a tube of length L open at both ends as shown in Figure 5.

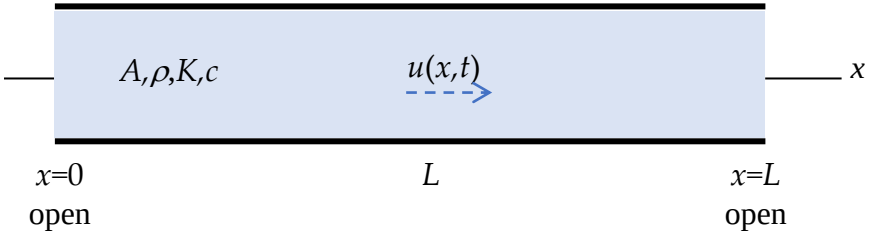


Figure 5 Air vibration in an open-ended tube of finite length L

The boundary conditions for this case are

$$p(0, t) = 0 \quad (\text{open-end boundary condition at the left}) \quad (75)$$

$$p(L, t) = 0 \quad (\text{open-end boundary condition at the right}) \quad (76)$$

Recall the displacement solution given in Eq. (36), i.e.,

$$u(x, t) = (C_1 \sin \gamma x + C_2 \cos \gamma x) e^{-i\omega t} \quad (36)$$

Recall Eq. (10) that connects pressure to displacement derivative, i.e.,

$$p = -K u' \quad (10)$$

Substitution of Eq. (36) into Eq. (10) yields the same pressure expression as in Section 3.2 Eq. (61), i.e.,

$$p(x, t) = -K(C_1 \gamma \cos \gamma x - C_2 \gamma \sin \gamma x) e^{-i\omega t} \quad (61)$$

Insertion of pressure expression Eq. (61) into the boundary conditions Eqs. (75), (76) yields

$$p(0, t) = -C_1 K \gamma e^{-i\omega t} = 0 \quad (77)$$

$$p(L, t) = -K(C_1 \gamma \cos \gamma L - C_2 \gamma \sin \gamma L) e^{-i\omega t} = 0 \quad (78)$$

Eq. (77) gives $C_1 = 0$. Substitution of $C_1 = 0$ into Eq. (78) yields, upon simplification,

$$C_2 \sin \gamma L = 0 \quad (79)$$

The trivial solution of Eq. (79) is $C_2 = 0$ which means no vibration. To find a nontrivial solution $C_2 \neq 0$ of Eq. (79), one needs to have $\sin \gamma L = 0$, i.e.,

$$\sin \gamma L = 0 \quad (80)$$

If Eq. (80) is satisfied, i.e., $\sin \gamma L = 0$, then the value of C_2 becomes arbitrary since any C_2 value will satisfy Eq. (79) as long as $\sin \gamma L = 0$. This means that one can take any value for C_2 and still satisfy the conditions of the problem, which means that the problem has one degree of indeterminacy.

3.3.1 EIGENVALUE EQUATION

Recall the notation of Eq. (46)

$$z = \gamma L \quad (46)$$

Substitution of Eq. (46) into Eq. (80) yields the eigenvalue equation, i.e.,

$$\sin z = 0 \quad (\text{eigenvalue equation}) \quad (81)$$

Equation (81) defines the values of z for which nontrivial solutions of Eq. (79) exist. These values of z are the *eigenvalues* of the system. Solution of Eq. (81) gives

$$z = \pi, 2\pi, 3\pi, \dots \quad (82)$$

$$z_j = j\pi \quad j = 1, 2, 3, \dots \quad (83)$$

3.3.2 WAVENUMBERS

Substitution of Eq. (83) into Eq. (46) yields the wavenumbers

$$\gamma_j = j\pi/L \quad j = 1, 2, 3, \dots (\text{wavenumbers}) \quad (84)$$

3.3.3 NATURAL FREQUENCIES

Recall Eq. (52) that gives angular frequency ω in terms of wavenumber γ , and wavespeed c , i.e.,

$$\omega = \gamma c \quad (52)$$

Substitution of Eq. (84) into Eq. (52) yields the *natural frequencies*

$$\omega_j = j\pi c/L \quad j = 1, 2, 3, \dots \quad (85)$$

The frequency ω_j is expressed in rad/sec. Since $\omega = 2\pi f$, the corresponding Hz frequencies f_j are given by the formula

$$f_j = j\pi c/2L \quad j = 1, 2, 3, \dots \quad (86)$$

We note that Eqs. (84), (85), (86) predict a denumerable infinite set of wavenumbers and natural frequencies. The first frequency, ω_1 , is called the *fundamental frequency*, whereas the other frequencies (f_j $j=2, 3, 4, \dots$) are called *overtones*. The overtones are multiples of the fundamental frequency. The fundamental frequency is also referred

to as the *fundamental harmonic*, whereas the overtones are also referred to as *higher harmonics*.

3.3.4 WAVELENGTH

Recall Eq. (55) giving the relationship between wavelength λ and wavenumber γ , i.e.,

$$\lambda = 2\pi/\gamma \quad (55)$$

Substitution of Eq. (84) into Eq. (55) yields

$$\lambda_j = \frac{2\pi}{\gamma_j} = \frac{2\pi}{j\pi/L} = \frac{2L}{j} \quad j = 1, 2, 3, \dots \quad (87)$$

It is apparent from Eq. (87) that higher frequencies have shorter wavelengths. The vibration of the air inside the tube can be interpreted in terms of standing waves fitting the boundary conditions. In the present case, the standing waves are such that the tube length L accommodates exact multiples of the half wavelengths, $\lambda_j/2$, $j=1, 2, 3, \dots$, i.e.,

$$L = j(\lambda_j/2) \quad j = 1, 2, 3, \dots \quad (88)$$

The higher the vibration frequency, the shorter the wavelength of these standing waves. The wavenumbers, resonant frequencies, and wavelength multiplicity for the vibration of air in an open-ended tube of length L are listed in Table 3.

3.3.5 MODESHAPES

Recall Eq. (33), i.e.,

$$U(x) = C_1 \sin \gamma x + C_2 \cos \gamma x \quad (33)$$

where the constants C_1 and C_2 are given by the solution of Eqs. (77) and (78). Eq. (77) gave $C_1=0$ whereas Eq. (78) yielded Eq. (79) i.e., $C_2 \sin \gamma L = 0$. The nontrivial solution of Eq. (79) was only possible for special values of γL that made $\sin \gamma L = 0$. These special values were called eigenvalues. Associated with the eigenvalues are the wavenumbers $\gamma_j = j\pi/L$. For each wavenumber $\gamma_j = j\pi/L$, one gets $\sin \gamma_j L = 0$ in Eq. (79) and hence the value of C_2 becomes arbitrary since any C_2 value will satisfy Eq. (79) as long as $\sin \gamma L = 0$. This means that one can take any value for C_2 and still satisfy the conditions of the problem, which means that the problem has one

degree of indeterminacy. Here, we take $C_2=1$. Thus, with $C_1=0$ and $C_2=1$, Eq. (33) becomes

$$U_j(x)=\cos\gamma_j x \quad j=1,2,3,\dots \quad (\text{modeshapes}) \quad (89)$$

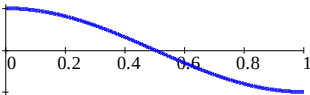
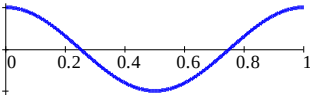
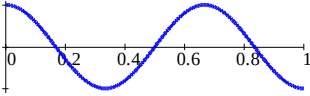
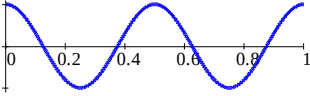
Eq. (89) defines *natural modes of vibration (modeshapes)*. The modeshapes of Eq. (89) may be scaled up or down arbitrarily. Whatever the scale, the modeshapes will still satisfy the boundary conditions of the problem.

Table 3 gives the first four natural frequencies and corresponding modeshapes. One notes that the air motion is maximum at the open ends of the tube, as expected.

3.3.6 PRESSURE MODESHAPES

Substitution of Eq. (89) into Eqs. (10), (18) yields the pressure modeshapes $p_j(x) = -\rho c^2 U'_j(x)$.

Table 3 Modeshapes of air vibration in an open-ended tube of length L

Mode #	Eigenvalue $z = \gamma L$ $z = j\pi$	Natural frequency	Modeshape		Wavelength multiplicity
1	$z_1 = \gamma_1 L = \pi$	$f_1 = \frac{c}{2L}$		$U_1 = \cos \pi \frac{x}{L}$	$L = \frac{\lambda_1}{2}$
2	$z_2 = \gamma_2 L = 2\pi$	$f_2 = 2 \frac{c}{2L}$		$U_2 = \cos 2\pi \frac{x}{L}$	$L = 2 \frac{\lambda_2}{2}$
3	$z_3 = \gamma_3 L = 3\pi$	$f_3 = 3 \frac{c}{2L}$		$U_3 = \cos 3\pi \frac{x}{L}$	$L = 3 \frac{\lambda_3}{2}$
4	$z_4 = \gamma_4 L = 4\pi$	$f_4 = 4 \frac{c}{2L}$		$U_4 = \cos 4\pi \frac{x}{L}$	$L = 4 \frac{\lambda_4}{2}$

4 MODESHAPE ORTHONORMALITY AND MODAL COEFFICIENTS

4.1 MODESHAPES ORTHONORMALITY

Modeshapes orthogonality is an important property that allows one to verify if some independently derived functions qualify to be modeshapes. For air vibration in a tube of length L , the modeshape orthogonality is expressed as

$$\int_0^L U_p(x) U_q(x) dx = \begin{cases} L/2 & \text{for } p = q \\ 0 & \text{otherwise} \end{cases} \quad (\text{orthogonality}) \quad (90)$$

Eq. (90) implies that the product between two different modeshapes is zero when integrated over the tube length L . This is the **orthogonality** property. If the two modeshapes are the same, i.e., the modeshape is multiplied upon itself, then the result is a finite number, in this case $L/2$. If one divides the integral by this finite number $L/2$, then the result is 1. This is the process of **normalization**. Thus, the **orthonormality** property of the modeshapes is expressed as

$$\int_0^L U_p(x) U_q(x) dx / (L/2) = \begin{cases} 1 & \text{for } p = q \\ 0 & \text{otherwise} \end{cases} \quad (\text{orthonormality}) \quad (91)$$

The modeshape orthonormality is usually verified through numerical integration, although closed-form analytical proof may also be possible.

4.2 MODAL COEFFICIENTS

The modal coefficients are used to analyze the modal behavior in response to an external excitation as described later in this course.

4.2.1 MODAL MASS AND MODAL STIFFNESS DEFINITION

The commonly used modal coefficients are the modal mass m_j and the modal stiffness k_j . Their general definition is

$$m_j = \int_0^L \rho A U_j^2(x) dx \quad (\text{modal mass}) \quad (92)$$

$$k_j = -\int_0^L K A U_j''(x) U_j(x) dx \quad (\text{modal stiffness}) \quad (93)$$

4.2.2 RELATIONSHIP BETWEEN MODAL MASS AND MODAL STIFFNESS

Modal mass and modal stiffness obey the following relationship

$$k_j = \omega_j^2 m_j \quad (94)$$

The relationship between modal mass and modal stiffness expressed by Eq. (94) is similar to the mass-stiffness relationship $k=\omega^2 m$ of a 1-dof mass-spring system. The modal coefficients allow us to treat each mode of vibration as a simple 1-dof vibration system.

Proof of Eq. (94)

The proof of Eq. (94) proceeds as follows. Recall Eq. (25), i.e.,

$$c^2 U''(x) = -\omega^2 U(x) \quad (95)$$

Recall $c^2=K/\rho$. Hence, Eq. (95) becomes

$$KU'''(x) = -\rho\omega^2 U(x) \quad (96)$$

For the j^{th} modeshape, Eq. (96) is written as

$$KU_j''(x) = -\rho\omega_j^2 U_j(x) \quad (97)$$

Substitution of Eq. (97) into the modal stiffness definition Eq. (93) and use of the modal mass definition Eq. (92) gives

$$\begin{aligned} k_j &= -\int_0^L KA U_j''(x) U_j(x) dx \\ &= -\int_0^L (-\rho\omega_j^2) A U_j(x) U_j(x) dx \quad \text{QED} \\ &= \omega_j^2 \int_0^L \rho A U_j^2(x) dx = \omega_j^2 m_j \end{aligned} \quad (98)$$

4.2.3 MODAL MASS AND MODAL STIFFNESS VALUES

In a uniform tube, the air density ρ and cross-section area A are constant and hence get outside the integral sign in Eq. (92) giving

$$m_j = \rho A \int_0^L U_j^2(x) dx \quad (\text{modal mass}) \quad (99)$$

In virtue of Eq. (90), the modal mass m_j take the value $\rho AL/2$. Substitution of this value in Eq. (94) yields the modal stiffness k_j as

$$k_j = \omega_j^2 \rho A L/2, \text{ i.e.,}$$

$$m_j = \rho AL/2 \quad (\text{modal mass}) \quad (100)$$

$$k_j = \omega_j^2 \rho A L/2 \quad (\text{modal stiffness}) \quad (101)$$

Note that the modal mass value of $\rho AL/2$ represents half of the mass of the air contained in the tube. Also note that the modal mass m_j has the same value for all the vibration modes.

4.3 CLOSED-FORM ANALYTICAL DERIVATION OF MODESHAPE ORTHONORMALITY

To study analytically modeshape orthonormality, consider the integrals

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx \quad \text{for } p \neq q \quad (102)$$

$$I_j = \int_0^L U_j^2(x) dx \quad \text{for } p=q=j \quad (103)$$

In virtue of Eq. (90), these equations should take the values

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = 0 \quad \text{for } p \neq q \quad (104)$$

$$I_j = \int_0^L U_j^2(x) dx = L/2 \quad \text{for } p=q=j \quad (105)$$

4.3.1 VIBRATION IN A CLOSED-CLOSED TUBE

The wavenumbers, natural frequencies, and modeshapes for a tube closed at both ends are given by Eqs. (50), (53), (58) as

$$\gamma_j = j\pi/L \quad \omega_j = jc\pi/L \quad U_j(x) = \sin \gamma_j x \quad j=1,2,3,\dots \quad (106)$$

4.3.1.1 Calculation of I_{pq}

Let j take two different values p and q and then substitute Eq. (106) into Eq. (104) to get

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = \int_0^L \sin \gamma_p x \sin \gamma_q x dx \quad (107)$$

Recall the “product of sines” identity

$$\sin \alpha \sin \beta = [\cos(\alpha - \beta) - \cos(\alpha + \beta)]/2 \quad (108)$$

The use of Eq. (108) into Eq. (107) yields, upon integration,

$$\begin{aligned} \int_0^L \sin \gamma_p x \sin \gamma_q x dx = \\ \frac{1}{2} \int_0^L \cos[(\gamma_p - \gamma_q)x] dx - \frac{1}{2} \int_0^L \cos[(\gamma_p + \gamma_q)x] dx = \\ \frac{\sin(\gamma_p - \gamma_q)L}{2(\gamma_p - \gamma_q)} - \frac{\sin(\gamma_p + \gamma_q)L}{2(\gamma_p + \gamma_q)} \end{aligned} \quad (109)$$

The wavenumbers in Eq. (109) are given by Eq. (106) as $\gamma_j = j\pi/L$. Hence, the $(\gamma_p - \gamma_q)L$ and $(\gamma_p + \gamma_q)L$ expressions showing up in Eq. (109) take the following values

$$(\gamma_p - \gamma_q)L = (p - q)\pi L/L = (p - q)\pi \quad (110)$$

$$(\gamma_p + \gamma_q)L = (p + q)\pi L/L = (p + q)\pi \quad (111)$$

Substitution of Eqs. (110), (111) into Eq. (109) followed by substitution into Eq. (121) yields

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = \frac{\sin(p - q)\pi}{2(p - q)\pi/L} - \frac{\sin(p + q)\pi}{2(p + q)\pi/L} = 0 \quad (112)$$

Eq. (112) is zero because sine of any integer multiple of π is zero. This holds true as long as $p \neq q$. QED.

4.3.1.2 Calculation of I_j

Substitution of Eq. (106) into Eq. (105) gives

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \sin^2(\gamma_j x) dx \quad (113)$$

Recall the sine squared trigonometric identity

$$\sin^2 \alpha = (1 - \cos 2\alpha)/2 \quad (114)$$

The use of Eq. (114) into Eq. (113) yields

$$\begin{aligned} I_j &= \int_0^L \sin^2(\gamma_j x) dx = \int_0^L \frac{1}{2} [1 - \cos(2\gamma_j x)] dx \\ &= \frac{1}{2} \int_0^L dx - \frac{1}{2} \int_0^L \cos(2\gamma_j x) dx \end{aligned} \quad (115)$$

The first integral in Eq. (115) gives

$$\int_0^L dx = L \quad (116)$$

The second integral in Eq. (115) gives

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2\gamma_j x) \Big|_0^L = \frac{1}{2\gamma_j} \sin(2\gamma_j L) \quad (117)$$

Recall from Eq. (106) that $\gamma_j = j\pi/L$; upon substitution in Eq. (117) one gets

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2j\pi \cancel{L}/\cancel{L}) = 0 \quad (118)$$

Eq. (118) is zero because sine of any integer multiple of π is zero. Substitution of Eqs. (116), (118) into Eq. (115) yields

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \sin^2(\gamma_j x) dx = L/2 \quad \text{QED} \quad (119)$$

4.3.2 VIBRATION IN A CLOSED-OPEN TUBE

The wavenumbers, natural frequencies, and modeshapes for a tube closed at one end and open at the other end are given by Eqs. (69), (70), (74) as

$$\gamma_j = (2j-1)\pi/2L \quad \omega_j = (2j-1)c\pi/2L \quad U_j(x) = \sin \gamma_j x \quad j=1,2,3,\dots \quad (120)$$

4.3.2.1 Calculation of I_{pq}

Let j take two different values p and q and substitute Eq. (120) into Eq. (104) to get

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = \int_0^L \sin \gamma_p x \sin \gamma_q x dx \quad (121)$$

Recall the “product of sines” identity

$$\sin \alpha \sin \beta = [\cos(\alpha - \beta) - \cos(\alpha + \beta)]/2 \quad (122)$$

The use of Eq. (122) into Eq. (121) yields, upon integration

$$\begin{aligned} \int_0^L \sin \gamma_p x \sin \gamma_q x dx = \\ \frac{1}{2} \int_0^L \cos[(\gamma_p - \gamma_q)x] dx - \frac{1}{2} \int_0^L \cos[(\gamma_p + \gamma_q)x] dx = \\ \frac{\sin(\gamma_p - \gamma_q)L}{2(\gamma_p - \gamma_q)} - \frac{\sin(\gamma_p + \gamma_q)L}{2(\gamma_p + \gamma_q)} \end{aligned} \quad (123)$$

Insertion of the γ_p, γ_q expressions from Eq. (120) into Eq. (123) yields terms like

$$(\gamma_p - \gamma_q)L = [(2p-1) - (2q-1)]\pi L/2L = (p-q)\pi \quad (124)$$

$$(\gamma_p + \gamma_q)L = [(2p-1) + (2q-1)]\pi L/2L = (p+q)\pi \quad (125)$$

Substitution of Eqs. (124), (125) into Eq. (123) followed by substitution into Eq. (121) yields

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = \frac{\sin(p-q)\pi}{2(p-q)\pi/L} - \frac{\sin(p+q)\pi}{2(p+q)\pi/L} = 0 \quad (126)$$

Eq. (126) is zero because sine of any integer multiple of π is zero. This holds true as long as $p \neq q$. QED.

4.3.2.2 Calculation of I_j

Substitution of Eq. (120) into Eq. (105) yields

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \sin^2(\gamma_j x) dx \quad (127)$$

Recall the sine squared trigonometric identity

$$\sin^2 \alpha = (1 - \cos 2\alpha)/2 \quad (128)$$

The use of Eq. (128) into Eq. (127) yields

$$\begin{aligned} I_j &= \int_0^L \sin^2(\gamma_j x) dx = \int_0^L \frac{1}{2} [1 - \cos(2\gamma_j x)] dx \\ &= \frac{1}{2} \int_0^L dx - \frac{1}{2} \int_0^L \cos(2\gamma_j x) dx \end{aligned} \quad (129)$$

The first integral in Eq. (129) gives

$$\int_0^L dx = L \quad (130)$$

The second integral in Eq. (129) gives

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2\gamma_j x) \Big|_0^L = \frac{1}{2\gamma_j} \sin(2\gamma_j L) \quad (131)$$

Recall from Eq. (120) that $\gamma_j = (2j-1)\pi/2L$; upon substitution in Eq. (131) one gets

$$\begin{aligned} \int_0^L \cos(2\gamma_j x) dx &= \frac{1}{2\gamma_j} \sin[\frac{1}{2}(2j-1)\pi] \\ &= \frac{1}{2\gamma_j} \sin(2j-1)\pi = 0 \end{aligned} \quad (132)$$

Eq. (132) is zero because sine of any integer multiple of π is zero. Substitution of Eqs. (130), (132) into Eq. (129) yields

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \sin^2(\gamma_j x) dx = L/2 \quad \text{QED} \quad (133)$$

4.3.3 VIBRATION IN AN OPEN-OPEN TUBE

The wavenumbers, natural frequencies, and modeshapes for a tube open at both ends are given by Eq. (84), (85), (89) as

$$\gamma_j = j\pi/L \quad \omega_j = jc\pi/L \quad U_j(x) = \cos \gamma_j x \quad j=1,2,3,\dots \quad (134)$$

4.3.3.1 Calculation of I_{pq}

Let j take two different values p and q in Eq. (134) and write

$$U_p(x) = \cos \gamma_p x \quad \gamma_p = p\pi/L \quad (135)$$

$$U_q(x) = \cos \gamma_q x \quad \gamma_q = q\pi/L \quad (136)$$

Substitution of Eqs. (135), (136) into Eq. (104) yields

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = \int_0^L \cos \gamma_p x \cos \gamma_q x dx \quad (137)$$

Recall the “product of cosines” identity

$$\cos \alpha \cos \beta = [\cos(\alpha + \beta) + \cos(\alpha - \beta)]/2 \quad (138)$$

The use of Eq. (138) into Eq. (137) yields, upon integration,

$$\begin{aligned} \int_0^L \cos \gamma_p x \cos \gamma_q x dx = \\ \frac{1}{2} \int_0^L \cos[(\gamma_p + \gamma_q)x] dx + \frac{1}{2} \int_0^L \cos[(\gamma_p - \gamma_q)x] dx = \\ \frac{\sin(\gamma_p + \gamma_q)L}{2(\gamma_p + \gamma_q)} + \frac{\sin(\gamma_p - \gamma_q)L}{2(\gamma_p - \gamma_q)} \end{aligned} \quad (139)$$

Insertion of the γ_p, γ_q expressions from Eqs. (135), (136) into Eq. (139) yields expressions of the form

$$(\gamma_p + \gamma_q)L = (p + q)\pi L/L = (p + q)\pi \quad (140)$$

$$(\gamma_p - \gamma_q)L = (p - q)\pi L/L = (p - q)\pi \quad (141)$$

Substitution of Eqs. (140), (141) into Eq. (139) gives

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = \frac{\sin(p + q)\pi}{2(p + q)\pi/L} + \frac{\sin(p - q)\pi}{2(p - q)\pi/L} = 0 \quad (142)$$

Eq. (142) is zero because sine of any integer multiple of π is zero. This holds true as long as $p \neq q$.

4.3.3.2 Calculation of I_j

Substitution of Eq. (134) into Eq. (105) yields

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \cos^2(\gamma_j x) dx \quad (143)$$

Recall the cos squared trigonometric identity

$$\cos^2 \alpha = (1 + \cos 2\alpha)/2 \quad (144)$$

The use of Eq. (144) into Eq. (143) yields

$$\begin{aligned} I_j &= \int_0^L \cos^2(\gamma_j x) dx = \int_0^L \frac{1}{2} [1 + \cos(2\gamma_j x)] dx \\ &= \frac{1}{2} \int_0^L dx + \frac{1}{2} \int_0^L \cos(2\gamma_j x) dx \end{aligned} \quad (145)$$

The first integral in Eq. (145) gives

$$\int_0^L dx = L \quad (146)$$

The second integral in Eq. (145) gives

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2\gamma_j x) \Big|_0^L = \frac{1}{2\gamma_j} \sin(2\gamma_j L) \quad (147)$$

Recall from Eq. (134) that $\gamma_j = j\pi/L$; upon substitution in Eq. (147) one gets

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2j\pi \cancel{L}/\cancel{L}) = 0 \quad (148)$$

Eq. (148) is zero because sine of any integer multiple of π is zero.

Substitution of Eqs. (146), (148) into Eq. (145) yields

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \cos^2(\gamma_j x) dx = L/2 \quad \text{QED} \quad (149)$$

5 SOUND INTENSITY

5.1 SOUND POWER AND SOUND INTENSITY

The sound intensity is the sound power per unit area. It is measured in W/m^2 . Power is force $\times$ velocity. Force per unit area is pressure. Hence the sound intensity, which is sound power per unit area, is pressure time velocity, i.e.,

$$I(x, t) = p(x, t)\dot{u}(x, t) \quad (\text{sound intensity}) \quad (150)$$

Recall the general solution Eq. (37) and assume

$$u(x, t) = \sin(\gamma x - \omega t) \quad (151)$$

Differentiation wrt t of Eq. (151) yields the particle velocity as

$$\dot{u}(x, t) = -\omega \cos(\gamma x - \omega t) \quad (152)$$

Recall Eq. (10) that gives pressure as

$$p(x, t) = -K u'(x, t) \quad (153)$$

Differentiation of Eq. (151) wrt x gives

$$u'(x, t) = \gamma \cos(\gamma x - \omega t) \quad (154)$$

Substitution of Eq. (154) into Eq. (153) yields

$$p(x, t) = -\gamma K \cos(\gamma x - \omega t) \quad (155)$$

Recall the wavenumber definition Eq. (26) which gives $\gamma = \omega/c$; upon substitution in Eq. (155) one gets

$$p(x, t) = -(\omega K/c) \cos(\gamma x - \omega t) \quad (156)$$

Elimination of $-\omega \cos(\gamma x - \omega t)$ between Eqs. (152), (156) yields pressure in terms of particle velocity, i.e.,

$$p(x, t) = \frac{K}{c} \dot{u}(x, t) \quad (157)$$

Recall from Eq. (16) that $K = \rho c^2$; upon substitution in Eq. (157) one gets

$$p(x, t) = \rho c \dot{u}(x, t) \quad (158)$$

Substitution of Eq. (158) into Eq. (150) give the sound intensity in terms of particle velocity, i.e.,

$$I(x, t) = \rho c \dot{u}^2(x, t) \quad (159)$$

Eq. (159) states that the sound intensity is proportional with the square of particle velocity.

The sound intensity can also be expressed in terms of sound pressure. To do so, solve Eq. (158) to get particle velocity in terms of sound pressure, i.e.,

$$\dot{u}(x,t) = \frac{1}{\rho c} p(x,t) \quad (160)$$

Substitution of Eq. (160) into Eq. (150) yields the sound intensity I in terms of sound pressure p , i.e.,

$$I(x,t) = \frac{1}{\rho c} p^2(x,t) \quad (161)$$

The sound intensity calculated so far represents power per unit area measured in W/m^2 .

5.2 SOUND PRESSURE CORRESPONDING TO A GIVEN SOUND INTENSITY

Eq. (161) can be solved to calculate the sound pressure corresponding to a given sound intensity, i.e.,

$$p = \sqrt{\rho c I} \quad (162)$$

5.3 TOTAL SOUND PRESSURE THROUGH A GIVEN AREA

To calculate the total sound power P of sound traversing a given area A one multiplies sound intensity by the area, i.e.,

$$P = IA \quad (\text{sound pressure through area } A) \quad (163)$$

5.4 ACOUSTIC IMPEDANCE

The quantity ρc in Eqs. (158), (159), (160), (161) is called *acoustic impedance* denoted by letter Z , i.e.,

$$Z = \rho c \quad (\text{acoustic impedance}) \quad (164)$$

The units of the acoustic impedance resulting from Eq. (164) are $\text{kg}/(\text{m}^2\text{sec})$. The acoustic impedance is used to relate particle velocity and acoustic pressure and to express sound intensity. Substitution of Eq. (164) into Eqs. (158), (159), (160), (161) yields, upon rearrangement,

$$p(x,t) = Z \dot{u}(x,t) \quad \dot{u}(x,t) = \frac{1}{Z} p(x,t) \quad (165)$$

$$I(x,t) = Z \dot{u}^2(x,t) = p^2(x,t)/Z \quad (166)$$

The units of Z resulting from Eq. (166) are $\text{Pa}/(\text{m/sec})$.

5.5 SOUND LOUDNESS

<https://www.nps.gov/teachers/classrooms/sound-intensity-and-loudness.htm#>

https://en.wikipedia.org/wiki/Absolute_threshold_of_hearing

https://en.wikipedia.org/wiki/Threshold_of_pain

<https://en.wikipedia.org/wiki/Decibel>

5.5.1 THRESHOLD OF HUMAN HEARING

The weakest sound that a human can hear is called the *threshold of human hearing*; it corresponds to a sound intensity of about $1 \times 10^{-12} \text{ W/m}^2 = 1 \text{ pW/m}^2$ (roughly the sound of a mosquito flying 3 meters away). Thus, the reference sound intensity, I_0 is defined as

$$I_0 = 1 \text{ pW/m}^2 \quad (\text{reference sound intensity}) \quad (167)$$

The sound pressure p_0 corresponding to the threshold of human hearing is usually taken as about $20 \times 10^{-6} \text{ Pa} = 20 \text{ } \mu\text{Pa}$. The exact value calculated with Eq (162) for $c=340 \text{ m/s}$ and $\rho=1.2 \text{ kg/m}^3$ is

$$p_0 = 20.199 \text{ } \mu\text{Pa} \quad (\text{reference sound pressure}) \quad (168)$$

5.5.2 PAIN THRESHOLD

The threshold of pain for humans has a sound intensity of $I_{\text{pain}}=1 \text{ W/m}^2$. It corresponds to $p_{\text{pain}} \approx 20 \text{ Pa}$ sound pressure, i.e.,

$$I_{\text{pain}} = 1 \text{ W/m}^2 \quad (\text{pain sound intensity}) \quad (169)$$

$$p_{\text{pain}} = 20 \text{ Pa} \quad (\text{pain sound pressure}) \quad (170)$$

5.5.3 NOISE

"Noise" is any unwanted sound or sound without function. Noise can adversely affect people. Loud or prolonged noise has been shown to cause hearing impairment, hypertension, sleep disturbance, annoyance, and ischemic heart disease (heart disease linked to reduced blood flow to the heart). Other complications include possible immune system changes and birth defects. Road noise has been shown to constrict arteries and elevate blood pressure. It may even lead to heart attacks. High noise levels at night increase stress hormone production which causes arteries to constrict and elevates blood pressure.

5.5.4 dB SCALE

Sound intensity is measured in bels and decibels. The decibel (dB) is the tenth part of a bel (B). The dB unit is used in most cases because using B would give too small values.

A whisper is between 20 and 30 dB; noisy conversation is about 50 dB; a vacuum cleaner is about 70 dB; a lawn mower is about 90 dB; a car horn at 1 m is about 110 dB.

The dB value of a sound is calculated by taking $\log_{10}$ (logarithm base 10) of the ratio between the sound intensity and the threshold of human hearing and multiplying the result by 10. To calculate the dB value of a sound of intensity I , one applies the following formula

$$I_{\text{dB}} = 10 \log_{10}(I/I_0) \quad (\text{sound intensity in dB}) \quad (171)$$

where I_0 is given by Eq. (167).

To calculate the dB value of a sound pressure p , one recalls Eq. (161) which indicates that the sound intensity is proportional to pressure squared. As $\log(p^2)=2\log(p)$, then the dB value of a sound pressure p is calculated with formula

$$p_{\text{dB}} = 20 \log_{10}(p/p_0) \quad (\text{sound pressure in dB}) \quad (172)$$

where p_0 is given by Eq. (162).

In view of Eqs. (171), (172), it becomes apparent that the threshold of human hearing corresponds to 0 dB since $I_0/I_0=1$, $p_0/p_0=1$ and $\log(1)=0$.

Note that the dB conversion formula for sound intensity Eq. (171) is different from the conversion formula for sound pressure Eq. (172). This ensures that a given dB value corresponds to same physical situation whether measured in sound intensity or sound pressure. The MATLAB function `mag2db` uses Eq. (172) to do the conversion, whereas the function `db2mag` does the inverse conversion.

5.5.5 PAIN THRESHOLD IN dB

The pain threshold was defined in Section 5.5.2 as $I_{\text{pain}} = 1 \text{ W/m}^2$. Its dB value is calculated as

$$(I_{\text{pain}})_{\text{dB}} = 10 \log_{10}(1\text{W}/1\text{pW})=10 \times 12=120 \text{ dB} \quad (173)$$

Eq. (173) indicates that the threshold of pain in sound intensity is reached at 120 dB, i.e.,

$$(I_{\text{pain}})_{\text{dB}} = 120\text{dB} \quad (\text{pain threshold of sound intensity in dB}) \quad (174)$$

†
F&F

dB scale decibel
1/10

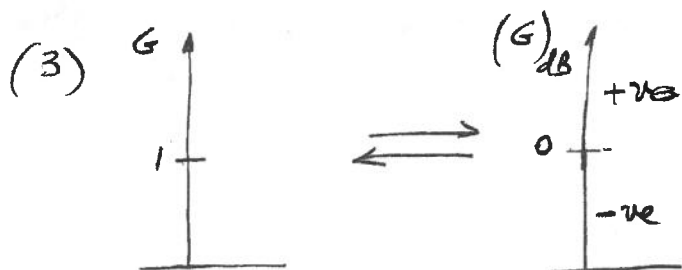
dB scale is a $\log_{10}$ scale magnified by 20

$$(G)_{dB} = 20 \log_{10} G$$

Properties

(1) Only +ve numbers have dB value
(cannot take log of -ve numbers!)

(2) $\left(\frac{1}{G}\right)_{dB} = -(G)_{dB}$ (reciprocal numbers)



$$G > 1$$

$(G)_{dB}$ is +ve

$$G = 1$$

$$(G)_{dB} = 0$$

$$G < 1$$

$(G)_{dB}$ is -ve

2
FRF

(4) Scale up or scale down in physical values means shift up or shift down in dB

→ (a) double (octave up) = +6 dB

(b) "half" (octave down) = -6 dB

(c) "ten times" (decade) = +20 dB

octave = ^{(A B C D E F G) A₁}
8th musical note has double the frequency $f_{A_1} = 2f_A$

Proof

(a) $\log_{10} 2 = 0.303$; $20 \log 2 \approx 6$

$G_2 = 2G_1$

$(G_2)_{dB} = 20 \log_{10} (2G_1) = 20 \log_{10} 2 + 20 \log_{10} G_1$

$= \frac{20 \times 0.303}{\approx 6} + (G_1)_{dB}$

$\gg \text{mag } 2 \text{ dB (2)}$

ans 6.0206

$(G_2)_{dB} = (G_1)_{dB} + 6 \text{ dB}$

(b) $G_2 = \frac{1}{2} G_1$

$(G_2)_{dB} = (G_1)_{dB} + 20 \log_{10} \left(\frac{1}{2}\right)$

$= (G_1)_{dB} + 20 (-\log_{10} 2)$

$= (G_1)_{dB} - 6 \text{ dB}$

3
FRF

$$(c) \quad G_2 = 10 G_1$$

$$(G_2)_{dB} = (G_1)_{dB} + 20 \log_{10}(10)^1$$

$$= (G_1)_{dB} + 20 \text{ dB}$$

6 HELMHOLTZ RESONATOR

https://en.wikipedia.org/wiki/Helmholtz_resonance

Helmholtz resonance or **wind throb** is the phenomenon of air [resonance](#) in a cavity, such as when one blows across the top of an empty bottle. The name comes from a device created in the 1850s by [Hermann von Helmholtz](#), the **Helmholtz resonator**, which he used to identify the various [frequencies](#) or [musical pitches](#) present in [music](#) and other complex sounds.^[1]

History [\[edit \]](#)

Helmholtz described in his 1862 book *On the Sensations of Tone* an apparatus able to pick out specific frequencies from a complex [sound](#). The Helmholtz [resonator](#), as it is now called, consists of a rigid container of a known volume, nearly spherical in shape, with a small neck and hole in one end and a larger hole in the other end to emit the sound.

When the resonator's 'nipple' is placed inside one's ear, a specific frequency of the complex sound can be picked out and heard clearly. In his book Helmholtz explains: When we "apply a resonator to the ear, most of the tones produced in the surrounding air will be considerably damped; but if the proper tone of the resonator is sounded, it brays into the ear most powerfully.... The proper tone of the resonator may even be sometimes heard cropping up in the whistling of the wind, the rattling of carriage wheels, the splashing of water."

A set of varied size resonators was sold to be used as discrete acoustic filters for the spectral analysis of complex sounds. There is also an adjustable type, called a universal resonator, which consists of two [cylinders](#), one inside the other, which can slide in or out to change the volume of the cavity over a continuous range. An array of 14 of this type of resonator has been employed in a mechanical [Fourier sound analyzer](#). This resonator can also emit a variable-frequency tone when driven by a stream of air in the "[tone variator](#)" invented by [William Stern](#), 1897.^[2]



A brass, spherical Helmholtz resonator based on his original design, circa 1890-1900. [\[edit \]](#)



A selection of Helmholtz resonators from 1870, at [Hunterian Museum and Art Gallery in Glasgow](#). [\[edit \]](#)

Resonance of a sphere of air (vented) [edit]

The resonant frequency of a rigid cavity of static volume V_0 with a necked sound hole of area A and length L is given by the [Helmholtz resonance](#) formula^{[6][7]}

$$f = \frac{v}{2\pi} \sqrt{\frac{A}{V_0 L_{eq}}}$$

where L_{eq} is the equivalent length of the neck with [end correction](#)

$$L_{eq} = L + 0.75d \text{ for an unflanged neck}^{[8]}$$

$$L_{eq} = L + 0.85d \text{ for a flanged neck}$$

For a spherical cavity, the resonant frequency formula becomes

$$f = \frac{vd}{\pi} \sqrt{\frac{3}{8L_{eq}D^3}}$$

where

D = diameter of sphere

d = diameter of sound hole

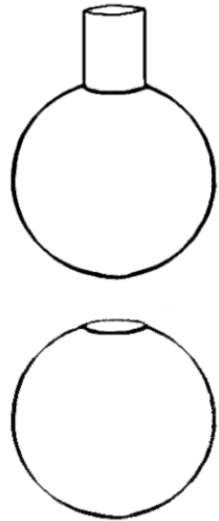
For a sphere with just a sound hole, $L=0$ and the surface of the sphere acts as a flange, so

$$f = \frac{v}{\pi} \sqrt{\frac{3d}{8(0.85)D^3}}$$

In dry air at 20 °C, with d and D in metres, f in [hertz](#), this becomes

$$f = 72.6 \sqrt{\frac{d}{D^3}}$$

Note: v is the sound speed



STRING VIBRATION

String vibration appears in taut strings, such as the strings of a violin or a guitar. The study of string vibration is associated with music.

1 ANALYSIS OF STRING VIBRATION

Consider a taut string stretched by a constant tension load T (Figure 1a). The string undergoes small-amplitude transverse vibration with frequency ω as described by the following formula:

$$u(x,t)=U(x)e^{-i\omega t} \quad (\text{vibration motion}) \quad (1)$$

Because the transverse motion is so small, the change in tension T due to this motion is negligible. Hence, the tension T will be assumed constant throughout this analysis.

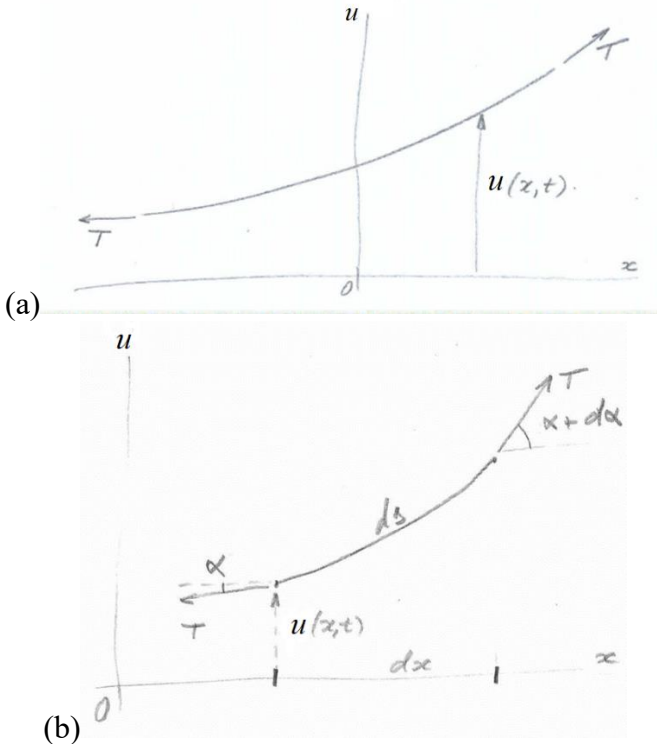


Figure 1 Taut string under tensile load T : (a) general view; (b) infinitesimal element

Consider an infinitesimally small element ds of the deformed string as shown in Figure 1b. The corresponding increment in the x direction is dx . The slope of the string grows from α at the left of the

element ds to $\alpha+d\alpha$ at the right of the element ds . However, the tension T in the string is assumed constant and it does not vary along the string element ds .

1.1 FREE-BODY DIAGRAM

Free body diagram analysis of the infinitesimal element gives the sum of forces in the transverse direction as

$$\Sigma F_y = -T \sin \alpha + T \sin(\alpha + d\alpha) \quad (2)$$

1.2 SMALL-DISPLACEMENT APPROXIMATION

Since u is assumed to be very small, we can apply the small-displacement approximation as follows:

(i) the infinitesimal string element ds can be approximated by its x -axis projection dx , i.e.,

$$ds \approx dx \quad (3)$$

(ii) the sine function is approximated by its argument in radians

$$\sin \alpha \approx \alpha \quad \text{and} \quad \sin(\alpha + d\alpha) \approx \alpha + d\alpha \quad (4)$$

Hence, Eq. (2) becomes

$$\Sigma F_y = -T\alpha + T(\alpha + d\alpha) = Td\alpha \quad (\text{linearized sum of } y \text{ forces}) \quad (5)$$

1.3 SLOPE IN TERMS OF DISPLACEMENT

The slope α is calculated as the gradient of the displacement $u(x,t)$, i.e.,

$$\alpha = u' \quad (\text{slope}) \quad (6)$$

where the dash signifies the derivative wrt space x , i.e.,

$$()' = \frac{\partial}{\partial x} () \quad (7)$$

Using Eq. (6), one writes the slope increment $d\alpha$ as

$$d\alpha \approx u'' dx \quad (8)$$

Substitution of Eq. (8) into Eq. (5) yields

$$\Sigma F_y = Tu'' dx \quad (\text{sum of } y \text{ forces}) \quad (9)$$

1.4 NEWTON'S LAW OF MOTION

The string has a mass density m per unit length. If the string is bare, then $m = \rho A$. However, some strings may have additional material wrapped on them in order to increase m and give a lower tone; in this case, m would be different from the ρA of the bare string. To keep things generic, we will consider that the string has a mass

density m per unit length that is determined by measuring the string mass and dividing it by the string length.

The mass of the infinitesimal string element dx is mdx . The acceleration of the infinitesimal element is $\ddot{u}$, where a dot over a variable represents derivative with respect to time, i.e.,

$$\left(\ddot{} \right) = \frac{\partial^2}{\partial t^2} \left(\right) \quad (10)$$

The two dots in $\ddot{u}$ represent the second derivative wrt time, i.e., the acceleration.

Newton's law of motion applied to the infinitesimal element states that the sum of forces should equal mass times acceleration, i.e.,

$$\Sigma F_y = (mdx)\ddot{u} \quad (\text{NLM equation}) \quad (11)$$

1.5 GOVERNING EQUATION FOR STRING VIBRATION

Substitution of Eq. (8) into Eq. (11) yields

$$T u'' dx = (m dx) \ddot{u} \quad (12)$$

or, upon simplification by dx ,

$$T u'' = m \ddot{u} \quad (\text{governing equation for string vibration}) \quad (13)$$

Eq. (13) is the governing equation for the vibration of a taught string.

Define the wavespeed c in a taught string as

$$c = \sqrt{\frac{T}{m}} \quad c^2 = \frac{T}{m} \quad (\text{wavespeed in a taught string}) \quad (14)$$

1.6 WAVE EQUATION

Substitution of Eq. (14) into Eq. (13) yields the wave equation

$$c^2 u'' = \ddot{u} \quad (\text{wave equation}) \quad (15)$$

2 GENERAL SOLUTION FOR STRING VIBRATION

Recall Eq. (1) describing the vibration motion as

$$u(x, t) = U(x) e^{-i\omega t} \quad (\text{vibration motion}) \quad (1)$$

Double differentiation of Eq. (1) wrt to space and time yields

$$u''(x, t) = \frac{\partial^2}{\partial x^2} \left(U(x) e^{-i\omega t} \right) = e^{-i\omega t} \frac{d^2}{dx^2} U(x) = U''(x) e^{-i\omega t} \quad (16)$$

$$\ddot{u}(x, t) = \frac{\partial^2}{\partial t^2} \left(U(x) e^{-i\omega t} \right) = U(x) \frac{d^2}{dt^2} e^{-i\omega t} = -\omega^2 U(x) e^{-i\omega t} \quad (17)$$

Substitution of Eqs. (16), (17) into Eq. (15) gives

$$c^2 U''(x) e^{-i\omega t} = -\omega^2 U(x) e^{-i\omega t} \quad (18)$$

Upon division by $e^{-i\omega t}$, Eq. (18) becomes

$$c^2 U''(x) = -\omega^2 U(x) \quad (19)$$

Recall the wavenumber γ defined as

$$\gamma = \omega/c \quad (\text{wavenumber}) \quad (20)$$

Substitution of Eq. (20) into Eq. (19) yields, upon rearrangement,

$$U''(x) + \gamma^2 U(x) = 0 \quad (21)$$

Eq. (20) is a linear order ordinary differential equation (ODE) in space. Assume exponential solution of the form

$$U(x) = e^{sx} \quad (22)$$

Upon differentiation, Eq. (22) becomes

$$U''(x) = s^2 e^{sx} = s^2 U(x) \quad (23)$$

2.1 CHARACTERISTIC EQUATION

Substitution of Eq. (23) into Eq. (21) yields the characteristic equation

$$s^2 + \gamma^2 = 0 \quad (\text{characteristic equation}) \quad (24)$$

The solutions of the characteristic equation Eq. (24) are

$$s_1 = +i\gamma \quad s_2 = -i\gamma \quad (25)$$

The complete solution of Eq. (21) is a linear superposition of the individual solutions, i.e.,

$$U(x) = A_1 e^{i\gamma x} + A_2 e^{-i\gamma x} \quad (26)$$

2.2 WAVE SOLUTION

Substitution of Eq. (26) into Eq. (1) yields the general solution as

$$u(x,t) = U(x) e^{-i\omega t} = A_1 e^{+i(\gamma x - \omega t)} + A_2 e^{-i(\gamma x - \omega t)} \quad (\text{wave solution}) \quad (27)$$

Eq. (27) is a formula that reflects the wave aspects of the phenomenon. Eq. (27) shows the wave propagation along the string. The first term is a forward wave whereas the second term is a backward wave. The wave propagation happens in the x direction along the string although the displacement of the string particles is transverse to the string.

2.3 VIBRATION SOLUTION

Using Euler's formulae, one can write the complex exponentials of Eq. (26) in terms of trigonometric functions and get the solution of Eq. (21) as

$$U(x)=C_1\sin\gamma x+C_2\cos\gamma x \quad (28)$$

Substitution of Eq. (28) into Eq. (1) yields the general solution as

$$u(x,t)=U(x)e^{-i\omega t}=(C_1\sin\gamma x+C_2\cos\gamma x)e^{-i\omega t} \quad (\text{vibration solution}) \quad (29)$$

Eq. (29) is a formula that reflects the vibration aspects of the phenomenon. The constants C_1 and C_2 are to be determined by matching the boundary conditions at the two ends of a finite-length string.

3 BOUNDARY CONDITIONS AT THE STRING ENDS

Assume the string of length L is fixed at both ends just like in a string instrument (Figure 2).



Figure 2 Taut string of length L under constant tension T

Let's examine the boundary conditions. At the fixed ends, the displacement is zero. Hence, the boundary conditions are

$$u(0,t)=0 \quad (\text{fixed boundary condition at the left}) \quad (30)$$

$$u(L,t)=0 \quad (\text{fixed boundary condition at the right}) \quad (31)$$

Insertion of Eq. (29) into the boundary conditions Eqs. (30), (31) and simplification by $e^{-i\omega t}$ yields

$$u(0,t) = C_2 = 0 \quad (32)$$

$$u(L,t) = (C_1 \sin \gamma L + C_2 \cos \gamma L) = 0 \quad (33)$$

Eq. (32) gives $C_2=0$. Introduction of this value $C_2=0$ into Eq. (33) yields, upon simplification,

$$C_1 \sin \gamma L = 0 \quad (34)$$

The trivial solution of Eq. (34) is $C_1=0$ which means no vibration. To find a nontrivial solution $C_1 \neq 0$ of Eq. (34), one needs to have $\sin \gamma L = 0$, i.e.,

$$\sin \gamma L = 0 \quad (35)$$

If Eq. (34) is satisfied, i.e., $\sin \gamma L = 0$, then the value of C_1 becomes arbitrary since any C_1 value will satisfy Eq. (34) as long as $\sin \gamma L = 0$. This means that one can take any value for C_1 and still satisfy the conditions of the problem, which means that the problem has one degree of indeterminacy.

3.1 EIGENVALUE EQUATION

Introduce the notation

$$z = \gamma L \quad (36)$$

Substitution of Eq. (36) into Eq. (35) yields the eigenvalue equation,

$$\sin z = 0 \quad (\text{eigenvalue equation}) \quad (37)$$

Equation (37) defines the values of z for which nontrivial solutions of Eq. (34) exist. These values of z are the *eigenvalues* of the system. Solution of Eq. (37) gives

$$z = \pi, 2\pi, 3\pi, \dots \quad (38)$$

$$z_j = j\pi \quad j = 1, 2, 3, \dots \quad (39)$$

3.2 WAVENUMBERS

Substitution of Eq. (39) into Eq. (36) yields the wavenumbers

$$\gamma_j = j\pi/L \quad j = 1, 2, 3, \dots \quad (40)$$

3.3 NATURAL FREQUENCIES

Recall the relationship between wavenumber γ , wavespeed c and angular frequency ω , i.e.,

$$\gamma = \omega/c \quad (41)$$

Eq. (41) can be solved for ω to yield

$$\omega = \gamma c \quad (42)$$

Substitution of Eq. (40) into Eq. (42) yields the *natural angular frequencies*

$$\omega_j = jc\pi/L \quad j = 1, 2, 3, \dots \quad (43)$$

Since $\omega = 2\pi f$, the corresponding Hz frequencies f_j are given by the formula

$$f_j = jc/2L \quad j = 1, 2, 3, \dots \quad (44)$$

We note that Eqs. (40), (43), (44) predict a denumerable infinite set of wavenumbers and natural frequencies. The first frequency, f_1 , is called the *fundamental frequency*, whereas the other frequencies (f_j , $j = 2, 3, 4, \dots$) are called *overtones*. The overtones are multiples of the fundamental frequency. The fundamental frequency is also referred to as the *fundamental harmonic*, whereas the overtones are also referred to as *higher harmonics*.

3.4 WAVELENGTHS

Recall the relationship between *wavelength* λ and wavenumber γ , i.e.,

$$\lambda = 2\pi/\gamma \quad (45)$$

Substitution of Eq. (40) into Eq. (45) yields

$$\lambda_j = \frac{2\pi}{\gamma_j} = \frac{2\pi}{j\pi/L} = \frac{2L}{j} \quad j = 1, 2, 3, \dots \quad (46)$$

It is apparent from Eq. (46) that higher frequencies have shorter wavelengths. The vibration of the bar can be interpreted in terms of standing waves fitting the boundary conditions. In the present, case the standing waves are such that the bar of length L accommodates multiples of the half wavelengths, $\lambda_j/2$, $j=1,2,3,\dots$, i.e.,

$$L=j\lambda_j/2 \quad j=1,2,3,\dots \quad (47)$$

The higher the vibration frequency, the shorter the wavelength of these standing waves. The wavenumbers, resonant frequencies, and wavelength multiplicity for the vibration of the bar of length L fixed at both ends are listed in Table 1.

3.5 MODESHAPES

Recall Eq. (28), i.e.,

$$U(x)=C_1\sin\gamma x+C_2\cos\gamma x \quad (28)$$

where the constants C_1 and C_2 are given by the solution of Eqs. (32) and (33). Eq. (32) gave $C_2=0$ whereas Eq. (33) yielded Eq. (34), i.e., $C_1\sin\gamma L=0$. The nontrivial solution of Eq. (34) was only possible for special values of γL that made $\sin\gamma L=0$. These special values were called eigenvalues. Associated with the eigenvalues are the wavenumbers $\gamma_j=j\pi/L$. For each wavenumber $\gamma_j=j\pi/L$, one gets $\sin\gamma_j L=0$ in Eq. (34) and hence the value of C_1 becomes arbitrary since any C_1 value will satisfy Eq. (34) as long as $\sin\gamma L=0$. This means that one can take any value for C_1 and still satisfy the conditions of the problem, which means that the problem has one degree of indeterminacy. Here, we take $C_1=1$. Thus, with $C_1=1$ and $C_2=0$, Eq. (28) becomes

$$U_j(x)=\sin\gamma_j x \quad j=1,2,3,\dots \quad (\text{modeshapes}) \quad (48)$$

where γ_j is given by Eq. (40).

Eq. (48) defines *natural modes of vibration (modeshapes)*. The modeshapes of Eq. (48) may be scaled up or down arbitrarily. Whatever the scale, the modeshapes will still satisfy the boundary conditions of the problem.

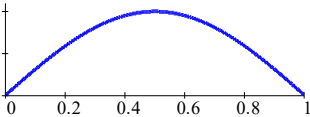
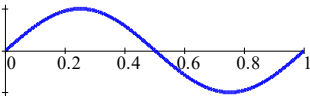
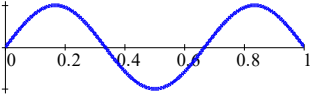
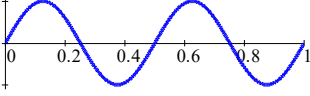
Table 1 gives the first four natural frequencies and corresponding modeshapes. One notes that the motion is zero at both ends as expected.

The complete vibration motion for each mode j is given by Eq. (29) as

$$u_j(x, t) = U_j(x) e^{-i\omega_j t} \quad (\text{vibration of mode } j) \quad (49)$$

where $U_j(x)$ is given by Eq. (48) and ω_j is given by Eq. (43). Note that each vibration mode has its own distinct frequency of vibration ω_j in rad/sec and the associated Hz frequency f_j .

Table 1 Vibration modeshapes of a taut string of length L fixed at both ends

Mode #	Eigenvalue $z = \gamma L$ $z = j\pi$	Resonant frequency	Modeshape		Wavelength multiplicity
1	$z_1 = \gamma_1 L = \pi$	$f_1 = \frac{c}{2L}$		$U_1 = \sin \pi \frac{x}{L}$	$L = \frac{\lambda_1}{2}$
2	$z_2 = \gamma_2 L = 2\pi$	$f_2 = 2 \frac{c}{2L}$		$U_2 = \sin 2\pi \frac{x}{L}$	$L = 2 \frac{\lambda_2}{2}$
3	$z_3 = \gamma_3 L = 3\pi$	$f_3 = 3 \frac{c}{2L}$		$U_3 = \sin 3\pi \frac{x}{L}$	$L = 3 \frac{\lambda_3}{2}$
4	$z_4 = \gamma_4 L = 4\pi$	$f_3 = 4 \frac{c}{2L}$		$U_4 = \sin 4\pi \frac{x}{L}$	$L = 4 \frac{\lambda_4}{2}$

4 DISCUSSION OF RELATIVE EFFECT OF T , m , AND L

Recall Eq. (44) that predicts the string frequency in terms of length L , mode number j and wavespeed c , i.e.,

$$f_j = jc/2L \quad j=1,2,3,\dots \quad (44)$$

Let's focus our attention on the fundamental frequency f_1 since the higher harmonics can be deduced from f_1 by multiplying by the mode number j . To calculate f_1 , make $j=1$ in Eq. (44) and write simply

$$f = c/2L \quad (\text{fundamental frequency}) \quad (50)$$

4.1 EFFECT OF LENGTH L

It is apparent from Eq. (50) that the frequency decreases when the length increases. Conversely, when the string is shortened, e.g., by placing the fingers on the fingerboard, then the frequency will increase resulting in a higher pitch sound.

The musical instruments with long strings, e.g., the harp or the bass, can produce sound of much lower pitch than the short-string instruments such as the violin.

4.2 EFFECT OF MASS

The distributed mass m of the string affects the wavespeed c which in turn affects the frequency f . Recall Eq. (14) defining the wavespeed as

$$c = \sqrt{T/m} \quad (\text{wavespeed in a taught string}) \quad (14)$$

As the distributed mass m increases, the wavespeed decreases. Eq. (50) shows that the frequency f is proportional to wavespeed c . Hence, as the wavespeed c decreases when mass m increase the frequency f will also decrease. This is the reason why the bass and double strings are overwound with material that adds to the mass but not to the stiffness of the string (Figure 4).

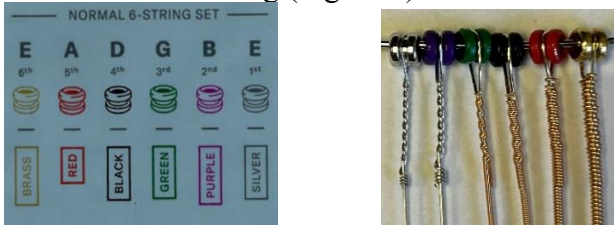


Figure 3 six light-gauge d'Addario guitar strings

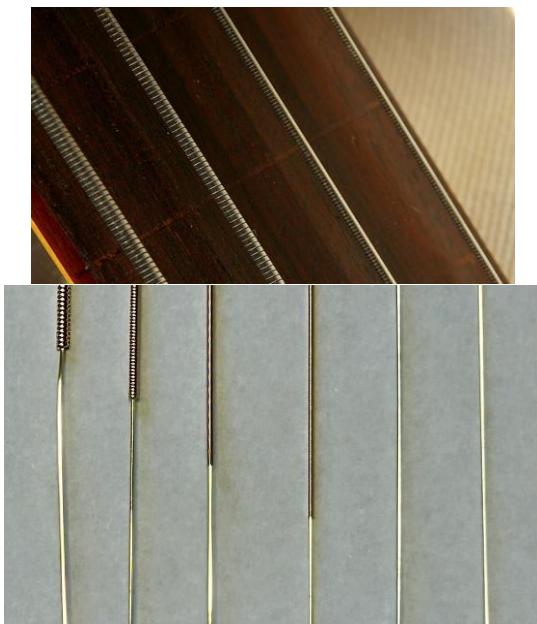


Figure 4 overwound strings have lower vibration frequency

4.3 EFFECT OF STRING TENSION

The string tension T affects the wavespeed c which in turn affects the frequency f . Recall Eq. (14) defining the wavespeed as

$$c = \sqrt{T/m} \quad (\text{wavespeed in a taught string}) \quad (14)$$

As the tension T increases, the wavespeed c also increases. Eq. (50) showed that the frequency f is proportional to wavespeed c . Hence, as the wavespeed c increases the frequency f will also increase. In summary, the frequency can be increased or decreased by tightening or loosening the string during tuning.

4.4 HOW TO LOWER OR RAISE THE PITCH OF A GIVEN STRING

The pitch of a given string can be lowered or raised by two methods:

- (i) during preperformance tuning, one adjust the string pitch by lowering or raising the tension in the string (Figure 5a). This method was previously discussed in Section 4.3.
- (ii) during the actual performance, the pitch of a string is modified by placing the finger on the fingerboard which

shortens the effective length L (Figure 5). This method was previously discussed in Section 4.1.



(a)



(b)

Figure 5 changing the pitch of a string: (a) preperformance tuning by adjusting string tension; (b) performance changing of the string pitch by changing finger position on the fingerboard

<https://www.youtube.com/watch?v=EM5GUmv7F2I>

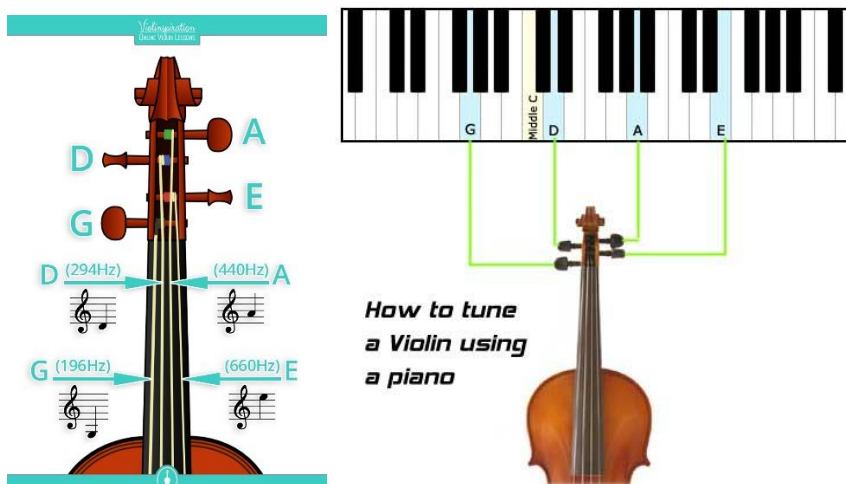


Figure 6 violin tuning

5 STRING MODESHAPE ORTHONORMALITY AND MODAL COEFFICIENTS

5.1 ORTHOGONALITY OF MODESHAPES

Modeshapes orthogonality is an important property that allows one to verify if some independently derived functions qualify to be considered as modeshapes. For vibration of a string of length L , the modeshape orthogonality is stated as

$$\int_0^L U_p(x)U_q(x)dx = \begin{cases} L/2 & \text{for } p = q \\ 0 & \text{otherwise} \end{cases} \quad (\text{orthogonality}) \quad (51)$$

Eq. (51) implies that the product between two different modeshapes is zero when integrated over the length L of the string. This is the **orthogonality** property. If the two modeshapes are the same, i.e., the modeshape is multiplied upon itself, then the result is a finite number, in this case $L/2$. If one divides the integral by this finite number $L/2$, then the result is 1. This is the process of **normalization**. Thus, the **orthonormality** property of the modeshapes is expressed as

$$\int_0^L U_p(x)U_q(x)dx / (L/2) = \begin{cases} 1 & \text{for } p = q \\ 0 & \text{otherwise} \end{cases} \quad (\text{orthonormality}) \quad (52)$$

The modeshape orthonormality is usually verified through numerical integration, although closed-form analytical proof may also be possible.

5.2 MODAL COEFFICIENTS

The modal coefficients are used to describe the modal behavior in response to an external excitation as described later in this course.

5.2.1 MODAL MASS AND MODAL STIFFNESS

Two commonly used modal coefficients are the modal mass m_j and the modal stiffness k_j . Their general definition is

$$m_j = \int_0^L m U_j^2(x)dx \quad (\text{modal mass}) \quad (53)$$

$$k_j = -\int_0^L T U_j''(x)U_j(x)dx \quad (\text{modal stiffness}) \quad (54)$$

5.2.2 RELATIONSHIP BETWEEN MODAL MASS AND MODAL STIFFNESS

Modal mass and modal stiffness obey the following relationship

$$k_j = \omega_j^2 m_j \quad (55)$$

The relationship between modal mass and modal stiffness expressed by Eq. (55) is similar to the mass-stiffness relationship $k=\omega^2 m$ observed in a 1-dof mass-spring system. The modal coefficients allow us to treat each mode of vibration as a simple 1-dof vibration system.

Proof of Eq. (55)

The proof of Eq. (55) proceeds as follows. Recall Eq. (19), i.e.,

$$c^2 U''(x) = -\omega^2 U(x) \quad (56)$$

Recall $c^2=T/m$. Hence, Eq. (56) becomes

$$T U''(x) = -m \omega^2 U(x) \quad (57)$$

For the j^{th} modeshape, Eq. (57) is written as

$$T U_j''(x) = -m \omega_j^2 U_j(x) \quad (58)$$

Substitution of Eq. (58) into the modal stiffness definition Eq. (54) and use of the modal mass definition Eq. (53) gives

$$\begin{aligned} k_j &= -\int_0^L T U_j''(x) U_j(x) dx \\ &= -\int_0^L (-m \omega_j^2) U_j(x) U_j(x) dx \quad \text{QED} \\ &= \omega_j^2 \int_0^L m U_j^2(x) dx = \omega_j^2 m_j \end{aligned} \quad (59)$$

5.2.3 MODAL MASS AND MODAL STIFFNESS VALUES

In a uniform string, the mass density m is constant and hence gets outside the integral sign in Eq. (53) giving

$$m_j = m \int_0^L U_j^2(x) dx \quad (\text{modal mass}) \quad (60)$$

In virtue of Eq. (51), the modal mass m_j take the value $mL/2$. Substitution of this value in Eq. (55) yields the modal stiffness k_j as

$$k_j = \omega_j^2 \rho A L/2, \text{ i.e.,}$$

$$m_j = mL/2 \quad (\text{modal mass}) \quad (61)$$

$$k_j = \omega_j^2 m L / 2 \quad (\text{modal stiffness}) \quad (62)$$

Note that the modal mass value of $mL/2$ represents half the mass of the string. Also note that the modal mass m_j has the same value for all the vibration modes.

5.3 CLOSED-FORM ANALYTICAL DERIVATION OF MODESHAPE ORTHONORMALITY

To study analytically modeshape orthonormality, consider the integrals

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx \quad \text{for } p \neq q \quad (63)$$

$$I_j = \int_0^L U_j^2(x) dx \quad \text{for } p=q=j \quad (64)$$

In virtue of Eq. (51), these equations should take the values

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = 0 \quad \text{for } p \neq q \quad (65)$$

$$I_j = \int_0^L U_j^2(x) dx = L/2 \quad \text{for } p=q=j \quad (66)$$

The wavenumbers, natural frequencies, and modeshapes are given by Eqs. (40), (43), (48) as

$$\gamma_j = j\pi/L \quad \omega_j = jc\pi/L \quad U_j(x) = \sin \gamma_j x \quad j=1,2,3,\dots \quad (67)$$

5.3.1 CALCULATION OF I_{pq}

Let j take two different values p and q and substitute Eq. (67) into Eq. (65) to get

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = \int_0^L \sin \gamma_p x \sin \gamma_q x dx \quad (68)$$

Recall the “product of sines” trigonometric identity

$$\sin \alpha \sin \beta = [\cos(\alpha - \beta) - \cos(\alpha + \beta)]/2 \quad (69)$$

The use of Eq. (69) into Eq. (68) yields, upon integration,

$$\begin{aligned} \int_0^L \sin \gamma_p x \sin \gamma_q x dx = \\ \frac{1}{2} \int_0^L \cos[(\gamma_p - \gamma_q)x] dx - \frac{1}{2} \int_0^L \cos[(\gamma_p + \gamma_q)x] dx = \\ \frac{\sin(\gamma_p - \gamma_q)L}{2(\gamma_p - \gamma_q)} - \frac{\sin(\gamma_p + \gamma_q)L}{2(\gamma_p + \gamma_q)} \end{aligned} \quad (70)$$

The wavenumbers in Eq. (70) are given by Eq. (67) as $\gamma_j = j\pi/L$. Hence, the $(\gamma_p - \gamma_q)L$ and $(\gamma_p + \gamma_q)L$ expressions showing up in Eq. (70) take the following values

$$(\gamma_p - \gamma_q)L = (p - q)\pi L/L = (p - q)\pi \quad (71)$$

$$(\gamma_p + \gamma_q)L = (p + q)\pi L/L = (p + q)\pi \quad (72)$$

Substitution of Eqs. (71), (72) into Eq. (70) followed by substitution into Eq. (68) yields

$$\int_0^L U_p(x) U_q(x) dx = \frac{\sin(p - q)\pi}{2(p - q)\pi/L} - \frac{\sin(p + q)\pi}{2(p + q)\pi/L} = 0 \quad \text{QED} \quad (73)$$

Eq. (73) is zero because sine of any integer multiple of π is zero. This holds true as long as $p \neq q$.

5.3.2 CALCULATION OF I_j

Substitution of Eq. (67) into Eq. (66) yields

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \sin^2(\gamma_j x) dx \quad (74)$$

Recall the “sine squared” trigonometric identity

$$\sin^2 \alpha = (1 - \cos 2\alpha)/2 \quad (75)$$

The use of Eq. (75) into Eq. (74) yields

$$\begin{aligned} I_j &= \int_0^L \sin^2(\gamma_j x) dx = \int_0^L \frac{1}{2} [1 - \cos(2\gamma_j x)] dx \\ &= \frac{1}{2} \int_0^L dx - \frac{1}{2} \int_0^L \cos(2\gamma_j x) dx \end{aligned} \quad (76)$$

The first integral in Eq. (76) gives

$$\int_0^L dx = L \quad (77)$$

The second integral in Eq. (76) gives

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2\gamma_j x) \Big|_0^L = \frac{1}{2\gamma_j} \sin(2\gamma_j L) \quad (78)$$

Recall from Eq. (67) that $\gamma_j = j\pi/L$; upon substitution into Eq. (78) one gets

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2j\pi \cancel{L}/\cancel{L}) = 0 \quad (79)$$

Eq. (79) is zero because sine of any integer multiple of π is zero. Substitution of Eqs. (77), (79) into Eq. (76) yields

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \sin^2(\gamma_j x) dx = L/2 \quad \text{QED} \quad (80)$$

6 INITIAL VALUE PROBLEM -- THE PLUCKED STRING

The plucking of a string is a most common method of generating musical notes from a variety of string instruments such as the harp the guitar, etc. In this section, we will analyze the initial value problem (IVP) represented by the plucking of a taut string. The schematic of the problem is depicted in Figure 7 which shows a string of length L being plucked by applying a small displacement $u_{\max}$ at its center.

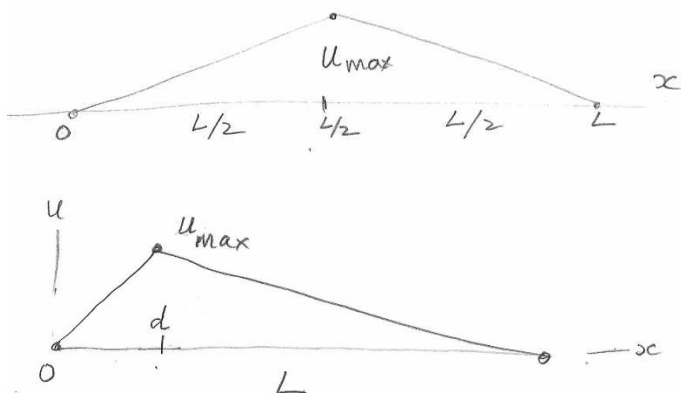


Figure 7 Taut string plucked at an arbitrary location d with displacement amplitude $u_{\max}$

6.1 MODAL EXPANSION METHOD

Recall Eq. (49) giving the vibration motion for each vibrating string mode as

$$u_j(x, t) = U_j(x) e^{-i\omega_j t} \quad (\text{vibration of mode } j) \quad (49)$$

where ω_j is given by Eq. (43) and $U_j(x)$ is given by Eq. (48) as

$$U_j(x) = \sin \gamma_j x \quad j=1, 2, 3, \dots \quad (\text{modeshapes}) \quad (48)$$

with γ_j given by Eq. (40) as

$$\gamma_j = j\pi/L \quad j=1, 2, 3, \dots \quad (40)$$

We will assume that the total vibrational motion is made up of a linear superposition of contributions from all the individual vibration modes, i.e.,

$$u(x, t) = \sum_{j=1}^{\infty} \eta_j u_j(x, t) = \sum_{j=1}^{\infty} \eta_j U_j(x) e^{-i\omega_j t} \quad (81)$$

where the constants η_j are the *modal participation factors* signifying how much of each mode $U_j(x)$ contributes to the summation. Depending on the initial displacement-velocity conditions, the modal participation factors may be complex numbers. The objective of the exercise is to find the values of η_j . For convenience, apply Euler's formula to Eq. (81) and write

$$\eta_j e^{-i\omega_j t} = (A_j \cos \omega_j t - B_j \sin \omega_j t) \quad (82)$$

Thus, we have replaced the complex number η_j with a pair of real numbers A_j, B_j . Thus, the objective of the exercise has become to find the real constants A_j, B_j .

Substitution of Eq. (82) into Eq. (81) yields

$$u(x, t) = \sum_{j=1}^{\infty} U_j(x) (A_j \cos \omega_j t - B_j \sin \omega_j t) \quad (83)$$

Upon differentiation with respect to time t , Eq. (83) becomes

$$\dot{u}(x, t) = - \sum_{j=1}^{\infty} \omega_j (A_j \sin \omega_j t + B_j \cos \omega_j t) U_j(x) \quad (84)$$

6.2 INITIAL VALUE PROBLEM

The initial value problem consists of finding the modal participation factors for given initial conditions expressed as initial displacement $u_0(x)$ and initial velocity $\dot{u}_0(x)$. Note that the initial displacement and initial velocity are functions of x and represent the initial shape of the string as well as the initial velocity of each point on the string. For the plucked string, the initial displacement is described by the shape of string with displacement $u_{\max}$ as shown in Figure 7. It is apparent from Figure 7 that the initial displacement consists of two straight lines that meet at location d . In addition, the initial velocity

$\dot{u}_0(x)$ is assumed zero since the string is released from rest. The corresponding mathematical expression is

$$u_0(x) = \begin{cases} \frac{x}{d} u_{\max} & \text{for } 0 < x \leq d \\ \frac{L-x}{L-d} u_{\max} & \text{for } d < x \leq L \end{cases} \quad (85)$$

$$\dot{u}_0(x) = 0 \quad (86)$$

Substitution of Eq. (83) into Eq. (85) yields

$$\begin{aligned} u(x, 0) &= \sum_{j=1}^{\infty} U_j(x) (A_j \cos \omega_j t - B_j \sin \omega_j t) \Big|_{t=0} \\ &= \sum_{j=1}^{\infty} A_j U_j(x) = u_0(x) \end{aligned} \quad (87)$$

Substitution of Eq. (84) into Eq (86) yields

$$\begin{aligned} \dot{u}(0, t) &= -\sum_{j=1}^{\infty} \omega_j U_j(x) (A_j \sin \omega_j t + B_j \cos \omega_j t) \Big|_{t=0} \\ &= \sum_{j=1}^{\infty} \omega_j U_j(x) B_j = \dot{w}_0(x) = 0 \end{aligned} \quad (88)$$

Eq. (88) gives $B_j=0$ which leaves A_j as the modal participation factors to be determined. Upon substitution in Eq. (87), yield

$$\sum_{j=1}^{\infty} A_j U_j(x) = u_0(x) \quad (89)$$

6.3 SOLUTION BY GALERKIN METHOD

To obtain a numerical solution, one needs to restrict the number of modes to a finite number N and write Eq. (83) as

$$u(x, t) = \sum_{j=1}^N U_j(x) A_j \cos \omega_j t \quad (90)$$

where $B_j=0$ in virtue of Eq. (88). The modal participation factors A_j , $j=1, \dots, N$, should satisfy Eq. (89), i.e.,

$$\sum_{j=1}^N A_j U_j(x) = u_0(x) \quad (91)$$

The application of the Galerkin method consists in multiplying Eq. (91) by each of the modes $U_j(x)$ and then integrating over the length of the beam. For convenience, we use index p instead of j in Eq. (91) and then multiply by mode $U_q(x)$ with $q=1, \dots, N$ to get

$$\sum_{p=1}^{\infty} A_p U_p(x) U_q(x) = u_0(x) U_q(x) \quad q=1, \dots, N \quad (92)$$

To apply the Galerkin method, integrate Eq. (92) over the length of the beam to obtain a system of N algebraic equations, i.e.,

$$\int_0^L \sum_{p=1}^{\infty} A_p U_p(x) U_q(x) dx = \int_0^L u_0(x) U_q(x) dx \quad q=1, \dots, N \quad (93)$$

Recall the orthonormality property of the modeshapes given by Eq. (51) as

$$\int_0^L U_p(x) U_q(x) dx = \begin{cases} L/2 & \text{if } p = q = j \\ 0 & \text{otherwise} \end{cases} \quad (51)$$

Implementation of Eq. (51) into Eqs. (93) yields

$$\frac{L}{2} A_j = \int_0^L u_0(x) U_j(x) dx \quad j=1, \dots, N \quad (94)$$

Eq. (94) can be easily solved for A_j to get

$$A_j = \frac{2}{L} \int_0^L u_0(x) U_j(x) dx \quad j=1, \dots, N \quad (95)$$

The integral in Eq. (95) may be calculated by using the u_0 expression given by Eq. (85) and then performing the integration numerically. However, a closed-form solution is also possible as discussed in the next section.

6.4 STRING RESPONSE

The coefficients A_j from Eq. (95) and $B_j=0$ from Eq. (88) are used in Eq. (83) to assemble the string response $u(x,t)$ as

$$u(x, t) = \sum_{j=1}^N A_j U_j(x) \cos \omega_j t \quad (96)$$

Substitution of Eq. (48) into Eq. (96) give the string response as

$$u(x, t) = \sum_{j=1}^N A_j \sin \gamma_j x \cos \omega_j t \quad (97)$$

6.4.1 STRING RESPONSE AT THE PLUCKING LOCATION

To calculate the string response at the plucking location, make $x=d$ in Eq. (97) to get

$$u(d, t) = \sum_{j=1}^N A_j \sin \gamma_j d \cos \omega_j t \quad (98)$$

6.4.2 RECONSTRUCTION OF STRING INITIAL SHAPE

A reconstruction of the string shape can be obtained by making $t=0$ in Eq. (97) which yields

$$u(x) = \sum_{j=1}^N A_j \sin \gamma_j x \quad (99)$$

Note that the cosine term is absent from Eq. (99) since the function $\cos \omega_j t = 1$ at $t=0$.

6.5 CLOSED-FORM SOLUTION FOR A_j

The integral in Eq. (95) can yield a closed-form expression by using the $u_0(x)$ expression given by Eq. (85) and then performing the integration analytically to obtain the modal coefficients A_j as

$$A_j = \frac{2u_{\max}}{\pi^2} \frac{1}{j^2} \frac{L^2}{d(L-d)} \sin(j\pi d/L) \quad j=1, \dots, N \quad (100)$$

Examination of Eq. (100) reveals that the ratio d/L dictates which modes will be rejected from the response. For example, if the string is plucked at a 3rd distance from the end, then $d/L=1/3$ i.e., $3d/L=1$ and the corresponding sines will vanish, which means that the corresponding coefficients will be zero, $A_3=A_6=A_9\dots$ Thus, the location where the string is plucked controls which modes will be

rejected from the resulting response, or, in other words, which frequencies will not appear in the resulting sound.

Another point to observe about Eq. (100) is that, as before, the A_1 value is much larger than all the other values because the formula decreases as j^2 . Thus, the fundamental frequency will be excited strongly and its pitch will dominate the resulting sound.

If the string is plucked in the middle with $d=L/2$, then the formula for A_j takes a simpler form, i.e.,

$$A_j = \frac{8u_{\max}}{\pi^2} \frac{1}{j^2} \sin \frac{j\pi}{2} \quad j=1,\dots,N \quad (\text{plucked at } d=L/2) \quad (101)$$

Examination of Eq. (101) reveals that $A_j=0$ for all the even modes $j=2,4,6,\dots$ since $\sin\pi=\sin2\pi=\sin3\pi=0$, etc. Recall that the even numbers are associated with modes that are antisymmetric about the midpoint. Since the excitation is symmetric about the midpoint, this means that a symmetric excitation cannot excite the antisymmetric modes.

Another point to observe about Eq. (101) is that the first term, A_1 , is much larger than all the other terms because the formula decreases as j^2 . Thus, the fundamental frequency will be excited strongly and its pitch will dominate the resulting sound.

Further discussion of the plucked string problem with interesting examples of shape reconstruction and experimental recordings can be found at the following web link:

<https://www.acs.psu.edu/drussell/Demos/Pluck-Fourier/Pluck-Fourier.html>

7 FORCED VIBRATION OF A TAUT STRING

Assume a taut string of length L subjected to a distributed time-dependent excitation force $f(x,t)$. The units of $f(x,t)$ are force per unit length. To perform the analysis, consider an infinitesimal element dx as shown in Figure 8.

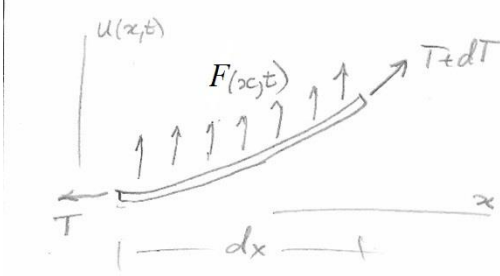


Figure 8 Infinitesimal element for the analysis of forced vibration of a taut string

7.1 ANALYSIS

Free-body analysis of the infinitesimal element dx performed under the same assumptions as in Section 1 yields the sum of y forces as

$$\Sigma F_y = Tu''dx + Fdx \text{ (sum of } y \text{ forces)} \quad (102)$$

Recall Newton's law of motion, i.e.,

$$\Sigma F_y = (mdx)\ddot{u} \quad (103)$$

Substitution of Eq. (102) into Eq. (103) gives, upon division by dx ,

$$Tu'' + F = m\ddot{u} \quad (104)$$

Upon rearrangement, Eq. (104) becomes

$$m\ddot{u}(x,t) - Tu''(x,t) = F(x,t) \quad (105)$$

Assume the external excitation to be time harmonic of the form

$$F(x,t) = \hat{F}(x)e^{i\omega t} \quad (106)$$

where ω is the excitation frequency and $\hat{f}(x)$ is the excitation amplitude distributed over the length of the string. Also assume the vibratory response to have same frequency ω as the excitation, i.e.,

$$u(x,t) = \hat{u}(x)e^{i\omega t} \quad (107)$$

where $\hat{u}(x)$ is the vibration amplitude at any location x along the string. Substitution of Eqs. (106), (107) into Eq. (105) gives

$$-\omega^2 m \hat{u}(x) e^{i\omega t} - T \hat{u}''(x) e^{i\omega t} = \hat{F}(x) e^{i\omega t} \quad (108)$$

Simplification of Eq. (108) by $e^{i\omega t}$ yields

$$-m\omega^2 \hat{u}(x) - T \hat{u}''(x) = \hat{F}(x) \quad (109)$$

7.2 MODAL EXPANSION METHOD

Assume that the function $\hat{u}(x)$ can be expressed as a series expansion in terms of vibration modes $U_j(x)$, i.e.,

$$\hat{u}(x) = \sum_{j=1}^{\infty} \eta_j U_j(x) \quad (110)$$

where the constants η_j are the *modal participation factors* signifying how much of each mode $U_j(x)$ contributes to the summation.

7.2.1 EXPANDED ODE

Substitution of Eq. (110) into Eq. (109) yields

$$-m\omega^2 \sum_{p=1}^{\infty} \eta_p U_p(x) - T \sum_{p=1}^{\infty} \eta_p U_p''(x) = \hat{F}(x) \quad (111)$$

where the index p is used instead of j for convenience.

7.2.2 PROJECTED ODE

Multiplication of Eq. (111) by the q^{th} mode $U_q(x)$ and integration over the length of the beam yields

$$\begin{aligned} -\omega^2 \sum_{p=1}^{\infty} \eta_p \int_0^L m U_q(x) U_p(x) dx - \sum_{p=1}^{\infty} \eta_p \int_0^L T U_p''(x) U_q(x) dx \\ = \int_0^L \hat{F}(x) U_q(x) dx \quad q = 1, 2, 3, \dots \end{aligned} \quad (112)$$

Recall the modeshape orthogonality property as given by Eq. (51), i.e.,

$$\int_0^L U_p(x) U_q(x) dx = 0 \quad \text{if } p \neq q \quad (113)$$

In virtue of Eq. (113), all the terms of Eq. (112) with $p \neq q$ vanish and the only remaining terms are those with $p=q$. Denoting by j the common index $p=q=j$, Eq. (112) becomes

$$\begin{aligned}
-\omega^2 \eta_j \int_0^L m U_j(x) U_j(x) dx - \eta_j \int_0^L T U_j''(x) U_j(x) dx \\
= \int_0^L \hat{F}(x) U_j(x) dx \quad j = 1, 2, 3, \dots
\end{aligned} \tag{114}$$

7.2.3 MODAL EXCITATION

Define the *modal excitation* F_j as

$$F_j = \int_0^L \hat{F}(x) U_j(x) dx \quad j = 1, 2, 3, \dots \tag{115}$$

7.2.4 USE OF MODAL COEFFICIENTS

Recall the definition of the modal mass m_j and modal stiffness k_j given by Eqs. (53), (54) as

$$m_j = \int_0^L m U_j^2(x) dx \quad (\text{modal mass}) \tag{53}$$

$$k_j = -\int_0^L T U_j''(x) U_j(x) dx \quad (\text{modal stiffness}) \tag{54}$$

Substitution of Eqs. (53), (54), (115) into Eq. (114) and factoring out of η_j yields

$$\eta_j (-\omega^2 m_j + k_j) = F_j \quad j = 1, 2, 3, \dots \tag{116}$$

Eq. (116) can be further simplified by recalling the relationship between modal stiffness and modal mass as given by Eq. (55), i.e.,

$$k_j = \omega_j^2 m_j \tag{55}$$

Substitution of Eq. (55) into Eq. (116) yields

$$\eta_j m_j (-\omega^2 + \omega_j^2) = F_j \quad j = 1, 2, 3, \dots \tag{117}$$

Solution of Eq. (117) yields the modal participation factors η_j as

$$\eta_j = \frac{F_j / m_j}{-\omega^2 + \omega_j^2} \quad j = 1, 2, 3, \dots \tag{118}$$

Define the mass-normalized modal excitation f_j as

$$f_j = F_j / m_j \quad j = 1, 2, 3, \dots \tag{119}$$

For a **point-load excitation** of amplitude F_e applied at excitation location x_e , substitution of Eq. (129) into Eq. (119) yields

$$f_j = F_e U_j(x_e) / m_j \quad (\text{point-load excitation at } x_e) \tag{120}$$

Substitution of Eq. (119) into Eq. (118) gives the modal participation factors in the form

$$\eta_j = \frac{f_j}{-\omega^2 + \omega_j^2} \quad j = 1, 2, 3, \dots \quad (121)$$

The expression of the modal participation factors given by Eq. (121) is similar to the expression of forced vibration amplitude developed earlier for undamped 1-dof systems, i.e.,

$$\hat{u}_{undamped}(\omega) = \frac{\hat{f}}{-\omega^2 + \omega_n^2} \quad (\text{undamped 1-dof vibration}) \quad (122)$$

Comparison of Eqs. (121) and (122) allows us to conclude that the modal coefficients allow us to treat each mode of vibration as a simple 1-dof vibration system.

7.3 DAMPING EFFECTS

So far, the vibration has been considered undamped. However, damping will always be present in the physical world. The simplest way to account for damping is to assume viscous modal damping ζ_j .

Recall from 1-dof damped vibration analysis the harmonic response of damped vibration systems as

$$\hat{u}(\omega) = \frac{f}{-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2} \quad (\text{damped 1-dof vibration}) \quad (123)$$

Comparison of Eqs. (122), (123) suggests that one can account for the modal damping ζ_j by adding the term $2i\zeta_j\omega_j\omega$ into the denominator of Eq. (121) to get

$$\eta_j = \frac{f_j}{-\omega^2 + 2i\zeta_j\omega_j\omega + \omega_j^2} \quad j = 1, 2, 3, \dots \quad (124)$$

7.4 RESPONSE BY MODAL ANALYSIS

Substitution of Eq. (124) into Eq. (110) and then into yields the vibration response of taut string in the form

$$u(x, t) = \left[\sum_{j=1}^{\infty} \frac{f_j}{-\omega^2 + 2i\zeta_j\omega_j\omega + \omega_j^2} U_j(x) \right] e^{i\omega t} \quad (125)$$

where f_j is the mass-normalized modal excitation

$$f_j = \frac{1}{m_j} \int_0^L \hat{F}(x) U_j(x) dx \quad j = 1, 2, 3, \dots \quad (126)$$

7.4.1 POINT-LOAD EXCITATION

Consider a point-load excitation of amplitude F_e applied at excitation location x_e , i.e.,

$$\hat{F}(x) = F_e \delta(x - x_e) \quad (127)$$

where $\delta(x)$ is the Dirac delta function which has the localization property

$$\int_{-\infty}^{+\infty} \delta(x - x_e) g(x) dx = g(x_e) \quad (\text{localization property}) \quad (128)$$

Substitution of Eq. (127) into Eq. (115) yields the point-load modal excitation as

$$F_j = \int_0^L \hat{F}(x) U_j(x) dx = \int_0^L F_e \delta(x - x_e) U_j(x) dx = F_e U_j(x_e) \quad (129)$$

Eq. (129) indicates that the point-load modal excitation equals the mode amplitude multiplied by excitation force. If the excitation location happens to be at a modal node, then that mode will not be excited.

7.4.2 RESPONSE FOR POINT-LOAD EXCITATION

For a point-load excitation of amplitude F_e applied at excitation location x_e , the expression of f_j takes the specific form of Eq. (120), and Eq. (125) becomes

$$u(x, t) = F_e \left[\sum_{j=1}^{\infty} \frac{1}{m_j} \frac{U_j(x_e)}{-\omega^2 + 2i\zeta_j \omega_j \omega + \omega_j^2} U_j(x) \right] e^{i\omega t} \quad (130)$$

When the excitation frequency ω is close to a modal frequency, ω_j , the factor multiplying the j^{th} mode $U_j(x)$ in Eq. (130) becomes very large and the j^{th} mode dominates the other modes. In other words, the continuous system will *resonate* when $\omega \rightarrow \omega_j$ and will vibrate predominantly in the natural mode $U_j(x)$. Around a natural frequency ω_j , the continuous system behaves more or less like a 1-dof system vibrating in the corresponding natural mode $U_j(x)$. If the excitation frequency ω covers a large frequency span, several

resonances can be encountered corresponding to several natural frequencies and natural modes.

7.5 FREQUENCY RESPONSE FUNCTION

Equation (125) represents a superposition of a number of terms, each term corresponding to a natural frequency and normal mode of vibration. Eq. (125) allows us to determine the response of a continuous system when excited harmonically with variable frequency. Thus, we obtain the *frequency response function* (FRF). The FRF can be obtained by measuring the response either at the point of excitation (auto FRF) or at another point (cross FRF).

7.5.1 FREQUENCY RESPONSE FUNCTION FOR POINT-LOAD EXCITATION

Assume the string is struck at location $x=x_e$ and the response is measured at location $x=x_0$ as shown in Figure 10.

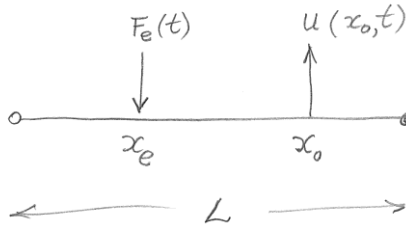


Figure 9 String struck at location $x=x_e$ with response measured at location $x=x_0$

Then, the $FRF(x_0, x_e)$ is defined as the ratio between the response $u(x_0, t)$ at location $x=x_0$ and the excitation $F_e(x_e, t)$ applied at location $x=x_e$. The time-dependent term $e^{-i\omega t}$ cancels out and one gets

$$FRF(\omega; x_e, x_0) = \frac{u(x_0, t)}{F_e e^{-i\omega t}} = \sum_{j=1}^{\infty} \frac{1}{m_j} \frac{U_j(x_0)U_j(x_e)}{-\omega^2 + 2i\zeta_j\omega_j\omega + \omega_j^2} \quad (131)$$

As the excitation frequency sweep passes through modal frequencies ω_j , the corresponding term in the series becomes very large and the structure experience a resonance. As the structure passes through a structural resonance, ω_j , the FRF plot displays a resonance peak. Over a wide frequency interval, the response can

display several peaks, corresponding to several resonances (Figure 10).

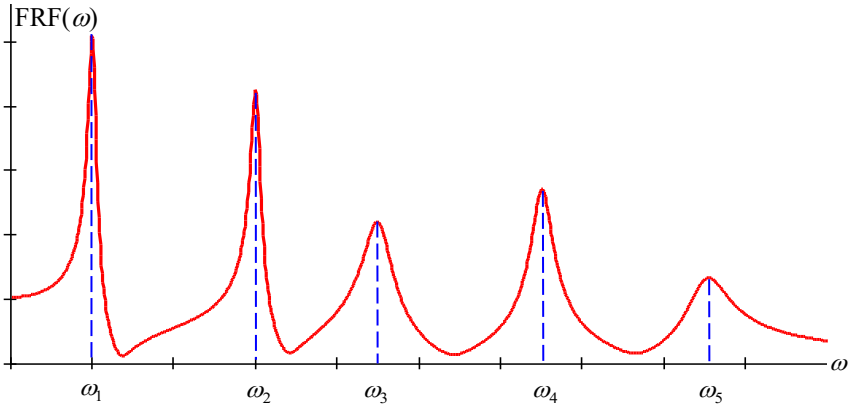


Figure 10 Multi-peak frequency response plot for a continuous system

7.5.2 SOUND PICKED UP AT A LOCATION x_0

Once the $FRF(\omega; x_0, x_e)$ has been determined, one can reconstruct the sound picked up at a location x_0 when the point-excitation of the string is at location x_e by using Eq. (131) to write

$$u(t; x_0, x_e) = FRF(\omega; x_0, x_e) F_e e^{-i\omega t} \quad (120)$$

Even if the excitation location x_e is constant, the sound picked up at various locations x_0 will be modified depending on the amplitude of various modes at that location x_0 . Some modes may respond stronger than others, hence the sound timbre will vary with pickup location x_0 .

7.6 EXAMPLE: THE HAMMERED DULCIMER

https://en.wikipedia.org/wiki/Hammered_dulcimer

The **hammered dulcimer** (also called the **hammer dulcimer**) is a percussion-stringed instrument which consists of strings typically stretched over a trapezoidal resonant sound board. The hammered dulcimer is set before the musician, who in more traditional styles may sit cross-legged on the floor, or in a more modern style may stand or sit at a wooden support with legs. The player holds a small spoon-shaped mallet hammer in each hand to strike the strings. The Graeco-Roman word *dulcimer* ("sweet song") derives from the Latin *dulcis* (sweet) and the Greek *melos* (song). The dulcimer, in which the strings are beaten with small hammers, originated from the psaltery, in which the strings are plucked.

The musician playing the hammered dulcimer excites the strings by percussion at selected location such that both fundamental and higher harmonics can be selectively excited.

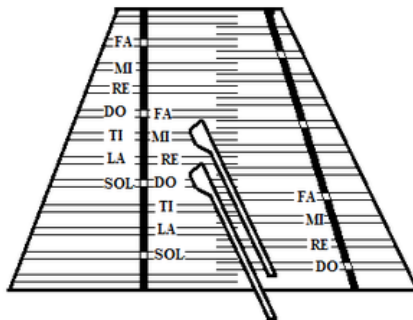


Figure 11 Hammered dulcimer

FLEXURAL VIBRATION

Flexural vibration is analyzed under the Bernoulli-Euler theory of bending which assumes that plane sections remain plane after the bending deformation is applied. This implies a linear distribution of axial displacement across the thickness. Shear deformation and rotary inertia effects are ignored.

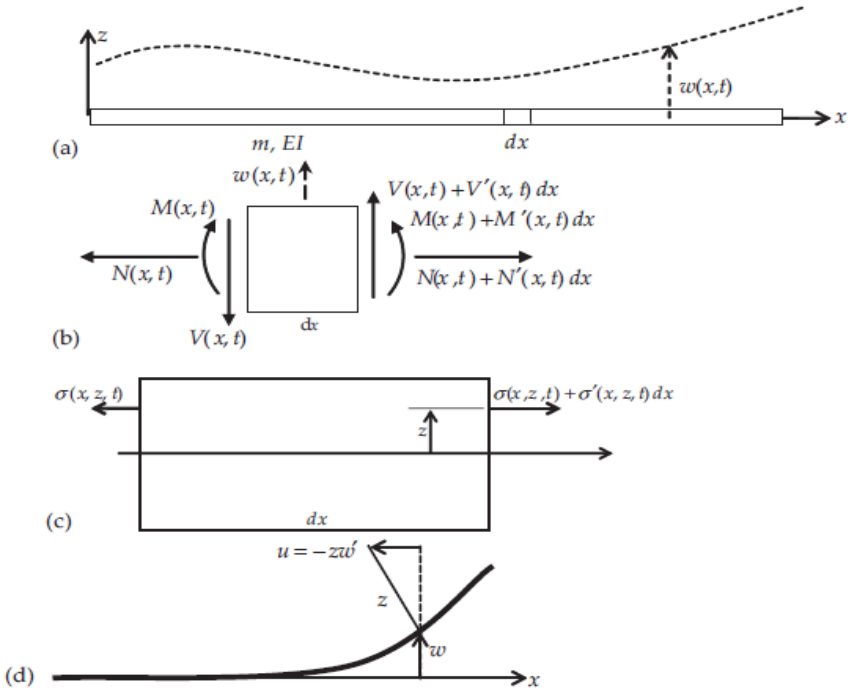


Figure 1 Beam undergoing flexural vibration: (a) general layout; (b) free-body diagram of an infinitesimal beam element; (c) stress distribution across thickness; (d) horizontal displacement induced by flexure

1 GOVERNING EQUATION

Consider a uniform beam of mass m per unit length and bending stiffness EI , undergoing flexural wave motion with vertical displacement $w(x,t)$ as shown in Figure 1a. Centroidal axes are assumed. An infinitesimal beam element of length dx is subjected to the action of axial forces $N(x,t)$, $N(x,t) + N'(x,t)dx$, bending

moments, $M(x,t)$, $M(x,t)+M'(x,t)dx$, and shear forces $V(x,t)$, $V(x,t)+V'(x,t)dx$ as shown in Figure 1b.

1.1 STRESS RESULTANTS

The $N(x,t)$ and $M(x,t)$ stress resultants are evaluated by integration of the normal stress $\sigma(x,z,t)$ across the cross-sectional area (Figure 1c), i.e.,

$$N(x,t) = \int_A \sigma(x,z,t) dA \quad (1)$$

$$M(x,t) = - \int_A \sigma(x,z,t) z dA \quad (2)$$

1.2 STRESS-STRAIN RELATION

Recall the stress-strain relation

$$\sigma = E \varepsilon \quad (3)$$

Substitution of Eq. (3) into Eqs. (1), (2) yields

$$N(x,t) = E \int_A \varepsilon(x,z,t) dA \quad (4)$$

$$M(x,t) = -E \int_A \varepsilon(x,z,t) z dA \quad (5)$$

1.3 KINEMATIC ANALYSIS OF THE ROTATED SECTION

Kinematic analysis of the rotated section (Figure 1d) yields

$$u(x,z,t) = -zw'(x,t) \quad (6)$$

Recall that the strain ε is the gradient of axial displacement u , i.e.,

$$\varepsilon(x,z,t) = u'(x,z,t) \quad (7)$$

Substitution of Eq. (6) into Eq. (7) yields the strain $\varepsilon(x,z,t)$ in terms of the flexural motion $w(x,t)$ and the thickness-wise location z , i.e.,

$$\varepsilon(x,z,t) = -zw''(x,t) \quad (8)$$

Substitution of Eq. (8) into Eqs. (4), (5) and integration over the area yields

$$N(x,t) = -Ew''(x,t) \int_A z dA = 0 \quad (9)$$

$$M(x,t) = Ew''(x,t) \int_A z^2 dA \quad (10)$$

Eq. (9) indicates that the axial stress resultant $N(x,t)$ is proportional to the first moment of area of the cross section which is zero because centroidal axes were assumed.

Eq. (10) yields

$$M(x,t) = EIw''(x,t) \quad (11)$$

where I is the second moment of area of the cross section, i.e.,

$$I = \int_A z^2 dA \quad (12)$$

1.4 FREE-BODY ANALYSIS

Free body analysis of the infinitesimal element of Figure 1b yields the sum of forces in the transverse direction z as

$$\Sigma F_z = -V(x,t) + V(x,t) + V'(x,t)dx = V'(x,t)dx \quad (13)$$

The sum of moments is calculated as

$$\Sigma M = M(x,t) + M'(x,t)dx - M(x,t) + V(x,t)dx = [M'(x,t) + V(x,t)]dx \quad (14)$$

1.5 NEWTON'S LAW OF MOTION

The mass of the infinitesimal element ds is $m dx$. The acceleration of the infinitesimal element is $\ddot{w}$. Newton's law of motion applied to the infinitesimal element states that sum of forces should equal mass times acceleration and sum of moments should equal rotational inertia times the rotational acceleration.

NLM in the x direction is not needed because N is identically zero as shown by Eq. (9).

NLM in the z direction yields

$$\Sigma F_z = (m dx) \ddot{w} \quad (15)$$

Substitution of Eq. (13) into Eq. (15) yields the NLM expression for the z direction as

$$V'(x,t) = m \ddot{w}(x,t) \quad (16)$$

NLM for rotation is applied under the assumption that rotational inertia is ignored. NLM for rotation is written as

$$\Sigma M = 0 \quad (17)$$

Eq. (17) indicates that the sum of moments should be zero.

1.6 GOVERNING EQUATION

Substitution of Eq. (14) into Eq. (17) yields, upon simplification

$$M'(x,t) + V(x,t) = 0 \quad (18)$$

Eq. (18) gives an expression of V in terms of M' , i.e.,

$$V(x,t) = -M'(x,t) \quad (19)$$

Substitution of Eq. (11) into Eq. (19) yields

$$V(x,t) = -EIw'''(x,t) \quad (20)$$

Differentiation of Eq. (20) with respect to x gives

$$V'(x,t) = -EIw''''(x,t) \quad (21)$$

Substitution of Eq. (21) into Eq. (16) yields, after rearrangement, the governing equation, i.e.,

$$EIw''''(x,t) + m\ddot{w}(x,t) = 0 \quad (\text{governing equation}) \quad (22)$$

1.7 SOLUTION OF THE GOVERNING EQUATION

Upon division by m Eq. (22) becomes

$$\frac{EI}{m}w''''(x,t) + \ddot{w}(x,t) = 0 \quad (23)$$

For convenience, define a constant a as

$$a^4 = \frac{EI}{m} \quad a = \left(\frac{EI}{m} \right)^{1/4} \quad (24)$$

The units of a are $\text{m}/\sqrt{\text{s}}$. Note that, for a beam of rectangular cross section with height h and width b , the mass per unit length is $m = \rho b h$, and the 2nd moment of area is $I = bh^3/12$. Hence, Eq. (24) becomes

$$a^4 = \frac{Eh^2}{12\rho} \quad a = \left(\frac{Eh^2}{12\rho} \right)^{1/4} \quad (25)$$

Substitution of Eq. (24) into Eq. (23) yields

$$a^4w'''' + \ddot{w} = 0 \quad (26)$$

Assume harmonic vibration and write

$$w(x,t) = W(x)e^{-i\omega t} \quad (\text{vibration motion}) \quad (27)$$

Substitution of Eq. (27) into Eq. (26) gives

$$a^4W''''(x)e^{-i\omega t} - \omega^2W(x)e^{-i\omega t} = 0 \quad (28)$$

The complex exponential $e^{-i\omega t}$ cancels out and Eq. (28) becomes

$$a^4W''''(x) - \omega^2W(x) = 0 \quad (29)$$

Upon division by a^4 , Eq. (29) becomes

$$W''''(x) - (\omega^2/a^4)W(x) = 0 \quad (30)$$

Define the wavenumber for flexural vibration γ as

$$\gamma^4 = \omega^2/a^4 \quad \gamma = \omega^{1/2}/a \quad (\text{flexural wavenumber}) \quad (31)$$

Substitution of Eq. (31) into Eq. (30) yields

$$W''''(x) - \gamma^4 W(x) = 0 \quad (32)$$

Eq. (32) is a 4th order linear ordinary differential equation (ODE) in x . Assume exponential solution of the form

$$W(x) = e^{sx} \quad (33)$$

Upon differentiation, Eq. (33) gives

$$W''''(x) = s^4 e^{sx} = s^4 W(x) \quad (34)$$

Substitution of Eqs. (33), (34) into Eq. (32) gives

$$s^4 W(x) - \gamma^4 W(x) = 0 \quad (35)$$

Division of Eq. (35) by $W(x)$ yields the characteristic equation, i.e.,

$$s^4 - \gamma^4 = 0 \quad (\text{characteristic equation}) \quad (36)$$

Eq. (36) has four roots, two imaginary $s_{1,2} = \pm i\gamma$ and two real $s_{3,4} = \pm \gamma$, i.e.,

$$s_1 = i\gamma \quad s_2 = -i\gamma \quad s_3 = \gamma \quad s_4 = -\gamma \quad (37)$$

Each of the four different roots shown in Eq. (37) leads to a different solution of Eq. (32). This means that Eq. (32) accepts four different solutions, i.e.,

$$e^{i\gamma x} \quad e^{-i\gamma x} \quad e^{\gamma x} \quad e^{-\gamma x} \quad (38)$$

The complete solution of Eq. (32) is a linear superposition of the four individual solutions, i.e.,

$$W(x) = A_1 e^{i\gamma x} + A_2 e^{-i\gamma x} + A_3 e^{\gamma x} + A_4 e^{-\gamma x} \quad (39)$$

where A_1, A_2, A_3, A_4 are arbitrary constants to be determined from the boundary conditions. Recall the definition of trigonometric and hyperbolic functions

$$\sin \alpha = (e^{i\alpha} - e^{-i\alpha})/2i \quad \cos \alpha = (e^{i\alpha} + e^{-i\alpha})/2 \quad (40)$$

$$\sinh \alpha = (e^{\alpha} - e^{-\alpha})/2 \quad \cosh \alpha = (e^{\alpha} + e^{-\alpha})/2 \quad (41)$$

hence

$$e^{i\alpha} = \cos \alpha + i \sin \alpha \quad e^{-i\alpha} = \cos \alpha - i \sin \alpha \quad (42)$$

$$e^{\alpha} = \cosh \alpha + \sinh \alpha \quad e^{-\alpha} = \cosh \alpha - \sinh \alpha \quad (43)$$

Eqs. (42), (43) are used to write Eq. (39) as

$$W(x) = C_1 \sin \gamma x + C_2 \cos \gamma x + C_3 \sinh \gamma x + C_4 \cosh \gamma x \quad (44)$$

The constants C_1, C_2, C_3, C_4 are to be determined from the boundary conditions. The complete vibration motion is given by Eq. (27), i.e.,

$$w(x, t) = W(x) e^{-i\omega t} \quad (\text{vibration motion}) \quad (27)$$

where $W(x)$ is given by Eq. (44).

2 SOLUTION FOR A CANTILEVER BEAM

A cantilever beam is fixed at one end and free at the other end (Figure 4). These boundary conditions are called *fixed-free boundary conditions*. The arms of a tuning fork can be modeled as cantilever beams. Other common examples of cantilever beams are vibrating reeds inside a music box (Figure 2) or a harmonica (Figure 3).

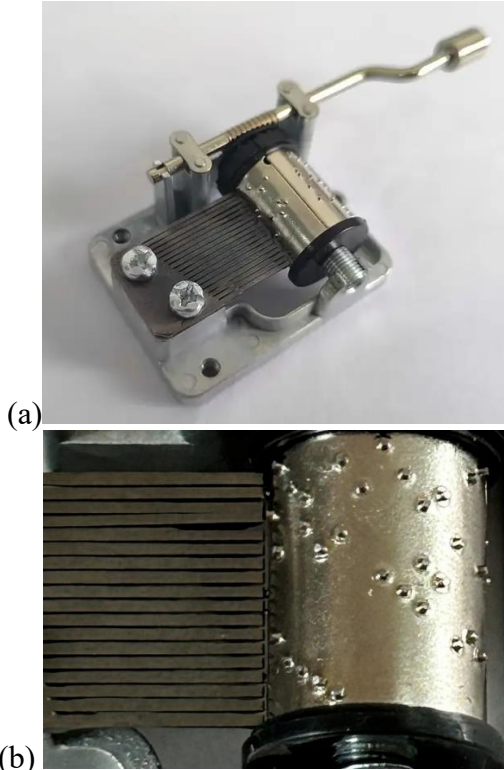
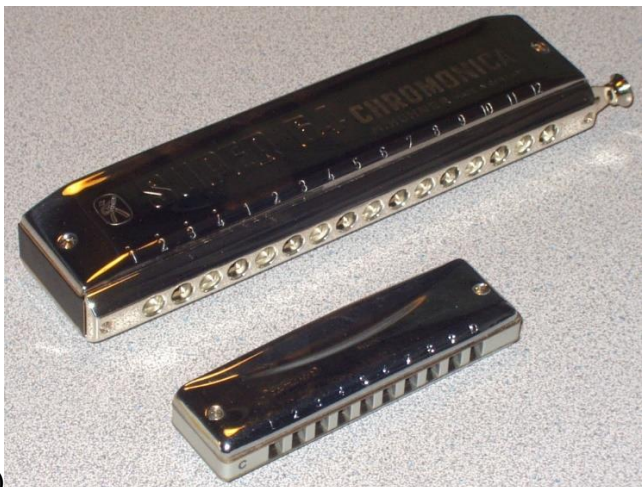
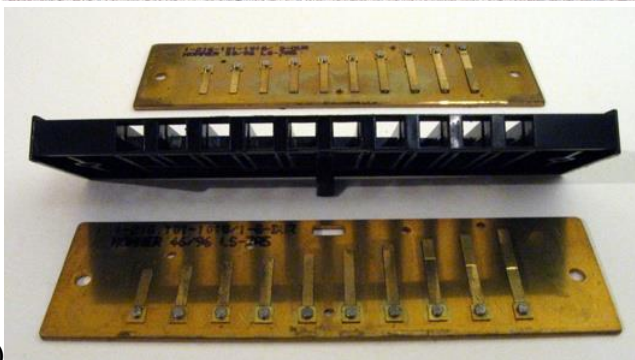


Figure 2 Music box: (a) overall view; (b) closeup view showing the vibrating reeds and the studded cylinder



(a)



(b)

Figure 3 Harmonica: (a) overall view; (b) inside view showing the vibrating reeds

The schematic representation of a cantilever beam is shown in Figure 4.

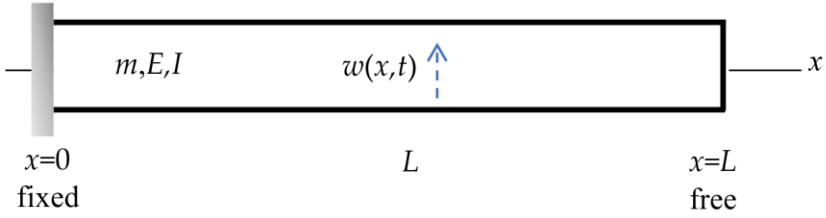


Figure 4 Vibration of a cantilever beam of length L . The cantilever beam is fixed at one end and free at the other end

2.1 BOUNDARY CONDITIONS

The boundary conditions of a cantilever beam are as follows. At the fixed end, the displacement and slope are zero. At the free end, the bending moment and shear force are zero. Hence, the *fixed-free boundary conditions* of a cantilever beam can be described as

$$\begin{array}{ll} w(0,t) = 0 & w'(0,t) = 0 \\ M(L,t) = 0 & V(L,t) = 0 \end{array} \quad \left(\begin{array}{c} \text{fixed-free} \\ \text{boundary conditions} \end{array} \right) \quad (45)$$

Recall Eqs. (11), (20), (27)

$$M(x,t) = EIw''(x,t) \quad (11)$$

$$V(x,t) = -EIw'''(x,t) \quad (20)$$

$$w(x,t) = W(x)e^{-i\omega t} \quad (27)$$

The use of Eqs. (11), (20), (27) into Eq. (45) yields the boundary conditions in terms of displacement $W(x)$ and its derivatives, i.e.,

$$\begin{array}{ll} W(0) = 0 & W'(0) = 0 \\ W''(L) = 0 & W'''(L) = 0 \end{array} \quad (46)$$

Recall the general solution Eq. (44), i.e.,

$$W(x) = C_1 \sin \gamma x + C_2 \cos \gamma x + C_3 \sinh \gamma x + C_4 \cosh \gamma x \quad (44)$$

Substitution of the general solution Eq. (44) into the boundary conditions (46) yields

$$\begin{aligned}
C_1 \sin \gamma x + C_2 \cos \gamma x + C_3 \sinh \gamma x + C_4 \cosh \gamma x \Big|_{x=0} &= 0 \\
\gamma \left[C_1 \cos \gamma x - C_2 \sin \gamma x + C_3 \cosh \gamma x + C_4 \sinh \gamma x \right] \Big|_{x=0} &= 0 \\
\gamma^2 \left[-C_1 \sin \gamma x - C_2 \cos \gamma x + C_3 \sinh \gamma x + C_4 \cosh \gamma x \right] \Big|_{x=L} &= 0 \\
\gamma^3 \left[-C_1 \cos \gamma x + C_2 \sin \gamma x + C_3 \cosh \gamma x + C_4 \sinh \gamma x \right] \Big|_{x=L} &= 0
\end{aligned} \tag{47}$$

Upon simplification, one gets

$$\begin{aligned}
C_2 + C_4 &= 0 \\
C_1 + C_3 &= 0 \\
-C_1 \sin \gamma L - C_2 \cos \gamma L + C_3 \sinh \gamma L + C_4 \cosh \gamma L &= 0 \\
-C_1 \cos \gamma L + C_2 \sin \gamma L + C_3 \cosh \gamma L + C_4 \sinh \gamma L &= 0
\end{aligned} \tag{48}$$

The first and second lines of Eq. (48) give

$$\begin{aligned}
C_4 &= -C_2 \\
C_3 &= -C_1
\end{aligned} \tag{49}$$

Substitution of Eq. (49) into the third and fourth lines of Eq. (48) yields

$$\begin{aligned}
-C_1 \sin \gamma L - C_2 \cos \gamma L - C_1 \sinh \gamma L - C_2 \cosh \gamma L &= 0 \\
-C_1 \cos \gamma L + C_2 \sin \gamma L - C_1 \cosh \gamma L - C_2 \sinh \gamma L &= 0
\end{aligned} \tag{50}$$

Eq. (50) is a homogenous linear algebraic system in C_1 and C_2 , i.e.,

$$\begin{aligned}
(\sin \gamma L + \sinh \gamma L)C_1 + (\cos \gamma L + \cosh \gamma L)C_2 &= 0 \\
(\cos \gamma L + \cosh \gamma L)C_1 + (-\sin \gamma L + \sinh \gamma L)C_2 &= 0
\end{aligned} \tag{51}$$

The homogenous algebraic system Eq. (51) accepts nontrivial solutions only if its determinant has value zero, i.e.,

$$\begin{vmatrix} \sin \gamma L + \sinh \gamma L & \cos \gamma L + \cosh \gamma L \\ \cos \gamma L + \cosh \gamma L & -\sin \gamma L + \sinh \gamma L \end{vmatrix} = 0 \tag{52}$$

Expansion and simplification of Eq. (52) yields

$$\cos \gamma L \cosh \gamma L + 1 = 0 \tag{53}$$

2.2 EIGENVALUE EQUATION

Introduce the notation

$$z = \gamma L \quad (54)$$

Substitution of Eq. (54) into Eq. (53) yields the *eigenvalue equation*

$$\cos z \cosh z + 1 = 0 \quad (55)$$

Equation (55) is a transcendental equation that yields the eigenvalues z corresponding to the free flexural vibration of a cantilever beam. The roots of Eq. (55) are obtained numerically, for example using the MATLAB function `fzero`. To this purpose, Eq. (55) is set up as a function $D(z)$ that needs to be made zero, i.e.,

$$D(z) = \cos z + 1/\cosh z \Rightarrow 0 \quad (56)$$

The formulation of Eq. (56) ensures better convergence than Eq. (55) because the second term quickly diminishes as $\cosh z$ increases. Numerical values of the first five eigenvalues z are shown in Table 1. As z becomes large, the numerical values approach a rational sequence. From the sixth eigenvalue onwards, they can be approximated to reasonable accuracy by

$$z_j = (2j-1)\frac{\pi}{2} \quad j = 6, 7, 8, \dots \quad (57)$$

Some numerical values of z_j are given in Table 1.

2.3 WAVENUMBERS

The eigenvalues z_j are used in Eq. (54) to solve for the wavenumbers γ_j , i.e.,

$$\gamma_j = z_j / L \quad (58)$$

where z_j are the eigenvalues given by the roots of Eq. (55).

Table 1 Eigenvalues, $z=\gamma L$, natural frequencies, ω , and the modal parameters β for the flexural vibration of a cantilever beam

j	$z_j=(\gamma L)_j$	ω_j , rad/s	f_j , Hz	β_j
1	1.87510407	$3.51601527 \sqrt{\frac{EI}{mL^4}}$	$0.55959121 \sqrt{\frac{EI}{mL^4}}$	0.73409551
2	4.6940911	$22.0344916 \sqrt{\frac{EI}{mL^4}}$	$3.5068983 \sqrt{\frac{EI}{mL^4}}$	1.01846732
3	7.8547574	$61.697214 \sqrt{\frac{EI}{mL^4}}$	$9.8194166 \sqrt{\frac{EI}{mL^4}}$	0.99922450
4	10.9955407	$120.901916 \sqrt{\frac{EI}{mL^4}}$	$19.2421376 \sqrt{\frac{EI}{mL^4}}$	1.00003355
5	14.1371684	$199.859530 \sqrt{\frac{EI}{mL^4}}$	$31.808632 \sqrt{\frac{EI}{mL^4}}$	0.99999855
6,7,8, ...	$(2j-1)\frac{\pi}{2}$	$\left[(2j-1)\frac{\pi}{2}\right]^2 \sqrt{\frac{EI}{mL^4}}$	$[(2j-1)]^2 \frac{\pi}{8} \sqrt{\frac{EI}{mL^4}}$	1.000000000

2.4 NATURAL FREQUENCIES

Recall Eq. (31), i.e.,

$$\gamma^4 = \omega^2 / a^4 \quad (31)$$

Eq. (31) can be solved for ω to get

$$\omega = \gamma^2 a^2 \quad (59)$$

Substitute Eq. (58) into Eq. (59) to get

$$\omega_j = z_j^2 a^2 / L^2 \quad (60)$$

Recall the definition of a given by Eq. (24), i.e.,

$$a^4 = \frac{EI}{m} \quad a = \left(\frac{EI}{m} \right)^{1/4} \quad (24)$$

Hence, one writes

$$a^2 / L^2 = \sqrt{\frac{EI}{mL^4}} \quad (61)$$

Substitute Eq. (61) into Eq. (60) and express the natural frequencies as

$$\omega_j = z_j^2 \sqrt{\frac{EI}{mL^4}} \quad f_j = \frac{1}{2\pi} z_j^2 \sqrt{\frac{EI}{mL^4}} \quad j = 1, 2, 3, \dots \quad (62)$$

Some numerical values for ω_j and f_j are given in Table 1.

2.5 MODESHAPES

For every value γ calculated with Eq. (58), one solves the algebraic system Eq. (51) and obtains the ratio C_1/C_2 either from the first or the second line of Eq. (51), i.e.,

$$\frac{C_1}{C_2} = -\frac{\cosh \gamma L + \cos \gamma L}{\sinh \gamma L + \sin \gamma L} = -\frac{\sinh \gamma L - \sin \gamma L}{\cosh \gamma L + \cos \gamma L} = -\beta \quad (63)$$

For the particular values of γ given by Eq. (58), the first and second parts of Eq. (63) should yield same numerical values. Eqs. (49), (63) allow one to express C_1 , C_3 , C_4 in terms of C_2 as

$$C_1 = -\beta C_2 \quad C_3 = \beta C_2 \quad C_4 = -C_2 \quad (64)$$

Substitution of Eq. (64) into the general solution Eq. (44) yields

$$W(x) = -\beta C_2 \sin \gamma x + C_2 \cos \gamma x + \beta C_2 \sinh \gamma x - C_2 \cosh \gamma x \quad (65)$$

After factoring out C_2 and β , Eq. (65) becomes

$$W(x) = -C_2 [(\cosh \gamma x - \cos \gamma x) - \beta (\sinh \gamma x - \sin \gamma x)] \quad (66)$$

The constant C_2 in Eq. (66) is arbitrary since Eq. (51) is homogeneous and hence has one degree of indeterminacy. Without any loss of generality, one can take $C_2 = -1$ and then use Eq. (66) to write the modeshapes for each value γ_j as given by Eq. (58). Hence, the *modeshapes for flexural vibration of a cantilever beam* are

$$W_j(x) = (\cosh \gamma_j x - \cos \gamma_j x) - \beta_j (\sinh \gamma_j x - \sin \gamma_j x), \quad j=1,2,3,\dots \quad (67)$$

where the modeshape coefficients β_j are given by

$$\beta_j = \frac{\sinh \gamma_j L - \sin \gamma_j L}{\cosh \gamma_j L + \cos \gamma_j L} = \frac{\sinh z_j - \sin z_j}{\cosh z_j + \cos z_j} \quad j=1,2,3,\dots \quad (68)$$

or by

$$\beta_j = \frac{\cosh \gamma_j L + \cos \gamma_j L}{\sinh \gamma_j L + \sin \gamma_j L} = \frac{\cosh z_j + \cos z_j}{\sinh z_j + \sin z_j} \quad j=1,2,3,\dots \quad (69)$$

where z_j are the cantilever beam eigenvalues and γ_j are the corresponding wavenumbers given by Eq. (58) as

$$\gamma_j = z_j / L \quad (58)$$

The first four modeshapes of a cantilever beam undergoing flexural vibration are shown in Figure 5.

The complete vibration motion for each mode j is given by Eq. (27) as

$$w_j(x,t) = W_j(x)e^{-i\omega_j t} \quad (\text{vibration motion of mode } j) \quad (70)$$

where $W_j(x)$ is given by Eqs. (67), (68) and ω_j is given by Eq. (60). Note that each vibration mode $w_j(x,t)$ has its own distinct frequency of vibration ω_j in rad/sec and the associated f_j in Hz.

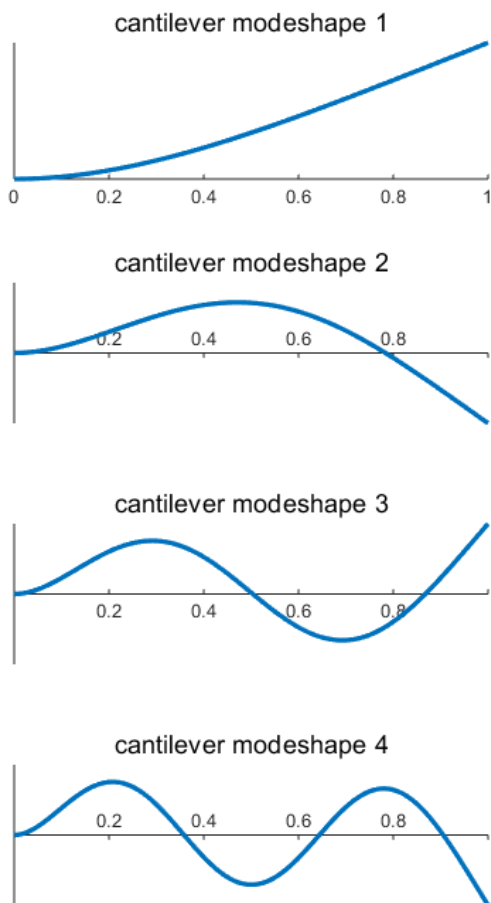


Figure 5 First four flexural vibration modeshapes of a cantilever beam

[See section 5 on page 199 for discussion of modeshape orthonormality](#)

2.6 EXAMPLE: THE TUNING FORK

Consider the commonly used 512 C aluminum-alloy tuning fork (Figure 6). This tuning fork is stated to give the middle C note (C5) according to the scientific pitch scale https://en.wikipedia.org/wiki/Scientific_pitch . In the scientific pitch scale, the middle C note C4 has the frequency $f_1=512\text{Hz}$. The dimensions of the tuning fork as measured by the author are also shown in Figure 6.

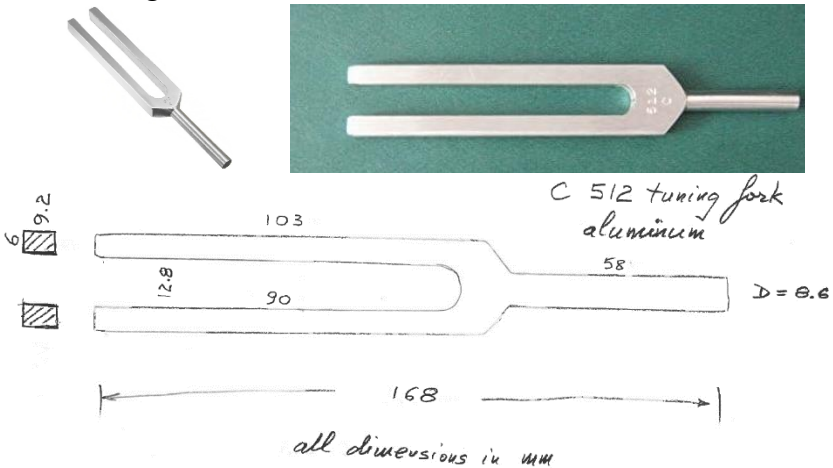


Figure 6 512 C tuning fork <https://www.amazon.com/C-512-Hz-Tuning-Fork/dp/B00BLBE8QE> .

In the first approximation, each arm of the tuning fork could be modeled as a cantilever beam. Using common values for aluminum elastic modulus and density $E=70\text{GPa}$ and $\rho=2700\text{kg/m}^3$ and an average beam length of $L=96.5\text{mm}$, one gets the fundamental frequency $f_1=530\text{Hz}$, which is different from the frequency of 512Hz stated in the product description. However, one realizes that the beam-length is subject to interpretation since the tuning fork is not exactly a cantilever beam (though it can be modeled as one). Hence, the beam length estimate can be adjusted to come closer to the product-description frequency. After iterative manual adjustments one finds that a new length estimation $L_{\text{new}}=98.2\text{mm}$ gives the desired frequency of 512Hz .

3 SOLUTION FOR A FREE-FREE BEAM

A free-free beam has both ends free (Figure 7). One common example is represented by the tubes of a wind chime.

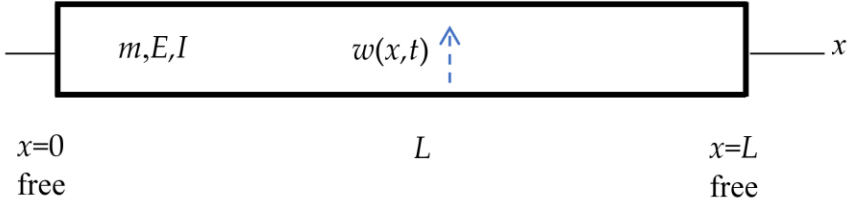


Figure 7 Vibration of a free-free beam of length L

3.1 BOUNDARY CONDITIONS

The boundary conditions for a free-free beam are expressed in terms of bending moment and shear force being zero at both ends, i.e.,

$$\begin{array}{ll} M(0, t) = 0 & V(0, t) = 0 \\ M(L, t) = 0 & V(L, t) = 0 \end{array} \quad \left(\begin{array}{c} \text{free-free} \\ \text{boundary conditions} \end{array} \right) \quad (71)$$

Recall Eqs. (11), (20), (27)

$$M(x, t) = EIw''(x, t) \quad (11)$$

$$V(x, t) = -EIw'''(x, t) \quad (20)$$

$$w(x, t) = W(x)e^{-i\omega t} \quad (27)$$

The use of Eqs. (11), (20), (27) into Eq. (71) yields the boundary conditions in terms of displacement $W(x)$ and its derivatives, i.e.,

$$\begin{array}{ll} W''(0) = 0 & W'''(0) = 0 \\ W''(L) = 0 & W'''(L) = 0 \end{array} \quad (72)$$

Recall the general solution Eq. (44), i.e.,

$$W(x) = C_1 \sin \gamma x + C_2 \cos \gamma x + C_3 \sinh \gamma x + C_4 \cosh \gamma x \quad (44)$$

Substitution of Eq. (44) into the boundary conditions Eq. (72) yields

$$\begin{array}{l} \gamma^2 \left[-C_1 \sin \gamma x - C_2 \cos \gamma x + C_3 \sinh \gamma x + C_4 \cosh \gamma x \right] \Big|_{x=0} = 0 \\ \gamma^3 \left[-C_1 \cos \gamma x + C_2 \sin \gamma x + C_3 \cosh \gamma x + C_4 \sinh \gamma x \right] \Big|_{x=0} = 0 \\ \gamma^2 \left[-C_1 \sin \gamma x - C_2 \cos \gamma x + C_3 \sinh \gamma x + C_4 \cosh \gamma x \right] \Big|_{x=L} = 0 \\ \gamma^3 \left[-C_1 \cos \gamma x + C_2 \sin \gamma x + C_3 \cosh \gamma x + C_4 \sinh \gamma x \right] \Big|_{x=L} = 0 \end{array} \quad (73)$$

Upon simplification, one gets

$$\begin{aligned}
 -C_2 + C_4 &= 0 \\
 -C_1 + C_3 &= 0 \\
 -C_1 \sin \gamma L - C_2 \cos \gamma L + C_3 \sinh \gamma L + C_4 \cosh \gamma L &= 0 \\
 -C_1 \cos \gamma L + C_2 \sin \gamma L + C_3 \cosh \gamma L + C_4 \sinh \gamma L &= 0
 \end{aligned} \tag{74}$$

The first and second lines of Eq. (74) give

$$\begin{aligned}
 C_4 &= C_2 \\
 C_3 &= C_1
 \end{aligned} \tag{75}$$

Substitution of Eq. (75) into the third and fourth lines of Eq. (74) yields

$$\begin{aligned}
 -C_1 \sin \gamma L - C_2 \cos \gamma L + C_1 \sinh \gamma L + C_2 \cosh \gamma L &= 0 \\
 -C_1 \cos \gamma L + C_2 \sin \gamma L + C_1 \cosh \gamma L + C_2 \sinh \gamma L &= 0
 \end{aligned} \tag{76}$$

Eq. (76) is a homogenous linear algebraic system in C_1 and C_2 , i.e.,

$$\begin{aligned}
 (-\sin \gamma L + \sinh \gamma L)C_1 + (-\cos \gamma L + \cosh \gamma L)C_2 &= 0 \\
 (-\cos \gamma L + \cosh \gamma L)C_1 + (\sin \gamma L + \sinh \gamma L)C_2 &= 0
 \end{aligned} \tag{77}$$

The homogenous algebraic system Eq. (77) accepts nontrivial solutions only if its determinant is zero, i.e.,

$$\begin{vmatrix} -\sin \gamma L + \sinh \gamma L & -\cos \gamma L + \cosh \gamma L \\ -\cos \gamma L + \cosh \gamma L & \sin \gamma L + \sinh \gamma L \end{vmatrix} = 0 \tag{78}$$

Expansion and simplification of Eq. (78) yields

$$\cos \gamma L \cosh \gamma L - 1 = 0 \tag{79}$$

3.2 EIGENVALUE EQUATION

Introduce the notation

$$z = \gamma L \tag{80}$$

Substitution of Eq. (80) into Eq. (79) yields the *eigenvalue equation*

$$\cos z \cosh z - 1 = 0 \tag{81}$$

Equation (81) is a transcendental equation that yields the eigenvalues z corresponding to the flexural vibration of a free-free beam. Numerical values of the first five eigenvalues z are shown in Table 2. As z becomes large, the numerical values approach a rational sequence. From the sixth eigenvalue onwards, they can be approximated to reasonable accuracy by

$$z_j = (2j+1)\frac{\pi}{2} \quad j = 6, 7, 8, \dots \quad (82)$$

3.3 WAVENUMBERS

The eigenvalues z_j are used in Eq. (80) to solve for the wavenumbers γ_j , i.e.,

$$\gamma_j = z_j/L \quad (83)$$

where z_j are the roots of Eq. (81).

3.4 NATURAL FREQUENCIES

Recall Eq. (62) that expresses the natural frequencies ω_j and f_j in terms of eigenvalues and beam properties, i.e.,

$$\omega_j = z_j^2 \sqrt{\frac{EI}{mL^4}} \quad f_j = \frac{1}{2\pi} z_j^2 \sqrt{\frac{EI}{mL^4}} \quad j = 1, 2, 3, \dots \quad (84)$$

where z_j are the roots of Eq. (81). Some ω_j and f_j values are given in Table 2.

Table 2 Eigenvalues, $z=\gamma L$, natural frequencies, ω , and the modal parameters β for the flexural vibration of a free-free beam

j	$z_j=(\gamma L)_j$	ω_j , rad/s	f_j , Hz	β_j
1	4.73004074	$22.373287 \sqrt{\frac{EI}{mL^4}}$	$3.5608190 \sqrt{\frac{EI}{mL^4}}$	0.982502215
2	7.85320462	$61.673823 \sqrt{\frac{EI}{mL^4}}$	$9.8155346 \sqrt{\frac{EI}{mL^4}}$	1.000777312
3	10.9956078	$120.903391 \sqrt{\frac{EI}{mL^4}}$	$19.2423723 \sqrt{\frac{EI}{mL^4}}$	0.999966450
4	14.1371655	$199.859448 \sqrt{\frac{EI}{mL^4}}$	$31.808619 \sqrt{\frac{EI}{mL^4}}$	1.000001450
5	17.2787597	$298.55554 \sqrt{\frac{EI}{mL^4}}$	$47.516591 \sqrt{\frac{EI}{mL^4}}$	0.999999937
6,7,8, ...	$(2j+1)\frac{\pi}{2}$	$\left[(2j+1)\frac{\pi}{2}\right]^2 \sqrt{\frac{EI}{mL^4}}$	$[(2j+1)]^2 \frac{\pi}{8} \sqrt{\frac{EI}{mL^4}}$	1.000000000

3.5 MODESHAPES

For every value γ_j calculated with Eq. (83), one solves the algebraic system Eq. (77) and obtains the ratio C_1/C_2 either from the first or the second line of Eq. (77).

$$\frac{C_1}{C_2} = -\frac{\cosh \gamma L - \cos \gamma L}{\sinh \gamma L - \sin \gamma L} = -\frac{\sinh \gamma L + \sin \gamma L}{\cosh \gamma L - \cos \gamma L} = -\beta \quad (85)$$

For the particular values of γ given from Eq. (83), the first and second parts of Eq. (85) should have identical numerical values. From Eqs. (75), (85) one can express C_1 , C_3 , C_4 in terms of C_2 as

$$C_1 = -\beta C_2 \quad C_3 = -\beta C_2 \quad C_4 = C_2 \quad (86)$$

Substitution of Eq. (86) into the general solution Eq. (44) yields

$$W(x) = -\beta C_2 \sin \gamma x + C_2 \cos \gamma x - \beta C_2 \sinh \gamma x + C_2 \cosh \gamma x \quad (87)$$

After factoring out C_2 and β , Eq. (87) becomes

$$W(x) = C_2 [(\cosh \gamma x + \cos \gamma x) - \beta (\sinh \gamma x + \sin \gamma x)] \quad (88)$$

The constant C_2 in Eq. (88) is arbitrary since Eq. (77) is homogeneous and hence has one degree of indeterminacy. Without any loss of generality, one can take $C_2=1$ and then use Eq. (88) to write the modeshapes for each value γ_j as given by Eq. (83). Hence, the *modeshapes for flexural vibration of a free-free beam* are

$$W_j(x) = (\cosh \gamma_j x + \cos \gamma_j x) - \beta_j (\sinh \gamma_j x + \sin \gamma_j x), \quad j=1,2,3,\dots \quad (89)$$

where the modeshape coefficients β_j are given by

$$\beta_j = \frac{\sinh \gamma_j L + \sin \gamma_j L}{\cosh \gamma_j L - \cos \gamma_j L} = \frac{\sinh z_j + \sin z_j}{\cosh z_j - \cos z_j} \quad j=1,2,3,\dots \quad (90)$$

The first four modeshapes of a free-free beam undergoing flexural vibration are shown in Figure 8. Please note that the first and third modes are symmetric wrt the middle of the beam whereas the second and fourth modes are antisymmetric.

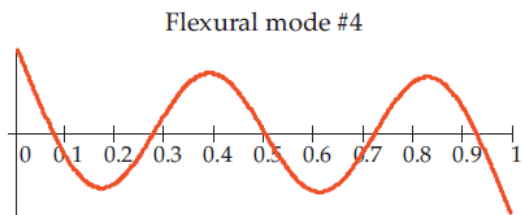
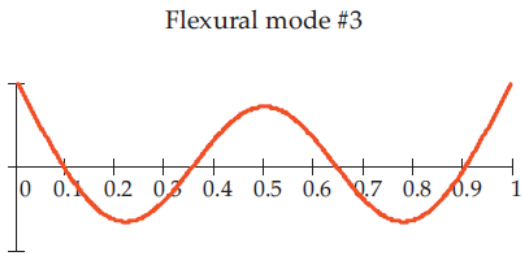
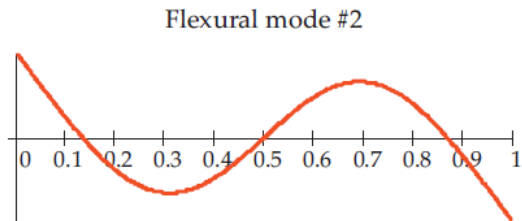
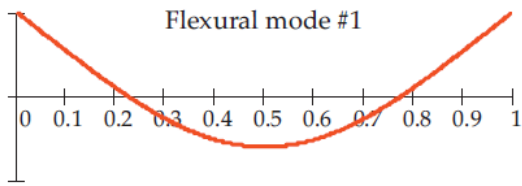


Figure 8 First four flexural vibration modes of a free-free beam

3.6 EXAMPLE: THE WIND CHIME

https://en.wikipedia.org/wiki/Wind_chime

A wind chime constructed from metal tubes may be modelled using the free-free beam analysis.



Figure 9 wind chimes

4 SOLUTION FOR A SIMPLY SUPPORTED BEAM

A simply supported beam has the zero moment and zero displacement at both ends (Figure 10). One common example is represented by the span of a bridge. The zero-moment and zero-displacement boundary condition also corresponds to a pinned case. In practice, as in the case of a bridge span, one end is pinned whereas the other end is just simply supported to permit sliding in response to longitudinal thermal expansion or small ground motion.

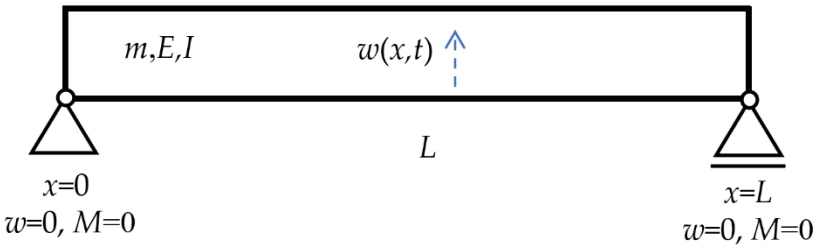


Figure 10 Vibration of a simply supported beam of length L

4.1 BOUNDARY CONDITIONS

The boundary conditions for a simply supported beam are expressed in terms of displacement and bending moment being zero at both ends, i.e.,

$$\begin{array}{ll} w(0, t) = 0 & M(0, t) = 0 \\ w(L, t) = 0 & M(L, t) = 0 \end{array} \quad \left(\begin{array}{l} \text{simply supported} \\ \text{boundary conditions} \end{array} \right) \quad (91)$$

Recall Eqs. (11) and (27)

$$M(x, t) = EIw''(x, t) \quad (11)$$

$$w(x, t) = W(x)e^{-i\omega t} \quad (27)$$

The use of Eqs. (11) and (27) into Eq. (91) yields the boundary conditions in terms of displacement $W(x)$ and its derivatives, i.e.,

$$\begin{array}{ll} W(0) = 0 & W''(0) = 0 \\ W(L) = 0 & W''(L) = 0 \end{array} \quad (92)$$

Recall the general solution Eq. (44), i.e.,

$$W(x) = C_1 \sin \gamma x + C_2 \cos \gamma x + C_3 \sinh \gamma x + C_4 \cosh \gamma x \quad (44)$$

Substitution of Eq. (44) into the boundary conditions Eq. (92) yields

$$\begin{aligned}
C_1 \sin \gamma x + C_2 \cos \gamma x + C_3 \sinh \gamma x + C_4 \cosh \gamma x \Big|_{x=0} &= 0 \\
\gamma^2 \left[-C_1 \sin \gamma x - C_2 \cos \gamma x + C_3 \sinh \gamma x + C_4 \cosh \gamma x \right] \Big|_{x=0} &= 0 \\
C_1 \sin \gamma x + C_2 \cos \gamma x + C_3 \sinh \gamma x + C_4 \cosh \gamma x \Big|_{x=L} &= 0 \\
\gamma^2 \left[-C_1 \sin \gamma x - C_2 \cos \gamma x + C_3 \sinh \gamma x + C_4 \cosh \gamma x \right] \Big|_{x=L} &= 0
\end{aligned} \tag{93}$$

Upon simplification, one gets

$$\begin{aligned}
C_2 + C_4 &= 0 \\
-C_2 + C_4 &= 0 \\
C_1 \sin \gamma L + C_2 \cos \gamma L + C_3 \sinh \gamma L + C_4 \cosh \gamma L &= 0 \\
-C_1 \sin \gamma L - C_2 \cos \gamma L + C_3 \sinh \gamma L + C_4 \cosh \gamma L &= 0
\end{aligned} \tag{94}$$

The first and second lines of Eq. (94) give

$$C_2 = C_4 = 0 \tag{95}$$

Substitution of Eq. (95) into the third and fourth lines of Eq. (74) yields

$$\begin{aligned}
C_1 \sin \gamma L + C_3 \sinh \gamma L &= 0 \\
-C_1 \sin \gamma L + C_3 \sinh \gamma L &= 0
\end{aligned} \tag{96}$$

Eq. (96) is a homogenous linear algebraic system in C_1 and C_3 which accepts nontrivial solutions only if its determinant is zero, i.e.,

$$\begin{vmatrix} \sin \gamma L & \sinh \gamma L \\ -\sin \gamma L & \sinh \gamma L \end{vmatrix} = 0 \tag{97}$$

Expansion and simplification of Eq. (97) yields

$$\sin \gamma L = 0 \tag{98}$$

where division by $2 \sinh \gamma L$ was applied.

If Eq. (98) is satisfied, i.e., $\sin \gamma L = 0$, then Eq. (96) predicts $C_3 = 0$ whereas C_1 becomes arbitrary since any C_1 value will satisfy Eq. (96) as long as $\sin \gamma L = 0$. This means that one can take any value for C_1 and still satisfy the conditions of the problem, which means that the problem has one degree of indeterminacy.

4.2 EIGENVALUE EQUATION

Introduce the notation

$$z = \gamma L \tag{99}$$

Substitution of Eq. (99) into Eq. (98) yields the *eigenvalue equation*

$$\sin z = 0 \quad (100)$$

The roots of Eq. (100) are the eigenvalues z corresponding to the flexural vibration of a simply supported beam. The eigenvalues z for which $\sin z = 0$ are expressed as multiples of π , i.e.,

$$z_j = j\pi \quad j=1,2,3,\dots \quad (101)$$

4.3 WAVENUMBERS

The eigenvalues z_j are used in Eq. (99) to solve for the wavenumbers γ_j , i.e.,

$$\gamma_j = j\pi/L \quad (102)$$

The eigenvalues for modes $j=1,\dots,4$ are given in Table 3.

4.4 NATURAL FREQUENCIES

Recall Eq. (31), i.e.,

$$\gamma^4 = \omega^2/a^4 \quad (31)$$

Eq. (31) can be solved for ω to get

$$\omega = \gamma^2 a^2 \quad (103)$$

Substitute Eq. (102) into Eq. (103) to get

$$\omega_j = j^2 \pi^2 a^2 / L^2 \quad (104)$$

Recall the definition of a given by Eq. (24), i.e.,

$$a^4 = \frac{EI}{m} \quad a = \left(\frac{EI}{m} \right)^{1/4} \quad (24)$$

Hence, one writes

$$a^2/L^2 = \sqrt{\frac{EI}{mL^4}} \quad (105)$$

Substitution of Eq. (105) into Eq. (104) yields

$$\omega_j = j^2 \pi^2 \sqrt{\frac{EI}{mL^4}} \quad f_j = j^2 \frac{\pi}{2} \sqrt{\frac{EI}{mL^4}} \quad j = 1, 2, 3, \dots \quad (62)$$

The natural frequencies for modes $j=1,\dots,4$ are given in Table 3.

4.5 MODESHAPES

Recall Eq. (44), i.e.,

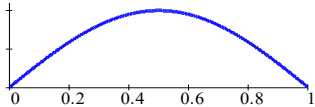
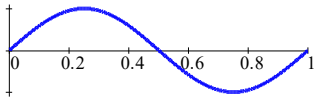
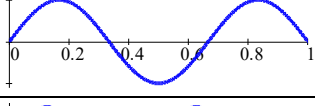
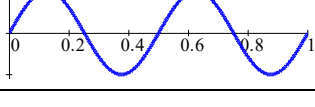
$$W(x) = C_1 \sin \gamma x + C_2 \cos \gamma x + C_3 \sinh \gamma x + C_4 \cosh \gamma x \quad (44)$$

where the constants C_1, C_2, C_3, C_4 are given by the solution of Eqs. (95), (96). Eq. (95) gave $C_2=C_4=0$ whereas the solution of Eq. (96) was only possible for special values of γL that made $\sin \gamma L=0$. These special values were called eigenvalues. Associated with the eigenvalues are the wavenumbers $\gamma_j=j\pi/L$. For each wavenumber $\gamma_j=j\pi/L$, one gets $\sin \gamma_j L=0$ in Eq. (96) which implies $C_3=0$ as well as the fact that C_1 becomes arbitrary since any C_1 value will satisfy Eq. (96) as long as $\sin \gamma L=0$. This means that one can take any value for C_1 and still satisfy the conditions of the problem because the problem has one degree of indeterminacy. Here, we take $C_1=1$. Thus, with $C_1=1$ and $C_2=C_3=C_4=0$, Eq. (44) gives the *modeshapes for flexural vibration of a simply supported beam* as

$$W_j(x) = \sin \gamma_j x \quad j=1,2,3,\dots \quad (106)$$

The modeshapes for modes $j=1,\dots,4$ are given in Table 3.

Table 3 Eigenvalues, natural frequencies, and vibration modeshapes of a beam of length L simply supported at both ends

Mode #	Eigenvalue $z=jL$ $z_j=j\pi$	Resonant frequency	Modeshape		Wavelength multiplicity
1	$z_1=\gamma_1 L=\pi$	$f_1 = \frac{\pi}{2} \sqrt{\frac{EI}{mL^4}}$		$W_1 = \sin \pi \frac{x}{L}$	$L = \frac{\lambda_1}{2}$
2	$z_2=\gamma_2 L=2\pi$	$f_1 = 4 \frac{\pi}{2} \sqrt{\frac{EI}{mL^4}}$		$W_2 = \sin 2\pi \frac{x}{L}$	$L = 2 \frac{\lambda_2}{2}$
3	$z_3=\gamma_3 L=3\pi$	$f_1 = 9 \frac{\pi}{2} \sqrt{\frac{EI}{mL^4}}$		$W_3 = \sin 3\pi \frac{x}{L}$	$L = 3 \frac{\lambda_3}{2}$
4	$z_4=\gamma_4 L=3\pi$	$f_1 = 16 \frac{\pi}{2} \sqrt{\frac{EI}{mL^4}}$		$W_4 = \sin 4\pi \frac{x}{L}$	$L = 4 \frac{\lambda_4}{2}$

5 MODESHAPE ORTHONORMALITY AND MODAL COEFFICIENTS

5.1 ORTHOGONALITY OF MODESHAPE

Orthogonality of modeshapes is an important property that allows one to verify if some independently derived functions qualify to be considered as modeshapes. For flexural vibration of a beam of length L , the modeshape orthogonality is stated as

$$\int_0^L W_p(x) W_q(x) dx = \begin{cases} \text{constant} & \text{for } p = q \\ 0 & \text{otherwise} \end{cases} \quad (\text{orthogonality}) \quad (107)$$

Eq. (107) implies that the product between two different modeshapes is zero when integrated over the beam length L . This is the **orthogonality** property. If the two modeshapes are the same, i.e., the modeshape is multiplied upon itself, then the result is a constant that is used to define the *norm of the modeshape*. If one divides the integral by this constant, then the result is 1. This is the process of **normalization**. Thus, the **orthonormality** property of the modeshapes is expressed as

$$\int_0^L W_p(x) W_q(x) dx / \text{const.} = \begin{cases} 1 & \text{for } p = q \\ 0 & \text{otherwise} \end{cases} \quad (\text{orthonormality}) \quad (108)$$

The modeshape orthonormality is usually verified through numerical integration, although closed-form analytical proof may also be possible in certain cases.

5.2 MODAL COEFFICIENTS

The modal coefficients are used to describe the modal behavior in response to an external excitation as described later in this course.

5.2.1 MODAL MASS AND MODAL STIFFNESS

Two commonly used modal coefficients are the modal mass m_j and the modal stiffness k_j . They are defined as follows:

$$m_j = \int_0^L m W_j^2(x) dx \quad (\text{modal mass}) \quad (109)$$

$$k_j = \int_0^L EI W_j''''(x) W_j(x) dx \quad (\text{modal stiffness}) \quad (110)$$

5.2.2 RELATIONSHIP BETWEEN MODAL MASS AND MODAL STIFFNESS

The relationship between modal mass and modal stiffness is expressed as

$$k_j = \omega_j^2 m_j \quad (111)$$

The relationship between modal mass and modal stiffness expressed by Eq. (111) is similar to the expression $k = \omega^2 m$ between the stiffness, mass, and natural frequency of a 1-dof mass-spring vibration system. The modal coefficients allow us to treat each mode of vibration as a simple 1-dof vibration system.

Proof of Eq. (111)

The proof of Eq. (111) proceeds as follows. Recall Eq. (146), i.e.,

$$a^4 W''''(x) - \omega^2 W(x) = 0 \quad (146)$$

Recall Eq. (24) defining a^4 as

$$a^4 = EI/m \quad (112)$$

Substitution of Eq. (112) into Eq. (146) gives

$$EI W''''(x) - \omega^2 m W(x) = 0 \quad (146)$$

For the j^{th} modeshape, Eq. (146) is written as

$$EI W_j''''(x) = \omega_j^2 m W_j(x) \quad (113)$$

Substitution of Eq. (113) into the modal stiffness definition Eq. (110) and use of the modal mass definition Eq. (109) gives

$$\begin{aligned} k_j &= \int_0^L EI W_j''''(x) W_j(x) dx \\ &= \int_0^L \omega_j^2 m W_j(x) W_j(x) dx \quad \text{QED} \\ &= \omega_j^2 \int_0^L m W_j^2(x) dx = \omega_j^2 m_j \end{aligned} \quad (114)$$

5.2.3 MODAL MASS AND MODAL STIFFNESS FOR UNIFORM BEAMS

In the case of a uniform beam, the distributed mass m is constant along the beam and it comes out in front of the integral sign such that the modal mass Eq. (109) becomes

$$m_j = m \int_0^L W_j^2(x) dx \quad (\text{modal mass for uniform beams}) \quad (115)$$

The modal mass calculated with Eq. (115) is used in Eq. (114) to calculate the modal stiffness as

$$k_j = \omega_j^2 m_j \quad (\text{modal stiffness}) \quad (116)$$

5.3 MODESHAPE ORTHONORMALITY AND MODAL COEFFICIENTS FOR DIFFERENT BOUNDARY CONDITIONS

5.3.1 CANTILEVER BEAM VIBRATION

The modeshapes for a cantilever beam undergoing flexural vibration are given by Eq. (67) as

$$W_j(x) = (\cosh \gamma_j x - \cos \gamma_j x) - \beta_j (\sinh \gamma_j x - \sin \gamma_j x), \quad j=1,2,3,\dots \quad (67)$$

where β_j is given by Eq. (68) as

$$\beta_j = \frac{\sinh \gamma_j L - \sin \gamma_j L}{\cosh \gamma_j L + \cos \gamma_j L} \quad j=1,2,3,\dots \quad (68)$$

In order to prove modeshape orthonormality, one has to substitute Eqs. (67), (68) into Eq. (107) and perform the integration. Upon numerical integration one finds numerically that the orthonormality of cantilever-beam modes can be expressed as

$$\int_0^L W_p(x) W_q(x) dx = \begin{cases} L & \text{if } p=q \\ 0 & \text{otherwise} \end{cases} \quad \begin{pmatrix} \text{cantilever} \\ \text{beam} \end{pmatrix} \quad (117)$$

Thus, Eq. (108) becomes

$$\int_0^L W_p(x) W_q(x) dx / L = \begin{cases} 1 & p=q \\ 0 & \text{otherwise} \end{cases} \quad \begin{pmatrix} \text{cantilever-beam} \\ \text{orthonormality} \end{pmatrix} \quad (118)$$

For example, the numerical integration of Eq. (118) for the cantilever-beam modes 1,2,3,4 gives the following 4×4 matrix:

$$\begin{array}{ccccc} \text{integral(WW)/L for cantilever beam =} & & & & \\ \begin{array}{ccccc} 1.0000 & 0 & 0 & 0 & 0 \\ 0 & 1.0000 & 0 & 0 & 0 \\ 0 & 0 & 1.0000 & 0 & 0 \\ 0 & 0 & 0 & 1.0000 & 0 \\ 0 & 0 & 0 & 0 & 1.0000 \end{array} & & & & (119) \end{array}$$

In Eq. (119), the zero off-diagonal terms confirm that the integration of dissimilar modes with $p \neq q$ gives zero thus confirming modes orthogonality. The diagonal terms of Eq. (119) confirm that for $p=q$ the integration is L as stated in Eq. (117).

Substitution of Eq. (117) into the modal mass formula Eq. (115) yields the modal mass of a uniform cantilever beam as

$$m_j = m \int_0^L W_j^2(x) dx = mL \quad \left(\begin{array}{c} \text{cantilever-beam} \\ \text{modal mass} \end{array} \right) \quad (120)$$

Note that the modal mass m_j given by Eq. (120) is independent of the mode number j which means that it has the same value for all modes. In the case of a cantilever beam, the modal mass value of mL equals the mass of the beam. However, this may not be the same for beams with other boundary conditions.

The modal stiffness is obtained by substitution of Eq. (120) into Eq. (116) to get

$$k_j = \omega_j^2 mL \quad (\text{cantilever-beam modal stiffness}) \quad (121)$$

5.3.2 FREE-FREE BEAM VIBRATION

The modeshapes for a free-free beam undergoing flexural vibration are given by Eq. (89) as

$$W_j(x) = (\cosh \gamma_j x + \cos \gamma_j x) - \beta_j (\sinh \gamma_j x + \sin \gamma_j x), \quad j=1,2,3,\dots \quad (89)$$

where β_j is given by Eq. (90) as

$$\beta_j = \frac{\sinh \gamma_j L + \sin \gamma_j L}{\cosh \gamma_j L - \cos \gamma_j L} = \frac{\sinh z_j + \sin z_j}{\cosh z_j - \cos z_j} \quad j=1,2,3,\dots \quad (90)$$

In order to prove modeshape orthogonality, one has to substitute Eqs. (89), (90) into Eq. (107) and perform the integration. Upon numerical integration it is found that the orthonormality of free-free beam modes can be expressed as

$$\int_0^L W_p(x) W_q(x) dx = \begin{cases} L & \text{if } p = q \\ 0 & \text{otherwise} \end{cases} \quad \left(\begin{array}{c} \text{cantilever} \\ \text{beam} \end{array} \right) \quad (122)$$

Thus, Eq. (108) becomes

$$\int_0^L W_p(x) W_q(x) dx / L = \begin{cases} 1 & p = q \\ 0 & \text{otherwise} \end{cases} \quad \left(\begin{array}{c} \text{cantilever-beam} \\ \text{orthonormality} \end{array} \right) \quad (123)$$

For example, the numerical integration of Eq. (118) for the modes 1,2,3,4 of a free-free beam gives the following 4×4 matrix:

$$\begin{array}{ccccc}
\text{integral}(\mathbf{W}\mathbf{W})/L \text{ for free-free beam} = & & & & \\
1.0000 & 0 & 0 & 0 & 0 \\
0 & 1.0000 & 0 & 0 & 0 \\
0 & 0 & 1.0000 & 0 & 0 \\
0 & 0 & 0 & 1.0000 & 0 \\
0 & 0 & 0 & 0 & 1.0000
\end{array} \quad (124)$$

In Eq. (124), the zero off-diagonal terms confirm that the integration of dissimilar modes with $p \neq q$ gives zero thus confirming modes orthogonality. The diagonal terms of Eq. (124) confirm that for $p=q$ the integration is L as stated in Eq. (122).

Substitution of Eq. (124) into the modal mass formula Eq. (115) yields the modal mass of a free-free beam as

$$m_j = m \int_0^L W_j^2(x) dx = mL \quad \left(\begin{array}{c} \text{free-free beam} \\ \text{modal mass} \end{array} \right) \quad (125)$$

Note that the modal mass m_j given by Eq. (125) is independent of the mode number j which means that it has the same value for all modes. In the case of a free-free beam, the modal mass value of mL equals the mass of the beam. However, this may not be the same for beams with other boundary conditions.

The modal stiffness is obtained by substitution of Eq. (125) into Eq. (116) to get

$$k_j = \omega_j^2 mL \quad (\text{free-free beam modal stiffness}) \quad (126)$$

5.3.3 SIMPLY SUPPORTED BEAM VIBRATION

The wavenumbers and modeshapes for a simply supported beam are given by Eqs. (102) and (106) as

$$\gamma_j = j\pi/L \quad W_j(x) = \sin \gamma_j x \quad j=1,2,3,\dots \quad (127)$$

In order to prove modeshape orthonormality, one has to substitute Eq. (127) into Eq. (107) and perform the integration. Upon numerical integration one finds that

$$\int_0^L W_p(x) W_q(x) dx = \begin{cases} L/2 & \text{if } p = q \\ 0 & \text{otherwise} \end{cases} \quad \left(\begin{array}{c} \text{simply supported} \\ \text{beam} \end{array} \right) \quad (128)$$

Thus, Eq. (108) becomes

$$\int_0^L W_p(x) W_q(x) dx / L/2 = \begin{cases} 1 & p = q \\ 0 & \text{otherwise} \end{cases} \begin{pmatrix} \text{simply} \\ \text{supported} \\ \text{orthonormality} \end{pmatrix} \quad (129)$$

Numerical integration confirms the theoretical results. For the first four modes, the numerical integration of Eq. (107) gives the following 4×4 matrix:

For example, the numerical integration of Eq. (129) for modes 1,2,3,4 of a simply supported beam gives the following 4×4 matrix:

$$W_p W_q / (L/2) = \begin{pmatrix} 1.0000 & -0.0000 & -0.0000 & -0.0000 & 0.0000 \\ -0.0000 & 1.0000 & 0.0000 & 0.0000 & -0.0000 \\ -0.0000 & 0.0000 & 1.0000 & -0.0000 & 0.0000 \\ -0.0000 & 0.0000 & -0.0000 & 1.0000 & -0.0000 \\ 0.0000 & -0.0000 & 0.0000 & -0.0000 & 1.0000 \end{pmatrix} \quad (130)$$

In Eq. (130), the zero off-diagonal terms confirm that the integration of dissimilar modes with $p \neq q$ gives zero thus confirming modes orthogonality. The diagonal terms of Eq. (130) confirm that for $p=q$ the integration is $L/2$ as stated in Eq. (128).

Substitution of Eq. (130) into the modal mass formula Eq. (115) yields the modal mass of a simply supported beam as

$$m_j = m \int_0^L W_j^2(x) dx = mL/2 \quad \begin{pmatrix} \text{simply supported beam} \\ \text{modal mass} \end{pmatrix} \quad (131)$$

Note that the modal mass m_j given by Eq. (131) is independent of the mode number j which means that it has the same value for all modes. In the case of a simply supported beam, the modal mass value of $mL/2$ equals half the mass of the beam. However, this may not be the same for beams with other boundary conditions.

The modal stiffness is obtained by substitution of Eq. (125) into Eq. (116) to get

$$k_j = \omega_j^2 mL/2 \quad (\text{free-free beam modal stiffness}) \quad (132)$$

5.3.4 ANALYTICAL FORMULAE FOR SIMPLY SUPPORTED BEAM

5.3.4.1 Orthogonality of a simply supported beam modes

Let j take two different values p and q and substitute Eq. (127) into Eq. (107) to get

$$\int_0^L U_p(x) U_q(x) dx = \int_0^L \sin \gamma_p x \sin \gamma_q x dx \quad (133)$$

Recall the “product of sines” identity

$$\sin \alpha \sin \beta = [\cos(\alpha - \beta) - \cos(\alpha + \beta)]/2 \quad (134)$$

The use of Eq. (134) into Eq. (133) yields, upon integration,

$$\begin{aligned} \int_0^L \sin \gamma_p x \sin \gamma_q x dx = \\ \frac{1}{2} \int_0^L \cos[(\gamma_p - \gamma_q)x] dx - \frac{1}{2} \int_0^L \cos[(\gamma_p + \gamma_q)x] dx = \end{aligned} \quad (135)$$

$$\frac{\sin(\gamma_p - \gamma_q)L}{2(\gamma_p - \gamma_q)} - \frac{\sin(\gamma_p + \gamma_q)L}{2(\gamma_p + \gamma_q)}$$

The wavenumbers in Eq. (135) are given by Eq. (127) as $\gamma_j = j\pi/L$. Hence, the $(\gamma_p - \gamma_q)L$ and $(\gamma_p + \gamma_q)L$ expressions showing up in Eq. (135) take the following values

$$(\gamma_p - \gamma_q)L = (p - q)\pi L/L = (p - q)\pi \quad (136)$$

$$(\gamma_p + \gamma_q)L = (p + q)\pi L/L = (p + q)\pi \quad (137)$$

Substitution of Eqs. (136), (137) into Eq. (135) followed by substitution into Eq. (133) yields

$$\int_0^L U_p(x) U_q(x) dx = \frac{\sin(p - q)\pi}{2(p - q)\pi/L} - \frac{\sin(p + q)\pi}{2(p + q)\pi/L} = 0 \quad (138)$$

Eq. (138) is zero because sine of any integer multiple of π is zero. This holds true as long as $p \neq q$.

5.3.4.2 Modal mass of a simply supported beam

Substitution of Eq. (127) into the modal mass definition Eq. (115) yields

$$m_j = \int_0^L m W_j^2(x) dx = m \int_0^L W_j^2(x) dx = m \int_0^L \sin^2(\gamma_j x) dx \quad (139)$$

Recall the sine squared trigonometric identity

$$\sin^2 \alpha = (1 - \cos 2\alpha)/2 \quad (140)$$

The use of Eq. (140) into the integral of Eq. (139) yields

$$\begin{aligned} \int_0^L \sin^2(\gamma_j x) dx &= \int_0^L \frac{1}{2} [1 - \cos(2\gamma_j x)] dx \\ &= \frac{1}{2} \int_0^L dx - \frac{1}{2} \int_0^L \cos(2\gamma_j x) dx \end{aligned} \quad (141)$$

The first integral in Eq. (141) gives

$$\int_0^L dx = L \quad (142)$$

The second integral in Eq. (141) gives

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2\gamma_j x) \Big|_0^L = \frac{1}{2\gamma_j} \sin(2\gamma_j L) \quad (143)$$

Recall from Eq. (127) that $\gamma_j = j\pi/L$; upon substitution in Eq. (143) one gets

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2j\pi \cancel{L}/\cancel{L}) = 0 \quad (144)$$

Eq. (144) is zero because sine of any integer multiple of π is zero. Substitution of Eqs. (142), (144) into Eq. (141) yields

$$\int_0^L W_j^2(x) dx = \int_0^L \sin^2(\gamma_j x) dx = L/2 \quad (145)$$

Substitution of Eq. (145) into the modal mass Eq. (139) yields

$$m_j = \int_0^L m W_j^2(x) dx = mL/2 \quad (146)$$

Note that the modal mass m_j given by Eq. (146) is independent of the mode number j which means that it has the same value for all modes. The modal mass value of $mL/2$ represents half the mass of the beam.

Numerical integration confirms these theoretical results as illustrated by Eq. (128).

5.4 GENERAL APPROACH TO MODESHAPE ORTHOGONALITY

Recall Eq. (107) expressing modeshape orthogonality as

$$\int_0^L W_p(x) W_q(x) dx = 0 \quad \text{if } p \neq q \quad (107)$$

One can prove the veracity of Eq. (107) through a generic theoretical approach that can be applied to any combination of boundary conditions. Recall the flexural vibration ODE Eq. (29), i.e.,

$$a^4 W''''(x) + \omega^2 W(x) = 0 \quad (147)$$

Consider two different modes p and q with the corresponding frequencies ω_p , ω_q and modeshapes $W_p(x)$ and $W_q(x)$. Hence, Eq. (147) is written as

$$a^4 W_p''''(x) + \omega_p^2 W_p(x) = 0 \quad (148)$$

$$a^4 W_q''''(x) + \omega_q^2 W_q(x) = 0 \quad (149)$$

Now, multiply Eq. (148) by $W_q(x)$ and Eq. (149) by $W_p(x)$ and then integrate over the length L of the beam to obtain

$$\begin{aligned} \int_0^L a^4 W_p''''(x) W_q(x) dx &= \omega_p^2 \int_0^L W_p(x) W_q(x) dx \\ \int_0^L a^4 W_q''''(x) W_p(x) dx &= \omega_q^2 \int_0^L W_q(x) W_p(x) dx \end{aligned} \quad (150)$$

Subtract the two lines of Eq. (150) to get

$$\begin{aligned} \int_0^L a^4 W_p''''(x) W_q(x) dx - \int_0^L a^4 W_q''''(x) W_p(x) dx \\ = (\omega_p^2 - \omega_q^2) \int_0^L W_p(x) W_q(x) dx \end{aligned} \quad (151)$$

Integration by parts of the left-hand side allows us to express Eq. (151) as

$$\begin{aligned} & \left[a^4 W_p''''(x) W_q(x) \right]_0^L - \left[a^4 W_p'''(x) W_q'(x) \right]_0^L \\ & + \int_0^L a^4 W_p''(x) W_q''(x) dx \\ & - \left[a^4 W_q''''(x) W_p(x) \right]_0^L + \left[a^4 W_q''(x) W_p'(x) \right]_0^L \\ & - \int_0^L a^4 W_p''(x) W_q''(x) dx = (\omega_p^2 - \omega_q^2) \int_0^L W_p(x) W_q(x) dx \end{aligned} \quad (152)$$

Upon rearrangement, Eq. (152) becomes

$$\begin{aligned} & \left(\omega_p^2 - \omega_q^2 \right) \int_0^L W_p(x) W_q(x) dx = \\ & a^4 \left[W_p''''(x) W_q(x) - W_q''''(x) W_p(x) \right]_0^L \\ & - a^4 \left[W_p''(x) W_q'(x) - W_q''(x) W_p'(x) \right]_0^L \end{aligned} \quad (153)$$

Expansion of the RHS of Eq. (153) gives

$$\begin{aligned} & \left(\omega_p^2 - \omega_q^2 \right) \int_0^L W_p(x) W_q(x) dx = \\ & a^4 \left[W_p''''(L) W_q(L) - W_q''''(L) W_p(L) \right] \\ & - a^4 \left[W_p''''(0) W_q(0) - W_q''''(0) W_p(0) \right] \\ & - a^4 \left[W_p''(L) W_q'(L) - W_q''(L) W_p'(L) \right] \\ & + a^4 \left[W_p''(0) W_q'(0) - W_q''(0) W_p'(0) \right] \end{aligned} \quad (154)$$

For a cantilever beam, the boundary conditions are given by Eq. (46) as

$$\begin{aligned} W(0) &= 0 & W'(0) &= 0 \\ W''(L) &= 0 & W'''(L) &= 0 \end{aligned} \quad (46)$$

The use of Eq. (46) in Eq. (154) leads to the cancellation of the right hand side, i.e.,

$$\begin{aligned} & \left(\omega_p^2 - \omega_q^2 \right) \int_0^L W_p(x) W_q(x) dx = \\ & a^4 \left[\cancel{W_p''''(L)} W_q(L) - \cancel{W_q''''(L)} W_p(L) \right] \\ & - a^4 \left[\cancel{W_p''''(0)} \cancel{W_q(0)} - \cancel{W_q''''(0)} \cancel{W_p(0)} \right] \\ & - a^4 \left[\cancel{W_p''(L)} W_q'(L) - \cancel{W_q''(L)} W_p'(L) \right] \\ & + a^4 \left[\cancel{W_p''(0)} \cancel{W_q'(0)} - \cancel{W_q''(0)} \cancel{W_p'(0)} \right] = 0 \end{aligned} \quad (155)$$

Hence, Eq. (155) simply becomes

$$(156)$$

$$\left(\omega_p^2 - \omega_q^2\right) \int_0^L W_p(x) W_q(x) dx = 0 \quad (157)$$

For a free-free beam, the boundary conditions are given by Eq. (72) as

$$\begin{aligned} W''(0) &= 0 & W'''(0) &= 0 \\ W''(L) &= 0 & W'''(L) &= 0 \end{aligned} \quad (72)$$

The use of Eq. (72) in Eq. (154) leads to the cancellation of the right hand side, i.e.,

$$\begin{aligned} &\left(\omega_p^2 - \omega_q^2\right) \int_0^L W_p(x) W_q(x) dx = \\ &a^4 \left[\cancel{W_p''''(L)} W_q(L) - \cancel{W_q''''(L)} W_p(L) \right] \\ &- a^4 \left[\cancel{W_p''''(0)} W_q(0) - \cancel{W_q''''(0)} W_p(0) \right] \\ &- a^4 \left[\cancel{W_p''(L)} W_q'(L) - \cancel{W_q''(L)} W_p'(L) \right] \\ &+ a^4 \left[\cancel{W_p''(0)} W_q'(0) - \cancel{W_q''(0)} W_p'(0) \right] = 0 \end{aligned} \quad (158)$$

For a simply supported beam, the boundary conditions are given by Eq. (92) as

$$\begin{aligned} W(0) &= 0 & W''(0) &= 0 \\ W(L) &= 0 & W''(L) &= 0 \end{aligned} \quad (92)$$

The use of Eq. (92) in Eq. (154) leads to the cancellation of the right hand side, i.e.,

$$\begin{aligned} &\left(\omega_p^2 - \omega_q^2\right) \int_0^L W_p(x) W_q(x) dx = \\ &a^4 \left[\cancel{W_p''''(L)} \cancel{W_q(L)} - \cancel{W_q''''(L)} \cancel{W_p(L)} \right] \\ &- a^4 \left[\cancel{W_p''''(0)} \cancel{W_q(0)} - \cancel{W_q''''(0)} \cancel{W_p(0)} \right] \\ &- a^4 \left[\cancel{W_p''(L)} W_q'(L) - \cancel{W_q''(L)} W_p'(L) \right] \\ &+ a^4 \left[\cancel{W_p''(0)} W_q'(0) - \cancel{W_q''(0)} W_p'(0) \right] = 0 \end{aligned} \quad (159)$$

Thus, after imposition of the boundary conditions, Eq. (154) becomes

$$\left(\omega_p^2 - \omega_q^2\right) \int_0^L W_p(x) W_q(x) dx = 0 \quad (160)$$

For distinct mode numbers, $p \neq q$, the frequencies are also distinct, $\omega_p^2 \neq \omega_q^2$, and hence their difference in Eq. (160) cannot be zero.

For Eq. (160), the integration must be zero, i.e.,

$$\int_0^L W_p(x) W_q(x) dx = 0 \quad \text{if } p \neq q \quad (161)$$

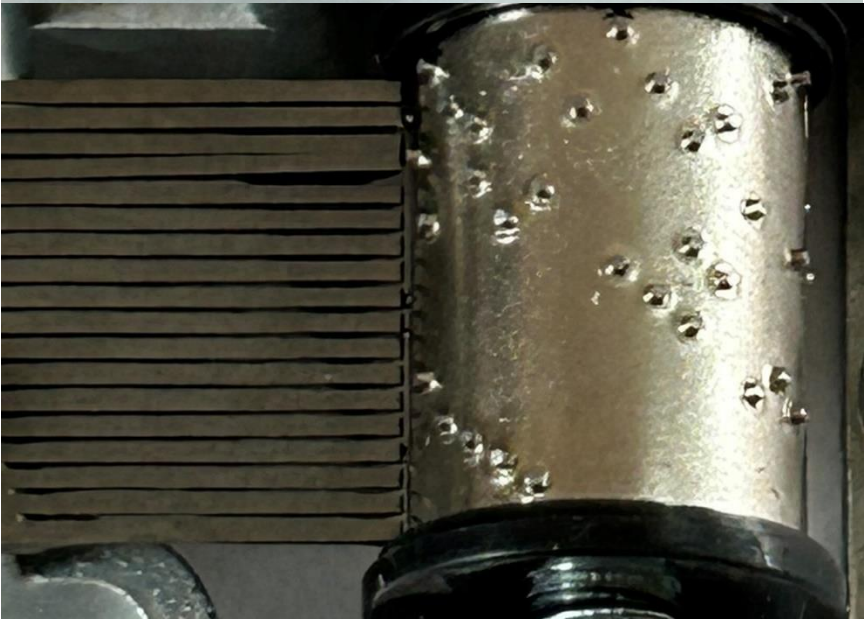
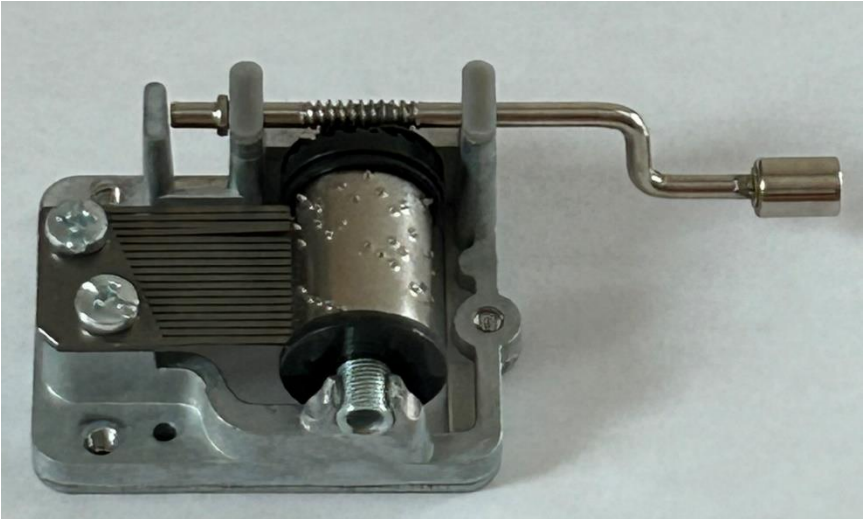
Eq. (161) is exactly the orthogonalization condition Eq. (107). Notice that the above derivation is quite general and that Eq. (161) always holds provided that the boundary conditions are such that the boundary-evaluated terms in Eq. (154) vanish.

6 INITIAL VALUE PROBLEM – THE PLUCKED REED

The vibrating reed is a typical application of vibrating cantilever beam analysis. The vibrating reed appears in most wind instruments as well as in simple devices such as the music box.

In this section, we will analyze the initial value problem (IVP) represented by the plucking of the music-box reeds by the rotating studded cylinder.





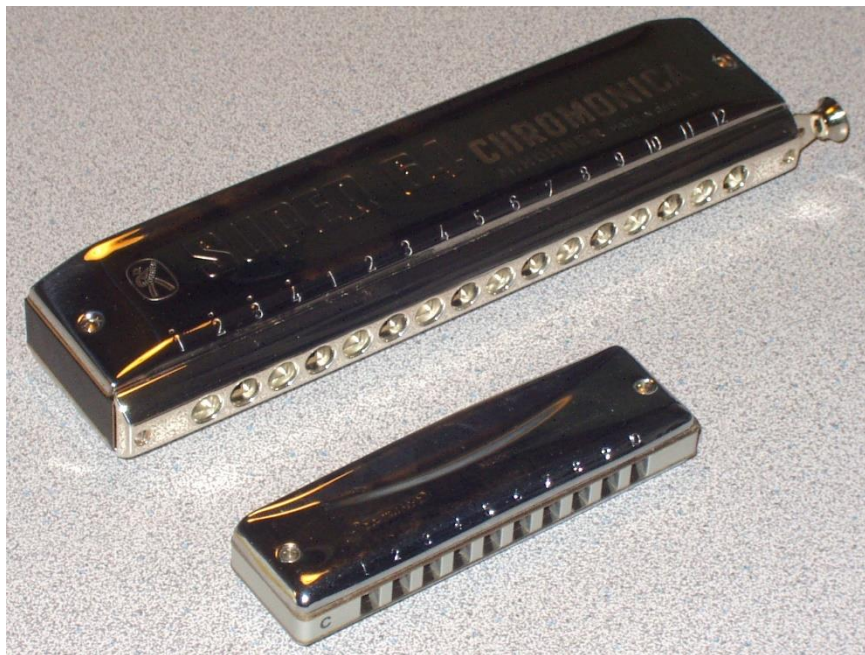


Figure 11 Examples of vibrating reeds in musical instruments

6.1 MODAL EXPANSION METHOD

Recall Eq. (70) giving the flexural vibration motion for each mode of a cantilever beam as

$$w_j(x, t) = W_j(x)e^{-i\omega_j t} \quad (\text{vibration motion of mode } j) \quad (70)$$

where $W_j(x)$ is given by Eqs. (67), (68) and ω_j is given by Eq. (60). We will now assume that the total vibrational motion is made up of a linear superposition of contributions from all the individual vibration modes, i.e.,

$$w(x, t) = \sum_{j=1}^{\infty} \eta_j w_j(x, t) = \sum_{j=1}^{\infty} \eta_j W_j(x) e^{-i\omega_j t} \quad (162)$$

where the constants η_j are the *modal participation factors* signifying how much of each mode $W_j(x)$ contributes to the summation. Depending on the displacement-velocity initial conditions, the modal participation factors may be complex numbers. The objective of the exercise is to find the values of η_j . For convenience, apply Euler's formula to Eq. (162) and write

$$\eta_j e^{-i\omega_j t} = (A_j \cos \omega_j t - B_j \sin \omega_j t) \quad (163)$$

Substitution of Eq. (163) into Eq. (162) yields

$$w(x, t) = \sum_{j=1}^{\infty} W_j(x) (A_j \cos \omega_j t - B_j \sin \omega_j t) \quad (164)$$

Upon differentiation with respect to t Eq. (164) becomes

$$\dot{w}(x, t) = - \sum_{j=1}^{\infty} \omega_j W_j(x) (A_j \sin \omega_j t + B_j \cos \omega_j t) \quad (165)$$

6.2 INITIAL VALUE PROBLEM

The initial value problem consists of finding the modal participation factors for given initial conditions expressed as initial displacement $w_0(x)$ and initial velocity $\dot{w}_0(x)$. Note that the initial displacement and initial velocity are functions of x and represent the initial shape of the beam as well as the initial velocity of each point on the beam. For the plucked cantilever reed case, the initial velocity is assumed

zero whereas the initial displacement is described by the shape of the beam with a known tip displacement $w_{\max}$, i.e.,

$$w_0(x) = \left[\frac{3}{2} \left(\frac{x}{L} \right)^2 - \frac{1}{2} \left(\frac{x}{L} \right)^3 \right] w_{\max} \quad (166)$$

$$\dot{w}_0(x) = 0 \quad (167)$$

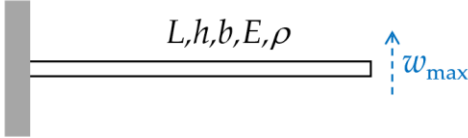


Figure 12 Initial value problem for a cantilever

Substitution of Eq. (164) into Eq. (166) yields

$$\begin{aligned} w(x, 0) &= \sum_{j=1}^{\infty} W_j(x) (A_j \cos \omega_j t - B_j \sin \omega_j t) \Big|_{t=0} \\ &= \sum_{j=1}^{\infty} A_j W_j(x) = w_0(x) \end{aligned} \quad (168)$$

Substitution of Eq. (165) into Eq. (167) yields

$$\begin{aligned} \dot{w}(x, 0) &= - \sum_{j=1}^{\infty} \omega_j W_j(x) (A_j \sin \omega_j t + B_j \cos \omega_j t) \Big|_{t=0} \\ &= \sum_{j=1}^{\infty} \omega_j W_j(x) B_j = \dot{w}_0(x) = 0 \end{aligned} \quad (169)$$

Eq. (169) gives $B_j=0$ which, upon substitution in Eq. (168), yield

$$\sum_{j=1}^{\infty} A_j W_j(x) = w_0(x) \quad (170)$$

6.3 SOLUTION BY GALERKIN METHOD

To obtain a numerical solution, one needs to restrict the number of modes to a finite number N , and write Eq. (164) as

$$w(x, t) = \sum_{j=1}^N A_j W_j(x) \cos \omega_j t \quad (171)$$

where $B_j=0$ in virtue of Eq. (169). The coefficients A_j , $j=1, \dots, N$, should satisfy Eq. (170), i.e.,

$$\sum_{j=1}^N A_j W_j(x) = w_0(x) \quad (172)$$

The application of the Galerkin method consists in multiplying Eq. (172) by each of the modes $W_j(x)$ and then integrating over the length of the beam. For convenience, we use index p instead of j in Eq. (172) and then multiply by mode $W_q(x)$ with $q=1, \dots, N$ to get

$$\sum_{p=1}^N A_p W_p(x) W_q(x) = w_0(x) W_q(x) \quad q=1, 2, 3, \dots, N \quad (173)$$

To apply the Galerkin method, integrate Eq. (173) over the length of the beam to obtain a system of N algebraic equations, i.e.,

$$\int_0^L \sum_{p=1}^N A_p W_p(x) W_q(x) dx = \int_0^L w_0(x) W_q(x) dx \quad q=1, 2, 3, \dots, N \quad (174)$$

Recall the orthonormality property of the modeshapes given by Eq. (117) as

$$\int_0^L W_p(x) W_q(x) dx = \begin{cases} L & \text{if } p = q = j \\ 0 & \text{otherwise} \end{cases} \quad \left(\begin{array}{c} \text{cantilever} \\ \text{beam} \end{array} \right) \quad (117)$$

Implementation of Eq. (117) into Eqs. (174) yields

$$A_j L = \int_0^L w_0(x) W_j(x) dx \quad j=1, 2, 3, \dots, N \quad (175)$$

Eq. (175) can be easily solved for A_j to get

$$A_j = \frac{1}{L} \int_0^L w_0(x) W_j(x) dx \quad j=1, 2, 3, \dots, N \quad (176)$$

The integral in Eq. (176) is calculated by using the w_0 expression given by Eq. (166) and then performing the integration numerically.

6.4 REED RESPONSE

The coefficients A_j of Eq. (176) are used in Eq. (171) to write the vibration response of the plucked reed as

$$w(x, t) = \sum_{j=1}^N A_j W_j(x) \cos \omega_j t \quad (171)$$

Waves and Vibration in a Slender Bar

1 WAVE EQUATION IN A SLENDER BAR

Assume a uniform slender bar of cross section A , density ρ , and elastic modulus E (Figure 1a). The bar undergoes time-varying axial displacement $u(x,t)$. The displacement is assumed uniform across the bar cross section. Consider an infinitesimal element dx as shown in Figure 1b.

The axial displacement $u(x,t)$ changes by an infinitesimal increment du given by

$$du = u'(x,t)dx \quad (1)$$

where the dash was used to designate derivative wrt space x , i.e.,

$$(\quad)' = \frac{\partial}{\partial x}(\quad) \quad (2)$$

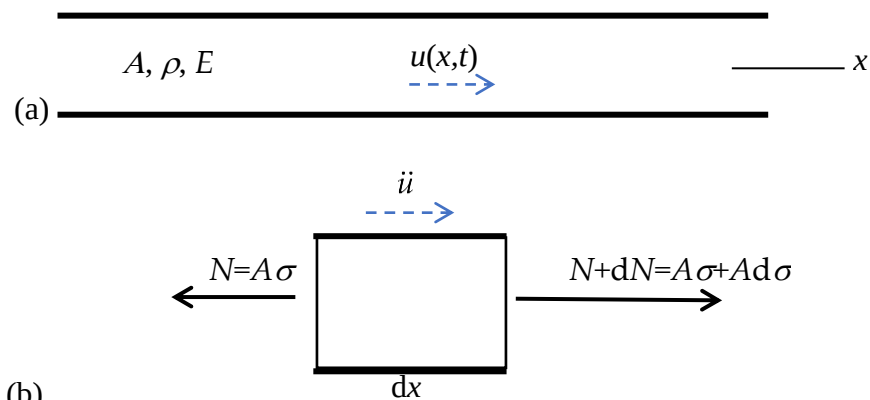


Figure 1 Wave motion in a uniform slender bar: (a) overview; (b) free-body diagram

1.1 FREE-BODY DIAGRAM AND NEWTON'S LAW OF MOTION

The free-body diagram of the infinitesimal element dx is shown in Figure 1b. The stress σ in the bar is assumed uniform across the cross section. The forces acting on the infinitesimal element are $N = \sigma A$ on the left and $N + dN = A\sigma + A d\sigma$ on the right. The net force in the positive x direction is

$$N + dN - N = (A\sigma + Ad\sigma) - A\sigma = Ad\sigma \quad (3)$$

The stress increment $d\sigma$ is given by the expression

$$d\sigma = \sigma' dx \quad (4)$$

Substitution of Eq. (4) in to Eq. (3) yields the net axial force on the infinitesimal element dx as

$$dN = A\sigma' dx \quad (\text{net axial force}) \quad (5)$$

The mass of the infinitesimal element is $\rho A dx$. The acceleration of the infinitesimal element is $\ddot{u}$, where a dot over a variable represents derivative with respect to time, i.e.,

$$\left(\ddot{} \right) = \frac{\partial}{\partial t} \left(\dot{} \right) \quad (6)$$

The two dots in $\ddot{u}$ represent the second derivative wrt time, i.e., the acceleration.

Newton's law of motion applied to the infinitesimal element states that net force should equal mass times acceleration, i.e.,

$$A\sigma' dx = (\rho A dx) \ddot{u} \quad (7)$$

Upon simplification, Eq. (7) becomes

$$\sigma' = \rho \ddot{u} \quad (8)$$

1.2 ELASTICITY EQUATION

Recall Hooke's law which connects the stress σ and strain ε through the elastic constant E , i.e.,

$$\sigma = E\varepsilon \quad (\text{Hooke's law}) \quad (9)$$

Recall the definition of strain ε as the space derivative of displacement, i.e.,

$$\varepsilon(x,t) = u'(x,t) \quad (10)$$

Substitution of Eq. (10) into Eq. (9) yields

$$\sigma(x,t) = Eu'(x,t) \quad (11)$$

Upon differentiation wrt x , Eq. (11) becomes

$$\sigma' = Eu'' \quad (12)$$

1.3 WAVE EQUATION

Substitution of Eq. (12) into Eq. (8) yields

$$Eu'' = \rho \ddot{u} \quad (13)$$

Upon division by ρ , Eq. (13) becomes

$$\frac{E}{\rho} u'' = \ddot{u} \quad (14)$$

Define the wavespeed c in the bar as

$$c = \sqrt{\frac{E}{\rho}} \quad c^2 = \frac{E}{\rho} \quad (\text{wavespeed in the bar}) \quad (15)$$

Eq. (15) gives the speed c with which sound waves propagate in a slender bar. Substitution of Eq. (15) into Eq. (14) yields the *wave equation*, i.e.,

$$c^2 u'' = \ddot{u} \quad (\text{wave equation}) \quad (16)$$

2 SOLUTIONS OF WAVE EQUATION

The wave equation Eq. (16) accepts two types of solutions:

- d'Alembert solutions that emphasize the propagating-disturbance character of the wave
- harmonic solutions that emphasize the oscillatory character of the wave

In this section we will develop the harmonic solution of the wave equation. Assume the time and space behaviors of the wave can be described by the multiplication of two separate functions, one function $U(x)$ that depends on space and the other function $T(t)$ that depends on time, i.e.,

$$u(x,t) = U(x)T(t) \quad (17)$$

Differentiation of Eq. (17) wrt x and t yields

$$\begin{aligned} u' &= U'(x)T(t) & \dot{u} &= U(x)\dot{T}(t) \\ u'' &= U''(x)T(t) & \ddot{u} &= U(x)\ddot{T}(t) \end{aligned} \quad (18)$$

Assume that the time behavior is vibratory with angular frequency ω , i.e.,

$$T(t) = e^{-i\omega t} \quad (19)$$

Substitution of Eq. (19) into Eq. (17) yields

$$u(x,t) = U(x) e^{-i\omega t} \quad (20)$$

Function $U(x)$ represents the amplitude of vibration at any location x along the bar. Substitution of Eq. (20) into Eq. (18) gives

$$u'' = U''(x)e^{-i\omega t} \quad \ddot{u} = -\omega^2 U(x)e^{-i\omega t} \quad (21)$$

Substitution of Eq. (21) into the wave equation Eq. (16) yields

$$c^2 U''(x) e^{-i\omega t} = -\omega^2 U(x) e^{-i\omega t} \quad (22)$$

Upon simplification, Eq. (22) becomes

$$c^2 U''(x) = -\omega^2 U(x) \quad (23)$$

Recall the wavenumber γ defined as

$$\gamma = \omega/c \quad (\text{wavenumber}) \quad (24)$$

Substitution of Eq. (24) into Eq. (23) yields, upon rearrangement,

$$U''(x) + \gamma^2 U(x) = 0 \quad (25)$$

Eq. (25) is a linear ordinary differential equation (ODE) in space variable x . Assume exponential solution of the form

$$U(x) = e^{sx} \quad (26)$$

Upon differentiation, Eq. (26) becomes

$$U''(x) = s^2 e^{sx} = s^2 U(x) \quad (27)$$

Substitution of Eq. (27) into Eq. (25) yields the characteristic equation

$$s^2 + \gamma^2 = 0 \quad (\text{characteristic equation}) \quad (28)$$

The solutions of the characteristic equation Eq. (28) are

$$s_1 = +i\gamma \quad s_2 = -i\gamma \quad (29)$$

The complete solution of the ODE Eq. (25) is a linear superposition of the individual solutions, i.e.,

$$U(x) = A_1 e^{i\gamma x} + A_2 e^{-i\gamma x} \quad (30)$$

Substitution of Eq. (30) into Eq. (20) yields the general solution of the wave equation as

$$u(x, t) = A_1 e^{i(\gamma x - \omega t)} + A_2 e^{-i(\gamma x + \omega t)} \quad (\text{general solution of wave eq.}) \quad (31)$$

Using Euler's formulae, one can write the complex exponentials of Eq. (30) in terms of trigonometric functions and get the solution of Eq. (25) as

$$U(x) = C_1 \sin \gamma x + C_2 \cos \gamma x \quad (32)$$

Hence, the general solution of the wave equation can be also written as

$$u(x, t) = U(x) e^{-i\omega t} = (C_1 \sin \gamma x + C_2 \cos \gamma x) e^{-i\omega t} \quad (33)$$

One may use either Eq. (31) or Eq. (33) as appropriate.

3 AXIAL VIBRATION OF A FINITE-LENGTH BAR

Consider a uniform slender bar of finite length L (Figure 2). Let's examine under what conditions the bar may undergo sustained vibration.

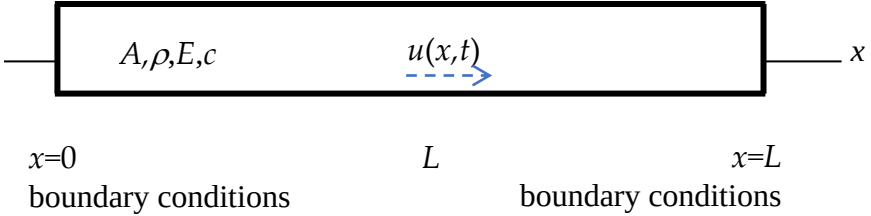


Figure 2 Vibration of a slender bar of finite length L

Recall Eq. (33) giving the harmonic solution of the wave equation in terms of trigonometric functions of x and a complex exponential function of t , i.e.,

$$u(x,t) = (C_1 \sin \gamma x + C_2 \cos \gamma x) e^{-i\omega t} \quad (34)$$

The constants C_1 and C_2 will now be determined by matching the boundary conditions at the two ends of the bar.

3.1 AXIAL VIBRATION OF A BAR FIXED AT BOTH ENDS

Consider a slender bar of length L fixed at both ends as shown in Figure 3. At a fixed end, the displacement is zero. Hence, the boundary conditions for a bar fixed at both ends are

$$u(0,t) = 0 \quad (\text{fixed-end boundary condition at the left}) \quad (35)$$

$$u(L,t) = 0 \quad (\text{fixed-end boundary condition at the right}) \quad (36)$$

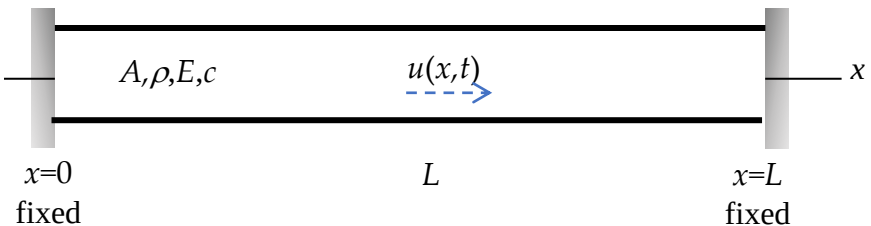


Figure 3 Vibration of a bar of length L fixed at both ends

Recall the displacement expression of Eq. (34), i.e.,

$$u(x,t)=(C_1\sin\gamma x+C_2\cos\gamma x)e^{-i\omega t} \quad (34)$$

Insertion of Eq. (34) into the boundary conditions Eqs. (35), (36), yields

$$u(0,t) = C_2 e^{-i\omega t} = 0 \quad (37)$$

$$u(L,t) = (C_1\sin\gamma L+C_2\cos\gamma L)e^{-i\omega t} = 0 \quad (38)$$

Eq. (37) gives $C_2=0$. Introduction of this value $C_2=0$ into Eq. (38) yields, upon simplification,

$$C_1\sin\gamma L = 0 \quad (39)$$

The trivial solution of Eq. (39) is $C_1=0$ which means no vibration. To find a nontrivial solution $C_1\neq 0$ of Eq. (39), one needs to have $\sin\gamma L=0$, i.e.,

$$\sin\gamma L = 0 \quad (40)$$

If Eq. (39) is satisfied, i.e., $\sin\gamma L=0$, then the value of C_1 becomes arbitrary since any C_1 value will satisfy Eq. (39) as long as $\sin\gamma L=0$. This means that one can take any value for C_1 and still satisfy the conditions of the problem, which means that the problem has one degree of indeterminacy.

3.1.1 EIGENVALUE EQUATION

Introduce the notation

$$z=\gamma L \quad (41)$$

Substitution of Eq. (41) into Eq. (40) yields the eigenvalue equation,

$$\sin z = 0 \quad (\text{eigenvalue equation}) \quad (42)$$

Equation (42) defines the values of z for which nontrivial solutions of Eq. (39) exist. These values of z are the *eigenvalues* of the system. Solution of Eq. (42) gives

$$z=\pi, 2\pi, 3\pi, \dots \quad (43)$$

$$z_j=j\pi \quad j=1,2,3,\dots \quad (44)$$

3.1.2 WAVENUMBERS

Substitution of Eq. (44) into Eq. (41) yields, upon solution, the wavenumbers

$$\gamma_j=j\pi/L \quad j=1,2,3,\dots \quad (45)$$

3.1.3 NATURAL FREQUENCIES

Recall Eq. (78) that gives the angular frequency ω in terms of wavenumber γ , and wavespeed c , i.e.,

$$\omega = \gamma c \quad (78)$$

Substitution of Eq. (45) into Eq. (78) yields the *natural angular frequencies*

$$\omega_j = j c \pi / L \quad j=1,2,3,\dots \quad (46)$$

Since $\omega=2\pi f$, the corresponding Hz frequencies f_j are given by the formula

$$f_j = j c / 2L \quad j=1,2,3,\dots \quad (47)$$

We note that Eqs. (45), (46), (47) predict a denumerable infinite set of wavenumbers and natural frequencies. The first frequency, f_1 , is called the *fundamental frequency*, whereas the other frequencies (f_j $j=2,3,4,\dots$) are called *overtones*. The overtones are multiples of the fundamental frequency. The fundamental frequency is also referred to as the *fundamental harmonic*, whereas the overtones are also referred to as *higher harmonics*.

3.1.4 WAVELENGTHS

Recall Eq. (81) that gives the relationship between wavelength, λ and wavenumber γ , i.e.,

$$\lambda = 2\pi / \gamma \quad (81)$$

Substitution of Eq. (45) into Eq. (81) yields

$$\lambda_j = \frac{2\pi}{\gamma_j} = \frac{2\pi}{j\pi/L} = \frac{2L}{j} \quad j = 1, 2, 3, \dots \quad (48)$$

It is apparent from Eq. (48) that higher frequencies have shorter wavelengths. The vibration of the bar can be interpreted in terms of standing waves fitting the boundary conditions. In the present, case the standing waves are such that the bar of length L accommodates multiples of the half wavelengths, $\lambda_j/2$, $j=1,2,3,\dots$, i.e.,

$$L = j\lambda_j/2 \quad j=1,2,3,\dots \quad (49)$$

The higher the vibration frequency, the shorter the wavelength of these standing waves. The wavenumbers, resonant frequencies, and wavelength multiplicity for the vibration of the bar of length L fixed at both ends are listed in Table 1.

3.1.5 MODESHAPES

Recall Eq. (32), i.e.,

$$U(x)=C_1\sin\gamma x+C_2\cos\gamma x \quad (32)$$

where the constants C_1 and C_2 are given by the solution of Eqs. (37) and (38). Eq. (37) gave $C_2=0$ whereas Eq. (38) yielded Eq. (39), i.e., $C_1\sin\gamma L=0$. The nontrivial solution of Eq. (39) was only possible for special values of γL that made $\sin\gamma L=0$. These special values were called eigenvalues. Associated with the eigenvalues are the wavenumbers $\gamma=j\pi/L$. For each wavenumber $\gamma=j\pi/L$, one gets $\sin\gamma L=0$ in Eq. (39) and hence the value of C_1 becomes arbitrary since any C_1 value will satisfy Eq. (39) as long as $\sin\gamma L=0$. This means that one can take any value for C_1 and still satisfy the conditions of the problem, which means that the problem has one degree of indeterminacy. Here, we take $C_1=1$. Thus, with $C_1=1$ and $C_2=0$, Eq. (32) becomes

$$U_j(x)=\sin\gamma_j x \quad j=1,2,3,\dots \quad (\text{modeshapes}) \quad (50)$$

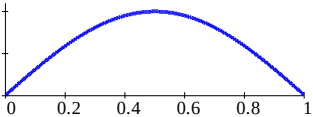
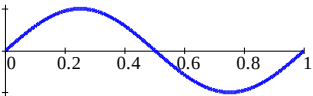
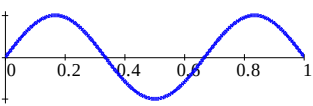
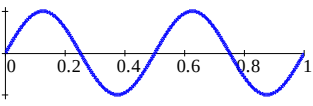
Eq. (50) defines a *natural mode of vibration (modeshape)*. The modeshapes of Eq. (50) can be scaled up or down arbitrarily. Whatever the scale, the modeshapes will still satisfy the boundary conditions of the problem.

Table 1 gives the first four natural frequencies and corresponding modeshapes. One notes that the motion is zero at both ends of the bar which is consistent with the fixed-end condition applying at both ends of the bar.

3.1.6 STRESS MODESHAPES

Substitution of Eq. (50) into Eq. (11) $\sigma=Eu'$ yields the stress modeshapes.

Table 1 Vibration modeshapes of a bar of length L fixed at both ends

Mode #	Eigenvalue $z = \gamma L$ $z = j\pi$	Natural frequency	Modeshape		Wavelength multiplicity
1	$z_1 = \gamma_1 L = \pi$	$f_1 = \frac{c}{2L}$		$U_1 = \sin \pi \frac{x}{L}$	$L = \frac{\lambda_1}{2}$
2	$z_2 = \gamma_2 L = 2\pi$	$f_2 = 2 \frac{c}{2L}$		$U_2 = \sin 2\pi \frac{x}{L}$	$L = 2 \frac{\lambda_2}{2}$
3	$z_3 = \gamma_3 L = 3\pi$	$f_3 = 3 \frac{c}{2L}$		$U_3 = \sin 3\pi \frac{x}{L}$	$L = 3 \frac{\lambda_3}{2}$
4	$z_4 = \gamma_4 L = 4\pi$	$f_4 = 4 \frac{c}{2L}$		$U_4 = \sin 4\pi \frac{x}{L}$	$L = 4 \frac{\lambda_4}{2}$

3.2 AXIAL VIBRATION OF A BAR FIXED AT ONE END AND FREE AT THE OTHER END

Consider a slender bar of length L fixed at one end and free at the other end as shown in Figure 4. At the fixed end, the displacement is zero. At the free end, the stress goes to zero. Hence, the boundary conditions for this case are

$$u(0,t)=0 \quad (\text{fixed-end boundary condition at the left}) \quad (51)$$

$$\sigma(L,t)=0 \quad (\text{free-end boundary condition at the right}) \quad (52)$$

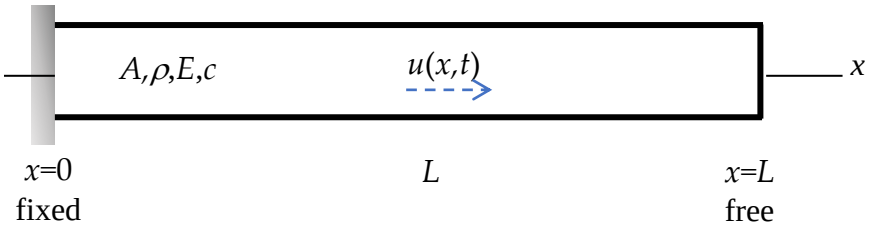


Figure 4 Vibration of a bar of length L fixed at one end and free at the other end

Recall the displacement solution Eq. (34), i.e.,

$$u(x,t)=(C_1 \sin \gamma x + C_2 \cos \gamma x) e^{-i\omega t} \quad (34)$$

Also recall the stress expression Eq. (68) developed in Section 3.3, i.e.,

$$\sigma(x,t)=E(C_1 \gamma \cos \gamma x - C_2 \gamma \sin \gamma x) e^{-i\omega t} \quad (68)$$

Insertion of Eq. (68) into the boundary conditions Eqs. (51), (52), yields

$$u(0,t) = C_2 e^{-i\omega t} = 0 \quad (53)$$

$$\sigma(L,t) = E(C_1 \gamma \cos \gamma L - C_2 \gamma \sin \gamma L) e^{-i\omega t} = 0 \quad (54)$$

Eq. (53) gives $C_2=0$. Introduction of this value $C_2=0$ into Eq. (54) yields, upon simplification,

$$C_1 \cos \gamma L = 0 \quad (55)$$

The trivial solution of Eq. (55) is $C_1=0$ which means no vibration. To find a nontrivial solution $C_1 \neq 0$ of Eq. (55), one needs to have $\cos \gamma L = 0$, i.e.,

$$\cos \gamma L = 0 \quad (56)$$

If Eq. (56) is satisfied, i.e., $\cos \gamma L = 0$, then the value of C_1 becomes arbitrary since any C_1 value will satisfy Eq. (56) as long as $\cos \gamma L = 0$. This means that one can take any value for C_1 and still satisfy the conditions of the problem, which means that the problem has one degree of indeterminacy.

3.2.1 EIGENVALUE EQUATION

Recall the notation of Eq. (41)

$$z = \gamma L \quad (41)$$

Substitution of Eq. (41) into Eq. (56) yields the eigenvalue equation,

$$\cos z = 0 \quad (\text{eigenvalue equation}) \quad (57)$$

Equation (57) defines the values of z for which nontrivial solutions of Eq. (55) exist. These values of z are the *eigenvalues* of the system. Solution of Eq. (57) gives

$$z = \pi/2, 3\pi/2, 5\pi/2, \dots \quad (58)$$

$$z_j = (2j-1)\pi/2 \quad j=1,2,3,\dots \quad (59)$$

3.2.2 WAVENUMBERS

Substitution of Eq. (59) into Eq. (41) yields the wavenumbers

$$\gamma_j = (2j-1)\pi/2L \quad j=1,2,3,\dots \quad (60)$$

3.2.3 NATURAL FREQUENCIES

Recall Eq. (78) that gives angular frequency ω in terms of wavenumber γ , and wavespeed c , i.e.,

$$\omega = \gamma c \quad (78)$$

Substitution of Eq. (60) into Eq. (78) yields the *natural angular frequencies*

$$\omega_j = (2j-1)c\pi/2L \quad j=1,2,3,\dots \quad (61)$$

Since $\omega = 2\pi f$, the corresponding Hz frequencies f_j are given by the formula

$$f_j = (2j-1)c/4L \quad j=1,2,3,\dots \quad (62)$$

We note that Eqs. (60), (61), (62) predict a denumerable infinite set of wavenumbers and natural frequencies. The first frequency, f_1 , is called the *fundamental frequency*, whereas the other frequencies (f_j $j=2,3,4,\dots$) are called *overtones*. The overtones are multiples of the fundamental frequency. The fundamental frequency is also referred

to as the *fundamental harmonic*, whereas the overtones are also referred to as *higher harmonics*.

3.2.4 WAVELENGTHS

Recall Eq. (81) that gives the relationship between wavelength, λ and wavenumber γ , i.e.,

$$\lambda = 2\pi/\gamma \quad (81)$$

Substitution of Eq. (60) into Eq. (81) yields

$$\lambda_j = \frac{2\pi}{\gamma_j} = \frac{2\pi}{(2j-1)\pi/2L} = \frac{4L}{(2j-1)} \quad j = 1, 2, 3, \dots \quad (63)$$

It is apparent from Eq. (63) that higher frequencies have shorter wavelengths. The vibration of the bar can be interpreted in terms of standing waves fitting the boundary conditions. In the present, case the standing waves are such that the bar length L accommodates odd multiples of the quarter wavelengths, $\lambda_j/4$, $j=1, 2, 3, \dots$, i.e.,

$$L = (2j-1)(\lambda_j/4) \quad j=1, 2, 3, \dots \quad (64)$$

The higher the vibration frequency, the shorter the wavelength of these standing waves. The wavenumbers, resonant frequencies, and wavelength multiplicity for the vibration of a bar of length L fixed at one end and free at the other end are listed in Table 2.

3.2.5 MODESHAPES

Recall Eq. (32), i.e.,

$$U(x) = C_1 \sin \gamma x + C_2 \cos \gamma x \quad (32)$$

where the constants C_1 and C_2 are given by the solution of Eqs. (53) and (54). Eq. (53) gave $C_2=0$ whereas Eq. (54) yielded Eq. (55), i.e., $C_1 \cos \gamma L = 0$. The nontrivial solution of Eq. (55) was only possible for special values of γL that made $\cos \gamma L = 0$. These special values were called eigenvalues. Associated with the eigenvalues are the wavenumbers $\gamma_j = (2j-1)\pi/2L$. For each wavenumber $\gamma_j = (2j-1)\pi/2L$, one gets $\sin \gamma_j L = 0$ in Eq. (55) and hence the value of C_1 becomes arbitrary since any C_1 value will satisfy Eq. (55) as long as $\cos \gamma L = 0$. This means that one can take any value for C_1 and still satisfy the conditions of the problem, which means that the problem has one degree of indeterminacy. Here, we take $C_1=1$. Thus, with $C_1=1$ and $C_2=0$, Eq. (32) becomes

$$U_j(x) = \sin \gamma_j x \quad j=1,2,3,\dots \quad (\text{modeshapes}) \quad (65)$$

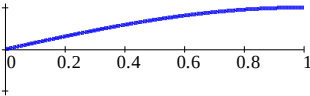
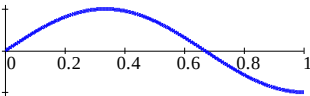
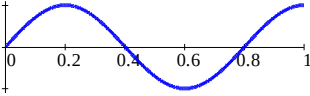
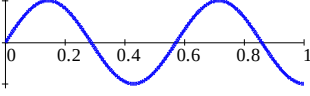
Eq. (65) defines a *natural mode of vibration (modeshape)*. The modeshapes of Eq. (65) can be scaled up or down arbitrarily. Whatever the scale, the modeshapes will still satisfy the boundary conditions of the problem.

Table 2 gives the first four natural frequencies and corresponding modeshapes. One notes that the motion is zero at the fixed end and maximum at the free end, as expected.

3.2.6 STRESS MODESHAPES

Substitution of Eq. (65) into Eq. (11) $\sigma = Eu'$ yields the stress modeshapes.

Table 2 Vibration modeshapes of a bar of length L fixed on the left and free at the right

Mode #	Eigenvalue $z_j = \gamma_j L$ $z_j = (2j-1)\pi/2$	Natural frequency	Modeshape		Wavelength multiplicity
1	$z_1 = \gamma_1 L = \pi/2$	$f_1 = \frac{c}{4L}$		$U_1 = \sin \frac{\pi x}{2L}$	$L = \frac{\lambda_1}{4}$
2	$z_2 = \gamma_2 L = 3\pi/2$	$f_2 = 3 \frac{c}{4L}$		$U_2 = \sin 3 \frac{\pi x}{2L}$	$L = 3 \frac{\lambda_2}{4}$
3	$z_3 = \gamma_3 L = 5\pi/2$	$f_3 = 5 \frac{c}{4L}$		$U_3 = \sin 5 \frac{\pi x}{2L}$	$L = 5 \frac{\lambda_3}{4}$
4	$z_4 = \gamma_4 L = 7\pi/2$	$f_4 = 7 \frac{c}{4L}$		$U_4 = \sin 7 \frac{\pi x}{2L}$	$L = 7 \frac{\lambda_4}{4}$

3.3 AXIAL VIBRATION OF A BAR FREE AT BOTH ENDS

Consider a slender bar of length L free at both ends as shown in Figure 5.

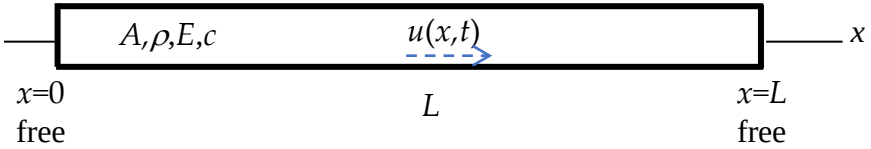


Figure 5 Vibration of a bar of length L free at both ends

At a free end, the stress is zero. Hence, the boundary conditions for this case are

$$\sigma(0,t)=0 \quad (\text{free-end boundary condition at the left}) \quad (66)$$

$$\sigma(L,t)=0 \quad (\text{free-end boundary condition at the right}) \quad (67)$$

Recall the displacement solution Eq. (34), i.e.,

$$u(x,t)=(C_1\sin\gamma x+C_2\cos\gamma x)e^{-i\omega t} \quad (34)$$

Recall Eq. (11) that connects stress and displacement derivative, i.e.,

$$\sigma=Eu' \quad (11)$$

Substitution of Eq. (34) into Eq. (11) yields

$$\sigma(x,t)=E(C_1\gamma\cos\gamma x-C_2\gamma\sin\gamma x)e^{-i\omega t} \quad (68)$$

Insertion of Eq. (68) into the boundary conditions Eqs. (66), (67) yields

$$\sigma(0,t)=C_1E\gamma e^{-i\omega t}=0 \quad (69)$$

$$\sigma(L,t)=E(C_1\gamma\cos\gamma L-C_2\gamma\sin\gamma L)e^{-i\omega t}=0 \quad (70)$$

Eq. (69) gives $C_1=0$. Introduction of this value $C_1=0$ into Eq. (70) yields, upon simplification,

$$C_2\sin\gamma L=0 \quad (71)$$

The trivial solution of Eq. (71) is $C_2=0$ which means no vibration. To find a nontrivial solution $C_2\neq 0$ of Eq. (71), one needs to have $\sin\gamma L=0$, i.e.,

$$\sin\gamma L=0 \quad (72)$$

If Eq. (72) is satisfied, i.e., $\sin\gamma L=0$, then the value of C_2 becomes arbitrary since any C_2 value will satisfy Eq. (71) as long as $\sin\gamma L=0$.

This means that one can take any value for C_2 and still satisfy the conditions of the problem, which means that the problem has one degree of indeterminacy.

3.3.1 EIGENVALUE EQUATION

Recall the notation of Eq. (41)

$$z = \gamma L \quad (41)$$

Substitution of Eq. (41) into Eq. (72) yields the eigenvalue equation, i.e.,

$$\sin z = 0 \quad (\text{eigenvalue equation}) \quad (73)$$

Equation (73) defines the values of z for which nontrivial solutions of Eq. (71) exist. These values of z are the *eigenvalues* of the system. Solution of Eq. (73) gives

$$z = \pi, 2\pi, 3\pi, \dots \quad (74)$$

$$z_j = j\pi \quad j = 1, 2, 3, \dots \quad (75)$$

3.3.2 WAVENUMBERS

Substitution of Eq. (75) into Eq. (41) yields, upon solution, the wavenumbers γ_j , i.e.,

$$\gamma_j = j\pi/L \quad j = 1, 2, 3, \dots \quad (76)$$

3.3.3 NATURAL FREQUENCIES

Recall the relationship between wavenumber γ , wavespeed c , and angular frequency ω , i.e.,

$$\gamma = \omega/c \quad (77)$$

Eq. (77) can be solved for ω to yield

$$\omega = \gamma c \quad (78)$$

Substitution of Eq. (76) into Eq. (78) yields the *natural angular frequencies* ω_j , i.e.,

$$\omega_j = j c \pi / L \quad j = 1, 2, 3, \dots \quad (79)$$

Since $\omega = 2\pi f$, the corresponding Hz frequencies f_j are given by the formula

$$f_j = j c / 2L \quad j = 1, 2, 3, \dots \quad (80)$$

We note that Eqs. (76), (79), (80) predict denumerable infinite sets of wavenumbers and natural frequencies. The first frequency, ω_1 , is called the *fundamental frequency*, whereas the other frequencies (f_j $j=2,3,4,\dots$) are called *overtones*. The overtones are multiples of the

fundamental frequency. The fundamental frequency is also referred to as the *fundamental harmonic*, whereas the overtones are also referred to as *higher harmonics*.

3.3.4 WAVELENGTH

Recall the relationship between *wavelength* λ and *wavenumber* γ , i.e.,

$$\lambda = 2\pi/\gamma \quad (81)$$

Substitution of Eq. (76) into Eq. (81) yields

$$\lambda_j = \frac{2\pi}{\gamma_j} = \frac{2\pi}{j\pi/L} = \frac{2L}{j} \quad j = 1, 2, 3, \dots \quad (82)$$

It is apparent from Eq. (82) that higher frequencies have shorter wavelengths. The vibration of the bar of length L can be interpreted in terms of standing waves fitting the boundary conditions. In the present case, the standing waves are such that the bar length L accommodates exact multiples of the half wavelengths, $\lambda_j/2$, $j=1, 2, 3, \dots$, i.e.,

$$L = j(\lambda_j/2) \quad j = 1, 2, 3, \dots \quad (83)$$

The higher the vibration frequency, the shorter the wavelength of these standing waves. The wavenumbers, resonant frequencies, and wavelength multiplicity for the vibration of bar of length L free at both ends are listed in Table 3.

3.3.5 MODESHAPES

Recall Eq. (32), i.e.,

$$U(x) = C_1 \sin \gamma x + C_2 \cos \gamma x \quad (32)$$

where the constants C_1 and C_2 are given by the solution of Eqs. (69) and (70). Eq. (69) gave $C_1=0$ whereas Eq. (70) yielded Eq. (71) i.e., $C_2 \sin \gamma L = 0$. The nontrivial solution of Eq. (71) was only possible for special values of γL that made $\sin \gamma L = 0$. These special values were called eigenvalues. Associated with the eigenvalues are the wavenumbers $\gamma = j\pi/L$. For each wavenumber $\gamma = j\pi/L$, one gets $\sin \gamma L = 0$ in Eq. (71) and hence the value of C_2 becomes arbitrary since any C_2 value will satisfy Eq. (71) as long as $\sin \gamma L = 0$. This means that one can take any value for C_2 and still satisfy the conditions of the problem, which means that the problem has one

degree of indeterminacy. Here, we take $C_2=1$. Thus, with $C_1=0$ and $C_2=1$, Eq. (32) becomes

$$U_j(x)=\cos\gamma_j x \quad j=1,2,3,\dots \quad (\text{modeshapes}) \quad (84)$$

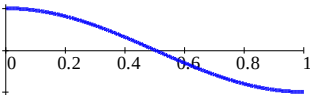
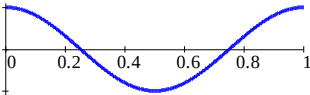
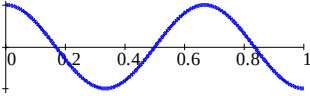
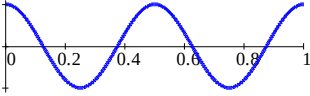
Eq. (84) defines a *natural mode of vibration (modeshape)*. The modeshapes of Eq. (84) can be scaled up or down arbitrarily. Whatever the scale, the modeshapes will still satisfy the boundary conditions of the problem.

Table 3 gives the first four natural frequencies and corresponding modeshapes. One notes that the motion is maximum at the free ends, as expected.

3.3.6 STRESS MODESHAPES

Substitution of Eq. (84) into Eq. (11) $\sigma=Eu'$ yields the stress modeshapes.

Table 3 Vibration modeshapes of bar of length L free at both ends

Mode #	Eigenvalue $z = \gamma L$ $z = j\pi$	Natural frequency	Modeshape		Wavelength multiplicity
1	$z_1 = \gamma_1 L = \pi$	$f_1 = \frac{c}{2L}$		$U_1 = \cos \pi \frac{x}{L}$	$L = \frac{\lambda_1}{2}$
2	$z_2 = \gamma_2 L = 2\pi$	$f_2 = 2 \frac{c}{2L}$		$U_2 = \cos 2\pi \frac{x}{L}$	$L = 2 \frac{\lambda_2}{2}$
3	$z_3 = \gamma_3 L = 3\pi$	$f_3 = 3 \frac{c}{2L}$		$U_3 = \cos 3\pi \frac{x}{L}$	$L = 3 \frac{\lambda_3}{2}$
4	$z_4 = \gamma_4 L = 4\pi$	$f_4 = 4 \frac{c}{2L}$		$U_4 = \cos 4\pi \frac{x}{L}$	$L = 4 \frac{\lambda_4}{2}$

4 MODESHAPE ORTHONORMALITY AND MODAL COEFFICIENTS

4.1 ORTHOGONALITY OF MODESHAPES

Modeshapes orthogonality is an important property that allows one to verify if some independently derived functions qualify to be considered as modeshapes. For axial vibration of a bar of length L , the modeshape orthogonality is stated as

$$\int_0^L U_p(x)U_q(x)dx = \begin{cases} L/2 & \text{for } p = q \\ 0 & \text{otherwise} \end{cases} \quad (\text{orthogonality}) \quad (85)$$

Eq. (85) implies that the product between two different modeshapes is zero when integrated over the bar length L . This is the **orthogonality** property. If the two modeshapes are the same, i.e., the modeshape is multiplied upon itself, then the result is a finite number, in this case $L/2$. If one divides the integral by this finite number $L/2$, then the result is 1. This is the process of **normalization**. Thus, the **orthonormality** property of the modeshapes is expressed as

$$\int_0^L U_p(x)U_q(x)dx / (L/2) = \begin{cases} 1 & \text{for } p = q \\ 0 & \text{otherwise} \end{cases} \quad (\text{orthonormality}) \quad (86)$$

The modeshape orthonormality is usually verified through numerical integration, although closed-form analytical proof may also be possible.

4.2 MODAL COEFFICIENTS

The modal coefficients are used to describe the modal behavior in response to an external excitation as described later in this course.

4.2.1 MODAL MASS AND MODAL STIFFNESS

The commonly used modal coefficients are the modal mass m_j and the modal stiffness k_j . They are defined as follows:

$$m_j = \int_0^L \rho A U_j^2(x) dx \quad (\text{modal mass}) \quad (87)$$

$$k_j = -\int_0^L EA U_j''(x)U_j(x) dx \quad (\text{modal stiffness}) \quad (88)$$

4.2.2 RELATIONSHIP BETWEEN MODAL MASS AND MODAL STIFFNESS

The relationship between modal mass and modal stiffness is expressed as

$$k_j = \omega_j^2 m_j \quad (89)$$

The relationship between modal mass and modal stiffness expressed by Eq. (89) is similar to the mass-stiffness relationship $k = \omega^2 m$ of a 1-dof mass-spring system. The modal coefficients allow us to treat each mode of vibration as a simple 1-dof vibration system.

Proof of Eq. (89)

The proof of Eq. (89) proceeds as follows. Recall Eq. (25), i.e.,

$$c^2 U''(x) = -\omega^2 U(x) \quad (90)$$

Recall that $c^2 = E/\rho$. Hence, Eq. (90) becomes

$$EU''(x) = -\rho\omega^2 U(x) \quad (91)$$

For the j^{th} modeshape, Eq. (91) is written as

$$EU_j''(x) = -\rho\omega_j^2 U_j(x) \quad (92)$$

Substitution of Eq. (92) into the modal stiffness definition Eq. (88) and use of the modal mass definition Eq. (87) gives

$$\begin{aligned} k_j &= -\int_0^L EA U_j''(x) U_j(x) dx \\ &= -\int_0^L (-\rho\omega_j^2) A U_j(x) U_j(x) dx \quad \text{QED} \\ &= \omega_j^2 \int_0^L \rho A U_j^2(x) dx = \omega_j^2 m_j \end{aligned} \quad (93)$$

4.2.3 MODAL MASS AND MODAL STIFFNESS VALUES

In a uniform bar, the density ρ and cross-section area A are constant and hence get outside the integral sign in Eq. (87) giving

$$m_j = \rho A \int_0^L U_j^2(x) dx \quad (\text{modal mass}) \quad (94)$$

In virtue of Eq. (85), the modal mass m_j take the value $\rho AL/2$. Substitution of this value in Eq. (89) yields the modal stiffness k_j as

$$k_j = \omega_j^2 \rho AL/2, \text{ i.e.,}$$

$$m_j = \rho AL/2 \quad (\text{modal mass}) \quad (95)$$

$$k_j = \omega_j^2 \rho A L / 2 \quad (\text{modal stiffness}) \quad (96)$$

Note that the modal mass value of $\rho A L / 2$ represents half of the bar mass. Also note that the modal mass m_j has the same value for all the vibration modes.

4.3 CLOSED-FORM ANALYTICAL DERIVATION OF MODESHAPE ORTHONORMALITY

To study analytically modeshape orthonormality, consider the integrals

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx \quad \text{for } p \neq q \quad (97)$$

$$I_j = \int_0^L U_j^2(x) dx \quad \text{for } p=q=j \quad (98)$$

In virtue of Eq. (85), these equations should take the values

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = 0 \quad \text{for } p \neq q \quad (99)$$

$$I_j = \int_0^L U_j^2(x) dx = L/2 \quad \text{for } p=q=j \quad (100)$$

4.3.1 FREE-FREE BAR VIBRATION

The wavenumbers, natural frequencies, and modeshapes for a bar free at both ends are given by Eq. (76), (79), (84) as

$$\gamma_j = j\pi/L \quad \omega_j = jc\pi/L \quad U_j(x) = \cos \gamma_j x \quad j=1,2,3,\dots \quad (101)$$

4.3.1.1 Calculation of I_{pq}

Let j take two different values p and q in Eq. (101) and write

$$U_p(x) = \cos \gamma_p x \quad \gamma_p = p\pi/L \quad (102)$$

$$U_q(x) = \cos \gamma_q x \quad \gamma_q = q\pi/L \quad (103)$$

Substitution of Eqs. (102), (103) into Eq. (99) yields

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = \int_0^L \cos \gamma_p x \cos \gamma_q x dx \quad (104)$$

Recall the “product of cosines” trigonometric identity

$$\cos \alpha \cos \beta = [\cos(\alpha + \beta) + \cos(\alpha - \beta)]/2 \quad (105)$$

The use of Eq. (105) into Eq. (104) yields, upon integration,

$$\begin{aligned}
\int_0^L \cos \gamma_p x \cos \gamma_q x \, dx = \\
\frac{1}{2} \int_0^L \cos[(\gamma_p + \gamma_q)x] dx + \frac{1}{2} \int_0^L \cos[(\gamma_p - \gamma_q)x] dx = \\
\frac{\sin(\gamma_p + \gamma_q)L}{2(\gamma_p + \gamma_q)} + \frac{\sin(\gamma_p - \gamma_q)L}{2(\gamma_p - \gamma_q)}
\end{aligned} \quad (106)$$

Insertion of the γ_p, γ_q expressions from Eqs. (102), (103) into Eq. (106) yields expression of the form

$$(\gamma_p + \gamma_q)L = (p+q)\pi L/L = (p+q)\pi \quad (107)$$

$$(\gamma_p - \gamma_q)L = (p-q)\pi L/L = (p-q)\pi \quad (108)$$

Substitution of Eqs. (107), (108) into Eq. (106) gives

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = \frac{\sin(p+q)\pi}{2(p+q)\pi/L} + \frac{\sin(p-q)\pi}{2(p-q)\pi/L} = 0 \quad (109)$$

Eq. (109) is zero because sine of any integer multiple of π is zero. This holds true as long as $p \neq q$. QED.

4.3.1.2 Calculation of I_j

Substitution of Eq. (101) into Eq. (100) yields

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \cos^2(\gamma_j x) dx \quad (110)$$

Recall the “cos squared” trigonometric identity

$$\cos^2 \alpha = (1 + \cos 2\alpha)/2 \quad (111)$$

The use of Eq. (111) into Eq. (110) yields

$$\begin{aligned}
I_j &= \int_0^L \cos^2(\gamma_j x) dx = \int_0^L \frac{1}{2} [1 + \cos(2\gamma_j x)] dx \\
&= \frac{1}{2} \int_0^L dx + \frac{1}{2} \int_0^L \cos(2\gamma_j x) dx
\end{aligned} \quad (112)$$

The first integral in Eq. (112) gives

$$\int_0^L dx = L \quad (113)$$

The second integral in Eq. (112) gives

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2\gamma_j x) \Big|_0^L = \frac{1}{2\gamma_j} \sin(2\gamma_j L) \quad (114)$$

Recall from Eq. (101) that $\gamma_j = j\pi/L$; upon substitution into Eq. (114) one gets

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2j\pi \mathcal{X}/\mathcal{L}) = 0 \quad (115)$$

Eq. (115) is zero because sine of any integer multiple of π is zero. Substitution of Eq. (148) into Eq. (112) yields

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \cos^2(\gamma_j x) dx = L/2 \quad \text{QED} \quad (116)$$

4.3.2 FIXED-FIXED BAR VIBRATION

The wavenumbers, natural frequencies, and modeshapes for a bar fixed at both ends are given by Eqs. (45), (46), (50) as

$$\gamma_j = j\pi/L \quad \omega_j = jc\pi/L \quad U_j(x) = \sin \gamma_j x \quad j=1,2,3,\dots \quad (117)$$

4.3.2.1 Calculation of I_{pq}

Let j take two different values p and q in Eq. (117) and write

$$U_p(x) = \sin \gamma_p x \quad \gamma_p = p\pi/L \quad (118)$$

$$U_q(x) = \sin \gamma_q x \quad \gamma_q = q\pi/L \quad (119)$$

Substitution of Eqs. (118), (119) into Eq. (99) yields

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = \int_0^L \sin \gamma_p x \sin \gamma_q x dx \quad (120)$$

Recall the “product of sines” trigonometric identity

$$\sin \alpha \sin \beta = [\cos(\alpha - \beta) - \cos(\alpha + \beta)]/2 \quad (121)$$

The use of Eq. (121) into Eq. (120) yields, upon integration,

$$\begin{aligned} \int_0^L \sin \gamma_p x \sin \gamma_q x dx = \\ \frac{1}{2} \int_0^L \cos[(\gamma_p - \gamma_q)x] dx - \frac{1}{2} \int_0^L \cos[(\gamma_p + \gamma_q)x] dx = \\ \frac{\sin(\gamma_p - \gamma_q)L}{2(\gamma_p - \gamma_q)} - \frac{\sin(\gamma_p + \gamma_q)L}{2(\gamma_p + \gamma_q)} \end{aligned} \quad (122)$$

The wavenumbers in Eq. (122) are given by Eq. (117) as $\gamma_j = j\pi/L$. Hence, the $(\gamma_p - \gamma_q)L$ and $(\gamma_p + \gamma_q)L$ expressions showing up in Eq. (122) take the following values

$$(\gamma_p - \gamma_q)L = (p - q)\pi/L = (p - q)\pi \quad (123)$$

$$(\gamma_p + \gamma_q)L = (p + q)\pi/L = (p + q)\pi \quad (124)$$

Substitution of Eqs. (123), (124) into Eq. (122) followed by substitution into Eq. (120) yields

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = \frac{\sin(p-q)\pi}{2(p-q)\pi/L} - \frac{\sin(p+q)\pi}{2(p+q)\pi/L} = 0 \quad (125)$$

Eq. (125) is zero because sine of any integer multiple of π is zero. This holds true as long as $p \neq q$. QED.

4.3.2.2 Calculation of I_j

Substitution of Eq. (117) into Eq. (100) yields

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \sin^2(\gamma_j x) dx \quad (126)$$

Recall the “sine squared” trigonometric identity

$$\sin^2 \alpha = (1 - \cos 2\alpha)/2 \quad (127)$$

The use of Eq. (127) into the integration part of Eq. (126) yields

$$\begin{aligned} I_j &= \int_0^L \sin^2(\gamma_j x) dx = \int_0^L \frac{1}{2} [1 - \cos(2\gamma_j x)] dx \\ &= \frac{1}{2} \int_0^L dx - \frac{1}{2} \int_0^L \cos(2\gamma_j x) dx \end{aligned} \quad (128)$$

The first integral in Eq. (128) gives

$$\int_0^L dx = L \quad (129)$$

The second integral in Eq. (128) gives

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2\gamma_j x) \Big|_0^L = \frac{1}{2\gamma_j} \sin(2\gamma_j L) \quad (130)$$

Recall from Eq. (117) that $\gamma_j = j\pi/L$; upon substitution into Eq. (130) one gets

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2j\pi \cancel{L}/\cancel{L}) = 0 \quad (131)$$

Eq. (131) is zero because sine of any integer multiple of π is zero. Substitution of Eqs. (129), (131) into Eq. (128) yields

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \sin^2(\gamma_j x) dx = L/2 \quad \text{QED} \quad (132)$$

4.3.3 FIXED-FREE BAR VIBRATION

The wavenumbers, natural frequencies, and modeshapes for a fixed-free bar are given by Eqs. (60), (61), (65) as

$$\gamma_j = (2j-1)\pi/2L \quad \omega_j = (2j-1)c\pi/2L \quad U_j(x) = \sin \gamma_j x \quad j=1,2,3,\dots \quad (133)$$

4.3.3.1 Calculation of I_{pq}

Let j take two different values p and q in Eq. (133) and write

$$U_p(x) = \sin \gamma_p x \quad \gamma_p = p\pi/L \quad (134)$$

$$U_q(x) = \sin \gamma_q x \quad \gamma_q = q\pi/L \quad (135)$$

Substitution of Eqs. (134), (135) into Eq. (99) yields

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = \int_0^L \sin \gamma_p x \sin \gamma_q x dx \quad (136)$$

Recall the “product of sines” trigonometric identity

$$\sin \alpha \sin \beta = [\cos(\alpha - \beta) - \cos(\alpha + \beta)]/2 \quad (137)$$

The use of Eq. (137) into Eq. (136) yields, upon integration,

$$\begin{aligned} \int_0^L \sin \gamma_p x \sin \gamma_q x dx = \\ \frac{1}{2} \int_0^L \cos[(\gamma_p - \gamma_q)x] dx - \frac{1}{2} \int_0^L \cos[(\gamma_p + \gamma_q)x] dx = \\ \frac{\sin(\gamma_p - \gamma_q)L}{2(\gamma_p - \gamma_q)} - \frac{\sin(\gamma_p + \gamma_q)L}{2(\gamma_p + \gamma_q)} \end{aligned} \quad (138)$$

Insertion of the γ_p, γ_q expressions from Eq. (133) into Eq. (138) yields terms like

$$(\gamma_p - \gamma_q)L = [(2p - 1) - (2q - 1)]\pi L/2L = (p - q)\pi \quad (139)$$

$$(\gamma_p + \gamma_q)L = [(2p - 1) + (2q - 1)]\pi L/2L = (p + q)\pi \quad (140)$$

Substitution of Eqs. (139), (140) into Eq. (138) followed by substitution into Eq. (136) yields

$$I_{pq} = \int_0^L U_p(x) U_q(x) dx = \frac{\sin(p - q)\pi}{2(p - q)\pi/L} - \frac{\sin(p + q)\pi}{2(p + q)\pi/L} = 0 \quad (141)$$

Eq. (141) is zero because sine of any integer multiple of π is zero. This holds true as long as $p \neq q$. QED.

4.3.3.2 Calculation of I_j

Substitution of Eq. (133) into Eq. (100) yields

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \sin^2(\gamma_j x) dx \quad (142)$$

Recall the “sine squared” trigonometric identity

$$\sin^2 \alpha = (1 - \cos 2\alpha)/2 \quad (143)$$

The use of Eq. (143) into Eq. (142) yields

$$\begin{aligned}
 I_j &= \int_0^L \sin^2(\gamma_j x) dx = \int_0^L \frac{1}{2} [1 - \cos(2\gamma_j x)] dx \\
 &= \frac{1}{2} \int_0^L dx - \frac{1}{2} \int_0^L \cos(2\gamma_j x) dx
 \end{aligned} \tag{144}$$

The first integral in Eq. (144) gives

$$\int_0^L dx = L \tag{145}$$

The second integral in Eq. (144) gives

$$\int_0^L \cos(2\gamma_j x) dx = \frac{1}{2\gamma_j} \sin(2\gamma_j x) \Big|_0^L = \frac{1}{2\gamma_j} \sin(2\gamma_j L) \tag{146}$$

Recall from Eq. (133) that $\gamma_j = (2j-1)\pi/2L$; upon substitution in Eq. (146) one gets

$$\begin{aligned}
 \int_0^L \cos(2\gamma_j x) dx &= \frac{1}{2\gamma_j} \sin[\frac{2(2j-1)\pi}{2} / \frac{2\pi}{2}] \\
 &= \frac{1}{2\gamma_j} \sin(2j-1)\pi = 0
 \end{aligned} \tag{147}$$

Eq. (147) is zero because sine of any integer multiple of π is zero. Substitution of Eqs. (145), (147) into Eq. (144) yields

$$I_j = \int_0^L U_j^2(x) dx = \int_0^L \sin^2(\gamma_j x) dx = L/2 \quad \text{QED} \tag{148}$$

MOMENTUM

IMPACT AND CONSERVATION OF MOMENTUM

CONSERVATION OF MOMENTUM

<https://en.wikipedia.org/wiki/Momentum>

$$p = \Sigma mv$$

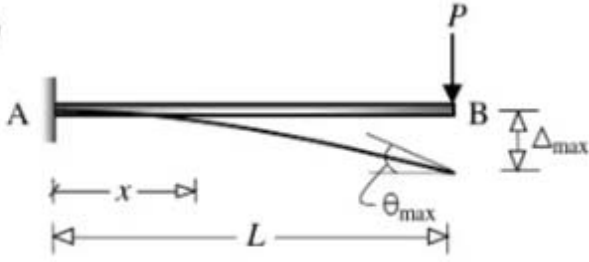
Consider two masses, m and M . The small mass m travels with initial velocity v . The large mass M is initially at rest. The small mass collides with the large mass and remains imbedded in it. After collision, the assembly travels with final speed v^* . Application of the conservation of momentum gives

$$mv + M \cdot 0 = (m + M)v^* \quad (1)$$

Hence, the final velocity is

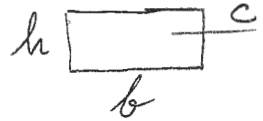
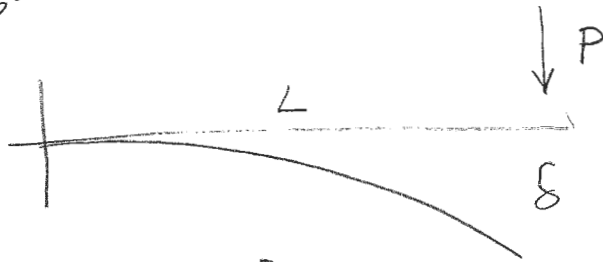
$$v^* = mv / (m + M) \quad (2)$$

BEAMS

 Scribd < > ⋮ ×		
Beam type	Slope at free end	Deflection at critical location(s)
a) 	$\theta_{\max} = \frac{PL^2}{2EI}$	$\Delta_{\max} = \frac{PL^3}{3EI}$

spring constant: $k = P/\Delta_{\max} = 3EI/L^3$

20240320



$$c = h/2$$

$$\delta = \frac{PL^3}{3EI} \rightarrow P = \frac{3EI\delta}{L^3}$$

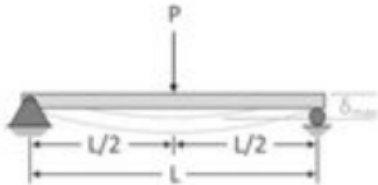
$$M = PL = \frac{3EI}{L^3} L\delta = \frac{3EI}{L^2} \delta$$

$$\sigma_{\max} = \frac{Mc}{I} = \frac{3EI}{L^2} \delta \frac{c}{I} = \frac{3c\delta}{L^2} E$$

Safety factor

$$SF = \frac{Y}{\sigma_{\max}}$$

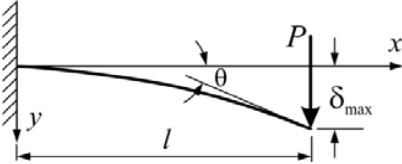
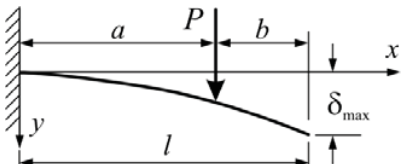
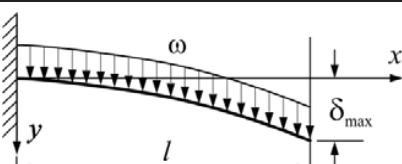
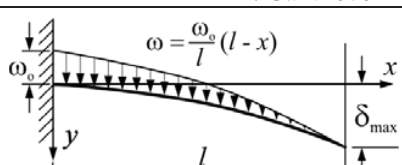
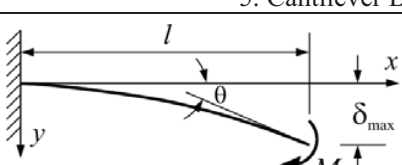
Y = yield stress



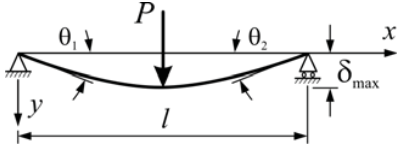
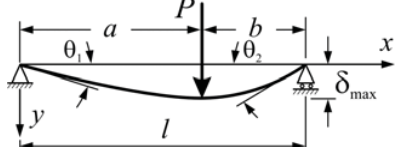
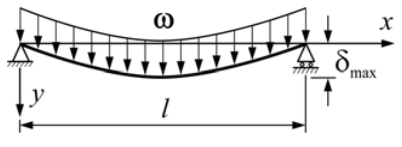
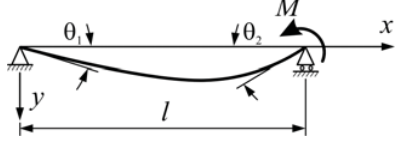
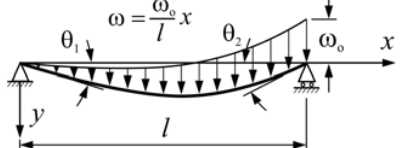
$$\delta_{max} = \frac{PL^3}{48EI}$$

249/265

BEAM DEFLECTION FORMULAE

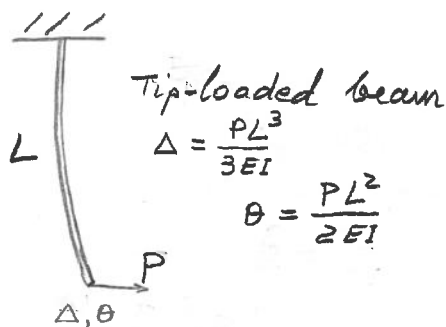
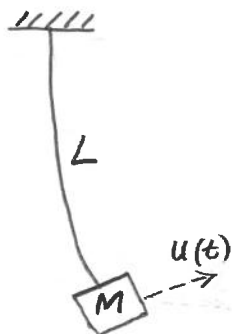
BEAM TYPE	SLOPE AT FREE END	DEFLECTION AT ANY SECTION IN TERMS OF x	MAXIMUM DEFLECTION
1. Cantilever Beam – Concentrated load P at the free end			
	$\theta = \frac{Pl^2}{2EI}$	$y = \frac{Px^2}{6EI}(3l - x)$	$\delta_{\max} = \frac{Pl^3}{3EI}$
2. Cantilever Beam – Concentrated load P at any point			
	$\theta = \frac{Pa^2}{2EI}$	$y = \frac{Px^2}{6EI}(3a - x) \text{ for } 0 < x < a$ $y = \frac{Pa^2}{6EI}(3x - a) \text{ for } a < x < l$	$\delta_{\max} = \frac{Pa^2}{6EI}(3l - a)$
3. Cantilever Beam – Uniformly distributed load ω (N/m)			
	$\theta = \frac{\omega l^3}{6EI}$	$y = \frac{\omega x^2}{24EI}(x^2 + 6l^2 - 4lx)$	$\delta_{\max} = \frac{\omega l^4}{8EI}$
4. Cantilever Beam – Uniformly varying load: Maximum intensity ω_o (N/m)			
	$\theta = \frac{\omega_o l^3}{24EI}$	$y = \frac{\omega_o x^2}{120EI}(10l^3 - 10l^2x + 5lx^2 - x^3)$	$\delta_{\max} = \frac{\omega_o l^4}{30EI}$
5. Cantilever Beam – Couple moment M at the free end			
	$\theta = \frac{Ml}{EI}$	$y = \frac{Mx^2}{2EI}$	$\delta_{\max} = \frac{Ml^2}{2EI}$

BEAM DEFLECTION FORMULAS

BEAM TYPE	SLOPE AT ENDS	DEFLECTION AT ANY SECTION IN TERMS OF x	MAXIMUM AND CENTER DEFLECTION
6. Beam Simply Supported at Ends – Concentrated load P at the center			
	$\theta_1 = \theta_2 = \frac{Pl^2}{16EI}$	$y = \frac{Px}{12EI} \left(\frac{3l^2}{4} - x^2 \right) \text{ for } 0 < x < \frac{l}{2}$	$\delta_{\max} = \frac{Pl^3}{48EI}$
7. Beam Simply Supported at Ends – Concentrated load P at any point			
	$\theta_1 = \frac{Pb(l^2 - b^2)}{6lEI}$ $\theta_2 = \frac{Pab(2l - b)}{6lEI}$	$y = \frac{Pbx}{6lEI} (l^2 - x^2 - b^2) \text{ for } 0 < x < a$ $y = \frac{Pb}{6lEI} \left[\frac{l}{b} (x - a)^3 + (l^2 - b^2)x - x^3 \right] \text{ for } a < x < l$	$\delta_{\max} = \frac{Pb(l^2 - b^2)^{3/2}}{9\sqrt{3}lEI} \text{ at } x = \sqrt{(l^2 - b^2)}/3$ $\delta = \frac{Pb}{48EI} (3l^2 - 4b^2) \text{ at the center, if } a > b$
8. Beam Simply Supported at Ends – Uniformly distributed load ω (N/m)			
	$\theta_1 = \theta_2 = \frac{\omega l^3}{24EI}$	$y = \frac{\omega x}{24EI} (l^3 - 2lx^2 + x^3)$	$\delta_{\max} = \frac{5\omega l^4}{384EI}$
9. Beam Simply Supported at Ends – Couple moment M at the right end			
	$\theta_1 = \frac{Ml}{6EI}$ $\theta_2 = \frac{Ml}{3EI}$	$y = \frac{Mlx}{6EI} \left(1 - \frac{x^2}{l^2} \right)$	$\delta_{\max} = \frac{Ml^2}{9\sqrt{3}EI} \text{ at } x = \frac{l}{\sqrt{3}}$ $\delta = \frac{Ml^2}{16EI} \text{ at the center}$
10. Beam Simply Supported at Ends – Uniformly varying load: Maximum intensity ω_o (N/m)			
	$\theta_1 = \frac{7\omega_o l^3}{360EI}$ $\theta_2 = \frac{\omega_o l^3}{45EI}$	$y = \frac{\omega_o x}{360lEI} (7l^4 - 10l^2x^2 + 3x^4)$	$\delta_{\max} = 0.00652 \frac{\omega_o l^4}{EI} \text{ at } x = 0.519l$ $\delta = 0.00651 \frac{\omega_o l^4}{EI} \text{ at the center}$

BP
2023/03/1

FLEX-BEAM PENDULUM



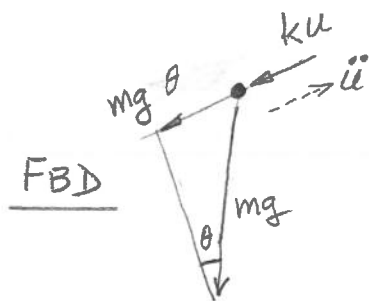
Tip-loaded beam

$$\Delta = \frac{PL^3}{3EI}$$

$$\theta = \frac{PL^2}{2EI}$$

small angle approximation

$$k \approx \frac{P}{\Delta} = \frac{3EI}{PL^3}$$



$$\theta = \frac{PL^2}{2EI}$$

$$u = \frac{PL^3}{3EI}$$

$$\frac{\theta}{u} = \frac{PL^2}{2EI} \cdot \frac{3EI}{PL^3} = \frac{3}{2L}$$

$$\theta = \frac{3}{2L} u$$

NLM

$$-mg\theta - ku = m\ddot{u}$$

$$m\ddot{u} + ku + mg\theta = 0 \quad \text{EOM}$$

$$m\ddot{u} + ku + mg\frac{3}{2L}u = 0$$

$$\ddot{u} + \left(\frac{k}{m} + \frac{3}{2L}g\right)u = 0$$

$$\ddot{u} + \omega_n^2 u = 0$$

governing eqⁿ

where

$$\omega_n = \sqrt{\frac{k}{m} + \frac{3}{2L}g}$$

E0

ENERGY METHODS
IN
VIBRATION ANALYSIS

ENERGY METHODS

IN

UNDAMPED VIBRATION

3.2.3 Energy Methods in 1-dof Vibration Analysis

In Sections 3.2.1 and 3.2.2, we derived the equation of motion of a 1-dof spring-mass system by applying Newton's second law using the free-body diagram approach. In this approach, we identified the resultant of all forces acting on the mass and then equated it with the product of mass and acceleration. An alternative way of deriving the equation of motion is by *energy methods*. We will illustrate the energy methods approach with a few simple examples. However, the true advantage of the energy methods approach becomes apparent when it is applied to complicated mechanical systems which, even if essentially 1-dof, are difficult to analyze in detail.

3.2.3.1 Undamped 1-dof Vibration Analysis by Energy Methods

Consider the undamped 1-dof horizontal vibration of a mass attached to a spring, as illustrated in Figure 3.10a. The mass undergoes horizontal displacement, $u(t)$, around the equilibrium position. The kinetic energy T of the moving mass is given by

$$T(t) = \frac{1}{2} m \dot{u}^2(t) \quad (125)$$

whereas the elastic energy V consisting of the energy stored in the displaced spring is given by

$$V(t) = \frac{1}{2} k u^2(t) \quad (126)$$

The total energy of the system is

$$E = T(t) + V(t) = \frac{1}{2} m \dot{u}^2(t) + \frac{1}{2} k u^2(t) \quad (127)$$

Substituting $u(t) = \hat{u} \cos \omega t$, we write

$$E = T(t) + V(t) = \frac{1}{2} m \omega^2 \hat{u}^2 \sin^2 \omega t + \frac{1}{2} k \hat{u}^2 \cos^2 \omega t \quad (128)$$

Recalling Eq. (15), i.e., $k = \omega_n^2 m$, we can write Eq. (128) as

$$E = T(t) + V(t) = \frac{1}{2} m \dot{u}^2 (\omega^2 \sin^2 \omega t + \omega_n^2 \cos^2 \omega t) \quad (129)$$

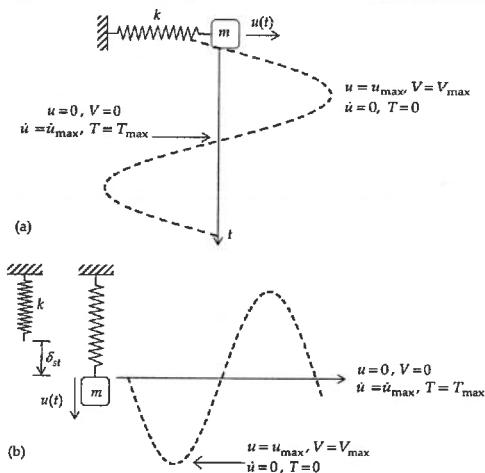


FIGURE 3.10 Energy analysis of the undamped vibration of a mass-spring system: (a) horizontal motion; (b) vertical motion.

DERIVATION OF THE EQUATION OF MOTION BY ENERGY METHODS

The principle of energy conservation stipulates that energy is conserved (i.e., it is stationary), which gives

$$E = E_0 = \text{const} \quad (130)$$

Hence, the derivative of energy with respect to time should be zero. Taking the time derivative of Eq. (127) and equating it to zero gives

$$\frac{d}{dt}[T(t) + V(t)] = \frac{d}{dt} \left[\frac{1}{2} m \dot{u}^2(t) + \frac{1}{2} k u^2(t) \right] = 0 \quad (131)$$

Upon differentiation, we get

$$m \ddot{u}(t) \dot{u}(t) + k \dot{u}(t) u(t) = 0 \quad (132)$$

Simplification by $\dot{u}(t)$ yields the equation of motion

$$m \ddot{u}(t) + k u(t) = 0 \quad (133)$$

Equation (133) is identical with Eq. (12) derived from Newton's law of motion.

ESTIMATION OF THE NATURAL FREQUENCY BY ENERGY METHODS

Energy methods can be expediently used to estimate the natural frequency of the vibrating system. As we analyze the oscillatory motion, we notice that there are two *salient points*: (1) the extreme position and (2) the equilibrium position. Assume the 1-dof system oscillates freely, i.e., at its natural frequency. Recall the expression of the oscillatory motion, $u(t) = C \cos \omega_n t$, and hence $\dot{u}(t) = -\omega_n C \sin \omega_n t$. When the mass passes through the extreme position, the displacement amplitude reaches the maximum, $u_{\max}$, while the velocity vanishes, because the sine function vanishes when the cosine function reaches an extreme (± 1). Hence the kinetic energy at the extreme position reaches the maximum, whereas the elastic energy is zero. The law of energy conservation (129) gives

$$E \Big|_{\substack{u=u_{\max} \\ \dot{u}=0}} = V_{\max} = \frac{1}{2} k u_{\max}^2 = \frac{1}{2} k \dot{u}^2 = E_0 \quad (134)$$

When the mass passes through the equilibrium position (i.e., through the center), its displacement is zero, hence the spring is undeformed and stores no energy. This means that the elastic energy is zero at the equilibrium position. At the same time, the mass passing through the equilibrium position has maximum velocity, because the sine function reaches an extreme (± 1) when the cosine vanishes. Hence the kinetic energy reaches a maximum as the mass passes through the equilibrium position. The law of energy conservation gives

$$E \Big|_{\substack{u=0 \\ \dot{u}=\dot{u}_{\max}}} = T_{\max} = \frac{1}{2} m \dot{u}_{\max}^2 = \frac{1}{2} m u_{\max}^2 \dot{u}^2 = E_0 \quad (135)$$

Comparison of Eqs. (134) and (135) yields

$$V_{\max} = T_{\max} \text{ i.e., } \frac{1}{2} m u_{\max}^2 \dot{u}^2 = \frac{1}{2} k \dot{u}^2 \quad (136)$$

Upon simplification, Eq. (136) yields

$$\omega_n^2 = \frac{k}{m} \text{ or } \omega_n = \sqrt{\frac{k}{m}} \quad (137)$$

Equation (137) has established a relationship between the natural frequency and the maximum values of the potential and kinetic energies. This concept leads to the *Rayleigh quotient* and the *Rayleigh-Ritz method* used in the estimation of the fundamental frequency of more complicated vibration systems (ref. [7], pp. 270, 274).

EFFECT OF GRAVITATIONAL FIELD ON ENERGY METHODS FORMULATION OF VIBRATION ANALYSIS

So far, we have considered the case of horizontal vibration in which the gravitational field does not intervene. Now, we consider the case of vertical vibration, which takes place in the same direction as the gravitational field (Figure 3.10b). We assume that the vibrational

motion takes place about a static equilibrium position δ_{st} . With this datum, the potential energy associated with the vibrational motion takes the form

$$V(t) = \frac{1}{2} k [\delta_{st} + u(t)]^2 - \frac{1}{2} k \delta_{st}^2 - mg u(t) \quad (138)$$

where the potential energy stored in the spring under static conditions, $\frac{1}{2} k \delta_{st}^2(t)$, was subtracted from the potential energy of the spring deformed during the vibrational motion, $\frac{1}{2} k [\delta_{st} + u(t)]^2$. The quantity $-mg u(t)$ is the gravitational potential. Upon expansion, Eq. (138) becomes

$$V(t) = \frac{1}{2} k \delta_{st}^2 + \frac{1}{2} 2k \delta_{st} u(t) + \frac{1}{2} k u(t)^2 - \frac{1}{2} k \delta_{st}^2 - mg u(t) \quad (139)$$

Recalling the equilibrium condition $k \delta_{st} = mg$, Eq. (138) simplifies to the form

$$V(t) = \frac{1}{2} k u^2(t) \quad (140)$$

which is identical with Eq. (126) derived for horizontal motion. Thus, it is apparent that the gravitational field can be eliminated from the energy method formulation of the vibration analysis by taking the vibration datum at the static equilibrium position.

ENERGY METHODS
IN
DAMPED VIBRATION

3.2.3.2 Damped 1-dof Vibration Analysis by Energy Methods

Energy methods can also be used for the analysis of damped systems. In this case, the total energy of the system is not conserved, as energy is dissipated by the damping forces. Two cases will be analyzed here: (1) the derivation of the damped 1-dof equation of motion via energy methods, and (2) the energy and power associated with the damped 1-dof response to forced excitation.

DERIVATION OF THE DAMPED 1-DOF EQUATION BY ENERGY METHODS

We use the first law of thermodynamics to express the fact that the incremental change in total energy equals the incremental work performed by the external forces acting on the system, i.e.,

$$\delta E = \delta W \quad (141)$$

The incremental change in the total energy is expressed by the incremental change in the kinetic and potential energies. Recalling Eq. (127), we get

$$\delta E = \delta T(t) + \delta V(t) = m\dot{u}(\ddot{u}\delta t) + ku(\dot{u}\delta t) \quad (142)$$

The incremental work δW is estimated by the product of the dissipation forces and the incremental displacement, i.e.,

$$\delta W = (-c\dot{u})\delta u = (-c\dot{u})\dot{u}\delta t \quad (143)$$

Combination of Eqs. (141) through (143) yields

$$m\dot{u}(\ddot{u}\delta t) + ku(\dot{u}\delta t) = (-c\dot{u})\dot{u}\delta t \quad (144)$$

Upon simplification and rearrangement, Eq. (144) yields Eq. (29) of damped free vibration, i.e.,

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = 0 \quad (145)$$

POWER AND ENERGY ASSOCIATED WITH DAMPED 1-DOF RESPONSE TO HARMONIC EXCITATION

Recall Eq. (85) describing the steady-state vibration response and Eq. (91) expressing the velocity in terms of displacement and excitation frequency, i.e.,

$$u(t) = \hat{u}(\omega)e^{i\omega t}, \quad \dot{u}(t) = i\omega u(t) \quad (146)$$

In virtue of the convention associated with the use of complex notation, we retain only the real parts and rewrite Eq. (146) as

$$u(t) = |\hat{u}(\omega)|\cos(\omega t + \psi), \quad \dot{u}(t) = -\omega|\hat{u}(\omega)|\sin(\omega t + \psi) \quad (147)$$

The instantaneous *power input* is defined as the product of force and velocity, i.e.,

$$P(t) = F(t)\dot{u}(t) = \hat{F}\cos(\omega t)[- \omega|\hat{u}(\omega)|\sin(\omega t + \psi)] \quad (148)$$

Upon simplification,

$$P(t) = -\omega\hat{F}|\hat{u}(\omega)|\cos(\omega t)\sin(\omega t + \psi) \quad (149)$$

Applying trigonometric formulae, we can express Eq. (149) as

$$P(t) = -\frac{1}{2}\omega\hat{F}|\hat{u}(\omega)|[\sin\psi + \sin(2\omega t + \psi)] \quad (150)$$

ENERGY METHOD IN CONTINUOUS SYSTEMS

where the generic functions $\eta_j(t)$ are time-dependent *modal (normal) coordinates*. Substitution of modal expansion (296) into the equation of motion (273) yields

$$EA \sum_{j=1}^{\infty} \eta_j(t) U_j'(x) + m \sum_{j=1}^{\infty} \ddot{\eta}_j(t) U_j(x) = f(x, t) \quad (297)$$

As before, multiply Eq. (297) by each of the modeshapes in turn, use the orthonormality conditions (245), (246) and get, upon rearrangement, a set of decoupled equations in the form

$$m_j \ddot{\eta}_j(t) + k_j \eta_j(t) = f_j(t), \quad j = 1, 2, 3, \dots \quad (298)$$

where $f_j(t)$ is the *time-dependent modal excitation* given by

$$f_j(t) = \int_0^l f(x, t) U_j(x) dx, \quad j = 1, 2, 3, \dots \quad (299)$$

Introducing the modal damping ratio, ζ_j , to account for the inherent dissipative losses encountered in practice, we obtain the modal equations for damped vibrations of a continuous system in the form

$$\ddot{\eta}_j(t) + 2\zeta_j \omega_j \dot{\eta}_j(t) + \eta_j(t) \omega_j^2 = f_j(t), \quad j = 1, 2, 3, \dots \quad (300)$$

Each of the modal equations (300) resembles the 1-dof Eq. (83). Since the modal equations are uncoupled, each of them can be solved independently, and then reassembled through modal summation to obtain the total response.

3.3.4 Axial Vibration Energy in a Bar

We have seen in the study of the 1-dof vibration of a mass-spring system that the vibration energy, E , contains a kinetic energy, T , and an elastic energy (potential energy), V , i.e.,

$$E = T(t) + V(t) = \frac{1}{2} m \dot{u}^2(t) + \frac{1}{2} k u^2(t) \quad (301)$$

To investigate these principles in the context of an elastic bar we will consider the kinetic and elastic energies stored in the bar, i.e.,

$$T(t) = \int_{\Omega} \frac{1}{2} \rho \dot{u}^2(x, t) d\Omega = \int_0^l \frac{1}{2} m \dot{u}^2(x, t) dx \quad (\text{axial vibration of a bar}) \quad (302)$$

$$V(t) = \int_{\Omega} \frac{1}{2} \sigma(x, t) \epsilon(x, t) d\Omega = \int_0^l \frac{1}{2} EA u'^2(x, t) dx$$

where Ω is the total volume of the bar. Using the modal expansion of Eq. (278), we write

$$T(t) = \int_0^l \frac{1}{2} \rho \left[\sum_{p=1}^{\infty} C_p U_p(x) (-\omega \sin \omega t) \right] \left[\sum_{q=1}^{\infty} C_q U_q(x) (-\omega \sin \omega t) \right] dx \quad (\text{kinetic energy}) \quad (303)$$

$$V(t) = \int_0^l \frac{1}{2} EA \left[\sum_{p=1}^{\infty} C_p U'_p(x) \cos \omega t \right] \left[\sum_{p=1}^{\infty} C_p U'_p(x) \cos \omega t \right] dx \quad (\text{elastic energy}) \quad (304)$$

Processing of Eqs. (303) and (304) through the use of the orthogonality conditions (209), (215) and of the modal mass and stiffness definitions (217), (224) yields

$$T(t) = \omega^2 \sin^2 \omega t \sum_{p=1}^{\infty} \frac{1}{2} C_p^2 m_j \quad (\text{kinetic energy}) \quad (305)$$

$$V(t) = \cos^2 \omega t \sum_{p=1}^{\infty} \frac{1}{2} k_j C_p^2 = \cos^2 \omega t \sum_{p=1}^{\infty} \frac{1}{2} m_j \omega_j^2 C_p^2 \quad (\text{elastic energy}) \quad (306)$$

Hence, the total energy is given by

$$\begin{aligned} E(t) &= T(t) + V(t) = \omega^2 \sin^2 \omega t \sum_{p=1}^{\infty} \frac{1}{2} C_p^2 m_j + \cos^2 \omega t \sum_{p=1}^{\infty} \frac{1}{2} C_p^2 k_j \\ &= \sum_{p=1}^{\infty} \frac{1}{2} m_j C_p^2 (\omega^2 \sin^2 \omega t + \omega_j^2 \cos^2 \omega t) \end{aligned} \quad (307)$$

If the system oscillates at one of the resonant frequencies, say $\omega = \omega_j$, then the kinetic and elastic energies are concentrated into just one mode of vibration, and resemble 1-dof behavior, i.e.,

$$T_j(t) = \frac{1}{2} m_j \omega_j^2 C_j^2 \sin^2 \omega_j t \quad (\text{modal kinetic energy}) \quad (308)$$

$$V_j(t) = \frac{1}{2} k_j C_j^2 \cos^2 \omega t = \frac{1}{2} m_j \omega_j^2 C_j^2 \cos^2 \omega_j t \quad (\text{modal elastic energy}) \quad (309)$$

The total energy of the system vibrating in the j th natural mode is given by the modal vibration energy,

$$\begin{aligned} E_j &= T_j(t) + V_j(t) = \frac{1}{2} m_j \omega_j^2 C_j^2 \sin^2 \omega_j t + \frac{1}{2} m_j \omega_j^2 C_j^2 \cos^2 \omega_j t \\ &= \frac{1}{2} m_j \omega_j^2 C_j^2 \end{aligned} \quad (\text{modal vibration energy}) \quad (310)$$

During the vibration cycle, the total vibration energy, E_j , is constant, but the contributions of the kinetic and elastic parts vary. At the point of maximum displacement, the elastic energy reaches a maximum while the kinetic energy is zero, whereas at the point of zero displacement (but maximum velocity), the elastic energy is zero while the kinetic energy reaches a maximum. Recall that the assumption of natural modes of vibrations is that the complete body undergoes the vibration in phase, i.e., all the points in the body reach the maximum amplitude and then the zero amplitude, etc., in the same time.

The energy input to the system per cycle of oscillation is obtained by integrating the power over a period of oscillation $\tau = 2\pi/\omega$, i.e.,

$$\begin{aligned}\Delta E_{\text{cyc}} &= \int_0^{2\pi/\omega} P(t)dt = -\frac{1}{2}\omega\hat{F}|\hat{u}(\omega)| \int_0^{2\pi/\omega} [\sin\psi + \sin(2\omega t + \psi)]dt \\ &= -\pi\hat{F}|\hat{u}(\omega)|\sin\psi\end{aligned}\quad (151)$$

Recall Eq. (88), i.e.,

$$\hat{u}(\omega) = \frac{\hat{F}}{-\omega^2 m + ic\omega + k} = \hat{F} \frac{(k - \omega^2 m) - ic\omega}{(k - \omega^2 m)^2 + (c\omega)^2} \quad (152)$$

The magnitude and phase angle of Eq. (152)

$$\begin{aligned}|\hat{u}(\omega)| &= \frac{\hat{F}}{\sqrt{(k - \omega^2 m)^2 + (c\omega)^2}} \\ \sin\psi &= \frac{\text{Im } \hat{u}(\omega)}{|\hat{u}(\omega)|} = \frac{-ic\omega}{\sqrt{(k - \omega^2 m)^2 + (c\omega)^2}}\end{aligned}\quad (153)$$

Substitution of Eq. (153) into Eq. (151) yields, upon manipulation,

$$\Delta E_{\text{cyc}}(\omega) = \pi c\omega |\hat{u}(\omega)|^2 \quad (154)$$

Equation (154) indicates that the required energy input per cycle (i.e., the energy dissipated by the damped system) increases linearly with frequency and quadratically with response amplitude. The average power is obtained by dividing the energy per cycle by the cycle duration, $\tau = 2\pi/\omega$, i.e.,

$$P_{av}(\omega) = \frac{1}{2}c\omega^2 |\hat{u}(\omega)|^2 \quad (155)$$

It is noted that power is proportional to the square of the vibration amplitude. It is also proportional to the system damping and the frequency squared. The response amplitude reaches the maximum at resonance. Hence, the power intake to the system also reaches a local maximum at resonance, i.e.,

$$P_{\text{max}} = \frac{1}{2}c\omega_r^2 |\hat{u}_r|^2 \quad (156)$$

For lightly damped systems, the resonance point can be sufficiently well approximated by the quadrature point, i.e., $\omega_r \cong \omega_{90}$, $\hat{u}_r \cong \hat{u}_{90}$, and

$$P_{\text{max}} = \frac{1}{2}c\omega_r^2 |\hat{u}_{90}|^2 \quad (157)$$

where the response amplitude at quadrature, $\hat{u}_{90}$, is given by Eq. (107). Upon substitution into Eq. (155), the power at resonance is found to be

$$P_{\text{max}} = \frac{1}{2} \frac{\hat{F}^2}{c} \quad (158)$$

At resonance, where the response amplitude reaches a maximum, the energy per cycle reaches a maximum too.

MECHANICAL - ELECTRICAL EQUIVALENTS

MECHANICAL-ELECTRICAL EQUIVALENTS

An important analogy exists between electrical RLC circuits and mechanical spring-mass-damper systems. Figure 3.9 shows a series circuit consisting of an a.c. voltage source, $v(t)$, of angular frequency ω , and typical circuit elements such as an electrical resistance, R , an inductance, L , and a capacitance, C . The electrical current flowing through the circuit is $i(t)$, and the electrical charge accumulated in the circuit is $q(t) = \int i(t) dt$. Basic electrical engineering analysis (ref. [6], p. 109) gives the following differential equation between the time-dependent charge, $q(t)$, the applied voltage $v(t)$, and the circuit elements R , L , C ,

$$L\ddot{q}(t) + R\dot{q}(t) + \frac{1}{C}q(t) = v(t) \quad (119)$$

This equation has the same form as Eq. (75), the forced damped vibration equation for the 1-dof mechanical system. Thus, by a simple interchange of symbols (Table 3.1), knowledge

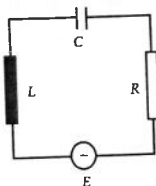


FIGURE 3.9 Typical a.c. electric circuit containing resistance, R , inductance, L , capacitance, C , and voltage source, E .

TABLE 3.1 Mechanical-electrical equivalents

Mechanical			Electrical		
Units	Symbol	Name	Name	Symbol	Units
m (meter)	u	Displacement	Charge	q	C (coulomb)
m/s (meter/sec)	$\dot{u}$	Velocity	Current	i	A (ampere)
N (Newton)	F	Force	Voltage	v	V (volts)
kg (kilogram)	m	Mass	Inductance	L	H (henry)
N/m (Newton/meter)	k	Stiffness	1/Capacitance	$1/C$	F (farad)
Ns/m (Newton-sec./meter)	c	Damping	Resistance	R	Ω (ohm)

about the behavior of the electrical circuit can be used to infer knowledge about the behavior of the mechanical system. This fact allows us to analyze the behavior of mechanical systems by using analytical models and software predictors developed for electrical systems. It also allows us to experimentally simulate the behavior of mechanical systems using electrical circuits built using the mechanical-electrical equivalents. Such circuits form the basis of *analog computers*.

If the applied voltage is harmonic, $v(t) = \hat{V}e^{j\omega t}$, then the response is also harmonic, $q(t) = \hat{Q}e^{j\omega t}$, and Eq. (119) becomes

$$\left(-\omega^2 L + i\omega R + \frac{1}{C}\right)\hat{Q} = \hat{V} \quad (120)$$

Recall that the current, i , is the time derivative of electric charge, i.e.,

$$\dot{i}(t) = \dot{q}(t) = i\omega i(t) \quad (121)$$

Then, Eq. (120) becomes

$$\left(i\omega L + R + \frac{1}{i\omega C}\right)\hat{I} = \hat{V} \quad (122)$$

where $i(t) = \hat{I}e^{j\omega t}$, $\hat{I} \in \mathbb{C}$

Equation (122) can be written in the usual form

$$Z(\omega)\hat{I} = \hat{V} \quad (123)$$

where $Z(\omega)$ is the *electrical impedance* given by

$$Z(\omega) = i\omega L + R + \frac{1}{i\omega C} \quad (124)$$

Comparison of Eq. (124) and Eq. (94) reveals the correspondence between the electrical impedance and the mechanical impedance.